CP PUBLICATION

SHORT TRICKS
Mathematics
for JEE (Main & Adv)

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- Career Point Editorial

SHORT TRICKS
Mathematics
for JEE (Main & Advanced)

CONTENTS

1

Quadratic Equation

KEY CONCEPTS

1. Polynomial

Algebraic expression containing many terms is called Polynomial.

e.g. : $4x^4 + 3x^3 - 7x^2 + 5x + 3$, $3x^3 + x^2 - 3x + 5$

(i) Real Polynomial : Let $a_0, a_1, a_2 .....a_n$ be real numbers and x is a real variable.
Then $f(x) = a_0 + a_1 x + a_2 x^2 +..... + a_n x^n$ is called real polynomial of real variable x with real coefficients.

(ii) Complex Polynomial: If $a_0, a_1, a_2 ... a_n$ be complex numbers and x is a varying complex number, then $f(x) = a_0 + a_1x + a_2x^2 + ... a_nx^n$ is called a complex polynomial of complex variable x with complex coefficients.
eg.- $3x^2 - (2 + 4i)x + (5i - 4)$,

(iii) Degree of Polynomial : Highest Power of variable x in a polynomial is called as a degree of polynomial.
e.g. $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ... a_{n-1}x^{n-1} + a_nx^n$ is n degree polynomial.

2. Quadratic Expression

A polynomial of degree two of the form $ax^2 + bx + c$ $(a \neq 0)$ is called a quadratic expression in x.

3. Quadratic Equation

A quadratic Polynomial f(x) when equated to zero is called Quadratic Equation.
$ax^2 + bx + c = 0$
Where, $a, b, c \in C$ and $a \neq 0$

4. Roots or Solution of Quadratic Equation

The values of variable x which satisfy the quadratic equation is called as Roots (also called solutions or zeros) of a Quadratic Equation.

(i) Factorization Method :
Let $ax^2 + bx + c = a(x - \alpha)(x - \beta) = 0$
Then $x = \alpha$ and $x = \beta$ will satisfy the given equation.
Hence factorize the equation and equating each to zero gives roots of equation.

(ii) Hindu Method (Sri Dharacharya Method) :
Quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ has two roots, given by

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

5. Nature of Roots

The term $b^2 - 4ac$ is called discriminant of the equation. It is denoted by Δ or D.

(i) Suppose a, b, c $\in$ R and $a \neq 0$ then

(a) If $D > 0 \Rightarrow$ roots are real and unequal

(b) If $D = 0 \Rightarrow$ roots are real and equal and each equal to $-b/2a$

(c) If $D < 0 \Rightarrow$ roots are imaginary and unequal or complex conjugate.

(ii) Suppose a, b, c $\in$ Q, $a \neq 0$ then

(a) If $D > 0$ & D is perfect square $\Rightarrow$ roots are unequal & rational

(b) If $D > 0$ & D is not perfect square $\Rightarrow$ roots are irrational & unequal

6. Conjugate Roots

The Irrational and complex roots of a quadratic equation are always occurs in pairs. Therefore (a, b, c, $\in$ Q)

If One Root	then Other Root
$\alpha + i\beta$	$\alpha - i\beta$
$\alpha + \sqrt{\beta}$	$\alpha - \sqrt{\beta}$

7. Sum and Product of Equation

(i) Quadratic Equation : If the roots of quadratic equation $ax^2 + bx + c$ $(a \neq 0)$ are α and β then

Sum of roots : $S = \alpha + \beta = \frac{-b}{a} = -\frac{\text{Coefficient of x}}{\text{cofficient of } x^2}$

and Product of Roots : $P = \alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{coefficient of } x^2}$

(ii) Cubic Equation : If α, β and γ are the roots of cubic equation $ax^3 + bx^2 + cx + d = 0$.

Then, $\Sigma\alpha = \alpha + \beta + \gamma = -\frac{b}{a}$

$\Sigma\alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

$\alpha\beta\gamma = -\frac{d}{a}$

(iii) Biquadratic Equation :

If α, β, γ and δ are the roots of the biquadratic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, then

$S_1 = \alpha + \beta + \gamma = -\frac{b}{a}$,

$S_2 = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = (-1)^2 \frac{c}{a} = \frac{c}{a}$

or $S_2 = (\alpha + \beta)(\gamma + \delta) + \alpha\beta + \gamma\delta = \frac{c}{a}$

$S_3 = \alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \alpha\beta\delta = (-1)^3 \frac{d}{a} = -\frac{d}{a}$

or $S_3 = \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = -\frac{d}{a}$

and $S_4 = \alpha\beta\gamma\delta = (-1)^4 = \frac{e}{a} = \frac{e}{a}$

8. Relation between Roots and Coefficients

If roots of quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ are α and β then

(i) $(\alpha - \beta) = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \pm \dfrac{\sqrt{b^2 - 4ac}}{a} = \dfrac{\pm\sqrt{D}}{a}$

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \dfrac{b^2 - 2ac}{a^2}$

(iii) $\alpha^2 - \beta^2 = (\alpha + \beta)\sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = -\dfrac{b\sqrt{b^2 - 4ac}}{a^2}$

(iv) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = -\dfrac{b(b^2 - 3ac)}{a^3}$

(v) $\alpha^3 - \beta^3 = (\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta) = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}\ \{(\alpha + \beta)^2 - \alpha\beta\}$

$$= \frac{(b^2 - ac)\sqrt{b^2 - 4ac}}{a^3}$$

(vi) $\alpha^4 + \beta^4 = \{(\alpha + \beta)^2 - 2\alpha\beta\}^2 - 2\alpha^2\beta^2 = \left(\dfrac{b^2 - 2ac}{a^2}\right)^2 - 2\dfrac{c^2}{a^2}$

(vii) $\alpha^4 - \beta^4 = (\alpha^2 - \beta^2)(\alpha^2 + \beta^2) = \dfrac{+b(b^2 - 2ac)\sqrt{b^2 - 4ac}}{a^4}$

(viii) $\alpha^2 + \alpha\beta + \beta^2 = (\alpha + \beta)^2 - \alpha\beta$

(ix) $\dfrac{\alpha}{\beta} + \dfrac{\beta}{\alpha} = \dfrac{\alpha^2 + \beta^2}{\alpha\beta} = \dfrac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$

(x) $\alpha^2\beta + \beta^2\alpha = \alpha\beta(\alpha + \beta)$

(xi) $\left(\dfrac{\alpha}{\beta}\right)^2 + \left(\dfrac{\beta}{\alpha}\right)^2 = \dfrac{\alpha^4 + \beta^4}{\alpha^2\beta^2} = \dfrac{(\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2}{\alpha^2\beta^2}$

(xii) $nb^2 = ac(1 + n)^2$ when one root is n times of another

9. Formation of an Equation with given Roots

(i) **Quadratic Equation :** A quadratic equation whose roots are α and β is given by
$x^2 -$ (sum of Roots)$x +$ Product of Roots $= 0$
$\therefore\ \ x^2 - (\alpha + \beta)x + \alpha\beta = 0$

(ii) **Cubic Equation :** α, β, γ are the roots of cubic equation then the equation is
$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma = 0$

10. Equation in terms of the Roots of another Equation

If are roots of the equation $ax^2 + bx + c = 0$ then the equation whose roots are

(i)	$-\alpha, -\beta \Rightarrow ax^2 - bx + c = 0$	(Replace x by $-x$)
(ii)	$1/\alpha, 1/\beta \Rightarrow\ cx^2 + bx + a = 0$	(Replace x by $1/x$)
(iii)	$\alpha^n, \beta^n;\ n \in N\ a(x^{1/n})^2 + b(x^{1/n}) + c = 0$	(Replace x by $x^{1/n}$)
(iv)	$k\alpha, k\beta \Rightarrow ax^2 + kbx + k^2c = 0$	(Replace x by x/k)
(v)	$k + \alpha, k + \beta \Rightarrow a(x - k)^2 + b(x - k) + c = 0$	(Replace x by $(x - k)$)
(vi)	$\dfrac{\alpha}{k}, \dfrac{\beta}{k}\ \ k^2ax^2 + kbx + c = 0$	(Replace x by kx)
(vii)	$\alpha^{1/n}, \beta^{1/n};\ n \in N \Rightarrow a(x^n)^2 + b(x^n) + c = 0$	(Replace x by x^n)

11. Roots under particular cases

For the quadratic equation $ax^2 + bx + c = 0$

(i) If $b = 0$ $\Rightarrow$ roots are of equal magnitude but of opposite sign

(ii) If $c = 0$ $\Rightarrow$ one root is zero other is $-b/a$

(iii) If $b = c = 0$ $\Rightarrow$ both root are zero

(iv) If $a = c$ $\Rightarrow$ roots are reciprocal to each other

(v) If $\left.\begin{matrix} a > 0 \;\; c < 0 \\ a < 0 \;\; c > 0 \end{matrix}\right\}$ $\Rightarrow$ Roots are of opposite signs

(vi) If $\left.\begin{matrix} a > 0, b > 0, c > 0 \\ a < 0, b < 0, c < 0 \end{matrix}\right\}$ $\Rightarrow$ Both roots are negative.

(vii) $\left.\begin{matrix} a > 0, b < 0, c > 0 \\ a < 0, b > 0, c < 0 \end{matrix}\right\}$ $\Rightarrow$ Both roots are positive.

(viii) If sign of a = sign of b $\neq$ sign of c
$\Rightarrow$ Greater root in magnitude is negative.

(ix) If sign of b = sign of c $\neq$ sign of a $\Rightarrow$ Greater root in magnitude is positive.

(x) If $a + b + c = 0 \Rightarrow$ one root is 1 and second root is c/a.

(xi) If $a = b = c = 0$ then equation will become an identity and will be satisfy by every value of x.

12. Condition for Common Roots

(i) Only One Root is Common : Let α be the common root of quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ then

$\therefore a_1\alpha^2 + b_1\alpha + c_1 = 0$

$a_2\alpha^2 + b_2\alpha + c_2 = 0$

$$\frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{a_2c_1 - a_1c_2} = \frac{1}{a_1b_2 - a_2b_1}$$

$\therefore$ The condition for only one Root common is

$(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$

(ii) Both roots are common : Then required conditions is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Note :

- To find the common roots of two equation make the coefficient of second degree term in the two equation equal and subtract. The value of x obtained is required common root.
- Two different quadratic equation with rational coefficient cannot have single common root which is complex or irrational, as imaginary and surd roots always occur in pair.

13. Graph of Quadratic Expression

An expression of the form $ax^2 + bx + c$, where $a, b, c \in R$ and $a \neq 0$ is called a quadratic expression in x.

We have $y = f(x) = ax^2 + bx + c \quad (a \neq 0)$

$$y = a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right]$$

$$y = a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a^2}\right]$$

$$y + \frac{D}{4a} = a\left(x + \frac{b}{2a}\right)^2$$

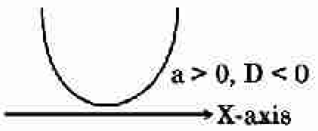
Let $y + \frac{D}{4a} = Y$ and $x + \frac{b}{2a} = X$

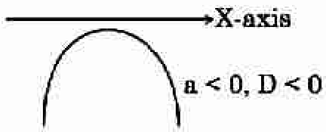
$$X^2 = \frac{Y}{a}$$

(i) The graph of the curve $y = f(x)$ is parabolic.

(ii) The axis of the parabola is $X = 0$ or $x + \frac{b}{2a} = 0$

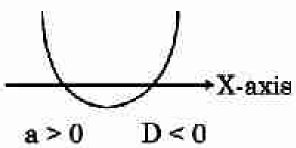
(iii) If $a > 0$, then the parabola opens upward.
If $a < 0$, then the parabola opens downward.

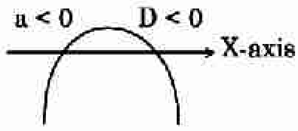




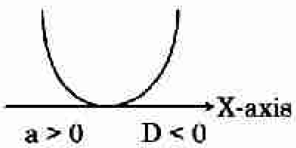
14. Position of Quadratic Equation with respect to axes

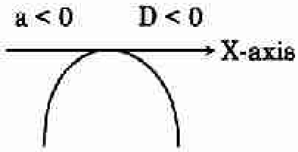
(i) For $D > 0$, parabola cuts X-axis and has two real and distinct points i.e. $x = \frac{-b \pm \sqrt{D}}{2a}$.



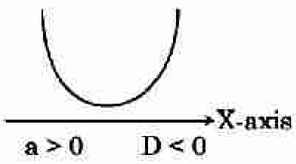


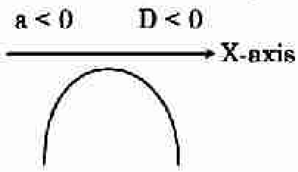
(ii) For $D = 0$ parabola touch X-axis in one point, $x = -\frac{b}{2a}$.





(iii) For $D < 0$, parabola does not cut X-axis (i.e., imaginary value of x).





Note :

D < 0 then,

- If $a > 0$, then $ax^2 + bx + c > 0$ for all x.
- If $a < 0$, then $ax^2 + bx + c < 0$ for all x.

15. Maximum & Minimum value of Quadratic Expression

In a quadratic expression $ax^2 + bx + c$

(i) If $a > 0$, quadratic expression has least value at $x = -\frac{b}{2a}$. This least value is given by $\frac{4ac - b^2}{4a} = -\frac{D}{4a}$

(ii) If $a < 0$, quadratic expression has greatest value at $x = -\frac{b}{2a}$. This greatest value is given by

$$\frac{4ac - b^2}{4a} = -\frac{D}{4a}$$

16. Range of an Rational Algebric Expression

To find the value of rational expression of the form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$ for real values of x.

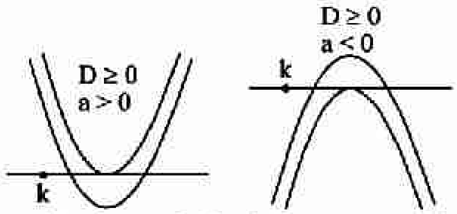
(i) Equate the given rational expression to y.
(ii) Obtain a quadratic equation in x by simplify the expression.
(iii) Obtain the discriminant of the quadratic equation.
(iv) Put discriminant ≥ 0 and solve the inequation for y.
The value of y, so obtained determines the set of values attained by the given rational expression.

17. Location of Roots of a Quadratic Equation $ax^2+bx+c=0$

Let $f(x) = ax^2 + bx + c$, where $a, b, c \in R$ and $a \neq 0$

(i) Condition for both the roots will be greater than k.

(a) $D \geq 0$ (b) $k < -\frac{b}{2a}$ (c) $af(k) > 0$

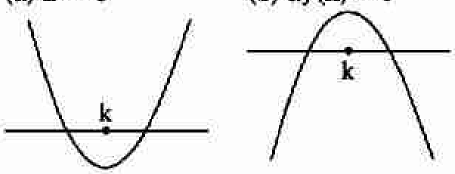


(ii) Condition for both the roots will be less than k.

(a) $D \geq 0$ (b) $k > -\frac{b}{2a}$ (c) $af(k) > 0$

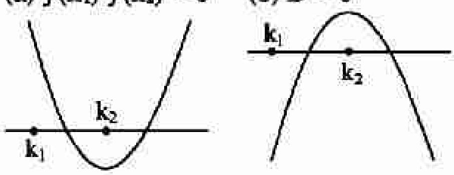
(iii) Condition for k lie between the roots

(a) $D > 0$ (b) $af(k) < 0$



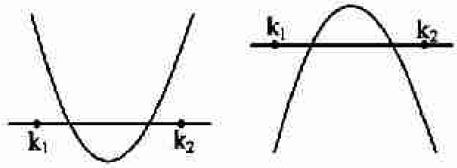
(iv) Condition for exactly one root lie in the interval (k_1, k_2) where $k_1 < k_2$

(a) $f(k_1)\,f(k_2) < 0$ (b) $D > 0$



(v) When both roots lie in the interval (k_1, k_2) where $k_1 < k_2$

(a) $D > 0$ (b) $f(k_1).\ f(k_2) > 0$ (c) $k_1 < -\dfrac{b}{2a} < k_2$



(vi) Any algebraic expression $f(x) = 0$ in interval $[a, b]$ if

(a) sign of $f(a)$ and $f(b)$ are of same then either no roots or even no. of roots exist.

(b) sign of $f(a)$ and $f(b)$ are opposite then $f(x) = 0$ has at least one real root or odd no. of roots.

18. Descartes Rule of Signs

(i) The maximum number of positive real roots of polynomial equation $f(x) = 0$ (arranged in decreasing order of the degree) is the number of changes of signs in $f(x) = 0$ as we move from left to right.

(ii) The maximum number of negative real roots of a polynomial equation $f(x) = 0$ is the number of changes of signs in $f(-x)$.

19. Quadratic Expression in Two Variables

The general form of a quadratic expression in two variables x & y is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$. The condition that this expression may be resolved into two linear rational factors is

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ and } h^2 - ab > 0$$

This expression is called discriminant of the above quadratic expression.

Important Points to be Remembered

(1) Every equation of n^{th} degree $(n \geq 1)$ has exactly n roots and if the equation has more than n roots, it is an identity.

(2) If quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and

$a_2x^2 + b_2x + c_2 = 0$ are in the same ratio $\left(\text{i.e.} \dfrac{\alpha_1}{\beta_1} = \dfrac{\alpha_2}{\beta_2}\right)$ then $\dfrac{b_1^2}{b_2^2} = \dfrac{a_1c_1}{a_2c_2}$

(3) If one root is k times the other root of quadratic equation

$a_1x^2 + b_1x + c_1 = 0$ then

$$\frac{(k+1)^2}{k} = \frac{b^2}{ac}$$

SHORT TRICKS

Trick -1 **If the roots of equation $Ax^2 + Bx + C = 0$ are real and equal and $A + B + C = 0$, then $A = C$**

Q.1 If roots of the equation $(a - b)x^2 + (c - a)x + (b - c) = 0$ are equal, then a, b, c are in -

(A) A.P. (B) H.P. (C) G.P. (D) None of these

Sol. **[A]**

Proper Method	**Short Trick**
Given $(a - b)x^2 + (c - a)x + (b - c) = 0$ Roots are equal $\therefore D = 0$ $(c - a)^2 - 4(a - b)(b - c) = 0$ $\Rightarrow c^2 + a^2 - 2ac - 4(ab - ac - b^2 + bc) = 0$ $\Rightarrow c^2 + a^2 + 4b^2 + 2ac - 4ab - 4bc = 0$ $\Rightarrow (c + a - 2b)^2 = 0$ $\Rightarrow c + a - 2b = 0$ $\Rightarrow 2b = a + c$ $\therefore$ a, b, c are in A.P.	$a - b + c - a + b - c = 0$ $\therefore a - b = b - c$ $\Rightarrow 2b = a + c$ So a, b, c, are in A.P.

Q.2 If the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are equal, then a, b, c are in -

(A) HP (B) GP (C) AP (D) None of these

Sol. **[A]**

Proper Method	**Short Trick**
Given : $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ Roots are equal $\therefore D = 0$ $\Rightarrow b^2(c - a)^2 - 4a(b - c)c(a - b) = 0$ $\Rightarrow b^2(c^2 + a^2 - 2ac) - 4ac(ba - b^2 - ca + bc) = 0$ $\Rightarrow b^2c^2 + b^2a^2 - b^2 2ac - 4a^2bc + 4acb^2 + 4a^2c^2 - 4abc^2 = 0$ $\Rightarrow b^2c^2 + b^2a^2 + 4a^2c^2 + 2acb^2 - 4a^2bc - 4abc^2 = 0$ $\Rightarrow (bc + ba - 2ac)^2 = 0$ $\Rightarrow bc + ba - 2ac = 0$ $\Rightarrow b = \frac{2ac}{a + c}$ $\therefore$ a, b, c are in H.P.	$a(b - c) + b(c - a) + c(a - b) = 0$ $\therefore a(b - c) = c(a - b)$ $\Rightarrow b = \frac{2ac}{a + c}$ So, a, b, c are in H.P.

Trick -2 **Method of substitution**

Q.3 If a, b, c are distinct positive real numbers such that $b(a + c) = 2ac$, then the roots of $ax^2 + 2bx + c = 0$ are:

(A) Real and equal (B) Real and distinct (C) Imaginary (D) None of these

Sol. **[C]**

Proper Method	Short Trick
$D = 4b^2 - 4ac = 4\left\{\frac{4a^2c^2}{(a+c)^2} - ac\right\}$ $= \frac{16ac}{(a+c)^2}\{4ac - (a+c)^2\}$ $= -\frac{16ac}{(a+c)^2}(a-c)^2 < 0\ [\because a, c > 0]$ $\Rightarrow$ roots are imaginary	$b = \frac{2ac}{a+c} \Rightarrow a, b, c$ are in H.P. let $a = 2, b = 3, c = 6$ now equation $2x^2 + 6x + 6 = 0$ $D = 36 - 48 < 0$ (imaginary roots)

Q.4 If a, b, c, $\in$ R and 1 is a root of the equation $ax^2 + bx + c = 0$, then the equation $4ax^2 + 3bx + 2c = 0$, $c \neq 0$ has roots which are :

(A) Real and equal (B) Real and distinct (C) Imaginary (D) Rational

Sol. **[B]**

Proper Method	Short Trick
1 is a root of $ax^2 + bx + c = 0 \Rightarrow a + b + c = 0$ D of $4ax^2 + 3bx + 2c = 0$ is $= 9b^2 - 32ac = 9(a+c)^2 - 32ac$ $= c^2\left\{9\left(\frac{a}{c}\right)^2 - 14\left(\frac{a}{c}\right) + 9\right\}$ $= c^2\left\{\left(3\left(\frac{a}{c}\right) - \left(\frac{7}{3}\right)\right)^2 + 9 - \frac{49}{9}\right\} > 0$ $c^2\left\{3\left(\frac{a}{c} - \frac{7}{3}\right)^2 + \frac{31}{9}\right\} > 0$ $\Rightarrow$ roots are real and distinct.	Let the roots are 1 and 2 then equation $ax^2 + bx + c = 0$ becomes $x^2 - 3x + 2 = 0$ $(a = 1, b = -3, c = 2)$ Now equation $4ax^2 + 3bx + 2c = 0$ $\Rightarrow 4x^2 - 9x + 4 = 0$ $D = 81 - 64 > 0$ (real and distinct)

Q.5 If α and β are the roots of the equation $x^2 - p(x+1) - q = 0$, then the value of $\frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+q} + \frac{\beta^2+2\beta+1}{\beta^2+2\beta+q}$ is :

(A) 2 (B) 1 (C) 0 (D) None

Sol. **[B]**

Proper Method	Short Trick
Equation is $x^2 - px - (p+q) = 0$ $\alpha + \beta = p, \alpha\beta = -(p+q)$ Now $(\alpha+1)(\beta+1) = \alpha\beta + (\alpha+\beta) + 1$ $= -(p+q) + p + 1 = 1 - q$ The given expression $= \frac{(\alpha+1)^2}{(\alpha+1)^2+(q-1)} + \frac{(\beta+1)^2}{(\beta+1)^2+(q-1)}$ $= \frac{[2(\alpha+1)^2(\beta+1)^2 + (q-1)\{(\alpha+1)^2+(\beta+1)^2\}]}{[(\alpha+1)^2(\beta+1)^2 + (q-1)\{(\alpha+1)^2+(\beta+1)^2\} + (q-1)^2]}$ $= \frac{2(1-q)^2 + (q-1)[(\alpha+1)^2+(\beta+1)^2]}{2(1-q)^2 + (q-1)[(\alpha+1)^2+(\beta+1)^2]} = 1$	Let $\alpha = 1$ and $\beta = 2$, then equation is $x^2 - 3x + 2 = 0$ so $p = 3$ and $-p - q = 2$ $\Rightarrow q = -5$ $\frac{\alpha^2+2\alpha+1}{\alpha^2+2\alpha+q} + \frac{\beta^2+2\beta+1}{\beta^2+2\beta+q}$ $= \frac{4}{-2} + \frac{9}{3} = 1$

Q.6 If tan A and tan B are the roots of $x^2 + ax + b = 0$, then the value of expression $\sin^2(A + B) + a \sin(A + B) \cos(A + B) + b \cos^2(A + B)$ is equal to -

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) a (D) b

Sol. [D]

Proper Method

$\tan A + \tan B = -a$ and $\tan A . \tan B = b$

$\Rightarrow \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A . \tan B} = \frac{-a}{1-b} = \frac{a}{b-1}$

$\Rightarrow \sin(A + B) = \frac{a}{\sqrt{a^2 + (b-1)^2}}$

and $\cos(A + B) = \frac{b-1}{\sqrt{a^2 + (b-1)^2}}$

$\sin^2(A + B) + a \sin(A + B) \cos(A + B) + b \cos^2(A + B)$

$= \frac{a^2}{a^2 + (b-1)^2} + \frac{a^2(b-1)}{a^2 + (b-1)^2} + \frac{b(b-1)^2}{a^2 + (b-1)^2}$

$= \frac{a^2 + a^2 b - a^2 + b^3 - 2b^2 + b}{a^2 + (b-1)^2}$

$= \frac{b(a^2 + b^2 - 2b + 1)}{(a^2 + b^2 - 2b + 1)}$

$= b$

Short Trick

Let $A = 30°$ and $B = 60°$, then roots are $\frac{1}{\sqrt{3}}, \sqrt{3}$

and equation is $x^2 - \frac{4}{\sqrt{3}}x + 1 = 0$

so, $a = -\frac{4}{\sqrt{3}}$ and $b = 1$

now $\sin^2(A + B) + a \sin(A + B) \cos(A + B) + b \cos^2(A + B) = 1 + 0 + 0 = 1 = b$

Trick -3 Combination of method of substitution and balancing

Q.7 If α, β are the roots of the equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$, then $a S_{n+1} + c S_{n-1} =$

(A) bS_n (B) b^2S_n (C) $2bS_n$ (D) $-bS_n$

Sol. [D]

Proper Method

Here α, β are roots

$\therefore \quad a\alpha^2 + b\alpha + c = 0 \quad$ ...(1)

$a\beta^2 + b\beta + c = 0 \quad$ ...(2)

Now let us consider (Keeping results (1), (2) in mind)

$a S_{n+1} + b S_n + c S_{n-1}$

$= a[\alpha^{n+1} + \beta^{n+1}] + b[\alpha^n + \beta^n] + c[\alpha^{n-1} + \beta^{n-1}]$

$= [a\alpha^{n+1} + b\alpha^n + c\alpha^{n-1}] + [a\beta^{n+1} + b\beta^n + c\beta^{n-1}]$

$= \alpha^{n-1}[a\alpha^2 + b\alpha + c] + \beta^{n-1}[a\beta^2 + b\beta + c]$

$= 0 + 0 = 0$

Hence $aS_{n+1} + cS_{n-1} = -bS_n$.

Short Trick

Let $\alpha = 1, \beta = 2$, then the equation

$x^2 - 3x + 2 = 0$

so, $a = 1, b = -3, c = 2$

and let

$n = 2 \Rightarrow S_2 = \alpha^2 + \beta^2 = 5$

now $aS_{n+1} + cS_{n-1}$

$= S_3 + 2S_1 = 9 + 6 = 15$

by option (D)

$-bS_n = (+3)(5) = 15$ is correct

Q.8 If $\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of the equation, $ax^2 + bx + 1 = 0$ $(a \neq 0,\ a,\ b \in R)$, then the equation $x(x + b^3) + (a^3 - 3abx) = 0$ has roots :

(A) $\alpha^{\frac{3}{2}}$ and $\beta^{\frac{3}{2}}$ (B) $\alpha\beta^{\frac{1}{2}}$ and $\alpha^{\frac{1}{2}}\beta$ (C) $\sqrt{\alpha\beta}$ and $\alpha\beta$ (D) $\alpha^{\frac{-3}{2}}$ and $\beta^{\frac{-3}{2}}$

[JEE Main Online-2014]

Sol. [A]

Proper Method

$\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of equation $ax^2 + bx + 1 = 0$

$\therefore \frac{1}{\sqrt{\alpha}} + \frac{1}{\sqrt{\beta}} = \frac{-b}{a}$

$\Rightarrow \frac{\sqrt{\alpha}+\sqrt{\beta}}{\sqrt{\alpha\beta}} = \frac{-b}{a}$ ....(1)

Product $\frac{1}{\sqrt{\alpha}} \cdot \frac{1}{\sqrt{\beta}} = \frac{1}{a}$

$\Rightarrow \frac{1}{\sqrt{\alpha\beta}} = \frac{1}{a}$ .... (2)

From (1) $\sqrt{\alpha} + \sqrt{\beta} = -b$ .... (3)

Now the equation

$x(x + b^3) + (a^3 - 3abx) = 0$

$\Rightarrow x^2 + x(b^3 - 3ab) + a^3 = 0$

$\Rightarrow x^2 - x(3ab - b^3) + a^3 = 0$

$\Rightarrow x^2 - x\{-3\sqrt{\alpha\beta}\ (\sqrt{\alpha}+\sqrt{\beta}) + (\sqrt{\alpha}+\sqrt{\beta})^3\} + (\alpha\beta)^{3/2} = 0$

$\Rightarrow x^2 - x\,(\alpha^{3/2} + \beta^{3/2}) + (\alpha\beta)^{3/2} = 0$

$\therefore$ roots are $\alpha^{3/2}$ and $\beta^{3/2}$

Short Trick

Let $\alpha = 4$ and $\beta = 9$ then the equation is

$6x^2 - 5x + 1 = 0$ and

$a = 6,\ b = -5,\ c = 1$

$x(x + b^3) + (a^3 - 3abx) = 0$ becomes

$x^2 - 35x = 216 = 0$

it roots are 8 and 27 satisfy the $\alpha^{3/2}, \beta^{3/2}$

Q.9 If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then the cubic equation whose roots are $\alpha(\beta + \gamma),\ \beta(\gamma + \alpha),\ \gamma(\alpha + \beta)$ is :

(A) $x^3 - 2qx^2 + (q^2 + pr)x + r^2 - pqr = 0$ (B) $x^3 - 2px^2 + (p^2 + qr)x + r^2 - pqr = 0$

(C) $x^3 - 2rx^2 + (r^2 + pq)x + r^2 - pqr = 0$ (D) None of these

Sol. [A]

Proper Method

α, β, γ are roots of $x^3 + px^2 + qx + r = 0$

$\therefore \alpha + \beta + \gamma = -p,\ \alpha\beta + \beta\gamma + \gamma\alpha = q,\ \alpha\beta\gamma = r$

$\alpha(\beta + \gamma),\ \beta(\gamma + \alpha),\ \gamma(\alpha + \beta)$ are symmetric.

Let $y = \alpha(\beta + \gamma) \Rightarrow y = \alpha\beta + \alpha\gamma + \beta\gamma - \beta\gamma$

$\Rightarrow y = q + r/\alpha \Rightarrow \alpha = \frac{r}{y-q}$

α is a solution of given equation

$\therefore \left(\frac{r}{y-q}\right)^3 + p\frac{r^2}{(y-q)^2} + q \cdot \frac{r}{y-q} + r = 0$

Short Trick

Let $\alpha = 1,\ \beta = 2,\ \gamma = 3$ are the roots then

$p = -6,\ q = 11,\ r = -6$

Now the equation whose roots

$\alpha(\beta + \gamma) = 5,\ \beta(\gamma + \alpha) = 8,\ \gamma(\alpha + \beta) = 9$

$x^3 - 22x^2 + 167x - 360 = 0$

now option (A) satisfy the condition.

$\Rightarrow \frac{r}{y-q}\left\{\frac{r^2}{(y-q)^2}+\frac{pr}{y-q}+y\right\}=0$ $\Rightarrow r^2 + pr(y-q) + y(y-q)^2 = 0$ $\Rightarrow y^3 - 2qy^2 + (q^2+pr)y + r^2 - pqr = 0$ Hence, required equation is $x^3 - 2qx^2 + (q^2+pr)x + r^2 - pqr = 0$	

Q.10 If α, β, γ are the roots of the equation, $x^3 + ax^2 + bx + c = 0$, then $(1-\alpha^2)(1-\beta^2)(1-\gamma^2)$ is equal to:
(A) $(1+b)^2 - (a+c)^2$ (B) $(1+b)^2 + (a+c)^2$ (C) $(1-b)^2 + (a+c)^2$ (D) None of these

Sol. **[A]**

Proper Method	**Short Trick**
α, β, γ are roots of $x^3 + ax^2 + bx + c = 0$ Putting $1 - x^2 = y$, $x^2 = 1 - y$, we have $x(1-y) + a(1-y) + bx + c = 0$ $\Rightarrow x[1-y+b] = -[c+a-ay]$ $\Rightarrow x^2[1+b-y]^2 = [c+a-ay]^2$ $\Rightarrow (1-y)[1+b-y]^2 - [c+a-ay]^2 = 0.$ This equation in y has roots $1-\alpha^2$, $1-\beta^2$ and $1-\gamma^2$ $\therefore (1-\alpha^2)(1-\beta^2)(1-\gamma^2) = (1+b)^2 - (a+c)^2$	Let $\alpha = 1, \beta = 2, \gamma = 3$ then $a = -6, b = 11, c = -6$ now $(1-\alpha^2)(1-\beta^2)(1-\gamma^2) = 0$ in options (A) $(1+b)^2 - (a+c)^2$ is correct option

Q.11 Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$. If α and β are non-zero complex number satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$ then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is : -
(A) $(p^3+q)x^2 - (p^3+2q)x + (p^3+q) = 0$ (B) $(p^3+q)x^2 - (p^3-2q)x + (p^3+q) = 0$
(C) $(p^3-q)x^2 - (5p^3-2q)x + (p^3-q) = 0$ (D) $(p^3-q)x^2 - (p^3+2q)x + (p^3-q) = 0$

Sol. **[B]**

Proper Method	**Short Trick**
$\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$ then $(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = q$ $\Rightarrow -p^3 - 3\alpha\beta(-p) = q$ $\Rightarrow \alpha\beta = \frac{p^3+q}{3p}$ Now equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is Sum $\Rightarrow \frac{\alpha}{\beta}+\frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$ $= \frac{p^2}{\left(\frac{p^3+q}{3p}\right)} - 2 = \frac{p^3-2q}{p^3+q}$ Product $= \frac{\alpha}{\beta}.\frac{\beta}{\alpha} = 1$ So equation $x^2 - x\left(\frac{p^3-2q}{p^3+q}\right) + 1 = 0$ $(p^3+q)x^2 - (p^3-2q)x + (p^3+q) = 0$	Let $\alpha = \omega$ and $\beta = \omega^2$ $\therefore \alpha + \beta = \omega + \omega^2 = -1 = -p \Rightarrow p = 1$ and $\alpha^3 + \beta^3 = \omega^3 + (\omega^2)^3 = 2 = q$ so required equation is $x^2\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right)x + 1 = 0$ $\Rightarrow x^2 + x + 1 = 0$ Put the value of p & q in option then (B) will give required result.

Q.12 If one root of equation $x^2 + px + q = 0$ is square of the other then : -

(A) $p^3 - q(3p - 1) + q^2 = 0$ (B) $p^3 - q(3p + 1) + q^2 = 0$

(C) $p^3 + q(3p - 1) + q^2 = 0$ (D) $p^3 + q(3p + 1) + q^2 = 0$

Sol. [A]

Proper Method	**Short Trick**
Let the roots of equation $x^2 + px + q = 0$ are α and α^2 then $\alpha + \alpha^2 = -p$ and $\alpha.\alpha^2 = q \Rightarrow \alpha^3 = q$ Now $(\alpha + \alpha^2)^3 = -p^3$ $\Rightarrow \alpha^3 + \alpha^6 + 3\alpha.\alpha^2(\alpha + \alpha^2) = -p^3$ $\Rightarrow q + q^2 + 3q(-p) = -p^3$ $\Rightarrow p^3 - q(3p - 1) + q^2 = 0$	Let $\alpha = 1, \beta = 1$ then the equation is $x^2 - 2x + 1 = 0$ $p = -2, q = 1$ put these value in options then (A) is correct option

Q.13 If α and β are the roots of equation $ax^2 + bx + c = 0$, then the sum of the roots of the equation $a^2x^2 + (b^2 - 2ac)x + b^2 - 4ac = 0$ is given by -

(A) $-(\alpha^2 - \beta^2)$ (B) $(\alpha + \beta)^2 - 2\alpha\beta$ (C) $\alpha^2\beta + \beta^2\alpha - 4\alpha\beta$ (D) $-(\alpha^2 + \beta^2)$

Sol. [D]

Proper Method	**Short Trick**
Let the roots of equation $ax^2 + bx + c = 0$ are α and β $\therefore \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$ and let the roots of equation $a^2x^2 + (b^2 - 2ac)x + b^2 - 4ac = 0$ are γ and δ then $\gamma + \delta = -\left(\frac{b^2 - 2ac}{a^2}\right), \gamma\delta = \frac{b^2 - 4ac}{a^2}$ $\gamma + \delta = \frac{b^2}{a^2} - \frac{2c}{a}$ $\gamma + \delta = -[(\alpha + \beta)^2 - 2\alpha\beta] = -(\alpha^2 + \beta^2)$	Let $\alpha = 1, \beta = 2$, then equation $x^2 - 3x + 2 = 0$ $\Rightarrow a = 1, b = -3, c = 2$ Now the required equation will reduce to $x^2 + 5x + 1 = 0$ $\Rightarrow$ sum of roots $= -5$ Put $\alpha = 1, \beta = 2$ in options then (D) is correct answer

Q.14 If (α, β) are roots of $ax^2 + 2bx - a = 0$ and quadratic equation whose roots are $\left(2\alpha - \frac{1}{\beta}\right)$ and $\left(2\beta - \frac{1}{\alpha}\right)$ is $px^2 + qx + r = 0$, then $p + q + r =$

(A) $2b$ (B) $6a - 8b$ (C) $6b - 8a$ (D) 0

Sol. [C]

Proper Method	**Short Trick**
$\alpha + \beta = -\frac{2b}{a}, \alpha\beta = -1$ Now sum and product of roots $\left(2\alpha - \frac{1}{\beta}\right)$ and $\left(2\beta - \frac{1}{\alpha}\right)$ Sum $\Rightarrow 2\alpha - \frac{1}{\beta} + 2\beta - \frac{1}{\alpha} = 2(\alpha + \beta) - \left(\frac{1}{\beta} + \frac{1}{\alpha}\right)$	Let $\alpha = 1, \beta = -1$, then equation is $x^2 - 1 = 0$ Compare with $ax^2 + 2bx - a = 0$, we get $a = 1$, $b = 0$ Roots of second equation are $(3, -3)$ then the equation $x^2 - 9 = 0$ $\therefore p + q + r = -8$

$= 2\left(\frac{-2b}{a}\right) - \left(\frac{\alpha+\beta}{\alpha\beta}\right)$ $= \frac{-4b}{a} - \left(\frac{+2b}{a}\right) = \frac{-6b}{a}$ Product $\Rightarrow \left(2\alpha - \frac{1}{\beta}\right)\left(2\beta - \frac{1}{\alpha}\right) = 4\alpha\beta - 4 + \frac{1}{\alpha\beta}$ $= -4 - 4 - 1 = -9$ $\therefore$ equation is $x^2 + \frac{6b}{a}x - 9 = 0 \Rightarrow ax^2 + 6bx - 9a = 0$ Compare with $px^2 + qx + r = 0$ We get $p = a$, $q = 6b$, $r = -9a$ Now $p + q + r = a + 6b - 9a = 6b - 8a$	

Q.15 If (α, β) are the roots of equation $x^2 - px + q = 0$ and (α', β') are that of $x^2 - p'x + q' = 0$, then $(\alpha - \alpha')^2 + (\beta - \alpha')^2 + (\alpha - \beta')^2 + (\beta - \beta')^2 =$

(A) $2(p^2 - 2q + p'^2 - 2q' - pp')$
(B) $2(p^2 - 2q + p'^2 - 2q' - qq')$
(C) $2(p^2 - 2q - p'^2 - 2q' - pp')$
(D) $2(p^2 - 2q - p'^2 - 2q' - qq')$

Sol. [A]

Proper Method	**Short Trick**
$\alpha + \beta = p$, $\alpha\beta = q$ and $\alpha' + \beta' = p'$, $\alpha'\beta' = q'$ Now $(\alpha - \alpha')^2 + (\beta - \alpha')^2 + (\alpha - \beta')^2 + (\beta - \beta')^2$ $= \alpha^2 + \alpha'^2 - 2\alpha\alpha' + \beta^2 + \alpha'^2 - 2\beta\alpha' + \alpha^2 + \beta'^2 - 2\alpha\beta' + \beta^2 + \beta'^2 - 2\beta\beta'$ $= 2(\alpha^2 + \beta^2) + 2(\alpha'^2 + \beta'^2) - 2\alpha'(\alpha + \beta) - 2\beta'(\alpha + \beta)$ $= 2(p^2 - 2q) + 2(p'^2 - 2q') - 2\alpha'(p) - 2\beta'(p)$ $= 2(p^2 - 2q + p'^2 - 2q' - pp')$	Let $\alpha = 1$, $\beta = 2$, then equation $x^2 - 3x + 2 = 0$ and $\alpha' = -1$, $\beta' = -2$, then equation $x^2 + 3x + 2 = 0$ so, $p = 3$, $q = 2$, $p' = -3$, $q' = -2$ now $(\alpha - \alpha')^2 + (\beta - \alpha')^2 + (\alpha - \beta')^2 + (\beta - \beta')^2$ $= 4 + 9 + 9 + 16 = 38$ Putting these values in options then (A) will correct answer

Trick -5 Method of substitution

Q.9 If S_n denotes the sum of n terms of an A.P., then $S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n$ is equal to -

(A) 0 (B) 1 (C) 1/2 (D) 2

Sol. [A]

Proper Method	**Short Trick**
In AP : S_n $\boxed{T_1, T_2, T_3........, T_n}$ $T_{n+1}, T_{n+2}, T_{n+3}$ Clearly $S_{n+3} = S_n + T_{n+1} + T_{n+2} + T_{n+3}$ $S_{n+2} = S_n + T_{n+1} + T_{n+2}$ $S_{n+1} = S_n + T_{n+1}$ Putting in $S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n$ We get $= T_{n+1} - 2T_{n+2} + T_{n+3}$ $= (T_{n+1} + T_{n+3}) - 2(T_{n+2}) = 0$	Let the A.P. is 1, 2, 3, 4, 5, 6, ... $S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n$ Put n = 1 $\Rightarrow S_4 - 3S_3 + 3S_2 - S_1$ $= 10 - 3(6) + 3(3) - 1 = 0$

Q.10 Let T_r be the r^{th} term of an A.P. whose first term is a and common difference is d. If for some positive integers m, n, $m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then a – d equals – **[AIEEE-2004]**

(A) 0 (B) 1 (C) 1/mn (D) $\frac{1}{m} + \frac{1}{n}$

Sol. [A]

Proper Method	**Short Trick**
$T_m = \frac{1}{n} \Rightarrow a + (m-1)d = \frac{1}{n}$ ......(i) & $T_n = \frac{1}{m} \Rightarrow a + (n-1)d = \frac{1}{m}$ ......(ii) (i) – (ii) $(m-n)d = \frac{1}{n} - \frac{1}{m} = \frac{m-n}{mn}$ $\therefore \boxed{d = \frac{1}{mn}}$ From (i) $a + \frac{(m-1)}{mn} = \frac{1}{n}$ $\Rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{n}$ $\therefore \boxed{a = \frac{1}{mn}}$ $\therefore a - d = 0$	Let m = 1, n = 2, then $T_1 = \frac{1}{2}$ and $T_2 = 1$ $\Rightarrow d = \frac{1}{2}$ $\Rightarrow a - d = 0$

Q.11 If a^2, b^2, c^2 are in A.P. then $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+d}$ are in-

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Proper Method	Short Trick
Let numbers are a & b $\therefore x = \dfrac{a+b}{2}$ & $\because$ a, y, z, b are in GP $\therefore y^2 = az$ & $z^2 = by$ Now, $\dfrac{y^3+z^3}{xyz} = \dfrac{1}{x}\left(\dfrac{y^2}{z}+\dfrac{z^2}{y}\right) = \dfrac{1}{x}(a+b) = 2$	Let the G.P is 1, 2, 4, 8 then y = 2, z = 4 and $x = \dfrac{9}{2}$ Now $\dfrac{y^3+z^3}{xyz}$ $= \dfrac{8+64}{(2)\left(\frac{9}{2}\right)(4)} = 2$

Q.20 Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are- **[IIT Sc.-1997]**

(A) Not in A.P./G.P./H.P. (B) in A.P. (C) in G.P. (D) in H.P.

Sol. [D]

Proper Method	Short Trick
a, b, c, d : AP Dividing by abcd, we get $\dfrac{1}{bcd}, \dfrac{1}{acd}, \dfrac{1}{abd}, \dfrac{1}{abc}$: AP $\therefore$ bcd, acd, abd, abc : HP	Let the A.P. is 1, 2, 3, 4, Now abc = 6, abd = 8, acd = 12, bcd = 24 are in H.P.

Trick -6 Combination of method of substitution and balancing

Q.21 Let the sequence $a_1, a_2, a_3, \ldots\ldots, a_n$ form an A.P., then $a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots\ldots + a_{2n-1}^2 - a_{2n}^2$ is equal to -

(A) $\dfrac{n}{2n-1}(a_1^2 - a_{2n}^2)$ (B) $\dfrac{2n}{n-1}(a_{2n}^2 - a_1^2)$ (C) $\dfrac{n}{n+1}(a_1^2 + a_{2n}^2)$ (D) None of these

Sol. [A]

Proper Method	Short Trick
Given $a_1, a_2, \ldots\ldots, a_{2n}$: AP $\therefore a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots.. + a_{2n-1}^2 - a_{2n}^2$ $= (a_1 - a_2)(a_1 + a_2) + (a_3 - a_4)(a_3 + a_4) + \ldots\ldots$ $+ (a_{2n-1} + a_{2n})(a_{2n-1} - a_{2n})$ But $a_1 - a_2 = a_3 - a_4 \ldots\ldots = a_{2n-1} - a_{2n} = -d$ $= -d[a_1 + a_2 + a_3 + a_4 \ldots\ldots\ldots\ldots + a_{2n}]$ $= -d \cdot \dfrac{2n}{2}[a_1 + a_{2n}]$ $= -nd(a_1 + a_{2n})$ But $a_{2n} = a_1 + (2n-1)d \Rightarrow d = \dfrac{a_{2n} - a_1}{2n-1}$ $= \dfrac{-n(a_{2n} - a_1)}{2n-1} \cdot (a_{2n} + a_1)$ $= \dfrac{n}{2n-1}(a_1^2 - a_{2n-1}^2)$	Let n = 2 $a_1 = 1, a_2 = 2, a_3 = 3, a_4 = 4$ $a_1^2 - a_2^2 + a_3^2 - a_4^2$ $= 1 - 4 + 9 - 16 = -10$ by option (A) $\dfrac{n}{2n-1}(a_1^2 - a_{2n}^2)$ $= \dfrac{2}{3}(1 - 16) = -10$

Q.24 Total number of four digit odd numbers that can be formed using 0, 1, 2, 3, 5, 7 are (repetition allowed) **[AIEEE-2002]**

(A) 216 (B) 375 (C) 400 (D) 720

Sol. **[D]**

Proper Method

0, 1, 2, 3, 5, 7 : Six digits

The last place can be filled in by 1, 3, 5, 7. *i.e.*, 4 ways as the number is to be odd. We have to fill in the remaining 3 places of the 4 digit number *i.e.* I, II, III place. Since repetition is allowed each place can be filled in 6 ways. Hence the 3 place can be filled in $6 \times 6 \times 6 = 216$ ways.

But in case of $0 = 216 - 36 = 180$ ways.

Hence by fundamental theorem, the total number will be $= 180 \times 4 = 720$

Short Trick

$$\boxed{}\ \boxed{}\ \boxed{}\ \boxed{1, 3, 5, 7}$$

$$5 \times 6 \times 6 \times 4 = 720$$

Q.25 How many numbers lying between 10 and 1000 can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 (repetition is allowed)

(A) 1024 (B) 810 (C) 2346 (D) 2549

Sol. **[B]**

Proper Method

The total number between 10 and 1000 are 989 but we have to form the numbers by using numerals 1, 2,........9, i.e. 0 is not occuring so the numbers containing any '0' would be excluded *i.e.*,

Required number of ways

$$= 989 - \begin{Bmatrix} 20, 30, 40, \ldots\ldots 100 = 9 \\ 101, 102, \ldots\ldots 200 = 19 \\ 201, \ldots\ldots 300 = 19 \\ \ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots \\ 901, \ldots\ldots 990 = 18 \end{Bmatrix}$$

$$= 989 - (9 + 18 + 19 \times 8) = 810.$$

Aliter : Between 10 and 1000, the numbers are of 2 digits and 3 digits.

Since repetition is allowed, so each digit can be filled in 9 ways.

Therefore number of 2 digit numbers $= 9 \times 9 = 81$

and number of 3 digit numbers $9 \times 9 \times 9 = 729$

Hence total ways $= 81 + 729 = 810$.

Short Trick

Two digit number

$$\boxed{}\ \boxed{}$$

$$9 \times 9 = 81$$

Three digit number

$$\boxed{}\ \boxed{}\ \boxed{}$$

$$9 \times 9 \times 9 = 729$$

Total = 810

2

Progression

KEY CONCEPTS

1. Definition

When the terms of a sequence or series are arranged under a definite rule then they are said to be in a Progression. Progression can be classified into 5 parts as-

(i) Arithmetic Progression (A.P.)
(ii) Geometric Progression (G.P.)
(iii) Arithmetic Geometric Progression (A.G.P.)
(iv) Harmonic Progression (H.P.)
(v) Miscellaneous Progression

2. Arithmetic Progression (A.P.)

Arithmetic Progression is defined as a series in which difference between any two consecutive terms is constant throughout the series. If 'a' is the first term and 'd' is the common difference, then an AP can be written as

$a + (a + d) + (a + 2d) + (a + 3d) + \ldots\ldots$

3. General term or nth term of an A.P.

(i) n^{th} term of an A.P. from beginning is given by $T_n = a + (n - 1)\, d$
(ii) n^{th} term from end is given by $= T_m - (n-1)\, d$
or $= (m - n + 1)^{th}$ term from beginning where m is total no. of terms.

4. Sum of n terms of an A.P.

The sum of first n terms of an A.P. is given by

$$S_n = \frac{n}{2}\,[2a + (n - 1)\, d]$$

or $S_n = \frac{n}{2}\,[a + T_n]$

Note:

- If sum of n terms S_n is given then general term $T_n = S_n - S_{n-1}$
- Common difference $d = S_2 - 2S_1$ where S_2 is sum of first two terms and S_1 is sum of first term or first term.
- Sum of n terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in n, in such case the common difference is twice the coefficient of n^2. i.e. 2A

- n^{th} term of an A.P. is of the form An + B i.e. a linear expression in n, in such a case the coefficient of n is the common difference of the A.P. i.e. A
- If for the different A.P's
  $\frac{S_n}{S'_n} = \frac{f_n}{\phi_n}$ then $\frac{T_n}{T'_n} = \frac{f(2n-1)}{\phi(2n-1)}$
- If for two A.P.'s $\frac{T_n}{T'_n} = \frac{An+B}{Cn+D}$ then $\frac{S_n}{S'_n} = \frac{A\left(\frac{n+1}{2}\right)+B}{C\left(\frac{n+1}{2}\right)+D}$

5. Arithmetic Mean (A.M.)

(i) If A is the A.M. between two given numbers a and b, then

$$A = \frac{a+b}{2} \Rightarrow 2A = a + b$$

(ii) A.M. of any n positive numbers $a_1, a_2 \ldots\ldots a_n$ is

$$A = \frac{a_1 + a_2 + a_3 + \ldots\ldots a_n}{n}$$

(iii) **n AM's between two given Numbers a and b**

If in between two numbers 'a' and 'b' we have to insert n AM $A_1, A_2, \ldots\ldots A_n$ then $a, A_1, A_2, A_3 \ldots A_n$, b will be in A.P. The series consist of (n+2) terms and the last term is b and first term is a.

$$\Rightarrow \quad a + (n + 2 - 1)\, d = b$$

$$\Rightarrow \quad d = \frac{b-a}{n+1}$$

$A_1 = a + d,\ A_2 = a + 2d, \ldots\ A_n = a + nd$ or $A_n = b - d$

(iv) Sum of n AM's inserted between a and b is $\frac{n}{2}(a + b)$

6. Supposition of terms in A.P.

(i) Three terms as : $a - d, a, a + d$

(ii) Five terms as : $a - 2d, a - d, a, a + d, a + 2d$

(iii) Four terms as : $a - 3d, a - d, a + d, a + 3d$

7. Some properties of an A.P.

(i) If each term of a given A.P. be increased, decreased, multiplied or divided by some non zero constant number then resulting series thus obtained will also be in A.P.

(ii) In an A.P., the sum of terms equidistant from the beginning and end is constant and equal to the sum of first and last term.

(iii) If for an A.P., sum of p terms is q, sum of q terms is p, then sum of (p + q) terms is (p + q).

(iv) If for an A.P., sum of p terms is equal to sum of q terms then sum of (p + q) terms is zero.

8. Geometric Progression (G.P.)

Geometric Progression is defined as a series in which ratio between any two consecutive terms is constant throughout the series. This constant ratio is called as a common ratio. If 'a' is the first term and 'r' is the common ratio, then a GP can be written as $a + ar + ar^2 + \ldots..$

9. General term of a G.P.

(i) If a is the first term and r is common ratio then general term (n^{th} term) from beginning is given by $T_n = ar^{n-1}$

(ii) The n^{th} term of a GP from the end is $T'_n = \dfrac{T_n}{r^{n-1}}$.

(iii) The n^{th} term from the end of a GP consisting of m terms is
$T'_n = ar^{m-n}$.

10. Sum of n terms of a G.P.

The sum of first n terms of an A.P. is given by

$S_n = \dfrac{a(1-r^n)}{1-r} = \dfrac{a-rT_n}{1-r}$ when $r < 1$

or $S_n = \dfrac{a(r^n-1)}{r-1} = \dfrac{rT_n - a}{r-1}$ when $r > 1$

and $S_n = an$ when $r = 1$

11. Sum of an Infinite G.P.

The sum of an infinite G.P. with first term a and common ratio r

If $-1 < r < 1$ then $S_\infty = \dfrac{a}{1-r}$

If $r \geq 1$ then $S_\infty \to \infty$

12. Geometrical Mean (G.M.)

(i) If G is the G.M. between two numbers a and b then
$G^2 = ab \Rightarrow G = \sqrt{ab}$

(ii) G.M. of any n positive numbers
$a_1, a_2, a_3, \ldots\ldots a_n$ is $(a_1.a_2.a_3\ldots\ldots a_n)^{1/n}$.

(iii) n GM's between two given Numbers a and b
If in between two numbers 'a' and 'b', we have to insert n GM $G_1, G_2 \ldots G_n$ then a, $G_1, G_2, \ldots G_n$, b will be in GP. The series consist of (n+2) terms and the last term is b and first term is a.

$\Rightarrow \; ar^{n+2-1} = b \qquad \Rightarrow r = \left(\dfrac{b}{a}\right)^{\frac{1}{n+1}}$

$G_1 = ar$, $G_2 = ar^2 \ldots\ldots G_n = ar^n$ or $G_n = b/r$

(iv) Product of n GM's inserted between a and b is $(ab)^{n/2}$

13. Supposition of terms in G.P.

(i) Three terms as : $\dfrac{a}{r}$, a, ar

(ii) Five terms as : $\dfrac{a}{r^2}, \dfrac{a}{r}$, a, ar, ar^2

(iii) Four terms as : $\dfrac{a}{r^3}, \dfrac{a}{r}$, ar, ar^3

14. Some Properties of G.P.

(i) If each term of a G.P. be multiplied or divided by the same non zero quantity, then resulting series is also a G.P.

(ii) In an G.P., the product of two terms which are at equidistant from the first and the last term, is constant and is equal to product of first and last term.

(iii) If each term of a G.P. be raised to the same power, then resulting series is also a G.P.

(iv) If a_1, a_2, a_3 ....a_n is a G.P. of non zero, non negative terms, then log a_1, log a_2,...... log a_n is an A.P. and vice-versa.

(v) If a_1, a_2, a_3....... and b_1, b_2, b_3 .... are two G.P.'s then $a_1 b_1$, a_2b_2, a_3b_3..... is also in G.P.

15. Arithmetico - Geometrical Progression (A.G.P.)

If each term of a Progression is the product of the corresponding terms of an A.P. and a G.P., then it is called arithmetic- geometric progression (A. G.P.)

e.g. : a, (a + d) r, (a + 2d) r^2 , ...

(i) The general term (n^{th} term) of an A.G.P. is
$T_n = [a + (n - 1) d] r^{n-1}$

(ii) Sum of n terms : $S_n = \dfrac{a}{1-r} + \dfrac{r.d(1-r^{n-1})}{(1-r)^2} - \dfrac{(a+(n-1)d)r^n}{(1-r)}$

(iii) Sum of infinite terms $S_\infty = \dfrac{a}{1-r} + \dfrac{dr}{(1-r)^2}$, $|r| < 1$

16. Harmonic Progression (H.P.)

Harmonical Progression is defined as a series in which reciprocal of its terms are in A.P. The standard form of a H.P. is

$$\frac{1}{a} + \frac{1}{a+d} + \frac{1}{a+2d} + ...$$

17. General Term of a H.P.

General term (n^{th} term) of a H.P. is given by

$$T_n = \frac{1}{a+(n-1)d}$$

Note: Sum of n terms in H.P. is not defined.

18. Harmonical Mean (H.M.)

(i) If H is the H.M. between a and b then $H = \dfrac{2ab}{a+b}$

(ii) H.M. of n numbers x_1, x_2, ... x_n is $H = \dfrac{n}{\left(\dfrac{1}{x_1} + \dfrac{1}{x_2} + ... + \dfrac{1}{x_n}\right)}$.

(iii) n H.M.'s between two given Numbers

To find n H.M.'s between a and b, we first find n A.M.'s between 1/a and 1/b then their reciprocals will be required H.M.'s.

19. Relation between A.M., G.M. & H.M.

If A, G, H are A.M., G.M. and H.M. between any two numbers

(i) $A \geq G \geq H$ (equality holds when all terms are equal)

(ii) $G^2 = AH$

(iii) If A and G are A.M., G.M. respectively between two positive numbers, then these numbers are $A + \sqrt{A^2 - G^2}$, $A - \sqrt{A^2 - G^2}$

20. Some Special Series

(i) Sum of first n natural numbers $= \sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

(ii) Sum of first n odd natural numbers $= \sum_{r=1}^{n} (2r-1) = n^2$

(iii) Sum of first n even natural numbers $= \sum_{r=1}^{n} 2r = n\,(n+1)$

(iv) Sum of squares of first n natural numbers $= \sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$

(v) Sum of cubes of first n natural numbers

$$= \sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2 = \left(\sum_{r=1}^{n} r\right)^2$$

SHORT TRICKS

Trick -1 If the ratio of sum of n terms of two arithmetic progression is f(n): g(n) then their n^{th} term will be in f(2n – 1): g(2n – 1)

Q.1 If the ratio of sum of n terms of two A.P's is (3n + 8) : (7n + 15), then the ratio of 12^{th} terms is-

(A) 16 : 7 (B) 7 :16 (C) 7 : 12 (D) 12 : 5

Sol. [B]

Proper Method

$$\frac{S_n}{S_n'} = \frac{3n+8}{7n+15}$$

$$\Rightarrow \frac{\frac{n}{2}[2a_1+(n-1)d_1]}{\frac{n}{2}[2a_2+(n-1)d_2]} = \frac{3n+8}{7n+15}$$

$$\Rightarrow \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2} = \frac{3n+8}{7n+15}$$

put $\frac{n-1}{2} = 11 \therefore n = 23$

$$\Rightarrow \frac{a_1+11d_1}{a_2+11d_2} = \frac{3.23+8}{7.23+15} = \frac{77}{176} = \frac{7}{16}$$

Short Trick

$$\frac{S_n}{S'_n} = \frac{3n+8}{7n+15}$$

Then $\frac{T_n}{T'_n} = \frac{3(2n-1)+8}{7(2n-1)+15}$

$$\frac{T_{12}}{T'_{12}} = \frac{7}{16}$$

Q.2 If the ratio of the sum of n terms of two AP's is 2n : (n + 1), then ratio of their 8^{th} terms is-

(A) 15 : 8 (B) 8 : 13 (C) n : (n – 1) (D) 5 : 17

Sol. [A]

Proper Method

$$\frac{S_n}{S_n'} = \frac{2n}{n+1}$$

$$\Rightarrow \frac{\frac{n}{2}[2a_1+(n-1)d_1]}{\frac{n}{2}[2a_2+(n-1)d_2]} = \frac{2n}{n+1}$$

$$\Rightarrow \frac{a_1+\left(\frac{n-1}{2}\right)d_1}{a_2+\left(\frac{n-1}{2}\right)d_2} = \frac{2n}{n+1}$$

put $\frac{n-1}{2} = 7 \therefore n = 15$

$$\Rightarrow \frac{a_1+7d_1}{a_2+7d_2} = \frac{2(15)}{15+1} = \frac{15}{8}$$

Short Trick

$$\frac{S_n}{S'_n} = \frac{2n}{n+1}$$

$$\Rightarrow \frac{T_n}{T'_n} = \frac{2(2n-1)}{(2n-1)+1}$$

$$\frac{T_8}{T'_8} = \frac{15}{8}$$

Trick -2 Sum of infinite series of type

$$\frac{1}{a_1b_1}+\frac{1}{a_2b_2}+\frac{1}{a_3b_3}+\ldots\ldots\infty=\frac{1}{a_1(b_1-a_1)}$$

Where a_1, a_2, a_3 .... are in A.P. and b_1, b_2, b_3 .... are in A.P and $b_1 - a_1 = b_2 - a_2 =$ ......

Q.3 The sum to infinity of the following series $\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+ \ldots$ shall be-

(A) ∞ (B) 1 (C) 0 (D) None of these

Sol. [B]

Proper Method

$$S_n=\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}\ldots\ldots\ldots+\frac{1}{n(n+1)}$$

$$S_n=\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)\ldots\ldots\ldots+\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$S_n=1-\frac{1}{n+1}=\frac{n}{n+1}$$

$$S_\infty=\lim_{n\to\infty}\left(\frac{n}{n+1}\right)=1$$

Short Trick

$$\text{Ans}=\frac{1}{(1)(2-1)}=1$$

Q.4 $\frac{1}{1.4}+\frac{1}{4.7}+\frac{1}{7.10}+\ldots\ldots$ upto ∞ terms is-

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{3}$ (D) None

Sol. [C]

Proper Method

$$S_n=\frac{1}{1.4}+\frac{1}{4.7}+\frac{1}{7.10}+\ldots\ldots+\frac{1}{(3n-2)(3n+1)}$$

$$S_n=\frac{1}{3}\left[\frac{4-1}{1.4}+\frac{7-4}{4.7}+\ldots\ldots+\frac{(3n+1)-(3n-2)}{(3n+1)(3n-2)}\right]$$

$$S_n=\frac{1}{3}\left[\left(\frac{1}{1}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{7}\right)+\left(\frac{1}{7}-\frac{1}{10}\right)+\ldots\ldots+\left(\frac{1}{3n-2}-\frac{1}{3n+1}\right)\right]$$

$$S_n=\frac{1}{3}\left[1-\frac{1}{3n+1}\right]\Rightarrow S_\infty=\lim_{n\to\infty}\frac{1}{3}\left[1-\frac{1}{3n+1}\right]$$

$$=\frac{1}{3}$$

Short Trick

$$\text{Ans}=\frac{1}{1(4-1)}=\frac{1}{3}$$

Trick -3 If arithmetic mean and geometric mean of two numbers a and b (a > b) are in ratio m : n, then

$$\frac{a}{b}=\frac{m+\sqrt{m^2-n^2}}{m-\sqrt{m^2-n^2}}$$

Q.5 If A.M. between p and q ($p \geq q$) is two times the GM, then p : q is-

(A) 1 : 1 (B) 2 : 1 (C) $(2+\sqrt{3}):(2-\sqrt{3})$ (D) 3 :1

Sol. [C]

Proper Method	Short Trick
Let AM of p and q is A and G.M of p and q is G then $A = \frac{p+q}{2}$, $G = \sqrt{pq}$ Given that $\frac{A}{G} = \frac{2}{1}$ $\Rightarrow \frac{p+q}{2\sqrt{pq}} = \frac{2}{1}$ Using componendo & dividendo $\Rightarrow \frac{p+q+2\sqrt{pq}}{p+q-2\sqrt{pq}} = \frac{2+1}{2-1}$ $\Rightarrow \left(\frac{\sqrt{p}+\sqrt{q}}{\sqrt{p}-\sqrt{q}}\right)^2 = \frac{3}{1}$ $\Rightarrow \frac{\sqrt{p}+\sqrt{q}}{\sqrt{p}-\sqrt{q}} = \frac{\sqrt{3}}{1}$ $\Rightarrow$ again using componendo & dividendo $\Rightarrow \frac{(\sqrt{p}+\sqrt{q})+(\sqrt{p}-\sqrt{q})}{(\sqrt{p}+\sqrt{q})-(\sqrt{p}-\sqrt{q})} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ $\Rightarrow \frac{\sqrt{p}}{\sqrt{q}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ $\Rightarrow \frac{p}{q} = \frac{2+\sqrt{3}}{2-\sqrt{3}}$	$\frac{A.M}{G.M} = \frac{2}{1}$ $\frac{p}{q} = \frac{2+\sqrt{3}}{2-\sqrt{3}}$

Q.6 If the arithmetic mean of two numbers a, b ; $a > b > 0$ is five times their geometric mean, then $\frac{a+b}{a-b}$ is equal to – **[JEE Main Online-2017]**

(A) $\frac{7\sqrt{3}}{12}$ (B) $\frac{3\sqrt{2}}{4}$ (C) $\frac{\sqrt{6}}{2}$ (D) $\frac{5\sqrt{6}}{12}$

Sol. [D]

Proper Method	Short Trick
Let AM of a and b is A and a.m of a and b is G then $A = \frac{a+b}{2}$, $G = \sqrt{ab}$ Given that $\frac{A}{G} = \frac{5}{1}$ $\Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{5}{1}$ Using compounds & dividends $\Rightarrow \frac{a+b+\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{5+1}{5-1}$	$\frac{A.M}{G.M} = \frac{5}{1}$ $\frac{a}{b} = \frac{5+\sqrt{24}}{5-\sqrt{24}}$ $\frac{a+b}{a-b} = \frac{5\sqrt{6}}{12}$

$\Rightarrow \left(\dfrac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}\right)^2 = \dfrac{6}{4}$ $\Rightarrow \dfrac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \dfrac{\sqrt{6}}{2}$ $\Rightarrow$ again using compounds & dividends $\Rightarrow \dfrac{(\sqrt{a}+\sqrt{b})+(\sqrt{a}-\sqrt{b})}{(\sqrt{a}+\sqrt{b})-(\sqrt{a}-\sqrt{b})} = \dfrac{\sqrt{6}+2}{\sqrt{6}-2}$ $\Rightarrow \dfrac{\sqrt{a}}{\sqrt{b}} = \dfrac{\sqrt{6}+2}{\sqrt{6}-2}$ $\Rightarrow \dfrac{a}{b} = \dfrac{12+4\sqrt{6}}{12-4\sqrt{6}}$ $\Rightarrow \dfrac{a+b}{a-b} = \dfrac{12}{8\sqrt{6}} = \dfrac{5\sqrt{6}}{12}$	

Questions based on recurring decimals

Trick -4 **Remove all decimals and then subtract that number which has no recurring decimals, then divide by as many 9's as many digits has recurring decimal and as many zero's as many digits has no recurring decimal**

Q.7 The fractional value of $0.\dot{1}\dot{2}\dot{5}$ is-

(A) 125/999 (B) 23/990 (C) 61/550 (D) None of these

Sol. **[A]**

Proper Method	**Short Trick**
Let $x = 0.\overline{125}$ $\therefore x = 0.125125125$ $\therefore 1000\,x = 125.125125125$ Now (ii) – (i); $999\,x = 125$ $\therefore x = 125/999$	$0.\dot{1}\dot{2}\dot{5} = \dfrac{125-0}{999} = \dfrac{125}{999}$

Q.8 The rational number which equals the number $2.\overline{357}$ with recurring decimal is -

(A) $\dfrac{2355}{1001}$ (B) $\dfrac{2379}{997}$ (C) $\dfrac{2355}{999}$ (D) None of these

Sol. **[C]**

Proper Method	**Short Trick**
$2.\overline{357} = 2 + 0.357 + 0.000357 + ... \infty$ $= 2 + \dfrac{357}{10^3} + \dfrac{357}{10^6} + ... \infty$ $= 2 + \dfrac{\frac{357}{10^3}}{1-\frac{1}{10^3}} = 2 + \dfrac{357}{999} = \dfrac{2355}{999}$	$2.\overline{357} = \dfrac{2357-2}{999}$ $= \dfrac{2355}{999}$

Sol. [A]

Proper Method

$\because a^2, b^2, c^2$ are in A.P.

$\therefore a^2 + ab + bc + ca,\ b^2 + bc + ca + ab,\ c^2 + ca + ab + bc$ ..... are also in A.P. [adding ab + bc + ca]

or $(a + c)(a + b), (b + c)(a + b), (c + a)(b + c)$ ... are also in A.P.

or $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+d}$ are in A.P.

[dividing by $(a + b)(b + c)(c + a)$]

Short Trick

Let $a^2 = 1$, $b^2 = 25$ and $c^2 = 49$

Now $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$

$= \frac{1}{12}, \frac{1}{8}, \frac{1}{6}$

Which are in A.P.

Q.12 If $a_1, a_2, a_3.....a_n$ are in AP where $a_i > 0$ for all i then the value of

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \_ + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} =$$

(A) $\frac{1}{\sqrt{a_1}+\sqrt{a_n}}$ (B) $\frac{1}{\sqrt{a_1}-\sqrt{a_n}}$ (C) $\frac{n}{\sqrt{a_1}-\sqrt{a_n}}$ (D) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

Sol. [D]

Proper Method

Let d be the c.d. of the A.P. Now L.H.S.

$$= \frac{\sqrt{a_1}-\sqrt{a_2}}{a_1-a_2} + \frac{\sqrt{a_2}-\sqrt{a_3}}{a_2-a_3} + ... \frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{a_{n-1}-a_n} \quad \text{(Note)}$$

$$= -\left(\frac{\sqrt{a_1}-\sqrt{a_2}+\sqrt{a_2}-\sqrt{a_3}+...\sqrt{a_n}}{d}\right)$$

$$= -\frac{(\sqrt{a_1}-\sqrt{a_n})}{d} = \frac{1}{d}\ \frac{(a_n-a_1)}{\sqrt{a_n}+\sqrt{a_1}}$$

$$= \frac{(n-1)d}{d\left[\sqrt{a_n}+\sqrt{a_1}\right]} \qquad [\because a_n = a_1 + (n-1)d]$$

$$= \frac{n-1}{\sqrt{a_n}+\sqrt{a_1}}$$

Short Trick

Let n = 3 and $a_1 = 1$, $a_2 = 25$, $a_3 = 49$

then $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}}$

$= \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$

by option (D) $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$

$= \frac{3-1}{1+7} = \frac{1}{4}$ is the correct answer

Q.13 The sum to n terms of the series $1 + 2\left(1+\frac{1}{n}\right) + 3\left(1+\frac{1}{n}\right)^2 + ...$ is given by-

(A) n^2 (B) $n(n + 1)$ (C) $n(1+1/n)^2$ (D) None of these

Sol. [A]

Proper Method

Let S be the sum of n terms of the given series and $x = 1 + 1/n$, Then,

$S = 1 + 2x + 3x^2 + 4x^3 + ...+ n x^{n-1}$

$\Rightarrow xS = x + 2x^2 + 3x^3 + ...+ (n-1) x^{n-1} + nx^n$

$\therefore S - xS = 1 + [x + x^2 + ...+ x^{n-1}] - nx^n$

$\Rightarrow S(1-x) = \frac{1-x^n}{1-x} - n x^n$

$\Rightarrow S(-1/n) = -n[1-(1+1/n)^n] - n(1+1/n)^n$

$\Rightarrow \frac{1}{n}.S = n[1-(1+1/n)^n + (1+1/n)^n] = n$

$\Rightarrow S = n^2$

Short Trick

Put n = 2

then sum of two terms

$= 1+2\left(1+\frac{1}{2}\right) = 4$

by option (A) n^2 is correct answer

Q.14 If p^{th}, q^{th} and r^{th} terms of H.P. are u, v, w respectively, then the value of the expression $(q-r)vw + (r-p)wu + (p-q)uv$ is-

(A) 1 (B) 0 (C) – 2 (D) – 1

Sol. [B]

Proper Method	Short Trick
Let H.P. be $\frac{1}{a} + \frac{1}{a+d} + \frac{1}{a+2d} + ...$ $u = \frac{1}{a+(p-1)d}$, $v = \frac{1}{a+(q-1)d}$, $w = \frac{1}{a+(r-1)d}$ $a + (p-1)d = \frac{1}{u}$, $a + (q-1)d = \frac{1}{v}$, $a + (r-1)d = \frac{1}{w}$ $\Rightarrow (q-r)\{a + (p-1)d\} + (r-p)\{a + (q-1)d\} + ...$ $= \frac{1}{u}(q-r) + \frac{1}{v}(r-p) + ...$ $\Rightarrow (q-r)vw + ... = 0$	Let p = 1, q = 2, r = 3 and u = 2, v = 3, w = 6 $(q-r)vw + (r-p)wu + (p-q)uv$ $= -18 + 24 - 6 = 0$

Q.15 If x, y, z are in A.P. and x, y, t are in G.P. then x, x – y, t – z are in -

(A) G.P. (B) A.P. (D) H.P. (D) A.P. and G.P. both

Sol. [A]

Proper Method	Short Trick
x, y, z are in A.P. $\Rightarrow 2y = x + z$ Or $2xy = x^2 + xz$ (multiplying with x) $\Rightarrow x^2 - 2xy = -xz$ ...(1) x, y, t are in G.P. $\Rightarrow y^2 = xt$ or $(x^2 - 2xy + y^2) = -xz + xt$ or $(x-y)^2 = x(t-z)$ x, x – y, t – z are in G.P.	Let x = 1, y = 2, z = 3 and t = 4 x, x – y, t – z = 1, – 1, 1 are in G.P.

Q.16 If a, b, c are in H.P. then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ will be in-

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Sol. [C]

Proper Method	Short Trick
a, b, c are in HP $\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP $\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ are in AP $\Rightarrow 1 + \frac{b+c}{a}, 1 + \frac{c+a}{b}, 1 + \frac{a+b}{c}$ are in AP $\Rightarrow \frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in AP $\Rightarrow \frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in HP.	Let a = 2, b = 3, c = 6 then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ $= \frac{2}{9}, \frac{3}{8}, \frac{6}{5}$ which are in H.P

Q.17 If $a_1, a_2, \ldots a_n$ are in H.P., then $\frac{a_1}{a_2+a_3+\ldots+a_n}, \frac{a_2}{a_1+a_3+\ldots+a_n}, \ldots\ldots, \frac{a_n}{a_1+a_2+\ldots+a_{n-1}}$ are in

(1) A.P. (2) G.P. (3) H.P. (4) None of these

Sol. **[C]**

Proper Method

$a_1, a_2, \ldots a_n$ are in H.P.

$\Rightarrow \frac{1}{a_1}, \frac{1}{a_2}, \ldots, \frac{1}{a_n}$ are in A.P.

$\Rightarrow \frac{a_1+a_2+a_3+\ldots+a_n}{a_1}, \frac{a_1+a_2+a_3+\ldots+a_n}{a_2}, \ldots,$

$\Rightarrow 1+\frac{a_2+a_3+\ldots+a_n}{a_1}, 1+\frac{a_1+a_3+\ldots+a_n}{a_2}, \ldots,$

$\frac{a_1+a_2+\ldots+a_{n-1}}{a_n}$ are in A.P.

$\Rightarrow \frac{a_2+a_3+\ldots+a_n}{a_1}, \frac{a_1+a_3+\ldots+a_n}{a_2}, \ldots,$

$\frac{a_1+a_2+\ldots+a_{n-1}}{a_n}$ are in A.P.

$\Rightarrow \frac{a_1}{a_2+a_3+\ldots+a_n}, \frac{a_2}{a_1+a_3+\ldots+a_n}, \ldots\ldots,$

$\frac{a_n}{a_1+a_2+\ldots+a_{n-1}}$ are in H.P.

Short Trick

Let $a_1 = 2$, $a_2 = 3$, $a_3 = 6$

$$\left.\begin{array}{l} \frac{a_1}{a_2+a_3} = \frac{2}{9} \\ \frac{a_2}{a_1+a_3} = \frac{3}{8} \\ \frac{a_3}{a_1+a_2} = \frac{6}{5} \end{array}\right\} \text{are in H.P.}$$

Q.18 If the sum of m terms of an A.P. is the same as the sum of its n terms, then the sum of its (m + n) terms is -

(1) mn (2) – mn (3) 1/mn (4) 0

Sol. **[D]**

Proper Method

Let a be the first term and d be the common difference the given AP then

$S_m = S_n \Rightarrow \frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$

$\Rightarrow 2a(m-n) + \{m(m-1) - n(n-1)\}d = 0$

$\Rightarrow 2a(m-n) + \{(m^2-n^2) - (m-n)\}d = 0$

$\Rightarrow (m-n)[2a + (m+n-1)d] = 0$

$\Rightarrow 2a + (m+n-1)d = 0$

Now $S_{m+n} = \frac{m+n}{2}[2a + (m+n-1)d] = \frac{m+n}{2} \times 0 = 0$

Short Trick

Let m = 2 and n = 3

– 2, – 1, 0, 1, 2 ...

sum of (m + n) terms = 0

Q.19 If x be the AM and y, z be two GM's between two positive numbers, then $\frac{y^3+z^3}{xyz}$ is equal to-

[IIT-1997]

(A) 1 (B) 2 (C) 3 (D) 4

Sol. **[B]**

Q.22 If S_r denotes sum of first r terms of A.P. and $\frac{S_a}{a^2} = \frac{S_b}{b^2} = c$ then S_c =

(A) c^3 (B) $\frac{c}{ab}$ (C) abc (D) a + b + c

Sol. [A]

Proper Method	Short Trick
$S_a = a^2.c$ $[\because S_n = A.n^2 + B.n$ of AP , $d = 2A]$ $S_b = b^2.c$ $\therefore S_a = c.\ a^2 + 0.a$ $S_b = c.\ b^2 + 0.b$ $\Rightarrow d = 2c$ $\Rightarrow S_c = c.c^2 + 0.c = c^3$	Let a = 1, b = 2, c = 3 $\Rightarrow \frac{S_1}{1} = \frac{S_2}{4} = 3$ $S_1 = 3, S_2 = 12$ $\therefore$ A.P 3, 9, 15 $S_3 = 27$ by option (B) $c^3 = 27$

Q.23 The first, second and middle terms of an AP are a, b, c respectively. The sum of A.P. is-

(A) $\frac{2(c-a)}{b-a}$ (B) $\frac{2c(c-a)}{b-a} + c$ (C) $\frac{2c(b-a)}{c-a}$ (D) $\frac{2b(c-a)}{b-a}$

Sol. [B]

Proper Method	Short Trick
We have first term = a, second term = b $\therefore$ d = common difference = b – a It is given that the middle term is c. This means that there are an odd number of terms in the AP. Let there be (2n + 1) terms in the AP. Then $(n + 1)^{th}$ term is the middle term. $\therefore$ middle term = c $\Rightarrow$ a + nd = c $\Rightarrow a + n(b-a) = c \Rightarrow n = \frac{c-a}{b-a}$ $\therefore \text{Sum} = \frac{2n+1}{2}[2a + (2n+1-1)d]$ $= \frac{1}{2}\left\{2\left(\frac{c-a}{b-a}\right)+1\right\}\left[2a+2\left(\frac{c-a}{b-a}\right)(b-a)\right]$ $= \frac{1}{2}\left\{\frac{2(c-a)}{b-a}+1\right\}\{2c\} = \frac{2c(c-a)}{b-a} + c$	Let the A.P. is 1, 2, 3, 4, 5 then a = 1, b = 2, c = 3 sum of A.P. = 15 by option (B) $\frac{2c(c-a)}{b-a} + c$ = 15 is correct answer

Q.24 The series of natural numbers is divided into groups as follows ; (1), (2, 3), (4, 5, 6), (7, 8, 9, 10) and so on. Find the sum of the numbers in the n^{th} group is-

(A) $\frac{1}{2}$ $[n(n^2 + 1)]$ (B) $\frac{n(n^2+1)}{4}$ (C) $\frac{2n(n+1)}{3}$ (D) $\frac{n^2(n+1)}{2}$

Sol. [A]

Proper Method	Short Trick
1 2 3 4 5 6 7 8 9 10 Let $S = 1 + 2 + 4 + 7 ------ T_n$	Let n = 2 then sum of number in 2^{nd} group = 2 + 3 = 5 Put n = 2 in option then option (A) $\frac{1}{2}n(n^2 + 1)$ is correct answer

$S = 1 + 2 + 4 ------ --- + T_n$ On subtracting $\therefore 0 = 1 + (1 + 2 + 3 ---- (n-1) \text{ terms}) - T_n$ $\therefore T_n = 1 + (1 + 2 + 3 --- (n-1) \text{ terms}$ $\therefore T_n = 1 + \frac{n(n-1)}{2} = \frac{2+n^2-n}{2}$ In the n^{th} group first term $a = \frac{n^2-n+2}{2}$ & $d = 1$ & No. of terms = n $\therefore S_n = \frac{n}{2}(2a+(n-1)d)$ $\therefore S_n = \frac{n}{2}[n^2-n+2+(n-1)]$ $\therefore S_n = \frac{n}{2}(n^2+1)$	

Q.25 The sum of the first n terms of the series $\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \ldots\ldots\ldots$ is -

(A) $\frac{6n}{n+1}$ (B) $\frac{9n}{n+1}$ (C) $\frac{12n}{n+1}$ (D) $\frac{15n}{n+1}$

Sol. [A]

Proper Method	Short Trick
$\because T_n = \frac{2n+1}{1^2+2^2 \ldots\ldots + n^2}$ $= \frac{2n+1}{\frac{n(n+1)(2n+1)}{6}}$ $\therefore T_n = \frac{6}{n(n+1)} = 6\left(\frac{1}{n} - \frac{1}{n+1}\right)$ $\therefore S_n = T_1 + T_2 \ldots\ldots + T_n$ $\therefore S_n = 6\left(\frac{1}{1}-\frac{1}{2}\right) + 6\left(\frac{1}{2}-\frac{1}{3}\right) + \ldots + 6\left(\frac{1}{n}-\frac{1}{n+1}\right)$ $\therefore S_n = 6\left(\frac{1}{1}-\frac{1}{n+1}\right) = \frac{6n}{n+1}$	Let n = 1, then sum = 3 Now, put n = 1 in option then option (A) $\frac{6n}{n+1}$ is correct answer

Q.26 If $(r)_n = \underbrace{r\ r\ r\ r \ldots\ldots\ r}_{n \text{ times}}$ and $a = (6)_n$, $b = (8)_n$, $c = (4)_{2n}$, then-

(A) $a^2 + b + c = 0$ (B) $a^2 + b - c = 0$ (C) $a^2 + b - 2c = 0$ (D) $a^2 + b - ac = 0$

Sol. [B]

Proper Method	Short Trick
$(r)_n = r\ r\ r \ldots\ldots\ r_{\text{(n times)}}$ $a = (6)_n$ $b = (8)_n$ $c = (4)_{2n}$	Let n = 1,then a = 6, b = 8 and c = 44 option (B) $a^2 + b - c = 0$ satisfy the condition

$(a)^2 = \left(\underbrace{66666.........6}_{n\ times}\right)^2 = 36\left[\underbrace{11111.........1}_{n\ times}\right]^2$ $= 36\left[1+10+10^2+.........+10^{n-1}\right]^2$ $= 36\left[\frac{(10^n-1)}{10-1}\right]^2$ $= 36\left[\frac{(10^n-1)^2}{81}\right] = \frac{4}{9}(10^n-1)^2$ $b = \underbrace{8888.........8}_{n\ times} = \frac{8(10^n-1)}{10-1} = \frac{8(10^n-1)}{9}$ $a^2 + b = \frac{4}{9}(10^n-1)^2 + \frac{8(10^n-1)}{9}$ $= \frac{4(10^n-1)}{9}[(10^n-1)+2] = \frac{4}{9}[10^{2n}-1] = 4\left[\frac{10^{2n}-1}{10-1}\right]$ $= \underbrace{44444............4}_{2n\ times} = c \Rightarrow a^2 + b = c$	

Q.27 $\underbrace{666.........6}_{n times}$ is equal to

(1) $\frac{4}{9}(10^n-1)$ (2) $\frac{2}{3}(10^n-1)$ (3) $\frac{2}{3}(10^{n-1}-1)$ (4) $\frac{4}{9}(10^{n-1}-1)$

Sol. [B]

Proper Method	**Short Trick**
$\underbrace{666.........6}_{n times} = 6[\underbrace{111.........1}_{n times}]$ $= 6[1 + 10 + 10^2 + ... 10^{n-1}] = 6\times1\frac{[10^n-1]}{10-1}$ $= \frac{6}{9}(10^n-1) = \frac{2}{3}[10^n-1]$	Let n = 1 Now we put n = 1 in options then (B) $\frac{2}{3}(10^n-1)$ is correct option.

Q.28 $1.2.3 + 2.3.4 + 3.4.5 + ... +$ n terms -

(1) $\frac{n(n+1)(n+2)(n+3)}{12}$ (2) $\frac{n(n+1)(n+2)(n+3)}{3}$

(3) $\frac{n(n+1)(n+2)(n+3)}{4}$ (4) $\frac{(n+2)(n+3)(n+4)}{6}$

Sol. [C]

Proper Method	**Short Trick**
$T_n = n(n+1)(n+2)$ $S = \sum_{n=1}^{n} T_n = \sum_{n=1}^{n} n(n+1)(n+2)$	Let n = 1, then sum = 6 Now, we put n = 1 in options, then option (C) $\frac{n(n+1)(n+2)(n+3)}{4}$ is correct option

$= \sum n^3 + 3n^2 + 2n$ $= \frac{n^2(n+1)^2}{4} + \frac{3n(n+1)(2n+1)}{6} + \frac{2n(n+1)}{2}$ $= \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + (2n+1) + 2\right]$ $= \frac{n(n+1)}{2}\left[\frac{n(n+1)+4n+6}{2}\right]$ $= \frac{n(n+1)(n+3)(n+2)}{4}$	

Q.29 If S_r denotes the sum of the first r terms of an A.P., then $\frac{S_{3r}-S_{r-1}}{S_{2r}-S_{2r-1}}$ is equal to

(1) $2r-1$ (2) $2r+1$ (3) $4r+1$ (4) $2r+3$

Sol. [B]

Proper Method	**Short Trick**
$\frac{S_{3r}-S_{r-1}}{S_{2r}-S_{2r-1}} = \frac{\frac{3r}{2}\{2a+(3r-1)d\} - \frac{(r-1)}{2}\{2a+(r-1-1)d\}}{T_{2r}}$ $= \frac{(2r+1)a + \frac{d}{2}\{3r(3r-1)-(r-1)(r-2)\}}{a+(2r-1)d}$ $= \frac{(2r+1)a + \frac{d}{2}\{8r^2-2\}}{a+(2r-1)d}$ $= \frac{(2r+1)a + d(4r^2-1)}{a+(2r-1)d}$ $= 2r+1$	Let the A.P. is 1, 2, 3, 4, 5, 6 .... let r = 2 then $\frac{S_6-S_1}{S_4-S_3}$ $= \frac{21-1}{10-6} = 5$ now we put r = 2 in option then (B) (2r + 1) is correct option

Q.30 The sum to n terms of the series 11 + 103 + 1005 + ... is

(1) $\frac{1}{9}(10^n-1)+n^2$ (2) $\frac{1}{9}(10^n-1)+2n$ (3) $\frac{10}{9}(10^n-1)+n^2$ (4) None of these

Sol. [C]

Proper Method	**Short Trick**
Let S_n denote the sum to n terms of given series $S_n = 11 + 103 + 1005 + ...$ to n terms. $\Rightarrow S_n = (10+1) + (10^2+3) + (10^3+5) ...+\{10^n+(2n-1)\}$ $\Rightarrow S_n = (10+10^2+.... 10^n) + \{1+3+5+...+(2n-1)\}$ $\Rightarrow S_n = \frac{10(10^n-1)}{(10-1)} + \frac{n}{2}(1+2n-1)$ $= \frac{10}{9}(10^n-1)+n^2$	Put n = 1 then sum = 11 Now, put n = 1 in option, then option (C) $\frac{10}{9}(10^n-1)+n^2$ is correct option

Q.31 The sum of the first n terms of the series $1^2 + 2.\ 2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...$ is $\frac{n(n+1)^2}{2}$ when n is even. When n is odd the sum is- **[AIEEE-2004]**

(A) $\frac{3n(n+1)}{2}$ (B) $\frac{n^2(n+1)}{2}$

(C) $\frac{n(n+1)^2}{4}$ (D) $\left[\frac{n(n+1)}{2}\right]^2$

Sol. **[B]**

Proper Method

Series is : $1^2 + 2.2^3 + 3^2 + 2.4^3 + 5^2 + 2.6^3$ .....

n is even : sum = $\frac{n(n+1)^2}{2}$

n is odd : Let the sum is S

$\therefore$ n + 1 is even :

$\therefore$ Sum = $\frac{(n+1)(n+2)^2}{2} = S + \underbrace{2\cdot(n+1)^2}_{(n+1)^{th}\ term}$

$\therefore S = \frac{(n+1)(n+2)^2}{2} - 2(n+1)^2$

$\therefore S = (n+1)\left[\frac{(n+2)^2}{2} - 2(n+1)\right]$

$\therefore S = (n+1)\left[\frac{n^2 + 4n + 4 - 4n - 4}{2}\right]$

$\therefore S = \frac{n^2(n+1)}{2}$

Short Trick

Put n = 5

$\Rightarrow S_5 = 1 + 8 + 9 + 32 + 25$

= 75

Now, put n = 5 in options

then option (B) $\frac{n^2(n+1)}{2}$

is correct answer

Q.32 The sum of the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ... + 2\,(2m)^2$ is **[AIEEE Online-2012]**

(A) $m^2(2m + 1)$ (B) $m(m + 2)^2$

(C) $m^2(m + 2)$ (D) $m(2m + 1)^2$

Sol. **[D]**

Proper Method

$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ... + 2(2m)^2$

$= (1^2 + 2^2 + 3^2 + 4^2 + ...$ 2m terms)

$+ (2^2 + 4^2 = 6^2 + ...$ m terms)

$= \frac{(2m)(2m+1)(4m+3)}{6} + 2^2(1^2 + 2^2 + 3^2 + ... +$ m terms)

$= \frac{(2m)(2m+1)(4m+3)}{6} + 4\frac{m(m+1)(2m+1)}{6}$

$= \frac{m(2m+1)}{6}[2(4m+3) + 4(m+1)]$

$= m(2m + 1)^2$

Short Trick

Put n = 2, then sum

= 1 + 8 + 9 + 32 = 50

now put n = 2 in options, then option (D) $m(2m + 1)^2$ is correct answer.

Q.33 The sum of the series $1+\frac{4}{3}+\frac{10}{9}+\frac{28}{27}+......$ upto n terms is : **[AIEEE Online-2012]**

(A) $n+\frac{1}{2}-\frac{1}{2.3^{n-1}}$ (B) $n-\frac{1}{3}+\frac{1}{3.2^{n-1}}$ (C) $\frac{5}{3}n-\frac{7}{6}+\frac{1}{2.3^{n-1}}$ (D) $\frac{7}{6}n+\frac{1}{6}-\frac{1}{3.2^{n-1}}$

Sol. **[A]**

Proper Method

Given series is

$$1+\frac{4}{3}+\frac{10}{9}+\frac{28}{27}+\ldots\text{n terms}$$

$$=1+\left(1+\frac{1}{3}\right)+\left(1+\frac{1}{9}\right)+\left(1+\frac{1}{27}\right)+\ldots\text{ n terms}$$

$$=(1+1+1+\ldots+\text{n terms})+\left(\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots\text{n terms}\right)$$

$$=n+\frac{\frac{1}{3}\left(1-\frac{1}{3^n}\right)}{1-\frac{1}{3}}$$

$$=n+\frac{1}{3}\times\frac{3}{2}[1-3^{-n}]$$

$$=n+\frac{1}{2}[1-3^{-n}]=n+\frac{1}{2}-\frac{1}{2.3^n}$$

Short Trick

Put n = 2 then sum

$$=1+\frac{4}{3}=\frac{7}{3}$$

Now, put n = 2 in options, then option (A)

$n+\frac{1}{2}-\frac{1}{2.3^{n-1}}$ is correct answer

Q.34 If the sum of n terms of an A.P. is cn^2, then the sum of squares of these n terms are :

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$ **[IIT-2009]**

Sol. **[C]**

Proper Method

$S_n = cn^2$

$\therefore T_n = c(2n-1)$

$\therefore T_n^2 = c^2(4n^2-4n+1)$

$S'_n = c^2[4\Sigma n^2 - 4\Sigma n + \Sigma 1]$

$$=c^2\left[\frac{4.n(n+1)(2n+1)}{6}-\frac{4.n(n+1)}{2}+n\right]$$

$$=\frac{nc^2}{6}[4(n+1)(2n+1)-12(n+1)+6]$$

$$=\frac{nc^2}{6}[4(2n^2+3n+1)-12n-12+6]$$

$$=\frac{nc^2}{6}[8n^2+12n+4-12n-6]$$

$$=\frac{nc^2}{6}[8n^2-2]=\frac{nc^2}{3}[4n^2-1]$$

Short Trick

$T_n = S_n - S_{n-1} = cn^2 - c(n-1)^2$

$= c(2n-1)$

$T_n^2 = c^2(2n-1)^2$

We put n = 1 then $T_1^2 = S_1^2 = c^2$

Now put n = 1 in the options, then (C)

$\frac{n(4n^2-1)c^2}{3}$ is correct option.

3

Complex Number

KEY CONCEPTS

1. Real Number System

(i) Natural Number (N) : N = {1, 2, 3, ...}

(ii) Whole Number (W) : W = {0, 1, 2, ... } = {N} + {0}

(iii) Integers (Z or I) : Z or I = {.....–3, –2, –1, 0, 1, 2, 3, ...}

(iv) Rational Numbers (Q) : The numbers which are in the form of p/q (Where p, q $\in$ I, q $\neq$ 0)

(v) Irrational Numbers (Q^c) : The numbers which are not rational i.e. which can not be expressed in p/q form or whose decimal part is non terminating non repeating but which may represent magnitude of physical quantities. e.g. $\sqrt{2}$, $5^{1/3}$, π, e,.....etc.

(vi) Real Numbers (R) : The set of Rational and Irrational Number is called as set of Real Numbers i.e. $N \subset W \subset Z \subset Q \subset R$

2. Imaginary Number

Square root of a negative real number is an imaginary number, while solving equation $x^2 + 1 = 0$ we get $x = \pm\sqrt{-1}$ which is imaginary. So the quantity $\sqrt{-1}$ is denoted by 'i' called 'iota' thus $i = \sqrt{-1}$

3. Integral Power of Iota

$i = \sqrt{-1}$ so $i^2 = -1$; $i^3 = -i$ and $i^4 = 1$

Hence $i^{4n+1} = i$; $i^{4n+2} = -1$; $i^{4n+3} = -i$; i^{4n} or $i^{4n+4} = 1$

4. Complex Number

A number of the form $z = x + iy$ where x, y $\in$ R and $i = \sqrt{-1}$ is called a complex number where x is called as real part and y is called imaginary part of complex number and they are expressed as

Re (z) = x, Im (z) = y.

Here if x = 0 the complex number is purely Imaginary and if y = 0 the complex number is purely Real.

Note :

- Inequalities in complex number are not defined because 'i' is neither positive, zero nor negative so $4 + 3i < 1 + 2i$ or $i < 0$ or $i > 0$ is meaning less.
- If two complex numbers are equal, then their real and imaginary parts are separately equal. Thus if $a + ib = c + id \Rightarrow a = c$ and $b = d$
  so if $z = 0 \Rightarrow x + iy = 0 \Rightarrow x = 0$ and $y = 0$
- The complex number 0 is purely real and purely imaginary both.

5. Algebra of Complex Number

If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ be any two complex number then

(i) Addition $(a + ib) + (c + id) = (a + c) + i\,(b + d)$

(ii) Subtraction $(a + ib)-(c + id) = (a - c) + i\,(b - d)$

(iii) Multiplication $(a + ib)\,(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc)$

(iv) Division $\dfrac{a+ib}{c+id} = \dfrac{(a+ib)(c-id)}{(c+id)(c-id)}$

(when at least one of c and d is non zero)

$$= \frac{(ac+bd)}{c^2+d^2} + i\frac{(bc-ad)}{c^2+d^2}$$

Note : Additive inverse of z is $-z$ and multiplicative inverse of z is $\dfrac{1}{z}$.

6. Conjugate of a Complex Number :

If $z = a + ib$ then its conjugate complex is obtained by changing the sign of its imaginary part and it is denoted by $\bar{z}$. i.e. $\bar{z} = a - ib$

7. Properties of Conjugate Complex Number

Let $z = a + ib$ and $\bar{z} = a - ib$ then

(i) $\overline{(\bar{z})} = z$

(ii) $z + \bar{z} = 2a = 2\,\text{Re}\,(z)$ = purely real

(iii) $z - \bar{z} = 2ib = 2i\,\text{Im}\,(z)$ = purely imaginary

(iv) $z\bar{z} = a^2 + b^2 = |z|^2$

(v) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

(vi) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

(vii) $\overline{re^{i\theta}} = re^{-i\theta}$

(viii) $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\bar{z}_1}{\bar{z}_2}$

(ix) $\overline{z^n} = (\bar{z})^n$

(x) $\overline{z_1 z_2} = \bar{z}_1\ \bar{z}_2$

(xi) $z + \bar{z} = 0$ or $z = -\bar{z} \Rightarrow z = 0$ or z is purely imaginary

(xii) $z = \bar{z} \Rightarrow$ z is purely real

8. Modulus of a Complex Number

If $z = x + iy$ then modulus of is denoted by $|z|$. Thus

$z = x + iy \Rightarrow |z| = \sqrt{x^2 + y^2}$

9. Properties of Modulus of a Complex Number

(i) $|z| \geq 0$

(ii) $-|z| \leq \text{Re}\,(z) \leq |z|$

(iii) $-|z| \leq \text{Im}\,(z) \leq |z|$

(iv) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

(v) $z\,\bar{z} = |z|^2$

(vi) $z^{-1} = \dfrac{\bar{z}}{|z|^2}$

(vii) $|z_1 z_2| = |z_1|\ |z_2|$

(viii) $|z^n| = |z|^n$

(ix) $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$, where $|z_2| \neq 0$

(ix) $|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\text{Re}\,(z_1 \bar{z}_2)$

(x) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2[|z_1|^2 + |z_2|^2]$

(xi) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2|z_1|\,|z_2|\cos(\theta_1 - \theta_2)$

where θ_1 and θ_2 are the argument of z_1 and z_2

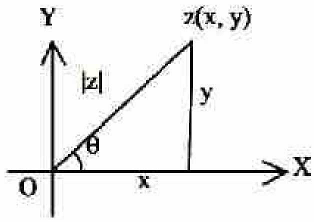
(xii) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Leftrightarrow \dfrac{z_1}{z_2}$ is purely imaginary.

(xiii) $|z_1 \pm z_2| \le |z_1| + |z_2|$ (triangle inequality)

(xiv) $|z_1 \pm z_2| \ge ||z_1| - |z_2||$ (triangle inequality)

10. Amplitude or Argument of Complex Number :

The amplitude or argument of a complex number z is the inclination of the directed line segment representing z, with real axis.



If $z = x + iy$ then $\text{amp}(z) = \tan^{-1}\left(\dfrac{y}{x}\right)$

Note :

- Principle value of any complex number lies between $-\pi < \theta \le \pi$.
- Principal value of argument.
  - (i) If complex number lies in first quadrant then $\arg(z) = \theta$.
  - (ii) If complex number lies in second quadrant then $\arg(z) = \pi - \theta$.
  - (iii) If complex number lies in third quadrant then $\arg(z) = -(\pi - \theta)$.
  - (iv) If complex number lies in fourth quadrant then $\arg(z) = -\theta$.

11. Properties of Argument of a Complex Number

(i) amp (any real positive number) = 0

(ii) amp (any real negative number) = π

(iii) $\text{amp}(z - \bar{z}) = \pm \dfrac{\pi}{2}$

(iv) $\text{amp}(z_1.z_2) = \text{amp}(z_1) + \text{amp}(z_2) + 2k\pi \ \forall \ k \in I$

(v) $\text{amp}\left(\dfrac{z_1}{z_2}\right) = \text{amp}(z_1) - \text{amp}(z_2) + 2k\pi \ \forall \ k \in I$

(vi) $\text{amp}(\bar{z}) = -\text{amp}(z) = \text{amp}\left(\dfrac{1}{z}\right)$

(vii) $\text{amp}(-z) = \text{amp}(z) \pm \pi$

(viii) $\text{amp}(z^n) = n\ \text{amp}(z) + 2k\pi$ (k = 0, 1 or −1)

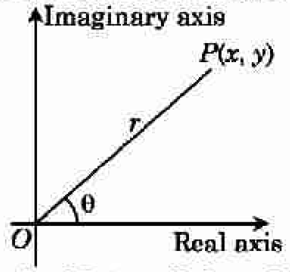
(ix) $\text{amp}(z) + \text{amp}(\bar{z}) = 2k\pi$

$\forall$ proper value of k must be chosen so that R.H.S. lies in $(-\pi, \pi]$.

12. Representation of a Complex Number in various forms

(i) Cartesian Form :

Every complex number $z = x + iy$ can be represented by a point on the Cartesian plane known as Complex plane (Argand diagram) by the ordered pair (x, y).



Length OP is called modulus of the complex number denoted by $|z|$ and θ is called the argument or amplitude.

$|z| = \sqrt{x^2 + y^2}$ & $\theta = \tan^{-1}\dfrac{y}{x}$

(ii) Trigonometric/Polar Form :

$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$;

(iii) Eulerian Form :

$z = re^{i\theta}$; $|z| = r$; $\arg z = \theta$; $\bar{z} = re^{-i\theta}$

$e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

13. Square Root of a Complex Number

The square root of $z = a + ib$ is

$$\sqrt{a+ib} = \pm\left[\sqrt{\frac{|z|+a}{2}} + i\sqrt{\frac{|z|-a}{2}}\right] \text{ for } b > 0 \text{ and}$$

$$= \pm\left[\sqrt{\frac{|z|+a}{2}} - i\sqrt{\frac{|z|-a}{2}}\right] \text{ for } b < 0.$$

14. Logarithm of a Complex Number

Let $z = x + iy$ be a complex number and in polar form of z is $re^{i\theta}$, then

$\log(x + iy) = \log(re^{i\theta}) = \log(r) + i\theta$

$= \log\left(\sqrt{x^2 + y^2}\right) + i\tan^{-1}\dfrac{y}{x}$

or $\log z = \log(|z|) + i\,\mathrm{amp}(z)$

In general, $z = re^{i(\theta + 2n\pi)}$

$\log z = \log|z| + i\arg z + 2n\pi i$

15. De- Moivre's Theorem

A simplest formula for calculating powers of complex numbers in the form of $\cos\theta$ and $\sin\theta$ is known as De-Moivre's theorem,.

If $n \in I$ (set of integers), then $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ and if $n \in Q$ (set of rational numbers), then $\cos n\theta + i\sin n\theta$ is one of the values of $(\cos\theta + i\sin\theta)^n$.

Note :

(i) $(\sin\theta + i\cos\theta)^n = \left[\cos\left(\dfrac{n\pi}{2} - n\theta\right) + i\sin\left(\dfrac{n\pi}{2} - n\theta\right)\right]$

(ii) $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)\ldots(\cos\theta_n + i\sin\theta_n) = \cos(\theta_1 + \theta_2 + \ldots + \theta_n) + i\sin(\theta_1 + \theta_2 + \ldots + \theta_n)$

(iii) $(\sin\theta \pm i\cos\theta)^n \neq \sin n\theta \pm i\cos n\theta$

16 Cube Root of Unity

Cube root of unity are 1, ω, ω^2, where $\omega = -\frac{1}{2} + \frac{i\sqrt{3}}{2} = e^{i2\pi/3}$ and $\omega^2 = \frac{-1-i\sqrt{3}}{2}$

$\arg(\omega) = \frac{2\pi}{3}$ and $\arg(\omega^2) = \frac{4\pi}{3}$

Properties of Cube Roots of Unity :

(i) $\omega^3 = 1$ or $\omega^{3r} = 1$

(ii) $1 + \omega + \omega^2 = 0$

(iii) Cube roots of unity lie on the unit circle $|z| = 1$ and divide its circumference into 3 equal parts.

(iv) It always forms an equilateral triangle.

(vi) Cube roots of –1 are $-1, -\omega, -\omega^2$

Important Identities

(a) $x^2 + x + 1 = (x - \omega)(x - \omega^2)$

(b) $x^2 - x + 1 = (x + \omega)(x + \omega^2)$

(c) $x^2 + xy + y^2 = (x - y\omega)(x - y\omega^2)$

(d) $x^2 - xy + y^2 = (x + \omega y)(x + y\omega^2)$

(e) $x^3 + y^3 = (x + y)(x + y\omega)(x + y\omega^2)$

(f) $x^3 - y^3 = (x - y)(x - y\omega)(x - y\omega^2)$

(g) $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + \omega y + \omega^2 z)(x + \omega^2 y + \omega z)$

17. The nth Roots of Unity :

The n^{th} roots of unity, it means any complex numebr z, which satisfies the equation $z^n = 1$ or $z = (1)^{1/n}$

or $z = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}$, where $k = 0, 1, 2, \ldots, (n - 1)$

18. Properties of n^{th} Roots of Unity :

(i) n^{th} roots of unity form a GP with common ratio $e^{\frac{i2\pi}{n}}$.

(ii) Sum of n^{th} roots of unity is always 0.

(iii) Sum of p^{th} powers of n^{th} roots of unity is n, if p is a multiple of n.

(iv) Sum of p^{th} powers of n^{th} roots of unity is zero, if p is not a multiple of n.

(v) Product of n^{th} roots of unity is $(-1)^{n-1}$.

(vi) The n^{th} roots of unity lie on the unit circle $|z| = 1$ and divide its circumference into n equal parts.

19. Some Important Results

If $z = \cos\theta + i\sin\theta$

(i) $z + \frac{1}{z} = 2\cos\theta$

(ii) $z - \frac{1}{z} = 2i\sin\theta$

(iii) $z^n + \frac{1}{z^n} = 2\cos n\theta$, $z^n - \frac{1}{z^n} = 2i\sin n\theta$

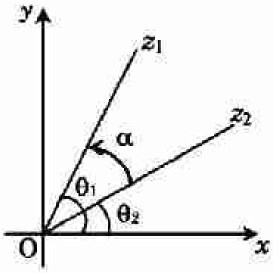
(iv) If $x = \cos\alpha + i\sin\alpha$, $y = \cos\beta + i\sin\beta$, $z = \cos\gamma + i\sin\gamma$ and given, $x + y + z = 0$, then

(a) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$ (b) $yz + zx + xy = 0$ (c) $x^2 + y^2 + z^2 = 0$ (d) $x^3 + y^3 + z^3 = 3xyz$

20. Concept of Rotation

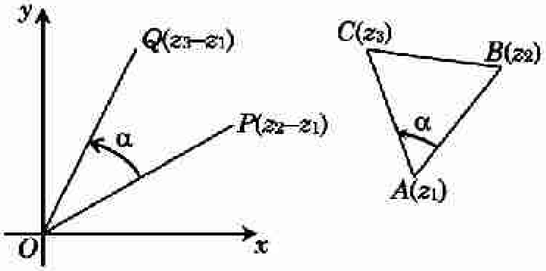
If z_1 and z_2 are two complex number then argument of

$$\frac{z_1}{z_2} = \frac{|z_1|e^{i\theta_1}}{|z_2|e^{i\theta_2}} = \left|\frac{z_1}{z_2}\right|e^{i(\theta_1-\theta_2)} = \left|\frac{z_1}{z_2}\right|e^{i\alpha}$$



In general, let z_1, z_2, z_3 be the three vertices of a ΔABC described in the counter clockwise sense. Draw OP and OQ parallel and equal to AB and AC respectively. Then P is $z_2 - z_1$ and Q is $z_3 - z_1$ then

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\alpha}$$



21. Application of Complex Number in Co-ordinate Geometry

(i) Distance between complex points :

(a) The distance between two points $P(z_1)$ and $Q(z_2)$ is given by
$PQ = |z_2 - z_1|$

(b) Section formula : If $R(z)$ divides the line segment joining $P(z_1)$ and $Q(z_2)$ in the ratio $m_1 : m_2$ (m_1, $m_2 > 0$). Then,

For internal division $z = \dfrac{m_1z_2 + m_2z_1}{m_1 + m_2}$

For external division $z = \dfrac{m_1z_2 - m_2z_1}{m_1 - m_2}$

(ii) Triangle in Complex Plane :

(a) If the vertices A, B, C of a Δ represent the complex nos. z_1, z_2, z_3 respectively, then

Centroid of the $\Delta ABC = \dfrac{z_1 + z_2 + z_3}{3}$

Incentre of the $\Delta ABC = \dfrac{az_1 + bz_2 + cz_3}{a + b + c}$

(b) Area of the triangle with vertices $A(z_1)$, $B(z_2)$ and $C(z_3)$ is given by

$$\Delta = \frac{1}{2}\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}$$

(c) The triangle whose vertices are the points represented by complex numbers z_1, z_2, z_3 is equilateral if $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$

or $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.

(iii) **Straight line in Complex Plane :**

(a) General equation of a line in complex plane is $\bar{a} z + a\bar{z} + b = 0$ where $b \in R$ and a is a fixed non zero complex number.

(b) The complex and real slope of the line $\bar{a}z + a\bar{z} = k$ are $\left(-\frac{a}{\bar{a}}\right)$ and $\left(-\frac{Re(a)}{Im(a)}\right)$.

(c) Equation of the line joining complex number z_1 and z_2 is

$$z = tz_1 + (1 - t) z_1, t \in R \text{ or } \frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} = \begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

(d) z_1z_2 are two fixed points, then $|z - z_1| = |z - z_2|$ represents perpendicular bisector of the line segment joining z_1 and z_2.

(e) The length of perpendicular form a point z_1 to $\bar{a}z + a\bar{z} + b = 0$ is given by $\frac{|\bar{a}z_1 + a\bar{z}_1 + b|}{2|a|}$

(f) The equation of a line parallel to the line $\bar{a}z + a\bar{z} + b = 0$ is $\bar{a}z + a\bar{z} + \lambda = 0$, where $\lambda \in R$.

(g) The equation of a line perpendicular to the line $\bar{a}z + a\bar{z} + b = 0$ is $a\bar{z} - \bar{a}z + i\lambda = 0$, where $\lambda \in R$.

(h) The equation of the perpendicular bisector of the line segment joining points A (z_1) and B (z_2) is $z(\bar{z}_1 - \bar{z}_2) + \bar{z}(z_1 - z_2) = |z_1|^2 - |z_2|^2$

(iii) **Circle in Complex Plane :**

(a) Equation of circle with centre z_0 and radius r is $|z - z_0| = r$ or $z\bar{z} + z_0\bar{z}_0 - z\bar{z}_0 - \bar{z}z_0 = r^2$

(b) General equation of a circle is $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$ where $b \in R$ and a is fixed complex number.

For this circle, centre is the points a and radius = $\sqrt{|a|^2 - b}$

(c) Condition for four given points z_1, z_2, z_3 & z_4 to be concyclic is, the number $\frac{z_3 - z_1}{z_3 - z_2} \cdot \frac{z_4 - z_2}{z_4 - z_1}$ is real.

(d) The equation of the circle described on the line segment joining z_1 & z_2 as diameter is $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$

22. Standard Loci

(i) $|z - z_1| + |z - z_2| = 2k$ (a constant) represent

(a) If $2k > |z_1 - z_2|$ $\Rightarrow$ An ellipse (z_1 and z_2 are foci)

(b) If $2k = |z_1 - z_2|$ $\Rightarrow$ A line segment

(c) If $2k < |z_1 - z_2|$ $\Rightarrow$ No solution

(ii) Equation $||z - z_1| - z - z_2|| = 2k$ (a constant) represent

(a) If $2k < |z_1 - z_2|$ $\Rightarrow$ A Hyperbola (z_1 and z_2 are foci)

(b) If $2k = |z_1 - z_2|$ $\Rightarrow$ A line break

(c) If $2k > |z_1 - z_2|$ $\Rightarrow$ No solution

(iii) If $\frac{|z - z_1|}{|z - z_2|} = k$ $(k \neq 1)$ then locus of z is a circle

(iv) If $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ then locus of z is a circle with z_1, z_2 are the end points of diameter.

(v) If $\arg\left(\frac{z - z_1}{z - z_2}\right) = \pm\frac{\pi}{2}$ then locus of z is a circle with z_1, z_2 are the end points of diameter.

(vi) If $|z - z_1|^2 + |z - z_2|^2 = k$ where $\frac{1}{2}|z_1 - z_2|^2 \le k$ then locus of z is a circle.

SHORT TRICKS

1. Square root of Complex Number

Formula :

$$\sqrt{z} = \sqrt{x+iy} = \pm\left[\sqrt{\frac{|z|+x}{2}} + i\sqrt{\frac{|z|-x}{2}}\right]$$

$$\sqrt{z} = \sqrt{x-iy} = \pm\left[\sqrt{\frac{|z|+x}{2}} - i\sqrt{\frac{|z|-x}{2}}\right]$$

Trick -1 To find $\sqrt{x+iy}$ first find the number $\frac{|y|}{2}$ and then factorised this number in such a way that difference of square of these factors is equal to real number (x)

Q.1 Square root of 7 + 24i is -

(A) ± (3 – 4i) (B) ± (4 + 3i) (C) ± (4 – 3i) (D) ± (–3 – 4i)

Sol. [B]

Proper Method

Let $\sqrt{7+24i} = a + ib$

$7 + 24i = (a + ib)^2$

$7 + 24i = a^2 - b^2 + 2iab$

$\Rightarrow a^2 - b^2 = 7$ ...(1)

and $2ab = 24$ ...(2)

$a^2 + b^2 = \pm\sqrt{(a^2-b^2)^2 + 4a^2b^2}$

$= \pm\sqrt{49+576}$

$a^2 + b^2 = \pm 25$ ...(3)

Solving (1) & (3)

$2a^2 = 7 \pm 25$

$a^2 = 16 \Rightarrow a \pm 4$

and $b = \pm 3$

$\therefore \sqrt{7+24i} = \pm (4 + 3i)$

Short Trick

$\sqrt{7+24i}$

↓

12

↙ ↘

4 3

Ans. = ± (4 + 3i)

Q.2 Square root of –5 – 12i is -

(A) ± (–2 – 3i) (B) ± (3 – 2i) (C) ± (2 – 3i) (D) ± (–3 – 2i)

Sol. [C]

Proper Method

$\sqrt{-5-12i} = a + ib$

$-5 - 12i = (a^2 - b^2) + 2iab$

$\Rightarrow a^2 - b^2 = -5$ ...(1)

and $2ab = -12$ ...(2)

$a^2 + b^2 = \pm\sqrt{(a^2-b^2)^2 + 4a^2b^2}$

$a^2 + b^2 = \pm\sqrt{25+144}$

$a^2 + b^2 = \pm 13$ ...(3)

Short Trick

$\sqrt{-5-12i}$

↓

6

↙ ↘

2 3

Ans. = ± (2 – 3i)

Solving (1) & (3) $\Rightarrow 2a^2 = -5 \pm 13$ $\Rightarrow a^2 = 4$ $\Rightarrow \boxed{a = \pm 2}$ and $b = \mp 3$ $\therefore \sqrt{5-12i} = \pm(2-3i)$	

2. Application of triangle inequality to find minimum or maximum value

(i) $|z_1 \pm z_2| \geq ||z_1| - |z_2||$

(ii) $|z_1 \pm z_2| \leq |z_1| + |z_2|$

Trick -2 If $|z + z_1| \leq x$ (Real number) then to find maximum and minimum value of $|z + z_2|$ we write $|z + z_2|$ such that $|z + z_2| = |(z + z_1) \pm z_3|$

Now, max. value of $|z + z_2| = |x| + |z_3|$

min. value of $|z + z_2| = ||x| - |z_3||$

Q.3 If $|z + 4| \leq 3$ then maximum and minimum value of $|z + 1|$ is -

(A) 4, 0 (B) 10, 2 (C) 6, 0 (D) 4, 2

Sol. **[C]**

Proper Method	**Short Trick**
$|z + 1| = |(z + 4) - 3|$ $|(z + 4) - 3| \leq |z + 4| + |3|$ $\leq 3 + |3|$ ≤ 6 and $|(z + 4) - 3| \geq |z + 4| - |3|$ $\geq 3 - |3|$ ≥ 0	$|z + 1| = |(z + 4) - 3|$ $|z + 1|_{max} = |3| + |-3| = 6$ $|z + 1|_{min} = |3| - |-3| = 0$

Q.4 If $|z + 4i| \leq 3$ then maximum and minimum value of $|z + 3|$ is -

(A) 8, 2 (B) 6, 2 (C) 8, 4 (D) 8, 0

Sol. **[A]**

Proper Method	**Short Trick**
$|z + 3| = |(z + 4i) + (3 - 4i)|$ $|z + 3| \leq |z + 4i| + |3 - 4i|$ $\leq 3 + 5$ ≤ 8 and $|z + 3| \geq ||z + 4i| - |3 - 4i||$ $\geq |3 - 5|$ ≥ 2	$|z + 3| = |(z + 4i) + (3 - 4i)|$ $|z + 3|_{max} = 3 + |3 - 4i| = 8$ $|z + 3|_{min} = |3 - |3 - 4i|| = 2$

Q.5 If $|z - 3 + 2i| \leq 4$ then the difference between the greatest and least value of $|z|$ is –

(A) $\sqrt{13}$ (B) $2\sqrt{13}$ (C) 8 (D) $4 + \sqrt{13}$

Sol. **[B]**

Proper Method	**Short Trick**
$|z| = |(z - 3 + 2i) + (3 - 2i)|$ $|z| \leq |z - 3 + 2i| + |3 - 2i|$ $|z| \leq 4 + \sqrt{13}$	$|z| = |z - 3 + 2i + (3 - 2i)|$ $|z|_{max} = 4 + \sqrt{13}$

and $\|z\| \geq \|z-3+2i\| - \|3-2i\|$ $\|z\| \geq 4-\sqrt{13}$ Difference between the greater and least value of $\|z\| = (4+\sqrt{13})-(4-\sqrt{13})$ $= 2\sqrt{13}$	$\|z\|_{min} = 4-\sqrt{13}$ Difference $= 2\sqrt{13}$

Trick -3 If $\left|z+\frac{1}{z}\right| = a$, then the greatest and least value of $|z|$ are respectively $\frac{a+\sqrt{a^2+4}}{2}$ and $\frac{-a+\sqrt{a^2+4}}{2}$

Q.6 If $\left|z-\frac{1}{z}\right| = 1$, then:

(A) $|z|_{max} = \frac{1+\sqrt{5}}{2}$ (B) $|z|_{min} = \frac{1+\sqrt{5}}{2}$

(C) $|z|_{max} = \frac{-1+\sqrt{5}}{2}$ (D) None of these

Sol. [A]

Proper Method	Short Trick
$\left|z-\frac{1}{z}\right| = 1 \Rightarrow \left\|\,|z|-\frac{1}{|z|}\right\| \leq 1$ $\Rightarrow -1 \leq \|z\| - 1/\|z\| \leq 1 \Rightarrow -\|z\| \leq \|z\|^2-1 \leq \|z\|$ $\therefore\ \|z\|^2+\|z\|-1 \geq 0$ and $\|z\|^2-\|z\|-1 \leq 0$ $\Rightarrow \frac{\sqrt{5}-1}{2} \leq \|z\| \leq \frac{\sqrt{5}+1}{2}$ $\therefore$ max. $\|z\| = \frac{\sqrt{5}+1}{2}$ and min. $\|z\| = \frac{\sqrt{5}-1}{2}$	$\|z\|_{max} = \frac{1+\sqrt{5}}{2}$

Trick -4 Method of substitution.

Q.7 If $z = x - iy$ and $z^{1/3} = p + iq$, then the value of $\frac{\left(\frac{x}{p}+\frac{y}{q}\right)}{(p^2+q^2)}$ is equal to :

(A) 2 (B) – 1 (C) 1 (D) – 2

Sol. [D]

Proper Method	Short Trick
$z = x - iy$ and $z^{1/3} = p + iq$ So, $z = (p+iq)^3$ $\Rightarrow x - iy = p^3 - iq^3 + 3p(iq)(p+iq)$ $\Rightarrow x - iy = p^3 - iq^3 + 3ip^2q - 3pq^2$ Comparing real and imaginary part $\Rightarrow x = p^3 - 3pq^2$ and $y = q^3 - 3p^2q$ $\Rightarrow x = p(p^2-3q^2)$ and $y = q(q^2-3p^2)$	Let $x = 1$, $y = 0$ then $z = 1$ and $z^{1/3} = 1^{1/3} = \omega$ (let) $\frac{-1}{2}+i\frac{\sqrt{3}}{2} = p + iq$ $\Rightarrow p = -\frac{1}{2}$ & $q = \frac{\sqrt{3}}{2}$

$\Rightarrow \dfrac{x}{p} = p^2 - 3q^2$ ...(1)

and $\dfrac{x}{q} = q^2 - 3p^2$ ...(2)

adding (1) & (2)

$\Rightarrow \dfrac{x}{p} = \dfrac{x}{q} = -2(p^2 + q^2)$

$\Rightarrow \dfrac{\frac{x}{p}+\frac{y}{q}}{p^2+q^2} = -2$

so $\dfrac{x}{p} + \dfrac{y}{q} + (p^2 + q^2) = -2$

Q.8 Let z and w be non-zero complex numbers such that $|z| = |w|$ and $\arg(z) + \arg(w) = \pi$. Then z =

(A) w (B) $-w$ (C) $\overline{w}$ (D) $-\overline{w}$

Sol. [D]

Proper Method

Let $\arg(z) = \theta$ then $\arg(w) = \pi - \theta$

$z = |z|(\cos\theta + i\sin\theta)$

$w = |w|\cos\{(\pi - \theta) + i\sin(\pi - \theta)\}$

$w = |w|(-\cos\theta + i\sin\theta)$

$w = -|w|(\cos\theta - i\sin\theta)$

$w = -|z|(\cos\theta - i\sin\theta)$

$w = -\overline{z}$

or $z = -\overline{w}$

Short Trick

$\therefore$ Let $z = 1 + i \Rightarrow \arg(z) = \dfrac{\pi}{4}$

and $w = -1 + i \Rightarrow \arg(w) = \dfrac{3\pi}{4}$.

$\therefore z = 1 + i = -(-1 - i) = -\overline{w}$

Q.9 If $|z| = 1$ and $z \neq \pm 1$ then all the values of $\dfrac{z}{1-z^2}$ lies on:

(A) A line which does not pass through origin. (B) $|z| = \sqrt{2}$

(C) x–axis (D) y–axis

Sol. [D]

Proper Method

Let $z = x + iy$

$\dfrac{z}{1-z^2} = \dfrac{x+iy}{1-(x+iy)^2}$

$= \dfrac{x+iy}{1-(x^2-y^2+2ixy)}$

$= \dfrac{x+iy}{(1-x^2+y^2-2ixy)}$

$= \dfrac{x+iy}{(1-x^2+y^2)-2ixy} \times \dfrac{(1-x^2+y^2)+2ixy}{(1-x^2+y^2)-2ixy}$

$= \dfrac{x(1-x^2+y^2)+2ix^2y+iy(1-x^2+y^2)-2xy^2}{(1-x^2+y^2)+4x^2y^2}$

$= \dfrac{x(1-x^2+y^2-2y^2)+iy(2x^2+1x^2+y^2)}{(1-x^2+y^2)+4x^2y^2}$

Short Trick

$\because |z| = 1$ therefore let $z = i$

then $\dfrac{z}{1-z^2} = \dfrac{i}{1-i^2} = \dfrac{i}{2}$ (purely imaginary)

so all the values of given expression lie on y–axis.

$$= \frac{x(1-(x^2+y^2))+iy(1+x^2+y^2)}{(1-x^2+y^2)+4x^2y^2}$$

Given that $|z| = 1 \Rightarrow x^2 + y^2 = 1$

$$= \frac{2iy}{(1-x^2+y^2)^2+4x^2y^2} \text{ purely imaginary}$$

$\therefore$ lies on y-axis

Q.10 If α and β are different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right|$ is equal to -

(A) 0 (B) 1/2 (C) 1 (D) 2

Sol. **[C]**

Proper Method

$|\beta| = 1 \Rightarrow \bar{\beta} = \frac{1}{\beta}$

$$\Rightarrow \frac{|\beta-\alpha|}{|1-\bar{\alpha}\beta|} = \frac{|\beta-\alpha|}{\left|1-\frac{\bar{\alpha}}{\bar{\beta}}\right|}$$

$$= |\bar{\beta}| \cdot \frac{|\beta-\alpha|}{|\bar{\beta}-\bar{\alpha}|} = |\bar{\beta}| = 1$$

Short Trick

Let $\alpha = i$ and $\beta = -i$

$$\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right| = \left|\frac{-i-i}{1-i^2}\right| = 1$$

Q.11 If $|z| = 1$, $z \neq -1$ and $w = \frac{z-1}{z+1}$ then real part of w = ?

(A) $\frac{-1}{|z+1|^2}$ (B) $\frac{1}{|z+1|^2}$ (C) $\frac{2}{|z+1|^2}$ (D) 0

Sol. **[D]**

Proper Method

$|z| = 1 \Rightarrow \bar{z} = \frac{1}{z}$

$w = \frac{z-1}{z+1}$

$$\bar{w} = \frac{\bar{z}-1}{\bar{z}+1} = \frac{\frac{1}{z}-1}{\frac{1}{z}+1} \Rightarrow \bar{w} = \frac{1-z}{1+z}$$

$\therefore w + \bar{w} = 0$

$\therefore$ Re(w) = 0

Short Trick

Let z = i

Then $w = \frac{i-1}{i+1} \times \frac{i-1}{i-1} = i$

$\therefore$ Re(w) = 0

Q.12 Let $z_1, z_2, \ldots, z_n$ be in G.P. with first term as unity such that $z_1 + z_2 + \ldots + z_n = 0$. Now if $z_1, z_2, \ldots z_n$ represents vertices of a polygon, then distance between the incentre and circumcentre of the polygon is :

(A) 2 (B) 1 (C) 0 (D) –1

Sol. **[C]**

Proper Method	**Short Trick**
Let r be the common ratio of the G.P. $z_1, z_2, \ldots, z_n$ with $z_1 = 1$. $z_1 + z_2 + \ldots + z_n = 0 \Rightarrow \frac{(r^n - 1)}{r-1} = 0\ (r \neq 1)$ $\Rightarrow r^n = 1 = \lvert r \rvert = 1$ $\Rightarrow \lvert z_k \rvert = \lvert r^k \rvert = \lvert r \rvert^k = 1$ $\Rightarrow$ The polygon $z_1, z_2, \ldots, z_n$ is regular $\Rightarrow$ incentre and circumcentre are same $\Rightarrow$ distance between incentre and circumcentre is zero	Let $z_1 = 1$, $z_2 = \omega$, $z_3 = \omega^2$ Now z_1, z_2, z_3 are the vertices of equilateral triangle then distance between incentre and circumcentre is 0

Trick -5 Combination of method of substitution and balancing.

Q.13 If ω is a imaginary roots of unity then the value of following series
$(2 - \omega)(2 - \omega^2) + 2(3 - \omega)(3 - \omega^2) + 3(4 - \omega)(4 - \omega^2) + \ldots\ldots\ldots + (n - 1)(n - \omega)(n - \omega^2)$ is :

(A) $\frac{n}{4}(n-1)(n^2+3n+4)$ (B) $\frac{n}{2}(n-1)(n^2+3n+4)$

(C) $\frac{n}{4}(n+1)(n^2+3n+4)$ (D) $\frac{n}{2}(n+1)(n^2+3n+4)$

Sol. **[A]**

Proper Method	**Short Trick**
$\text{sum} = \sum_{r=1}^{n-1} r(r+1-\omega)(r+1-\omega^2)$ $= \sum_{r=1}^{n-1} r[(r+1)^2 - (r+1)(\omega+\omega^2) + \omega^3]$ $= \sum_{r=1}^{n-1} (r^3 + 3r^2 + 3r)$ $= \sum_{r=1}^{n-1} r^3 + 3\sum_{r=1}^{n-1} r^2 + 3\sum_{r=1}^{n-1} r$ $= \left[\frac{1}{2}n(n-1)\right]^2 + 3\,\frac{(n-1)(n)(2n-1)}{6} + \frac{3}{2}(n-1)(n)$ $= \frac{n(n-1)}{2}\left(\frac{n(n-1)}{2} + (2n-1) + 3\right)$ $= \frac{n(n-1)}{2}\left\{\frac{n^2 - n + 4n - 2 + 6}{2}\right\}$ $= \frac{n(n-1)}{4}(n^2 + 3n + 4)$	Let $n = 2$, L.H.S $= (2 - \omega)(2 - \omega^2)$ $= 4 - 2(\omega^2 + \omega) + \omega^3 = 4 - 2(-1) + 1 = 7$ Now put $n = 2$ in options then option (A) will give $\frac{2}{4}(2 - 1)(2^2 + 3.2 + 4) = 7$

Q.14 The set of all $\alpha \in R$, for which $w = \dfrac{1+(1-8\alpha)z}{1-z}$ is a purely imaginary number, for all $z \in C$ satisfying $|z| = 1$ and $Re(z) \neq 1$, is

(A) {0} (B) an empty set (C) $\left\{0, \frac{1}{4}, -\frac{1}{4}\right\}$ (D) equal to R

Sol. [A]

Proper Method

$w = \dfrac{1+(1-8\alpha)z}{1-z}$

$w = \dfrac{1+(1-8\alpha)(x+iy)}{1-(x+iy)}$

$w = \dfrac{1+x(1-8\alpha)+iy(1-8\alpha)}{(1-x)-iy} \times \dfrac{(1-x)+iy}{(1-x)+iy}$

$w = \dfrac{\{1+x(1-8\alpha)\}(1-x)-y^2(1-8\alpha) + iy\{1+x(1-8\alpha)\}+iy(1-8\alpha)(1-x)}{(1-x)^2+y^2}$

given that w is purely imaginary $\therefore Re(w) = 0$

$\Rightarrow \{1+x(1-8\alpha)\}(1-x)-y^2(1-8\alpha) = 0$

$\Rightarrow 1-x+x(1-8\alpha)-x^2(1-8\alpha)-y^2(1-8\alpha) = 0$

$\Rightarrow 1-x+x-8\alpha x-(1-8\alpha)(x^2+y^2) = 0$

[given that $|z| = 1 \therefore x^2+y^2 = 1$]

$\Rightarrow 1-8\alpha x-1+8\alpha = 0$

$\Rightarrow 8\alpha(-x+1) = 0$

$\Rightarrow 8\alpha = 0 \quad (\because x \neq 1)$

$\Rightarrow \alpha = 0$

Short Trick

$|z| = 1$ and $Re(z) \neq 1$

So assume $z = i$

$w = \dfrac{1+(1-8\alpha)i}{1-i} \times \dfrac{1+i}{1+i}$

$= \dfrac{(1-1+8\alpha)+i(1+1-8\alpha)}{2}$

w is purely imaginary

$\therefore Re(w) = 0$

$\Rightarrow 8\alpha = 0 \Rightarrow \alpha = 0$

Q.15 If $c^2 + s^2 = 1$, then $\dfrac{1+c+is}{1+c-is} =$

(A) $c + is$ (B) $s + ic$ (C) $c - is$ (D) $s - ic$

Sol. [A]

Proper Method

$\because c^2 + s^2 = 1$

$\therefore (c + is)(c - is) = 1$

$\therefore c - is = \dfrac{1}{c+is}$

$\therefore \dfrac{1+(c+is)}{1+(c-is)} = \dfrac{1+(c+is)}{1+\left(\frac{1}{c+is}\right)}$

$= (c + is)$

Short Trick

Let $c + is = i$ ($\because c^2 + s^2 = 1$)

$\dfrac{1+c+is}{1+c-is} = \dfrac{1+i}{1-i} = i$

Q.16 If $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$, then xyz equals-

(A) $(a + b)^3$ (B) $a^3 - b^3$ (C) $(a + b)^3 + 3ab(a + b)$ (D) $a^3 + b^3$

Sol. [D]

Proper Method

$xyz = (a + b)(a\omega + b\omega^2)(a\omega^2 + b\omega)$

$= (a + b)[a^2 + b^2 + ab(\omega^2 + \omega)]$

$= (a + b)(a^2 + b^2 - ab) = a^3 + b^3$

Short Trick

Let $a = 1$ and $b = 1$

Then $x = 2$, $b = \omega + \omega^2 = -1$,

$c = \omega^2 + \omega = -1$

Now $xyz = 2$

Q.17 If $|z_1| = |z_2| .... = |z_n| = 1$, then $\left|\dfrac{z_1 + z_2 + ... + z_n}{z_1^{-1} + z_2^{-1} + ... + z_n^{-1}}\right|$ equals-

(A) 1/n (B) n (C) 1 (D) $|z_1 + z_2 + ...+ z_n|$

Sol. **[C]**

Proper Method	**Short Trick**
$\|z_1\| = \|z_2\| .... = \|z_n\| = 1$ $\therefore\ \bar{z}_1 = \dfrac{1}{z_1},\ \bar{z}_2 = \dfrac{1}{z_2}$ .... $= \dfrac{\|z_1 + z_2 .... + z_n\|}{\|\bar{z}_1 + \bar{z}_2 + ... + \bar{z}_n\|}$ = 1 (they are conjugate)	Let n = 2, then $\|z_1\| = \|z_2\| = 1$ also $z = 1$ and $z_2 = i$ $\left\|\dfrac{z_1 + z_2}{z_1^{-1} + z_2^{-1}}\right\| = \left\|\dfrac{1+i}{1-i}\right\| = 1$

Q.18 The value of the expression

$\left(1+\frac{1}{\omega}\right)\left(1+\frac{1}{\omega^2}\right) + \left(2+\frac{1}{\omega}\right)\left(2+\frac{1}{\omega^2}\right) + \left(3+\frac{1}{\omega}\right)\left(3+\frac{1}{\omega^2}\right) + ... + \left(n+\frac{1}{\omega}\right)\left(n+\frac{1}{\omega^2}\right)$, where ω is an imaginary cube root of unity is-

(A) $\dfrac{n(n^2+3)}{3}$ (B) $\dfrac{n(n^2+2)}{3}$ (C) $\dfrac{n(n^2+1)}{3}$ (D) None of these

Sol. **[B]**

Proper Method	**Short Trick**
$T_r = \left(r+\dfrac{1}{\omega}\right)\left(r+\dfrac{1}{\omega^2}\right) = (r + \omega^2)(r + w)$ $\therefore\ T_r = r^2 + 1 - r$ $\therefore\ \sum\limits_{r=1}^{n} T_r = \Sigma r^2 - \Sigma r + \Sigma 1$ $= \dfrac{n(n+1)(2n+1)}{6} - \dfrac{n(n+1)}{2} + n$ $= \dfrac{n}{6}[2n^2 + 3n + 1 - 3n - 3 + 6]$ $= \dfrac{n}{6}(2n^2 + 4)$ $= \dfrac{n}{3}(n^2 + 2)$	Put n = 1 $\Rightarrow \left(1+\dfrac{1}{\omega}\right)\left(1+\dfrac{1}{\omega^2}\right)$ $= (1 + \omega^2)(1 + \omega)$ $= (-\omega)(-\omega^2) = 1$ Now put n = 1 in option then option (B) $\dfrac{n(n^2+2)}{3}$ is the correct answer.

Q.19 If 1, $a_1, a_2, a_3, ... a_{n-1}$ are n^{th} roots of unity then $\dfrac{1}{1-a_1} + \dfrac{1}{1-a_2} + ... + \dfrac{1}{1-a_{n-1}}$ equal to -

(A) $\dfrac{n-1}{2}$ (B) $\dfrac{n}{2}$ (C) $\dfrac{2^n-1}{n}$ (D) none

Sol. [A]

Proper Method

$z = (1)^{1/n}$

$z^n - 1 = 0$ have roots $1, a_1, a_2, \ldots, a_{n-1}$

Let $z = \frac{1}{1-z}$

$\frac{1}{z} = 1 - z$

$z = 1 - \frac{1}{z} = \frac{z-1}{z}$

Now replace z with $\frac{z-1}{z}$ equation becomes

$\left(\frac{z-1}{z}\right)^n - 1 = 0$

$(z-1)^n - z^n = 0$

$-{}^nC_1(z)^{n-1} + {}^nC_2(z)^{n-2} \ldots = 0$

$\therefore$ sum of roots

$= \frac{1}{1-a_1} + \frac{1}{1-a_2} + \ldots\ldots + \frac{1}{1-a_{n-1}}$

$= (-1)^1 \frac{{}^nC_2}{-{}^nC_1} = \frac{n(n-1)/2!}{n}$

$= \frac{n-1}{2}$

Short Trick

Let n = 3 then $a_1 = \omega$, $a_2 = \omega^2$

$\frac{1}{1-a_1} + \frac{1}{1-a_2} = \frac{1}{1-\omega} + \frac{1}{1-\omega^2}$

$= \frac{1-\omega^2+1-\omega}{(1-\omega^2-\omega+\omega^3)} = \frac{2-(\omega+\omega^2)}{2-(\omega+\omega^2)} = 1$

Now, put n = 3 in option, then option (A) $\frac{n-1}{2}$ is the correct answer.

Q.20 $1, z_1, z_2, z_3, \ldots z_{n-1}$ are nth roots of unity then the value of $\frac{1}{3-z_1} + \frac{1}{3-z_2} + \ldots + \frac{1}{3-z_{n-1}}$ is equal to:

(A) $\frac{n3^{n-1}}{3^n-1} + \frac{1}{2}$ (B) $\frac{n3^{n-1}}{3^n-1} - 1$ (C) $\frac{n3^{n-1}}{3^n-1} + 1$ (D) None of these

Sol. [D]

Proper Method

$1, z_1, z_2, \ldots, z_n$ n, nth roots of unity

$\Rightarrow z^n - 1 = (z-1)(z-z_1)(z-z_2)\ldots(z-z_{n-1})$

$\Rightarrow \log(z^n - 1) = \log(z-1) + \log(z-z_2) + \ldots + \log(z-z_{n-1})$

Differentiating both sides, we get

$\frac{nz^{n-1}}{z^n-1} = \frac{1}{z-1} + \frac{1}{z-z_1} + \frac{1}{z-z_2} + \ldots + \frac{1}{z-z_{n-1}}$

Putting z = 3, we get

$\frac{n3^{n-1}}{3^n-1} = \frac{1}{2} + \frac{1}{3-z_1} + \frac{1}{3-z_2} + \ldots + \frac{1}{3-z_{n-1}}$

$\Rightarrow \frac{1}{3-z_1} + \frac{1}{3-z_2} + \ldots + \frac{1}{3-z_{n-1}} = \frac{n3^{n-1}}{3^n-1} - \frac{1}{2}$

Short Trick

Let n = 3, then $z_1 = \omega$, $z_2 = \omega^2$

Now $= \frac{1}{3-\omega} + \frac{1}{3-\omega^2}$

$= \frac{3-\omega^2+3-\omega}{9-3(\omega+\omega^2)+\omega^3} = \frac{7}{13}$

in option we put n = 3 then (D) is correct

4

Binomial Theorem

KEY CONCEPTS

1. Binomial Expression & Theorem

An algebraic expression containing two terms is called a **binomial expression.**
For example, $2x + 3$, $x^2 - x/3$, $x + a$ etc. are
The rule by which any power of a binomial can be expanded is called the **Binomial Theorem.**

2. Binomial Theorem for positive Integral Index

If x and a are two real numbers and n is a positive integer then
$(x + a)^n = {}^nC_0 x^n a^0 + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + ... + {}^nC_r x^{n-r} a^r + ... + {}^nC_n x^0 a^n$.
Where ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, ... {}^nC_r ....$ are called **binomial coefficients** which can be denoted by $C_0, C_1, C_2, C_3 ... C_r$

3. General Term

In the expansion of $(x + a)^n$ the $(r + 1)^{th}$ term is called the **general term** which can be represented by T_{r+1}.
$T_{r+1} = {}^nC_r x^{n-r} a^r$

(i) General term in the expansion of $(x - a)^n$ is
$T_{r+1} = (-1)^r \ {}^nC_r x^{n-r} a^r$

(ii) General term in the expansion of $(1 + x)^n$ is
$T_{r+1} = {}^nC_r x^r$

(iii) General term in the expansion of $(1 - x)^n$ is
$T_{r+1} = (-1)^r \ {}^nC_r x^r$

4. Properties of Binomial Expansion $(x + a)^n$

(i) There are $(n + 1)$ terms in the expansion

(ii) The binomial coefficients of term equidistant from the beginning and the end are equal as ${}^nC_r = {}^nC_{n-r}$

(iii) The p^{th} term from the end of the expansion is the $(n - p + 2)^{th}$ term from the beginning.

5. Middle term in the Expansion of $(x + a)^n$

(i) If n is even then middle term $= \left(\frac{n}{2}+1\right)^{th}$ term.

(ii) If n is odd then middle terms are $= \left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ term. Binomial coefficient of middle term is the greatest Binomial coefficient.

6. Greatest term in the Expansion of $(x + a)^n$

Calculate $\dfrac{n+1}{\left|\dfrac{x}{a}\right|+1} = k$ (say)

(i) If k is an integer then T_k and T_{k+1} are the numerically greatest terms.

(ii) If k is not an integer. Let m is its integral part then T_{m+1} is the numerically greatest term.

7. Binomial Coefficients & Their Properties

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_rx^r + ... + {}^nC_nx^n$.

Let us denote ${}^nC_0, {}^nC_1, {}^nC_2 ... {}^nC_n$ by $C_0, C_1, C_2 ... C_n$.

These quantities are called binomial coefficients and have the following properties.

(i) $C_0 + C_1 + C_2 + ... + C_n = \sum_{r=0}^{n} {}^nC_r = 2^n$

(ii) $C_0 - C_1 + C_2 - C_3 + ...(-1)^n C_n = \sum_{r=0}^{n} (-1)^r \; {}^nC_r = 0$

(iii) $C_0 + C_2 + C_4 + ... = 2^{n-1}$

(iv) $C_1 + C_3 + C_5 + ... = 2^{n-1}$

(v) $C_0 + 2C_1 + 3C_2 + ... + (n+1)C_n = 2^{n-1}(n+2)$

(ix) $C_0 + \dfrac{C_1}{2} + \dfrac{C_2}{3} + ... + \dfrac{C_n}{n+1} = \dfrac{2^{n+1}-1}{n+1}$

(x) $C_0 - \dfrac{C_1}{2} + \dfrac{C_2}{3} - \dfrac{C_3}{4} + ... + \dfrac{(-1)^n.C_n}{n+1} = \dfrac{1}{(n+1)}$

(xi) $C_0C_r + C_1C_{r+1} + ... + C_{n-r}C_n = {}^{2n}C_{n-r}$

(xii) $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2 = {}^{2n}C_n$.

(xiii) $C_0C_1 + C_1C_2 + ... + C_{n-1}C_n = {}^{2n}C_{n-1}$

8. Binomial Theorem for any Index

When n is a negative integer or a fraction then the expansion of the binomial is possible only where

(i) Its first term is 1

(ii) Its second term is numerically less than 1

i.e. $|x| < 1$. Then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + ... + \frac{n(n-1).....(n-r+1)}{r!}x^r + ... \infty$$

General term :

$$T_{r+1} = \frac{n(n-1)(n-2)........(n-r+1)}{r!}x^r$$

9. Some Important Expansions

(i) $(1-x)^{-1} = 1 + x + x^2 + x^3 + ... + x^r + ...$ General term $T_{r+1} = x^r$

(ii) $(1+x)^{-1} = 1 - x + x^2 - x^3 + ... (-x)^r + ...$ General term $T_{r+1} = (-x)^r$

(iii) $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + ... + (r+1)x^r + ...$ General term $T_{r+1} = (r+1)x^r$

(iv) $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + ... + (r+1)(-x)^r + ...$ General term $T_{r+1} = (r+1)(-x)^r$.

10. R-f Factor Relation

If $(\sqrt{P}+Q)^n = I + f$ where I and n are the positive integers, and $0 \le f < 1$ then

(i) $(I + f)f = k^n$, If n is odd and $P - Q^2 = k > 0$

(ii) $(I + f)(1 - f) = k^n$, If n is even and $\sqrt{P} - Q < 1$

11. Multinomial Expansion

The number of distinct terms in the expansion of $(x_1 + x_2 + x_3 + ... + x_r)^m$ is given by ${}^{m+r-1}C_{r-1}$.

If $n \in N$ then the general terms of the multinomial expansion $(x_1 + x_2 + ...x_k)^n$ is

$$\frac{\lfloor n}{\lfloor a_1 \lfloor a_2 \lfloor a_3 ..... \lfloor a_k} \cdot x_1^{a_1} . x_2^{a_2} ......x_k^{a_k}$$

where $a_1 + a_2 + a_3 + ...a_k = n$.

12. Number of terms in the Expansion of $(x + y + z)^n$

$= {}^{n+2}C_2$

SHORT TRICKS

Trick -1 If the coefficient of T_r, T_{r+1}, T_{r+2} terms in the expansion of $(1 + x)^n$ are in A.P., then the value of r is $r = \frac{n \pm \sqrt{n+2}}{2}\ \forall\ n \in N$

Q.1 If the coefficients of T_r, T_{r+1}, T_{r+2} terms of $(1+x)^{14}$ are in A.P., then r -

(A) 6 (B) 7 (C) 8 (D) 9

Sol. [D]

Proper Method	Short Trick
Given ${}^nC_{r-1}, {}^nC_r, {}^nC_{r+1}$: AP $\because 2.{}^nC_r = nC_{r-1} + nC_{r+1}$ $\Rightarrow 2 = \frac{{}^nC_{r-1}}{{}^nC_r} + \frac{{}^nC_{r+1}}{{}^nC_r}$ $\Rightarrow 2 = \frac{1}{\left(\frac{n-r+1}{r}\right)} + \left(\frac{n-r-1+1}{r+1}\right)$ $\Rightarrow 2 = \frac{r}{n-r+1} + \frac{n-r}{r+1}$ Here n = 14 $\Rightarrow 2 = \frac{r}{15-r} + \frac{14-r}{r+1}$ $\Rightarrow 2(14r - r^2 + 15) = r^2 + r + 210 - 15r - 14r + r^2$ $\Rightarrow 4r^2 - 56r + 180 = 0 \Rightarrow r^2 - 14r + 45 = 0$ $\therefore r = 5, 9$	$r = \frac{14 \pm \sqrt{14+2}}{2} = 9$ or 5

Q.2 If the coefficients of r^{th}, $(r + 1)^{th}$ and $(r + 2)^{th}$ terms in the binomial expansion of $(1 + y)^m$ are in A.P., then m and r satisfy the equation - **[AIEEE-2005]**

(A) $m^2 - m(4r - 1) + 4r^2 - 2 = 0$ (B) $m^2 - m(4r + 1) + 4r^2 + 2 = 0$

(C) $m^2 - m(4r + 1) + 4r^2 - 2 = 0$ (D) $m^2 - m(4r - 1) + 4r^2 + 2 = 0$

Sol. [C]

Proper Method	Short Trick
Given ${}^mC_{r-1}, {}^mC_r, {}^mC_{r+1}$ are in AP $\Rightarrow 2.{}^mC_r = {}^mC_{r-1} + {}^mC_{r+1}$ $\Rightarrow 2 = \left(\frac{{}^mC_{r-1}}{{}^mC_r}\right) + \left(\frac{{}^mC_{r+1}}{{}^mC_r}\right)$ $\Rightarrow 2 = \frac{1}{\left(\frac{m-r+1}{r}\right)} + \left(\frac{m-r-1+1}{r+1}\right)$ $\Rightarrow 2 = \frac{1}{\left(\frac{m-r+1}{r}\right)} + \left(\frac{m-r-1+1}{r+1}\right)$	$r = \frac{m \pm \sqrt{m+2}}{2}$ $\Rightarrow (2r - m)^2 = (m + 2)$ $\Rightarrow m^2 - m(4r + 1) + 4r^2 - 2 = 0$

$\Rightarrow 2 = \frac{r}{m-r+1} + \frac{m-r}{r+1}$ $\Rightarrow 2(m-r+1)(r+1) = r(r+1) + (m-r)(m-r+1)$ $\Rightarrow 2mr + 2m - 2r^2 - 2r + 2r + 2$ $= r^2 + r + m^2 - mr + m - mr + r^2 - r$ $\Rightarrow m^2 - 4mr - m + 4r^2 - 2 = 0$	

Q.3 If in the expansion of $(1+y)^n$, the coefficient of 5^{th}, 6^{th} and 7^{th} terms are in A.P., then n is equal to-

(A) 7, 11 (B) 7, 14 (C) 8, 16 (D) 7, 15

Sol. [B]

Proper Method	**Short Trick**
As given ${}^nC_4$, ${}^nC_5$, ${}^nC_6$ are in AP. $\Rightarrow {}^nC_4 + {}^nC_6 = 2.\,{}^nC_5$ $\Rightarrow \frac{n!}{(n-4)!\,4!} + \frac{n!}{(n-6)!\,6!} = 2\,\frac{n!}{(n-5)!\,5!}$ $\Rightarrow 30 + (n-5)(n-4) = 2.6\,(n-4)$ $\Rightarrow n^2 - 21n + 98 = 0$ $\Rightarrow (n-7)(n-14) = 0$ $\therefore n = 7, 14$	$5 = \frac{n \pm \sqrt{n+2}}{2}$ $\Rightarrow (10-n)^2 = n + 2$ $\Rightarrow n^2 - 21n + 98 = 0$ $\Rightarrow n = 7, 14$

Trick -2 Method of substitution

Q.4 The fractional part of $\frac{2^{4n}}{15}$ is:

(A) $\frac{1}{15}$ (B) $\frac{2}{15}$ (C) $\frac{4}{15}$ (D) $\frac{7}{15}$

Sol. [A]

Proper Method	**Short Trick**
$\frac{2^{4n}}{15} = \frac{16^n}{15} = \frac{(1+15)^n}{15}$ $= \frac{1 + {}^nC_1 15 + {}^nC_2 15^2 + ... + {}^nC_n 15^n}{15}$ $= \frac{1+15k}{15}$, where $k \in N$ $= \frac{1}{15} + k$ $\therefore \left\{\frac{2^{4n}}{15}\right\} = \frac{1}{15}$	Put $n = 1$ $\Rightarrow$ fractional part of $\frac{16}{15}$ is $\frac{1}{15}$

Q.5 If the coefficients of four consecutive terms in the expansion of $(1+x)^n$ are a_1, a_2, a_3 and a_4, then $\frac{a_1}{a_1+a_2}, \frac{a_2}{a_2+a_3}, \frac{a_3}{a_3+a_4}$ are in -

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Sol. [A]

Proper Method	Short Trick
$a_1 = {}^nC_r$, $a_2 = {}^nC_{r+1}$, $a_3 = {}^nC_{r+2}$, $a_4 = {}^nC_{r+3}$ then $\frac{a_1}{a_1+a_2}, \frac{a_2}{a_2+a_3}, \frac{a_3}{a_3+a_4}$ $\Rightarrow \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}}, \frac{{}^nC_{r+1}}{{}^nC_{r+1} + {}^nC_{r+2}}, \frac{{}^nC_{r+2}}{{}^nC_{r+2} + {}^nC_{r+3}}$ $\Rightarrow \frac{{}^nC_r}{{}^{n+1}C_{r+1}}, \frac{{}^nC_{r+1}}{{}^{n+1}C_{r+2}}, \frac{{}^nC_{r+2}}{{}^{n+1}C_{r+3}}$ using ${}^nC_r = \frac{n}{r} . {}^{n-1}C_{r-1}$ we get $\Rightarrow \frac{1}{\left(\frac{n+1}{r+1}\right)}, \frac{1}{\left(\frac{n+1}{r+2}\right)}, \frac{1}{\left(\frac{n+1}{r+3}\right)}$ $\Rightarrow \frac{r+1}{n+1}, \frac{r+2}{n+1}, \frac{r+3}{n+1}$ $\Rightarrow (r + 1), (r + 2), (r + 3)$: AP	Put $n = 3$ $\Rightarrow (1+x)^n = 1 + 3x + 3x^2 + x^3$ Here $a_1 = 1$, $a_2 = 3$, $a_3 = 3$, $a_4 = 1$ So, $\frac{a_1}{a_1+a_2} = \frac{1}{4}$, $\frac{a_2}{a_2+a_3} = \frac{3}{6}$, $\frac{a_3}{a_3+a_4} = \frac{3}{4}$ $\frac{1}{4}, \frac{3}{6}, \frac{3}{4}$ are in A.P.

Trick -3 Combination of method of substitution and balancing

Q.6 If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $C_0 + 2C_1 + 3C_2 + ... + (n+1)\ C_n$ is equal to-
(A) $2^{n-1}\ (n+2)$ (B) $2^n(n+1)$ (C) $2^{n-1}\ (n+1)$ (D) $2^n\ (n+2)$

Sol. [A]

Proper Method	Short Trick
Putting $x = 1$ in the given expansion, we get $C_0 + C_1 + C_2 + C_3 + ...C_n = 2^n$ ...(1) Now, differentiating the given expansion with respect to x and then putting $x = 1$, we get $C_1 + 2C_2 + 3C_3 + ... + nC_n = n.2^{n-1}$ Given Exp. $= C_0 + 2C_1 + 3C_2 + ... + (n + 1)\ C_n$ $= (C_0 + C_1 + C_2 + ... + C_n) + (C_1 + 2C_2 + 3C_3 + ... + nC_n)$ $= 2^n + n.\ 2^{n-1}$ [from (1) and (2)] $= 2^{n-1}\ (n + 2)$	Put $n = 2$ $\Rightarrow C_0 + 2.C_1 + 3.C_2$ $= 1 + 2(2) + 3(1) = 8$ Now put $n = 2$ in options then (A) $2^{n-1}(n + 2)$ is correct option

Q.7 If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $\frac{(C_0+C_1)(C_1+C_2)...(C_{n-1}+C_n)}{C_1C_2...C_n}$ equals-

(A) $\frac{n^n}{(n+1)!}$ (B) $\frac{(n+1)^n}{n!}$ (C) $\frac{n^n}{n!}$ (D) None of these

Sol. [B]

Proper Method	Short Trick
The given expression $= \frac{C_0C_1C_2....C_{n-1}}{C_1C_2....C_n} \cdot \left(1+\frac{C_1}{C_0}\right)\left(1+\frac{C_2}{C_1}\right)\left(1+\frac{C_3}{C_2}\right) \cdots \left(1+\frac{C_n}{C_{n-1}}\right)$ $= \left(1+\frac{n}{1}\right)\left(1+\frac{n-1}{2}\right)\left(1+\frac{n-2}{3}\right)...\left(1+\frac{1}{n}\right)$ $(\because C_0 = C_n)$ $= \frac{(n+1)^n}{n!}$	Put $n = 2$ $\Rightarrow \frac{(C_0+C_1)(C_1+C_2)}{C_1C_2}$ $= \frac{(1+2)(2+1)}{(2)(1)} = \frac{9}{2}$ Now put $n = 2$ in option then (B) $\frac{(n+1)^n}{n!}$ is correct option.

Q.8 If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then $3C_0 - 5C_1 + 7C_2 + \ldots + (-1)^n (2n+3) C_n$ equals-

(A) 1 (B) $2(2n+3)\,2^n$ (C) $(2n+3)\,2^{n-1}$ (D) 0

Sol. [D]

Proper Method

We have

$3C_0 - 5C_1 + 7C_2 + \ldots + (-1)^n (2n+3) C_n$

$= 3C_0 - 3C_1 + 3C_2 + \ldots + (-1)^n 3C_n - 2C_1 + 4C_2 + \ldots + (-1)^n 2nC_n$

$= 3(C_0 - C_1 + C_2 + \ldots + (-1)^n C_n) - 2(C_1 - 2C_2 + \ldots (-1)^n nC_n)$

$= 3 \times 0 - 2 \times 0 = 0.$

Short Trick

Put $n = 2$

$\Rightarrow 3C_0 - 5C_1 + 7C_2$

$= 3(1) - 5(2) + 7(1) = 0$

So, (D) is the correct option.

Q.9 If $(1+x)^n = 1 + C_1x + C_2x^2 + \ldots + C_nx^n$, then $C_1 + C_3 + C_5 + \ldots$ is equal to-

(A) 2^n (B) $2^n - 1$ (C) $2^n + 1$ (D) 2^{n-1}

Sol. [D]

Proper Method

put $x = 1 \Rightarrow 2^n = 1 + C_1 + C_2 + C_3 \ldots$

put $x = -1 \Rightarrow 0 = 1 - C_1 + C_2 - C_3 \ldots$

subtracting, we get

$2^n = 2\,(C_1 + C_3 + C_5 \ \ldots)$

$\therefore C_1 + C_3 + C_5 \ldots = 2^{n-1}$

Short Trick

Put $n = 3$

$\Rightarrow C_1 + C_3 = 3 + 1 = 4$

Now put $n = 3$ in options, then (D) 2^{n-1} is correct option

Q.10 $n!\left(\frac{1}{n!} + \frac{1}{2!(n-2)!} + \frac{1}{4!(n-4)!} + \ldots + \frac{1}{n!}\right)$ is equal to -

(A) 2^n (B) 2^{n-1} (C) 2^{n+1} (D) 2^{-n+1}

Sol. [B]

Proper Method

$n!\left(\frac{1}{0!\,n!} + \frac{1}{2!\,n-2!} + \frac{1}{4!\,n-4!} + \ldots\right)$

$= \frac{n!}{0!\,n!} + \frac{n!}{2!\,n-2!} + \frac{n!}{4!\,n-4!} \ldots$

$= C_0 + C_2 + C_4 \ldots$

$= 2^{n-1}$

Short Trick

Put $n = 2$

$\Rightarrow 2!\left(\frac{1}{2!} + \frac{1}{2!\,(2-2)!}\right)$

$= 2\left(\frac{1}{2} + \frac{1}{2}\right) = 2$

Now put $n = 2$ in options then (B) 2^{n-1} is correct option

Q.11 $\frac{1}{1!(n-1)!} + \frac{1}{3!(n-3)!} + \frac{1}{5!(n-5)!} + \ldots =$

(A) $\frac{2^n}{n!}$ (B) $\frac{2^{n-1}}{n!}$ (C) 0 (D) $\frac{2^n}{n}$

Sol. [B]

Proper Method

$= \frac{1}{1!\,n-1!} + \frac{1}{3!\,n-3!} + \frac{1}{5!\,n-5!} + \ldots$

$= \frac{C_1}{n!} + \frac{C_3}{n!} + \frac{C_5}{n!} \ldots$

$= \frac{1}{n!}\,(C_1 + C_3 + C_5 \ldots)$

$= \frac{2^{n-1}}{n!}$

Short Trick

Put $n = 3$

$\Rightarrow \frac{1}{(1!)\,(2!)} + \frac{1}{(3!)\,(0!)}$

$= \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$

Now put $n = 3$ in options then (B) $\frac{2^{n-1}}{n!}$ is correct option

Q.12 If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then $C_0C_1 + C_1C_2 + C_2C_3 + \ldots + C_{n-1}C_n$ is equal to-

(A) $\frac{2n!}{n!\,n!}$ (B) $\frac{2n!}{n!(n+1)!}$ (C) $\frac{2n!}{(n-1)!\,(n+1)!}$ (D) $\frac{2n!}{(n-1)!\,n!}$

Sol. **[C]**

Proper Method

$C_0C_1 + C_1C_2 \ldots\ldots + C_{n-1}\,C_n$

$= C_0C_{n-1} + C_1C_{n-2} \ldots\ldots + C_{n-1}C_0$

= Coefficient of x^{n-1} in $(1+x)^n\,(1+x)^n$

= Coefficient of x^{n-1} is $(1+x)^{2n}$

$= {}^{2n}C_{n-1}$

Short Trick

Put n = 2

$\Rightarrow C_0C_1 + C_1C_2$

$= (1)\,(2) + (2)\,(1) = 4$

Now put n = 2 in options then (C) $\frac{(2n)!}{(n-1)!\,(n+1)!}$ is correct option

Q.13 ${}^nC_0 - \frac{1}{2}\,{}^nC_1 + \frac{1}{3}\,{}^nC_2 - \ldots\ldots + (-1)^n\,\frac{{}^nC_n}{n+1} =$

(A) n (B) 1/n (C) $\frac{1}{n+1}$ (D) $\frac{1}{n-1}$

Sol. **[C]**

Proper Method

$\because (1+x)^n = C_0 + C_1x + C_2x^2 \ldots\ldots + C_nx^n$

Integrating w.r.t. x from 0 to x

$$\int_0^x (1+x)^n\,dx = C_0\int_0^x dx + C_1\int_0^x x\,dx \;\ldots\; C_n\int_0^x x^n dx$$

$$\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1}\right]_0^x = C_0[x]_0^x + \frac{C_1}{2}[x^2]_0^x \ldots + \frac{C_n\left[x^{n+1}\right]_0^x}{n+1}$$

$$\Rightarrow \frac{(1+x)^{n+1}-1}{n+1} = C_0x + \frac{C_1x^2}{2} + \ldots + \frac{C_0x^{n+1}}{n+1}$$

put $\boxed{x=-1}$ in (i)

$$\Rightarrow \frac{-1}{n+1} = -C_0 + \frac{C_1}{2} - \frac{C_2}{3} + \frac{C_3}{4} \ldots$$

$$\therefore C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} \ldots\ldots = \frac{1}{n+1}$$

Short Trick

Put n = 2

$\Rightarrow {}^2C_0 - \frac{1}{2}\,{}^2C_1 + \frac{1}{3}\,{}^2C_2$

$= 1 - \frac{2}{2} + \frac{1}{3} = \frac{1}{3}$

Now put n = 2 in options then (C) $\frac{1}{n+1}$ is correct option

Q.14 In the expansion of $(1+x)^n\left(1+\frac{1}{x}\right)^n$, the term independent of x is-

(A) $C_0^2 + 2C_1^2 + \ldots\ldots + (n+1)C_n^2$ (B) $(C_0 + C_1 + \ldots\ldots + C_n)^2$

(C) $C_0^2 + C_1^2 + \ldots\ldots + C_n^2$ (D) None of these

Sol. **[C]**

Proper Method

$(1+x)^n\left(1+\frac{1}{x}\right)^n$

$= (C_0 + C_1x + C_2x^2 \ldots\ldots + C_nx^n)\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} \ldots\ldots + \frac{C_n}{x^n}\right)$

$\therefore$ Coefficient of independent term

$= C_0^2 + C_1^2 + C_2^2 \ldots\ldots + C_n^2$

Short Trick

Put n = 1

$\Rightarrow (1+x)\left(1+\frac{1}{x}\right)$

$= 2 + x + \frac{1}{x}$ term in dependent of x is 2

Now put n = 1 in options then (C)

$(C_0^2 + C_1^2 + \ldots + C_n^2)$

Q.15 If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $C_0C_r + C_1C_{r+1} + C_2C_{r+2} + ... + C_{n-r}C_n$ is equal to-

(A) $\frac{2n!}{(n-r)!(n+r)!}$ (B) $\frac{2n!}{n!(n+r)!}$ (C) $\frac{2n!}{n!(n-r)!}$ (D) $\frac{2n!}{(n-1)!(n+1)!}$

Sol. [A]

Proper Method	**Short Trick**
$C_0C_r + C_1C_{r+1} + ... + C_{n-r}C_n$ $= C_0C_{n-r} + C_1C_{n-r-1} .... + C_{n-r}C_0$ = Coefficient of x^{n-r} in $(1+x)^n.(1+x)^n$ = Coefficient of x^{n-r} in $(1+x)^{2n}$ $= {}^{2n}C_{n-r}$	Put n = 2 and r = 2 $\Rightarrow C_0C_2 = (1)(1) = 1$ Now put n = 2 and r = 2 in options, then option (A) $\frac{(2n)!}{(n-r)!\,(n+r)!}$ is correct option

Q.16 If $C_0, C_1, C_2 .... C_n$ are binomial coefficients in the expansion of $(1+x)^n$, $n \in N$, then $\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + ... + \frac{C_n}{n+1}$ is equal to -

(A) $\frac{2^{n+1}-1}{n+1}$ (B) $(n+1) \, . \, 2^{n+1}$ (C) $\frac{2^n-1}{n+1}$ (D) $\frac{2^n-1}{n}$

Sol. [C]

Proper Method	**Short Trick**
$\because (1+x)^n = C_0 + C_1x + C_2x^2 ..... + C_nx^n$ Integrating w.r.t. x from 0 to x $\int_0^x (1+x)^n dx = C_0\int_0^x dx + C_1\int_0^x x\,dx \; ... \; C_n\int_0^x x^n dx$ $\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1}\right]_0^x$ $= C_0[x]_0^x + \frac{C_1}{2}[x^2]_0^x ... + \frac{C_n[x^{n+1}]_0^x}{n+1}$ $\Rightarrow \frac{(1+x)^{n+1}-1}{n+1} = C_0x + \frac{C_1x^2}{2} + ... + \frac{C_0x^{n+1}}{n+1}$ ...(i) put $\boxed{x=-1}$ in (i) $\Rightarrow \frac{-1}{n+1} = -C_0 + \frac{C_1}{2} - \frac{C_2}{3} + \frac{C_3}{4} ...$ $\therefore C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} ...... = \frac{1}{n+1}$ ... (ii) Now put x = 1 in (i) $\Rightarrow \frac{2^{n+1}-1}{n+1} = C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \frac{C_3}{4}$ .... (iii) Now subtracting (ii) from (iii), we get $\Rightarrow 2\left(\frac{C_1}{2} + \frac{C_3}{4} .....\right) = \frac{2^{n+1}-2}{n+1}$ $\therefore \frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} ...... = \frac{2^n-1}{n+1}$	Put $n = 3 \Rightarrow \frac{C_1}{2} + \frac{C_3}{4}$ $= \frac{3}{2} + \frac{1}{4} = \frac{7}{4}$ Now put n = 3 in options, then (C) $\frac{2^n-1}{n+1}$ is correct option

Q.17 If $(1 + x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$, then for n odd, $C_1^2 + C_3^2 + C_5^2 + ...+ C_n^2$ is equal to

(A) 2^{2n-2} (B) 2^n (C) $\frac{(2n)!}{2(n!)^2}$ (D) $\frac{(2n)!}{(n!)^2}$

Sol. **[C]**

Proper Method	Short Trick
$\because C_0^2 + C_1^2 + C_2^2 ... + C_n^2 = C_0.C_0 + C_1.C_1 + ... + C_n.C_n$ $= C_0.C_n + C_1.C_{n-1} ... + C_nC_0$ = Coefficient of x^n in $(1 + x)^n .(1 + x)^n$ = Coefficient of x^n in $(1 + x)^{2n}$ $\therefore \boxed{C_0^2 + C_1^2 + C_2^2 + C_3^2 .... = {}^{2n}C_n}$ & $C_0^2 - C_1^2 + C_2^2 - C_3^2 .....$ $= C_0C_n - C_1C_{n-1} + C_2C_{n-2} - C_3C_{n-3} .....$ $= C_0.C_0 - C_1.C_1 + C_2.C_2 - C_3.C_3 .....$ = Coefficient of x^n in $(1 + x)^n (1 - x)^n$ = Coefficient of x^n in $(1 - x^2)^n$ n is odd so not possible $\therefore \boxed{C_0^2 - C_1^2 + C_2^2 - C_3^2 ...... = 0}$ Subtracting (ii) from (i), we get $2(C_1^2 + C_3^2 + C_5^2 ...) = {}^{2n}C_n$ $\therefore C_1^2 + C_3^2 + C_5^2 ... = \frac{1}{2}.{}^{2n}C_n$	**Short Trick** Put $n = 3 \Rightarrow C_1^2 + C_3^2$ $= 9 + 1 = 10$ Now put n = 3 in options then (C) $\frac{(2n)!}{2(n!)^2}$ is correct option

Q.18 If $(1+ x)^n = C_0 + C_1x + C_2x^2 + ...+ C_nx^n$, then the value of $1^2 C_1 + 2^2 C_2 + 3^2 C_3 + ...+ n^2 C_n$ is

(A) $n(n + 1) 2^{n-2}$ (B) $n(n + 1) 2^{n-1}$ (C) $n(n + 1) 2^n$ (D) None of these

Sol. **[A]**

Proper Method	Short Trick
$\because C_0 + C_1x + C_2x^2 .... + C_nx^n = (1 + x)^n$ Differentiating w.r.t. x, we get - $\Rightarrow 1C_1 + 2C_2x ...... + nC_nx^{n-1} = n(1 + x)^{n-1}$ Multiplying with w.r.t x, we get - $\Rightarrow 1C_1x + 2C_2x^2 ...... + nC_nx^n = nx(1 + x)^{n-1}$ Differentiating w.r.t. x, we get - $\Rightarrow 1^2C_1 + 2^2C_2x .... + n^2C_nx^{n-1}$ $= n [(1 + x)^{n-1} + (n - 1)x (1 + x)^{n-2}]$ Put x = 1 $\Rightarrow 1^2C_1 + 2^2C_2 ..... + n^2C_n = n(2^{n-1} + (n - 1).2^{n-2})$ $= n.2^{n-2} (2 + n - 1) = n (n + 1)2^{n-2}$	**Short Trick** Put n = 2 $\Rightarrow 1^2 C_1 + 2^2 C_2$ $= (1) (2) + 4(1) = 6$ Now, put n = 2 in options, then option (A) $n(n + 1)2^{n-2}$ is correct option.

Q.19 The sum of the terms of the series $[3.{}^nC_0 - 8.{}^nC_1 + 13.{}^nC_2 - 18.{}^nC_3 + ...+ (n + 1)]$ is $(n \in N, n \geq 2)$-

(A) $3. 2^n - 5n.2^{n-1}$ (B) 0 (C) $3.2^n + 5n.2^{n-1}$ (D) $n.2^n$

Sol. **[B]**

Proper Method	Short Trick
$3C_0 - 8C_1 + 13C_2 - 18C_3 ....$ $= 3 [C_0 - C_1 + C_2 - C_3 ...] - 5[C_1 - 2C_2 + 3C_3 ...]$ $= 0 - 0 = 0$	**Short Trick** Put n = 2 $\Rightarrow 3.{}^2C_0 - 8.{}^2C_1 + 13\ {}^2C_2$ $= 3(1) - 8(2) + 13(1) = 0$ Hence (B) is correct option

Q.20 If $(1 + x)^n = C_0 + C_1x + C_2x^2 + ... + C_n.x^n$ then the value of $C_0C_n + C_1C_{n-1} + C_2C_{n-2} + ... + C_nC_0$ is -

(A) 1 (B) $\frac{(2n)!}{(n)!\ (n)!}$ (C) $\left(\frac{(2n)!}{n!}\right)^2$ (D) $(2^n)^2$

Sol. [B]

Proper Method	Short Trick
$C_0C_n + C_1C_{n-1} .... + C_nC_0$ = Coefficient of x^n in $(1 + x)^n (1 + x)^n$ = Coefficient of x^n in $(1 + x)^{2n}$ $= {}^{2n}C_n$	Put n = 2 $\Rightarrow C_0C_2 + C_1C_1 + C_2C_0$ $= (1)(1) + (2)(2) + (1)(1) = 6$ Now put n = 2 in options, then option (B) $\frac{(2n)!}{(n)!\ (n)!}$ is correct option

Q.21 The middle term in the expansion of $\left(x+\frac{1}{2x}\right)^{2n}$ is-

(A) $\frac{1.3.5.....(2n-3)}{n!}$ (B) $\frac{1.3.5.....(2n-1)}{n!}$ (C) $\frac{1.3.5.....(2n+1)}{n!}$ (D) None of these

Sol. [B]

Proper Method	Short Trick
$\because$ 2n is even $\therefore \left(\frac{2n+2}{2}\right)^{th} = (n + 1)^{th}$ is single middle term $\therefore T_{n+1} = {}^{2n}C_n.(x)^n \left(\frac{1}{2x}\right)^n$ $\therefore \boxed{T_{n+1} = \frac{{}^{2n}C_n}{2^n}}$ ... (i) But ${}^{2n}C_n = \frac{2n!}{n!.n!} = \frac{1.2.3.4....(2n-1).2n}{n!.n!}$ $= \frac{[1.3.5.....(2n-1)].(2.4.6......2n)}{n!.n!}$ $= \frac{[1.3.5......(2n-1)].2^n(1.2.3...n)}{n!(1.2.3.....n)}$ ${}^{2n}C_n = \frac{1.3.5.....(2n-1)2^n}{n!}$ Putting in (i) : $T_{n+1} = \frac{1.3.5....(2n-1)}{n!}$	Put $n = 1 \Rightarrow \left(x+\frac{1}{2x}\right)^2$ $= x^2 + 1 + \frac{1}{4x^2}$ middle term is 1 Now put n = 1 in options, then (B) $\frac{1.3.5....(2n-1)}{(n)!}$ is correct option.

Q.22 If $(1 + x)^n = \sum_{r=0}^{n} {}^nC_r.(x)^r$, then $\left(1+\frac{{}^nC_1}{{}^nC_0}\right).\left(1+\frac{{}^nC_2}{{}^nC_1}\right).....\left(1+\frac{{}^nC_n}{{}^nC_{n-1}}\right) =$

(A) $\frac{n^{n-1}}{(n-1)!}$ (B) $\frac{(n+1)^{n-1}}{(n-1)!}$ (C) $\frac{(n+1)^n}{n!}$ (D) $\frac{(n+1)^{n+1}}{n!}$

Sol. **[C]**

Proper Method	**Short Trick**
$\left(1+\frac{n}{1}\right)\left(1+\frac{\frac{n(n-1)}{2}}{n}\right)\ldots\ldots\left(1+\frac{1}{n}\right)$ $= (1+n)\,\frac{(1+n)}{2}\,\frac{(1+n)}{3}\ldots\ldots\frac{(1+n)}{n}$ $= \frac{(1+n)^n}{n!}$	Put $n = 2 \Rightarrow \left(1+\frac{^2C_1}{^2C_0}\right)\left(1+\frac{^2C_2}{^2C_1}\right)$ $= \left(1+\frac{2}{1}\right)\left(1+\frac{1}{2}\right) = \frac{9}{2}$ Now put n = 2 in options, then (C) $\frac{(n+1)^n}{(n)!}$ is correct option

Q.23 If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \ldots\ldots + \frac{nC_n}{C_{n-1}} =$ **[AIEEE-2002]**

(A) $\frac{n}{2}$ (B) $n(n+1)$ (C) $\frac{n(n+1)}{12}$ (D) $\frac{n(n+1)}{2}$

Sol. **[D]**

Proper Method	**Short Trick**
Let $S = \frac{1C_1}{C_0} + \frac{2C_2}{C_1} + \ldots\ldots + \frac{nC_n}{C_{n-1}}$ $\because T_r = r.\frac{C_r}{C_{r-1}} = r.\frac{n-r+1}{r}$ $\therefore T_r = (n+1) - r$ Now $S = \Sigma T_r = (n+1)\Sigma 1 - \Sigma r$ $= (n+1).n - \frac{(n+1)}{2}$ $\therefore S = \frac{n(n+1)}{2}$	Put n = 2 $\Rightarrow \frac{C_1}{C_0} + \frac{2.C_2}{C_1} = \frac{2}{1} + \frac{2(1)}{2} = 3$ Now, put n = 2 in options, then (D) $\frac{n(n+1)}{2}$ is correct option

Q.24 If $S_n = \sum_{r=0}^{n}\frac{1}{^nC_r}$ and $t_n = \sum_{r=0}^{n}\frac{r}{^nC_r}$, then $\frac{t_n}{s_n}$ is equal to- **[AIEEE-2004]**

(A) $\frac{1}{2}n$ (B) $\frac{1}{2}n - 1$ (C) $n - 1$ (D) $\frac{2n-1}{2}$

Sol. **[A]**

Proper Method	**Short Trick**
$S_n = \sum_{r=0}^{n}\frac{1}{^nC_r} = \frac{1}{^nC_0} + \frac{1}{^nC_1} \ldots\ldots + \frac{1}{^nC_n}$ Now $t_n = \sum_{r=0}^{n}\frac{r}{^nC_r}$ $\therefore t_n = \frac{0}{^nC_0} + \frac{0}{^nC_1} \ldots\ldots + \frac{n-1}{^nC_{n-1}} + \frac{n}{^nC_n}$ $\oplus\, t_n = \frac{0}{^nC_n} + \frac{1}{^nC_{n-1}} \ldots\ldots + \frac{n-1}{^nC_{n-1}} + \frac{n}{^nC_0}$ $2t_n = n\left[\frac{1}{^nC_0} + \frac{1}{^nC_1} \ldots\ldots + \frac{1}{^nC_n}\right] = nS_n$ $\therefore \frac{t_n}{S_n} = \frac{n}{2}$	Put n = 3 $\Rightarrow S_3 = \frac{1}{^3C_0} + \frac{1}{^3C_1} + \frac{1}{^3C_2} + \frac{1}{^3C_3}$ $= \frac{1}{1} + \frac{1}{3} + \frac{1}{3} + 1 = \frac{8}{3}$ $t_3 = \frac{0}{^3C_0} + \frac{1}{^3C_1} + \frac{2}{^3C_2} + \frac{3}{^3C_3}$ $= 0 + \frac{1}{3} + \frac{2}{3} + \frac{3}{1} = 4$ So, $\frac{t_n}{S_n} = \frac{3}{2}$ Put n = 3 in options then (A), $\frac{n}{2}$ is correct option

Q.25 Sum of the series

$2C_0 + \frac{C_1}{2}2^2 + \frac{C_2}{3}2^3 + \frac{C_3}{4}2^4 + \ldots\ldots \frac{C_n}{n+1}2^{n+1}$ equals-

(A) $\frac{3^{n+1}-1}{n-1}$ (B) $\frac{3^{n+1}-1}{n+1}$ (C) $\frac{3^{n+1}+1}{n+1}$ (D) $\frac{3^{n-1}-1}{n+1}$

Sol. **[B]**

Proper Method

$2C_0 + \frac{C_1}{2}2^2 + \frac{C_2}{3}2^3 + \ldots\ldots \frac{C_n}{n+1}2^{n+1}$

$= \sum_{r=0}^{n} \frac{1}{r+1}\,{}^nC_r\, 2^{r+1} = \frac{1}{n+1} \sum_{r=0}^{n} \frac{n+1}{r+1}\,{}^nC_r\, 2^{r+1}$

$= \frac{1}{n+1} \sum_{r=0}^{n} {}^{n+1}C_{r+1}\, 2^{r+1} = \frac{1}{n+1} \sum_{s=1}^{n+1} {}^{n+1}C_s\, 2^s$

$= \frac{1}{n+1}\left[\sum_{s=0}^{n+1} {}^{n+1}C_s 2^s - {}^{n+1}C_0 2^0\right]$

$= \frac{1}{n+1}[(1+2)^{n+1}-1] = \frac{3^{n+1}-1}{n+1}$

Short Trick

Put n = 2

$\Rightarrow 2C_0 + \frac{C_1}{2}.2^2 + \frac{C_2}{3}.2^3$

$= 2(1) + \frac{2}{2}2^2 + \frac{1}{3}2^3$

$= 2 + 4 + \frac{8}{3} = \frac{26}{3}$

Put n = 2 in options, then (B) $\frac{3^{n+1}-1}{n+1}$ is correct option.

Q.26 Coefficient of x^n in the expansion of $\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\ldots\ldots\frac{x^n}{n!}\right)^2$ is-

(A) $\frac{2^{n+1}}{n!}$ (B) $\frac{2^{n-1}}{n!}$ (C) $\frac{2^n}{n!}$ (D) $\frac{2^n}{(n+1)}$

Sol. **[C]**

Proper Method

$\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\ldots\ldots\frac{x^n}{n!}\right)\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\ldots\ldots\frac{x^n}{n!}\right)$

Coefficient of $x^n = 1.\frac{1}{n!} + \frac{1}{1!}.\frac{1}{(n-1)!} + \frac{1}{2!}\frac{1}{(n-2)!} + \ldots \frac{1}{n!}.1$

$= \frac{1}{n!}\left[\frac{n!}{0!\,n!} + \frac{n!}{1!(n-1)!} + \frac{n!}{2!(n-2)!} + \ldots \frac{n!}{n!\,0!}\right]$

$= \frac{1}{n!}[{}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots {}^nC_n]$

$= \frac{2^n}{n!}$

Short Trick

Put n = 1 $\Rightarrow \left(1+\frac{x}{(1)!}\right)^2$

$= 1 + 2x + x^2$

Coeff. of x is 2

Now, put n = 1 in options, then, (C) $\frac{2^n}{(n)!}$ is correct option

Q.27 If a_1, a_2, a_3, a_4 are the coefficients of any four consecutive terms in the expansion of $(1+x)^n$, then $\frac{a_1}{a_1+a_2} + \frac{a_3}{a_3+a_4}$ is equal to-

(A) $\frac{a_2}{a_2+a_3}$ (B) $\frac{1}{2}\frac{a_2}{a_2+a_3}$ (C) $\frac{2a_2}{a_2+a_3}$ (D) $\frac{2a_3}{a_2+a_3}$

Sol. **[C]**

Proper Method

Let $a_1 = {}^nC_r$

$\therefore\ a_2 = {}^nC_{r+1}$

$a_3 = {}^nC_{r+2}$

$a_4 = {}^nC_{r+3}$

$\therefore\ \dfrac{a_1}{a_1+a_2} + \dfrac{a_3}{a_3+a_4}$

$= \dfrac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} + \dfrac{{}^nC_{r+2}}{{}^nC_{r+2} + {}^nC_{r+3}}$

$= \dfrac{{}^nC_r}{{}^{n+1}C_{r+1}} + \dfrac{{}^nC_{r+2}}{{}^{n+1}C_{r+3}}$

$= \dfrac{1}{\left(\dfrac{n+1}{r+1}\right)} + \dfrac{1}{\left(\dfrac{n+1}{r+3}\right)}$

$= \dfrac{1}{n+1}(2r+4) = \dfrac{2}{n+1}(r+2) = 2\left(\dfrac{r+2}{n+1}\right)$

$= 2.\ \dfrac{{}^nC_{r+1}}{{}^{n+1}C_{r+2}} = 2.\ \dfrac{a_2}{{}^nC_{r+1} + {}^nC_{r+2}}$

$= 2.\ \dfrac{a_2}{a_2+a_3}$

Short Trick

Put n = 3

$\Rightarrow (1+x)^3 = 1 + 3x + 3x^2 + x^3$

Here $a_1 = 1$, $a_2 = 3$, $a_3 = 3$,

$a_4 = 1$

So, $\dfrac{a_1}{a_1+a_2} + \dfrac{a_3}{a_3+a_4}$

$= \dfrac{1}{4} + \dfrac{3}{4} = 1$

After putting value of a_2 and a_3 (C) is correct option

Q.28 If $\dfrac{C_0}{1} - \dfrac{C_1}{3} + \dfrac{C_2}{5} - \ldots + \dfrac{(-1)^n C_n}{2n+1} = k\displaystyle\int_0^1 x(1-x^2)^{n-1}dx$, then k is :

(A) $\dfrac{2^{2n+1}n(n!)^2}{(2n+1)!}$ (B) $\dfrac{2^{2n+1}(n!)^2}{(2n+1)!}$ (C) $\dfrac{2^{n+1}n(n!)^2}{(2n+1)!}$ (D) None of these

Sol. **[A]**

Proper Method

$\displaystyle\int_0^1 (1-x^2)^n dx = \int_0^1 (C_0 - C_1x^2 + C_2x^4 - \ldots)dx$

$= \dfrac{C_0}{1} - \dfrac{C_1}{3} + \dfrac{C_2}{5} - \ldots$ ....(1)

$\therefore\ \dfrac{C_0}{1} - \dfrac{C_1}{3} + \dfrac{C_2}{5} - \ldots = k\displaystyle\int_0^1 x(1-x^2)^{n-1}dx$(say)

$= \dfrac{k}{2}.\left[\dfrac{t^n}{n}\right]_0^1 = \dfrac{k}{2n}$

Now $\displaystyle\int_0^1 (1-x^2)^n\,dx = \int_0^{\pi/2} \cos^{2n+1}\theta\, d\theta$

$[\sin\theta = x]$

Short Trick

Put n = 2

$\Rightarrow \dfrac{C_0}{1} - \dfrac{C_1}{3} + \dfrac{C_2}{5} = k\displaystyle\int_0^1 x(1-x^2)dx$

$\Rightarrow 1 - \dfrac{2}{3} + \dfrac{1}{5} = k\left(\dfrac{x^2}{2} - \dfrac{x^4}{4}\right)_0^1$

$\Rightarrow k = \dfrac{+2}{15}$

Put n = 2 in options, then (A) $\dfrac{2^{2n+1}n(n!)^2}{(2n+1)!}$ is correct option

$= \dfrac{\Gamma(n+1)\Gamma(1/2)}{2\Gamma\left(\frac{2n+3}{2}\right)}$ $\dfrac{n!\sqrt{\pi}}{2\left(\frac{2n+1}{2}\right)\left(\frac{2n-1}{2}\right)...(1/2)\sqrt{\pi}}$ $\Rightarrow \dfrac{k}{2n} = \dfrac{2^{2n}.n!n!}{(2n+1)!}$ $\Rightarrow k = \dfrac{2^{2n+1}.n(n!)^2}{(2n+1)!}$	

Q.29 If $C_r = {}^nC_r$ and $(C_0 + C_1)(C_1 + C_2) .... (C_{n-1} + C_n) = k\dfrac{(n+1)^n}{n!}$, then the value of k is:

(A) $C_0C_1C_2...C_n$ (B) $C_1^2C_2^2...C_n^2$ (C) $C_1 + C_2 + ... + C_n$ (D) None of these

Sol. [A]

Proper Method	**Short Trick**
$(C_0 + C_1)(C_1 + C_2)(C_2 + C_3)...(C_{n-1} + C_n)$ $= C_1C_2...C_n\left(1+\frac{C_0}{C_1}\right)\left(1+\frac{C_1}{C_2}\right)...\left(1+\frac{C_{n-1}}{C_n}\right)$ $= C_1C_2...C_n\left(\frac{n+1}{n}\right)\left(\frac{n+1}{n}\right)...\left(\frac{1+n}{1}\right)$ $= C_1C_2...C_n, \dfrac{(n+1)^n}{n!}$ $= C_0C_1C_2...C_n.(n+1)^n/n!$ $[C_0 = 1]$ $\therefore k = C_0C_1C_2...C_n$	Put n = 2 $\Rightarrow (C_0 + C_1)(C_1 + C_2)$ $= k\,\dfrac{3^2}{(2)!}$ $\Rightarrow (1 + 2)(2 + 1) = k\dfrac{9}{(2)!}$ $\Rightarrow k = 2$ Put n = 2 in options then (A) $C_0C_1C_2.....C_n$ is correct option

Q.30 If x + y = 1, then $\sum_{r=0}^{n} r^2\,{}^nC_r x^r y^{n-r}$ equals -

(A) nxy (B) nx (x + yn) (C) nx (nx + y) (D) None of these.

Sol. [C]

Proper Method	**Short Trick**
Given (x + y) = 1 $\because (x + y)n = \Sigma^nC_r.x^{n-r}.y^r$ or $(x + y)^n = \Sigma^nC_r.x^r.y^{n-r}$ * Differentiating w.r.t. x, we get $\Rightarrow n(x + y)^{n-1} = \Sigma r.{}^nC_rx^{r-1}.y^{n-r}$ * Multiplying x $\Rightarrow nx(x + y)^{n-1} = \Sigma r.{}^nC_r.x^r.y^{n-r}$ *Differentiating w.r.t. x, we get $\Rightarrow n[(x + y)^{n-1} + (n-1)x(x + y)^{n-2}] = \Sigma r^2.{}^nC_r.x^{r-1}.y^{n-r}$ Put x + y = 1 $\Rightarrow \Sigma r^2.{}^nC_r.x^{r-1}.y^{n-r} = n[1 + (n-1)x]$ $= n[1 + nx - x]$ $= n(nx + y)$ Multiplying by x, we get $\Rightarrow \Sigma r^2.{}^nC_r.x^r.y^{n-r} = nx(nx + y)$	Put $x = \frac{1}{4}$, $y = \frac{3}{4}$, n = 2 $= 0 + {}^2C_1\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) + 2^2.{}^2C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^0$ Option (C) nx (nx + y) is correct option

Q.31 Coefficients of x^r $[0 < r < (n-1)]$ in the expansion of

$(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \ldots\ldots + (x+2)^{n-1}$

(A) ${}^nC_r(3^r - 2^n)$ (B) ${}^nC_r(3^{n-r} - 2^{n-r})$ (C) ${}^nC_r(3^r + 2^{n-r})$ (D) ${}^nC_r(3^n - 2^n)$

Sol. [B]

Proper Method	**Short Trick**
Let $a = (x+3)$ & $b = (x+2)$ Now given series is : $a^{n-1} + a^{n-2}.b + a^{n-3}.b^2 \ldots\ldots + b^{n-1}$ $= a^{n-1}\left[1 + \frac{b}{a} + \left(\frac{b}{a}\right)^2 \ldots\ldots + \left(\frac{b}{a}\right)^{n-1}\right]$ $= a^{n-1}.1.\left[\frac{1-\left(\frac{b}{a}\right)^n}{1-\left(\frac{b}{a}\right)}\right] = \frac{a^n - b^n}{a-b}$ $= (x+3)^n - (x+2)^n = (3+x)^n - (2+x)^n$ $\therefore$ Coefficient of $x^r = {}^nC_r.3^{n-r} - {}^nC_r.2^{n-r}$	Put $r = 1$, $n = 3$ $= (x+3)^2 + (x+3)(x+2) + (x+2)^2$ $= 3x^2 + 15x + 19$ Coeff of $x = 15$ Now in option (B) ${}^nC_r(3^{n-r} - 2^{n-r})$ is correct option

Q.32 $1 - {}^nC_1\frac{1+x}{1+nx} + {}^nC_2\frac{1+2x}{(1+nx)^2} - {}^nC_3\frac{1+3x}{(1+nx)^3} + \ldots =$

(A) 0 (B) x^n (C) n^n (D) n

Sol. [A]

Proper Method	**Short Trick**
$1 - {}^nC_1\frac{1+x}{1+nx} + {}^nC_2\frac{1+2x}{(1+nx)^2} - {}^nC_3\frac{1+3x}{(1+nx)^3} + \ldots$ $= 1 - {}^nC_1\frac{1}{1+nx} + {}^nC_2\frac{1}{(1+nx)^2} - {}^nC_3\frac{1}{(1+nx)^3} + \ldots$ $-\frac{nx}{1+nx}\left\{1 - {}^{n-1}C_1\frac{1}{1+nx} + {}^{n-1}C_2\frac{1}{(1+nx)^2} - \ldots\right\}$ $= \left(1 - \frac{1}{1+nx}\right)^n - \frac{nx}{1+nx}\left(1 - \frac{1}{1+nx}\right)^{n-1}$ $= \frac{(nx)^n}{(1+nx)^n} - \frac{(nx)^n}{(1+nx)^n} = 0$	Put $n = 1 \Rightarrow 1 - \frac{1+x}{1+x} = 0$ By option (A) is correct option

Q.33 The sum of the series $\frac{2(n/2)!\,(n/2)!}{n!}\{C_0^2 - 2C_1^2 + 3C_2^2 - 4C_3^2 + \ldots + (-1)^n(n+1)C_n^2\}$ where n is an even positive integer, is equal to: **[IIT-1993]**

(A) $(-1)^{n/2}(n+1)$ (B) $(-1)^n(n+2)$ (C) $(-1)^{n/2}(n+2)$ (D) $(-1)^n(n-1)$

Sol. **[C]**

Proper Method	**Short Trick**
$C_0 + C_1x + C_2x^2 + C_3x^3 + ... + C_nx^n = (1 + x)^n$ ..... (1) Diff. (1) after multiplying by x, we get $C_0 + 2C_1x + 3C_2x^2 + ... + (n + 1)C_nx^n$ $= (1 + x)^n + nx(1 + x)^{n-1}$ .... (2) Also, $C_0x^n - C_1x^{n-1} + C_2x^{n-2} - .... + x^n = (x - 1)^n = (1 - x)^n$ $\therefore\ C_0^2 - 2C_1^2 + 3C_2^2 - ... + (n+1)C_n^2$ [$\because$ n is even] = coeff. of x^n in $(1 - x)^n \{(1 + x)^n + nx(1 + x)^{n-1}\}$ = coeff. of x^n in $\{(1 - x^2)^n + nx(1 - x)(1 - x^2)^{n-1}\}$ $= (-1)^{n/2}\,{}^nC_{n/2} - n.(-1)^{n-2/2}\,{}^{n-1}C_{(n-2)/2}$ $= (-1)^{n/2}\left[\frac{n!}{(n/2)!(n/2)!} + \frac{n.(n-1)!(n/2)}{\{(n-2)/2\}!(n/2)!(n/2)}\right]$ $= (-1)^{n/2}.\frac{(n+2)}{2}.\frac{n!}{(n/2)!(n/2)!}$ $\therefore \frac{2(n/2)!(n/2)!}{n!}$ $\{C_0^2 - 2C_1^2 + 3C_2^2 - 4C_3^2 + ... + (-1)^n(n+1)C_n^2\}$ $= (-1)^{n/2}(n + 2)$.	Put n = 2 $\Rightarrow \frac{2(1)!.(1)!}{(2)!}$ $(C_0^2 - 2C_1^2 + 3C_2^2)$ $= (1 - 2(4) + 3) = -4$ Put n = 2 in option (C) $(-1)^{n/2}(n + 2)$ is correct option

5

Permutations & Combinations

KEY CONCEPTS

1. Factorial Notation

The continuous product of first n natural numbers is called factorial and it can be represented by notation

$\underline{|n}$ or n!, where $n \in W$

(i) $n! = 1.2.3 \ldots\ldots (n-1).n$

(ii) $n! = n\,(n-1)! = n\,(n-1)\,(n-2)! = n\,(n-1)\,(n-2)\,(n-3)!$

(iii) $n\,(n-1) \ldots\ldots\ldots\ldots (n-r+1) = \dfrac{n!}{(n-r)!}$

(iv) $0! = 1$

2. Fundamental Principle of Counting

(i) Fundamental principle of Multiplication :

Let there are two parts A and B of an operation and if these two parts can be performed in m and n different number of ways respectively, then that operation can be completed in $m \times n$ ways.

(ii) Fundamental Principle of Addition :

If there are two operations such that they can be done independently in m and n ways respectively, then any one of these two operations can be done by $(m + n)$ number of ways.

3. Permutations

An arrangement of some given things taking some or all of them, is called a **permutation** of these things.

The number of permutations of **n** **different** things taken r at a time is ${}^nP_r$, where

$${}^nP_r = \frac{n!}{(n-r)!} = n\,(n-1)\,(n-2) \ldots\ldots (n-r+1)$$

4. Properties of Permutation

(i) ${}^nP_n = n(n-1)\,(n-2) \ldots\ldots 1 = n!$

(ii) ${}^nP_0 = 1$

(iii) ${}^nP_1 = n$

(iv) ${}^nP_{n-1} = n!$

(v) $^nP_r = n \cdot {}^{n-1}P_{r-1} = n(n-1) \cdot {}^{n-2}P_{r-2}$

(vi) $^{n-1}P_r + r \cdot {}^{n-1}P_{r-1} = {}^nP_r$

(vii) $\dfrac{^nP_r}{^nP_{r-1}} = n - r + 1$

5. Important Results on Permutation

(i) The number of permutations of n dissimilar things taken all at a time $= {}^nP_n = n!$

(ii) The number of permutations of n things taken all at a time when p of them are alike and of one kind, q of them are alike and of second kind, r of them are alike and of third kind and all remaining being different is $\dfrac{n!}{p!\,q!\,r!}$.

(iii) The number of ways of n distinct objects taking r of them at a time where any object may be Repeated any number of times is n^r.

(iv) In above case when n things are taken at a time then total number of permutation is n^n.

6. Restricted Permutations

(i) The number of permutations of n dissimilar things taken r at a time, when m particular things always occupy definite places $= {}^{n-m}P_{r-m}$

(ii) The number of permutations of n different things taken altogether when r particular things are to be placed at some r given places $= {}^{n-r}P_{n-r} = (n-r)!$

(iii) The number of permutations of n different things taken r at a time, when m particular things are always to be excluded $= {}^{n-m}P_r$

(iv) The number of permutations of n different things taken r at a time, when m particular things are always to be included $= {}^{n-m}C_{r-m} \times r!$

7. Number of Circular Permutations

(i) The number of circular permutations of n *different* things taking r at a time $= \dfrac{^nP_r}{r}$, when clockwise and anti-clockwise orders are treated as different.

(ii) The number of circular permutations of n *different* things taking altogether $= \dfrac{^nP_n}{n} = (n-1)!$, when clockwise and anti clockwise orders are treated as different.

(iii) The number of circular permutations of n *different* things taking r at a time $= \dfrac{^nP_r}{2r}$, when the above two orders are treated as same.

(iv) The number of circular permutations of n *different* things taking altogether $= \dfrac{^nP_n}{2n} = \dfrac{1}{2}(n-1)!$, when above two orders are treated as same.

Note :

➢ In a garland of flowers or a necklace of beads, it is difficult to distinguish clockwise and anticlockwise orders of things, so a circular permutation under both these orders is considered to be the same.

(iii) Given, n distinct points in the plane, no three of which are collinear, then the number of triangle formed = ${}^nC_3$.

(iv) Given n distinct points in a plane, in which m are collinear ($m \geq 3$), then the number of triangle formed = ${}^nC_3 - {}^mC_3$.

(v) The number of diagonals in a n-sided closed polygon = ${}^nC_2 - n$

(vi) Given, n points on the circumference of a circle, then

(a) number of straight line = ${}^nC_2$

(b) number of triangles = ${}^nC_3$

(c) number of quadrilaterals = ${}^nC_4$

(vii) Number of rectangles of any size in a square of $n \times n$ is $\sum_{r=1}^{n} r^3$ and number of square of any size is $\sum_{r=1}^{n} r^2$.

(viii) In a rectangle of $n \times p (n < p)$, numbers of rectangles of any size is ${}^{n+1}C_2 . {}^{P+1}C_2$ or $\frac{np}{4}(n+1)(p+1)$ and number of squares of any size is $\sum_{r=1}^{n}(n+1-r)(p+1-r)$.

13. Divisors

Let $N = P_1^{\alpha_1} . P_2^{\alpha_2} . P_3^{\alpha_3} \ldots\ldots P_k^{\alpha_k}$, where $P_1, P_2, P_3, \ldots\ldots P_k$ are different primes and $\alpha_1, \alpha_2, \alpha_3, \ldots\ldots, \alpha_k$, are natural numbers then :

(i) The total number of divisors of N including 1 and N is $(\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1) \ldots (\alpha_k + 1)$

(ii) The total number of divisors of N excluding 1 and N is $(\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1) \ldots (\alpha_k + 1) - 2$

(iii) The total number of divisors of N excluding 1 or N is $(\alpha_1 + 1)(\alpha_2 + 1)(\alpha_3 + 1) \ldots (\alpha_k + 1) - 1$

14. Division into Groups

(i) The number of ways in which n distinct objects can be split into three groups containing respectively r, s and t objects, r, s and t are distinct and $r + s + t = n$, is given by

$${}^nC_r . {}^{n-r}C_s . {}^{n-r-s}C_t = \frac{n!}{r!\,s!\,t!}$$

If these 3 groups containing r, s and t objects are distributed among 3 persons (i.e. permutation of these group is also considered) the ways are

$${}^nC_r . {}^{n-r}C_s . {}^{n-r-s}C_t . 3! = \frac{n!.3!}{r!\,s!\,t!}$$

(ii) If $3n$ things are to be divided equally between 3 persons (i.e. division of $3n$ things into 3 equal groups with permutation of groups) then the number of ways $\frac{(3n)!}{(n!)^3}$

(iii) If $3n$ things are divided into three equal groups, then the number of ways

$$= \frac{(3n)!}{n!\,n!\,n!\,3!} = \frac{(3n)!}{3!(n!)^3}.$$

Since for any one way the three groups can be placed in 3! ways without obtaining the new division. So it is divided by 3!.

15. Derangement Theorem

(i) If n items are arranged in a row, then the number of ways in which they can be rearranged so that no one of them occupies the place assigned to it is

$$n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}+.....+(-1)^n\frac{1}{n!}\right]$$

(ii) If n things are arranged at n places then the number of ways to rearrange exactly r things at right places is

$$\frac{n!}{r!}\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}+.....+(-1)^{n-r}\frac{1}{(n-r)!}\right]$$

16. Sum of numbers

(i) For given n different digits $a_1, a_2, a_3 ...., a_n$ the sum of the digits in the unit place of all numbers formed (if numbers are not repeated) is

$(a_1 + a_2 + a_3 + ...+ a_n)\ (n-1)!$

i.e. (sum of the digits) $(n-1)!$

(ii) Sum of the total numbers which can be formed with given n different digits $a_1, a_2, a_3 .....a_n$ is **$(a_1 + a_2 + a_3 + ...+ a_n)(n-1)!.(111 ...n$ times)**

(iii) The sum of all digits of numbers that can be formed by using the digits $a_1, a_2, ...a_n$ (repetition of digits is not allowed) is

$$(n-1)!\ (a_1 + a_2 + ... + a_n)\left(\frac{10^n-1}{9}\right)$$

17. Integral solution

(i) The number of non negatives integral solutions of

$x_1 + x_2 + ... + x_r = n$, where $x_1, x_2, ... x_r, \geq 0$ is ${}^{n+r-1}C_r$ or ${}^{n+r-1}C_{n-1}$.

(ii) Number of positive integral solutions of

$x_1 + x_2 + ... + x_r = n$, where $x_1, x_2, ... x_r, \geq 1$, is ${}^{n-1}C_{r-1}$.

18. Multinomial theorem

(i) If there are ℓ object of one kind, m object of second kind, n object of third kind and so on; then the number of ways of choosing r objects out of these (i. e., $\ell + m + n +.....$) is the coefficient of x^r in the expansion of $(1 + x + x^2 + x^3 + ...+ x^\ell)\ (1 + x + x^2 + ...+ x^m)\ (1 + x + x^2 + x^3 + ...+ x^n)$

Further if one object of each kind is to be included, there the number of ways of choosing r objects out of these objects (i. e., $\ell + m + n +....$) is the coefficient of x^r in the expansion of $(x + x^2 + x^3 +...+ x^\ell)\ (x + x^2 + x^3 +...+ x^m)\ (x + x^2 + x^3 +.....+ x^n).....$

(ii) If there are ℓ object of one kind, m object of second kind, n object of third kind and so on; then the number of possible arrangements / permutations of r object out of these object (i. e., $\ell + m + n +.......$) is the coefficient of x^r in the expansion of

$$r!\left(1+\frac{x}{1!}+\frac{x}{2!}+......+\frac{x^\ell}{\ell!}\right)\left(1+\frac{x}{1!}+\frac{x^2}{2!}+......+\frac{x^m}{m!}\right)\left(1+\frac{x}{1!}+\frac{x^2}{2!}+......+\frac{x^n}{n!}\right)$$

19. Exponent of prime p in n!

$$E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots\ldots\ldots + \left[\frac{n}{p^r}\right]$$

(where [.] represents greatest integer)

20. Arrangements

(i) The number of ways in which m (one type of different things) and n (another type of different things) can be arranged in a row so that all the second type of things come together is $n!\,(m+1)!$

(ii) The number of ways in which m (one type of different things) and n (another type of different things) can be arranged in row so that no two things of the same type come together is $2 \times m!\,n!$

(iii) The number of ways in which m (one type of different things) and n (another type of different things) $(m \geq n)$, can be arranged in a circle so that no two things of second type come together is $(m-1)!\,{}^mP_n$ and when things of second type come together $= m!\,n!$

(iv) The number of ways in which m things of one type and n things of another type (all different) can be arranged in the form of a garland so that all the second type of things comes together, is $\frac{m!\,n!}{2}$ and if no two things of second type come together, is $\frac{(m-1)!\;{}^mP_n}{2}$

21. Selection

(i) The total number of ways in which it is possible to make a selection by taking some or all the given n different object is

${}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n - 1$

(ii) If there are m items of one kind, n items of another kind and so on. Then, the number of ways of choosing r items out of these items = coefficient of x^r in

$(1 + x + x^2 + \ldots\ldots + x^m)(1 + x + x^2 + \ldots\ldots + x^n) \ldots\ldots\ldots$

(iii) If there are m items of one kind, n items of another kind and so on. Then, the number of ways of choosing r items out of these items such that atleast one item of each kind is included in every selection = coefficient of x^r in

$(x + x^2 + \ldots\ldots + x^m)\,(x + x^2 + \ldots\ldots + x^n) \ldots\ldots\ldots$

(iv) The number of ways of selecting r items from a group of n items in which p are identical, is ${}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \ldots\ldots + {}^{n-p}C_0$, if $r \leq p$ and ${}^{n-p}C_r + {}^{n-p}C_{r-1} + {}^{n-p}C_{r-2} + \ldots\ldots + {}^{n-p}C_{r-p}$ + if $r > p$

(v) The number of ways in which n identical things can be distributed into r different groups is ${}^{n+r-1}C_{r-1}$ or ${}^{n-1}C_{r-1}$.

According is blanks groups are or are not admissible.

(vi) The number of ways of answering one or more of n question is $2^n - 1$.

(vii) The number of ways of answering one or more n question when each question has an alternative $= 2^n$.

SHORT TRICKS

Trick -1 Rank of word when arranging letters according to dictionary.

Method

(i) write numbering of each letter according to alphabetical order over the word.

(ii) Below each letter write the number of digits which are right side on a particular letter and less than the assigned number (let $a_1, a_2, a_3 \ldots a_n$)

(iii) From the right side write 0!, 1!, 2! .... (n – 1)! Below each letter

(iv) Then rank of word = $(a_n 0! + a_{n-1} 1! + \ldots + a_n(n-1)!) + 1$

Q.1 The letters of the word RANDOM are written in all possible orders and these words are written out as in a dictionary then the rank of the word RANDOM is -

(A) 614 (B) 615 (C) 613 (D) 616

Sol. [A]

Proper Method

Alphabetically: A, D, M, N, O, R

so words before random are :

A _ _ _ _ _ → 5! = 120

D _ _ _ _ _ → 5! = 120

M _ _ _ _ _ → 5! = 120

N _ _ _ _ _ → 5! = 120

O _ _ _ _ _ → 5! = 120

R A D _ _ _ → 3! = 6

R A M _ _ _ → 3! = 6

R A N D M O → = 1

R A N D O M → = 1

∴ Rank = 614

Short Trick

6	1	4	2	5	3
R	A	N	D	O	M
5	0	2	0	1	0
×	×	×	×	×	×
5!	4!	3!	2!	1!	0!

$= (5 \times 5! + 2 \times 3! + 1 \times 1!) + 1 = 614$

Q.2 If the letters of the word MOTHER are written in all possible orders and these words are written out as in dictionary, then the rank of the word MOTHER is -

(A) 240 (B) 261 (C) 308 (D) 309

Sol. [D]

Proper Method

Alphabetically: E, H, M, O, R, T

∴ words before MOTHER are :

E _ _ _ _ _ ⇒ 5! = 120

H _ _ _ _ _ ⇒ 5! = 120

M E _ _ _ _ ⇒ 4! = 24

M H _ _ _ _ ⇒ 4! = 24

M O E _ _ _ ⇒ 3! = 6

M O H _ _ _ ⇒ 3! = 6

M O R _ _ _ ⇒ 3! = 6

M O T E _ _ ⇒ 2! = 2

M O T H E R ⇒ 1

∴ Rank = 309

Short Trick

3	4	6	2	1	5
M	O	T	H	E	R
2	2	3	1	0	0
×	×	×	×	×	×
5!	4!	3!	2!	1!	0!

$= (2 \times 5! + 2 \times 4! + 3 \times 3! + 1 \times 2!) + 1 = 309$

Q.3 If the letters of the word SACHIN are arranged in all possible ways and these words are written out as in dictionary, then the word SACHIN appears at serial number - **[AIEEE-2005]**

(A) 601 (B) 600 (C) 603 (D) 602

Sol. [A]

Proper Method

In order: A, C, H, I, N, S

In dictionary:

A _ _ _ _ _ $\to 5! = 120$

C _ _ _ _ _ $\to 5! = 120$

H _ _ _ _ _ $\to 5! = 120$

I _ _ _ _ _ $\to 5! = 120$

N _ _ _ _ _ $\to 5! = 120$

SACHIN

$\therefore$ Rank of Sachin = 601

Short Trick

6	1	2	3	4	5
S	A	C	H	I	N
5	0	0	0	0	0
×	×	×	×	×	×
5!	4!	3!	2!	1!	0!

$= (5 \times 5!) + 1 = 601$

Q.4 If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is : **[JEE Main -2016]**

(A) 46^{th} (B) 59^{th} (C) 52^{nd} (D) 58^{th}

Sol. [D]

Proper Method

(1) Words start with A $\Rightarrow \boxed{A}$ _ _ _ _ (S, M, L, L) $= \frac{4!}{2!} = 12$

(2) Words start with L $\Rightarrow \boxed{L}$ _ _ _ _ (A, S, L, M) $= 4! = 24$

(3) Words start with M $\Rightarrow \boxed{M}$ _ _ _ _ (A, S, L, L) $= \frac{4!}{2!} = 12$

(4) Words start with SA $\Rightarrow \boxed{S}\,\boxed{A}$ _ _ _ (M, L, L) $= \frac{3!}{2!} = 3$

(5) Words start with SL $\Rightarrow \boxed{S}\,\boxed{L}$ _ _ _ (A, M, L) $= 3! = 6$

(6) Words start with SMALL $\Rightarrow$

Total rank = 12 + 24 + 12 +3 + 6 +1 = 58

Short Trick

4	3	1	2	2
S	M	A	L	L
$\frac{4}{2!}$	$\frac{3}{2!}$	$\frac{0}{2!}$	$\frac{0}{2!}$	$\frac{0}{1!}$
×	×	×	×	×
4!	3!	2!	1!	0!

$= (48 + 9) + 1$

$= 58$

Q.5 The letters of word 'ZENITH' are written in all possible ways. If all these words are written in the order of a dictionary, then the rank of the word 'ZENITH' is-

(A) 716 (B) 692 (C) 698 (D) 616

Sol. [D]

Proper Method

Alphabetically : E, H, I, N, T, Z

words start with E $\Rightarrow 5! = 120$

words start with H $\Rightarrow 5! = 120$

words start with I $\Rightarrow 5! = 120$

words start with N $\Rightarrow 5! = 120$

words start with T $\Rightarrow 5! = 120$

ZEH — — — $\Rightarrow 3! = 6$

Short Trick

6	1	4	3	5	2
Z	E	N	I	T	H
5	0	2	1	1	0
×	×	×	×	×	×
5!	4!	3!	2!	1!	0!

$= (600 + 12 + 2 + 1) + 1$

$= 616$

ZEI ——— $\Rightarrow 3! = 6$ ZENH —— $\Rightarrow 2! = 2$ ZENIH — $\Rightarrow 1! = 1$ ZENITH $\underline{= 1}$ Rank = 616	

Q.6 If the letters of the word 'RACHIT' are arranged in all possible ways and these words are written out as in a dictionary, then the rank of this word is-

(A) 365 (B) 702 (C) 481 (D) 512

Sol. **[C]**

Proper Method	**Short Trick**
The number of words beginning with A (i.e. in which A comes in first place) is $^5P_5 = 5!$. Similarly number of words beginning with C is 5!, beginning with H is 5! and beginning with I is also 5!. Now letters of R, four letters A, C, H, I can occur in 4 (5!) = 480 ways. Now word 'RACHIT' happens to be the first word beginning with R. Therefore the rank of this word = 480 + 1 = 481	5 1 2 3 4 6 R A C H I T 4 0 0 0 0 0 × × × × × × 5! 4! 3! 2! 1! 0! = 480 + 1 = 481

Q.7 The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetica order as in an english dictionary. The number of words that appear before the word COCHIN is-

(A) 360 (B) 192 (C) 96 (D) 48

Sol. **[C]**

Proper Method	**Short Trick**
According to dictionary, CC HINO $\longrightarrow$ 4! CH CINO $\longrightarrow$ 4! CI CHNO $\longrightarrow$ 4! CN CHIO $\longrightarrow$ 4! CO CHIN $\longrightarrow$ it self $\because$ Total ways = $4 \times 4! = 96$	1 5 1 2 3 4 C O C H I N 0 4 0 0 0 0 × × × × × × $\frac{5!}{2!}$ 4! 3! 2! 1! 0! Words before COCHIN is = $4 \times 4! = 96$

Trick -2 Box Method

Q.8 The number of different words (meaningful or meaningless) can be formed by taking four different letters from English alphabets is-

(A) $(26)^4$ (B) 358800 (C) $(25)^4$ (D) 15600

Sol. **[B]**

Proper Method	**Short Trick**
The first letter of four letter word can be chosen by 26 ways, second by 25 ways, third by 24 ways and fourth by 23 ways. So number of four letter words $= 26 \times 25 \times 24 \times 23 = 358800$	$\boxed{}\ \boxed{}\ \boxed{}\ \boxed{}$ $26 \times 25 \times 24 \times 23 = 358800$

Q.9 The number of numbers lying between 100 and 1000 which can be formed with the digits 0, 1, 2, 3, 4, 5, 6 is (repetition is not allowed)-

(A) 180 (B) 216 (C) 200 (D) None of these

Sol. **[A]**

Proper Method

Required numbers will have 3 digits so their total number = $^7P_3 - {}^6P_2 = 180$

Short Trick

$\boxed{\neq 0}\ \boxed{}\ \boxed{}$

$6 \times 6 \times 5 = 180$

Q.10 How many numbers between 1000 and 4000 (including 4000) can be formed with the digits 0, 1, 2, 3, 4 if each digit can be repeated any number of times?

(A) 125 (B) 275 (C) 375 (D) 500

Sol. **[C]**

Proper Method

Required number will have 4 digits and their thousand digit will be 1 or 2 or 3 or 4. The number of such numbers will be 124, 125, 125 and 1 respectively.

$\therefore$ Total numbers = 124 + 125 + 125 + 1 = 375

Short Trick

$\boxed{\neq 0, \neq 4}\ \boxed{}\ \boxed{}\ \boxed{}$

$3 \times 5 \times 5 \times 5 = 375$

Q.11 The number of 4 digit numbers divisible by 5 which can be formed by using the digits 0, 2, 3, 4, 5 is-

(A) 36 (B) 42 (C) 48 (D) None of these

Sol. **[B]**

Proper Method

For a number to be divisible by 5, unit place should be occupied by 0 or 5-

(i) If unit place is 0 then remaining 3 places can be filled by $^4P_3$ ways = 24

(ii) If unit place is 5 then no. of ways $= 4! - 3! = 18$

$\therefore$ Total number of ways $= 24 + 18 = 42$

Short Trick

(i) $\boxed{\neq 0}\ \boxed{}\ \boxed{}\ \boxed{5}$

$3 \times 3 \times 2 \times 1 = 18$

(ii) $\boxed{\neq 0}\ \boxed{}\ \boxed{}\ \boxed{0}$

$4 \times 3 \times 2 \times 1 = 24$

Total = 42

Q.12 How many numbers can be formed with the digits 0,1,2, 3,4,5 which are greater than 3000?

(A) 180 (B) 360 (C) 1380 (D) 1500

Sol. **[C]**

Proper Method

The numbers greater than 3000 may contain 4, 5 or 6 digits. Now the 6 digit numbers can be formed by using all 6 given digits which will be $^6P_6$. These numbers include the numbers which starts with 0. Such type of numbers are $^5P_5$.

Therefore 6 digit numbers greater than 3000 $= {}^6P_6 - {}^5P_5 = 600$

Similarly 5 digit numbers greater than 3000

Short Trick

(i) 4 digit number

$\boxed{3, 4, 5}\ \boxed{}\ \boxed{}\ \boxed{}$

$3 \times 5 \times 4 \times 3 = 180$

(ii) 5 digit number

$\boxed{\neq 0}\ \boxed{}\ \boxed{}\ \boxed{}\ \boxed{}$

$5 \times 5 \times 4 \times 3 \times 2 = 600$

(iii) 6 digit number

$= {}^6P_5 - {}^5P_4 = 600$ Now 4 digit numbers which are greater than 3000 should begin with the digit 3, 4 or 5. If the first place of the number is occupied by the digit 3, 4 or 5, then the remaining three places of the number can be filled in ${}^5P_3 = 60$ ways. Therefore numbers of 4 digits greater than 3000 $= 3 \times 60 = 180$ Hence required numbers = 600 + 600 + 180 = 1380	$\boxed{\neq 0}\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }$ $5 \times 5 \times 4 \times 3 \times 2 \times 1$ $= 600$ Total = 1380

Q.13 How many numbers divisible by 5 and lying between 3000 and 4000 can be formed from the digits 1, 2, 3, 4, 5, 6 (repetition is not allowed)

(A) ${}^6P_2$ (B) ${}^5P_2$ (C) ${}^4P_2$ (D) ${}^6P_3$

Sol. **[C]**

Proper Method	**Short Trick**
To be divisible by 5, the digit 5 must be at unit place, so now to be between 3000 and 4000, 3 must be at thousand place. Hence the required number of ways $= {}^4P_2$	$\boxed{3}\ \boxed{\ }\ \boxed{\ }\ \boxed{5}$ $1 \times 4 \times 3 \times 1 = {}^4P_2$

Q.14 How many even numbers of 3 different digits can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 (repetition is not allowed)

(A) 224 (B) 280 (C) 324 (D) 56

Sol. **[A]**

Proper Method	**Short Trick**
The number will be even if last digit is 2, 4, 6 or 8 *i.e.*, the last digit can be filled in 4 ways and remaining two digits can be filled in ${}^8P_2$ ways. Hence required number of numbers are ${}^8P_2 \times 4 = 224$	$\boxed{\ }\ \boxed{\ }\ \boxed{2, 4, 6, 8}$ $8 \times 7 \times 4 = 224$

Q.15 The figures 4, 5, 6, 7, 8 are written in every possible order. The number of numbers greater than 56000 is

(A) 72 (B) 96 (C) 90 (D) 98

Sol. **[C]**

Proper Method	**Short Trick**
Required number of ways $= 5! - 4! - 3!$ $120 - 24 - 6 = 90$. [Number will be less than 56000 only if either 4 occurs on the first place or 5, 4 occurs on the first two places]	$\boxed{5}\ \boxed{6, 7, 8}\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }$ $1 \times 3 \times 3 \times 2 \times 1 = 18$ and $\boxed{6, 7, 8}\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }\ \boxed{\ }$ $3 \times 4 \times 3 \times 2 \times 1 = 72$ Total = 90

Q.16 The number of 3 digit odd numbers, that can be formed by using the digits 1, 2, 3, 4, 5, 6 when the repetition is allowed, is

(A) 60 (B) 108 (C) 36 (D) 30

Sol. **[B]**

Proper Method	Short Trick
Extreme left place can be filled in 6 ways, the middle place can be filled in 6 ways and extreme right place in only 3 ways. ($\because$ number to be formed is odd) $\therefore$ Required number of numbers = 6 × 6 × 3 = 108.	$\boxed{}\ \boxed{}\ \boxed{1,3,5}$ $6 \times 6 \times 3 = 108$

Q.17 How many numbers of five digits can be formed from the numbers 2, 0, 4, 3, 8 when repetition of digits is not allowed

(A) 96 (B) 120 (C) 144 (D) 14

Sol. **[A]**

Proper Method	Short Trick
Numbers are 2, 0, 4, 3, 8 Numbers can be formed = (Total) – (those beginning with 0) = 5! – 4! = 120 – 24 = 96.	$\boxed{\neq 0}\ \boxed{}\ \boxed{}\ \boxed{}\ \boxed{}$ $4 \times 4 \times 3 \times 2 \times 1 = 96$

Q.18 How many numbers lying between 500 and 600 can be formed with the help of the digits 1, 2, 3, 4, 5, 6 when the digits are not to be repeated

(A) 20 (B) 40 (C) 60 (D) 80

Sol. **[A]**

Proper Method	Short Trick
There must be 5 at hundred place, now 2 numbers to be chosen from 5 numbers i.e., ${}^5P_2 = 5 \times 4 = 20$	$\boxed{5}\ \boxed{}\ \boxed{}$ $1 \times 5 \times 4 = 20$

Q.19 How many numbers, lying between 99 and 1000 be made from the digits 2, 3, 7, 0, 8, 6 when the digits occur only once in each number

(A) 100 (B) 90 (C) 120 (D) 80

Sol. **[A]**

Proper Method	Short Trick
Required number of ways $= {}^6P_3 - {}^5P_2 = 120 - 20 = 100$. {Since between 99 and 1000, the numbers are of 3 digits, but those numbers should be omit in which '0' comes at hundred place}	$\boxed{\neq 0}\ \boxed{}\ \boxed{}$ $5 \times 5 \times 4 = 100$

Q.20 How many numbers can be made with the digits 3, 4, 5, 6, 7, 8 lying between 3000 and 4000 which are divisible by 5 while repetition of any digit is not allowed in any number

(A) 60 (B) 12 (C) 120 (D) 24

Sol. **[B]**

Proper Method	Short Trick
3 must be at thousand place and since the number should be divisible by 5, so 5 must be at unit place. Now we have to filled two place (ten and hundred) *i.e.*, ${}^4P_2 = 12$	$\boxed{3}\ \boxed{}\ \boxed{}\ \boxed{5}$ $1 \times 4 \times 3 \times 1 = 12$

Q.21 How many numbers greater than 24000 can be formed by using digits 1, 2, 3, 4, 5 when no digit is repeated

(A) 36 (B) 60 (C) 84 (D) 120

Sol. [C]

Proper Method

The digits are 1, 2, 3, 4, 5. We have to form number greater than 24000.

Required number will be = (Total) – (Those beginning with 1) – (Those beginning with 21) – (Those beginning with 23)

$5! - 4! - 3! - 3! = 120 - 24 - 6 - 6 = 84$

Short Trick

2	4, 5			
1 ×	2 ×	3 ×	2 ×	1 = 12

and

3, 4, 5				
3 ×	4 ×	3 ×	2 ×	1 = 72

Total = 84

Q.22 How many numbers greater than hundred and divisible by 5 can be made from the digits 3, 4, 5, 6, if no digit is repeated

(A) 6 (B) 12 (C) 24 (D) 30

Sol. [B]

Proper Method

Numbers which are divisible by 5 have '5' fixed in extreme right place

3 Digit Numbers

H	T	U
×	×	5

$^3P_2$ ways

$= \frac{3!}{1!} = 3 \times 2$

4 Digit Numbers

Th	H	T	U
×	×	×	5

$^3P_3$ ways

$= \frac{3!}{0!} = 3 \times 2$

$\Rightarrow$ Total ways = 12.

Short Trick

3 digit number

		5
3 ×	2 ×	1 = 6

4 digit number

			5
3 ×	2 ×	1 ×	1 = 6

Total = 12

Q.23 The number of 4 digit even numbers that can be formed using 0, 1, 2, 3, 4, 5, 6 without repetition is

(A) 120 (B) 300 (C) 420 (D) 20

Sol. [C]

Proper Method

The units place can be filled in 4 ways as any one of 0, 2, 4 or 6 can be placed there. The remaining three places can be filled in with remaining 6 digits in $^6P_3$ = 120 way. So, total number of ways = 4 × 120 = 480. But, this includes those numbers in which 0 is fixed in extreme left place. Numbers of such numbers $= 3 \times {}^5P_2 = 3 \times 5 \times 4 = 60$

0	×	×	×
Fix	$^5P_2$ ways		3 ways (only 2, 4 or 6)

$\therefore$ Required number of ways = 480 – 60 = 420.

Short Trick

			0
6 ×	5 ×	4 ×	1 = 120

and

≠ 0			2, 4, 6
5 ×	5 ×	4 ×	3 = 300

Total = 420

Q.26 How many numbers between 5000 and 10,000 can be formed using the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 each digit appearing not more than once in each number

(A) $5 \times {}^8P_3$ (B) $5 \times {}^8C_3$ (C) $5! \times {}^8P_3$ (D) $5! \times {}^8C_3$

Sol. **[A]**

Proper Method	**Short Trick**
A number between 5000 and 10,000 can have any of the digits 5, 6, 7, 8, 9 at thousand's place. So thousand's place can be filled in 5 ways. Remaining 3 places can be filled by the remaining 8 digits in ${}^8P_3$ ways. Hence required number = $5 \times {}^8P_3$	$\boxed{5, 6, 7, 8, 9}\ \boxed{}\ \boxed{}\ \boxed{}$ $5 \times 8 \times 7 \times 6 = 5 \times {}^8P_3$

Q.27 How many numbers can be formed between 20000 and 30000 by using digits 2, 3, 5, 6, 9 when digits may be repeated?

(A) 125 (B) 24 (C) 625 (D) 1250

Sol. **[C]**

Proper Method	**Short Trick**
First digit between 20000 and 30000 will be 2 which can be chosen by one way. Every number will be of five digits and all the digits can be anything from the given five digit except first digit. So each digit of the remaining four digits can be chosen in 5 ways $\therefore$ required numbers = $1 \times 5 \times 5 \times 5 \times 5 = 625$	$\boxed{2}\ \boxed{}\ \boxed{}\ \boxed{}\ \boxed{}$ $1 \times 5 \times 5 \times 5 \times 5 = 625$

6

Determinant

KEY CONCEPTS

1. Definition

An expression expressed in equal number of rows and column and put between two vertical lines is named as determinant.

e.g. $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ are determinant of second and third order.

2. Expansion of Determinant

(i) Expansion of two order : $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$

(ii) Expansion of three order : (with respect to first row)

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - b_2a_3)$$

3. Minor and Cofactor

(i) Minor :

If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ then

Minor of a_{11} is $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$,

Similarly minor of a_{12} is $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

(ii) Cofactor :

The cofactor of an element a_{ij} is denoted by F_{ij} & is equal to

$(-1)^{i+j} M_{ij}$ where M is a Minor of element a_{ij}.

If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

then $F_{11} = (-1)^{1+1} M_{11} = M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

and $F_{12} = (-1)^{1+2} M_{12} = -M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

4. Properties of minor and cofactor

(i) The sum of products of the element of any row with their corresponding cofactor is equal to the value of determinant i.e.
$\Delta = a_{11}F_{11} + a_{12}F_{12} + a_{13}F_{13}$

(ii) The sum of the product of element of any row with corresponding cofactor of another row is equal to zero i.e. $a_{11}F_{21} + a_{12}F_{22} + a_{13}F_{23} = 0$

(iii) If order of a determinant (Δ) is 'n' then the value of the determinant formed by replacing every element by its cofactor is Δ^{n-1}.

5. Properties of Determinants

(i) The value of the determinant remains unaltered,if two rows and columns are interchanged. This is denoted by 'and is also called transpose.

e.g. $A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = A'$

(ii) If any two rows (or columns of a determinant be interchanged, the determinant is unaltered in numerical value, but is changed in sign only.

e.g. $A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $B = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ then $B = -A$

(iii) If all the element of a row in Δ are zero or two rows (columns) are identical (or proportional), then the value of Δ is zero.

(iv) If all the elements of any row (or column) be multiplied by the same number, then the determinant is multiplied by that number.

e.g. $A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $B = \begin{vmatrix} ka_1 & kb_1 & kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ then $B = kA$

(v) The value of a determinant is not altered by adding the elements of any row (or column) to the same multiple of the corresponding elements of any other row (or column).

e.g. let $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + mb_1 - c_1 & b_1 & c_1 \\ a_2 + mb_2 - c_2 & b_2 & c_2 \\ a_3 + mb_3 - c_3 & b_3 & c_3 \end{vmatrix}$

(vi) If Δ becomes zero on putting $n = \alpha$ then we say that $(n - \alpha)$ is a factor of Δ.

If each element of any row (column) can be expressed as a sum of two terms, then the determinant can be expressed as the sum of two determinants.

e.g. $\begin{vmatrix} a_1+x & b_1+y & c_1+z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(vii) $\begin{vmatrix} a_1 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & 0 & c_3 \end{vmatrix} = a_1b_2c_3 = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & 0 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ 0 & b_2 & 0 \\ 0 & 0 & c_3 \end{vmatrix}$

(viii) If all the elements below leading diagonal or above leading diagonal or except leading elements are zero then the value of the determinant equal to multiplied of all diagonal element.

e.g. $\begin{vmatrix} a_1 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & 0 & c_3 \end{vmatrix} = a_1b_2c_3$

6. Multiplication of Two Determinants

Multiplication of two determinants is possible if they are of same order.

Multiplication of two third order Determinants is defined as follows

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1\ell_1 + b_1\ell_2 + c_1\ell_3 & a_1m_1 + b_1m_2 + c_1m_3 & a_1n_1 + b_1n_2 + c_1n_3 \\ a_2\ell_1 + b_2\ell_2 + c_2\ell_3 & a_2m_1 + b_2m_2 + c_2m_3 & a_2n_1 + b_2n_2 + c_2n_3 \\ a_3\ell_1 + b_3\ell_2 + c_3\ell_3 & a_3m_1 + b_3m_2 + c_3m_3 & a_3n_1 + b_3n_2 + c_3n_3 \end{vmatrix}$$

Note :

- In above case the order of Determinant is same, if the order is different then for their multiplication first of all they should be expressed in the same order.

7. Symmetric Determinant

If the elements of a determinant are such that $a_{ij} = a_{ji}$, then the determinant is said to be a symmetric determinant e.g. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

8. Skew Symmetric Determinant

If $a_{ij} = -a_{ji}$ then the determinant is said to be a skew symmetric determinant.

e.g. $\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$

Note :

- Every diagonal element of a skew symmetric determinant is always zero.
- The value of a skew symmetric determinant of even order is always a perfect square and that of odd order is always zero.

9. Differentiation of a Determinant

There are two way's of differentiating a determinant

(i) $$\frac{d}{dx}\begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix} = \begin{vmatrix} R'_1 \\ R_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R'_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2 \\ R'_3 \end{vmatrix}$$

R_1, R_2, R_3 represent row's of determinant

(ii) $$\frac{d}{dx}|C_1\ C_2\ C_3| = |C'_1\ C_2\ C_3| + |C_1\ C'_2\ C_3| + |C_1\ C_2\ C'_3|$$

where C_1, C_2, C_3 represents column of determinant.

10. Integration of Determinant

If Δ any single row or column contain elements which are functions of x, then integration can be done in determinant format only as following.

If $\Delta = \begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ p & q & r \end{vmatrix}$ where a, b, c, p, q, r all are constants, then

$$\int_a^b \Delta dx = \begin{vmatrix} \int_a^b f(x)dx & \int_a^b g(x)dx & \int_a^b h(x)dx \\ a & b & c \\ p & q & r \end{vmatrix}$$

If elements are not like above it is advisable to simplify determinants then integrate it.

11. Crammer's Rule (System of equation)

(i) System of equation involving two variable :

$$a_1x + b_1y + c_1 = 0$$
$$a_2x + b_2y + c_2 = 0$$

Consistent
(System of equation has solution)

- unique solution
  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ or $a_1b_2 - a_2b_1 \neq 0$
  (Equations represents intersecting lines)
- Infinite solution
  $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
  (Equations represents coincident lines)

Inconsistent
(System of equation has no solution)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
(Equation represents parallel disjoint lines)

If $\Delta_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$, $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ then $x = \frac{\Delta_1}{\Delta}$, $y = \frac{\Delta_2}{\Delta}$

(ii) System of equations involving three variables :

$a_1x + b_1y + c_1z = d_1$
$a_2x + b_2y + c_2z = d_2$
$a_3x + b_3y + c_3z = d_3$

To solve this system we first define following determinants

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

then $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

Now following algorithm is followed to solve the system.

Check value of Δ

- $\Delta \neq 0$ → Consistent system and has unique solution $x = \frac{\Delta_1}{\Delta}; y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$
- $\Delta = 0$ → Check the values of Δ_1, Δ_2 and Δ_3
  - atleast one of Δ_1, Δ_2, and Δ_3 is not zero → Inconsistent system
  - $\Delta_1 = \Delta_2 = \Delta_3 = 0$ → Put $z = t$ and solve any two equations to get the values of x & y in terms of t

Note :

➢ If $\Delta = 0$ and $\Delta_1 = \Delta_2 = \Delta_3 = 0$, then system of equation may or may not be consistent :

(i) If values of x, y and z in terms of t satisfy third equation then system is said to be consistent and will have infinite solutions.

(ii) If the values of x, y, z doesn't satisfy third equation, then system is said to be inconsistent and will have no solution.

➢ **Trivial solution :** In the solution set of system of equation if all the variables assumes zero, then such a solution set is called trivial solution otherwise the solution are called non-trivial solution.

➢ If $d_1 = d_2 = d_3 = 0$ then system of linear equation are known as system of Homogeneous linear equation which always posses atleast one solution (0, 0, 0)

➢ If system of homogeneous linear equation posses non-zero/non-trivial solution then D = 0. In such case given system has infinite solutions.

12. Application of Determinants in Geometry

(i) The lines : $a_1x + b_1y + c_1 = 0$ .....(1)

$a_2x + b_2y + c_2 = 0$ ..... (2)

$a_3x + b_3y + c_3 = 0$ ......(3)

are concurrent if, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0,$

(ii) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

(iii) Area of a triangle whose vertices are (x_r, y_r); $r = 1, 2, 3$ is $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

If $\Delta = 0$ then the three points are collinear.

(iv) Equation of a straight line passing through (x_1, y_1) & (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

13. Important Determinants

(i) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

(ii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$

(iii) $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)\ (b-c)(c-a)(ab+bc+ca)$

(iv) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2+b^2+c^2-ab-bc-ca)$

(v) $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)\ (b-c)\ (c-a)$

(vi) $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)\ (b-c)(c-a)\ (ab+bc+ca)$

(vii) $\begin{vmatrix} a & bc & abc \\ b & ca & abc \\ c & ab & abc \end{vmatrix} = \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} = abc\ (a-b)\ (b-c)(c-a)$

(viii) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -a^3-b^3-c^3+3abc$

SHORT TRICKS

Trick -1 Method of substitution

Q.1 If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ then

(A) $\Delta_1 = 3\Delta_2^2$ (B) $\frac{d}{dx}(\Delta_1) = 3\Delta_2^2$ (C) $\frac{d}{dx}(\Delta_1) = 3\Delta_2$ (D) $\Delta_2 = 3\Delta_1^2$

Sol. **[C]**

<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
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Since we know that if $\Delta = f(x) = \begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix}$, then

$$\frac{d}{dx}(\Delta) = \begin{vmatrix} \frac{d}{dx}(R_1) \\ R_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ \frac{d}{dx}(R_2) \\ R_3 \end{vmatrix} + \begin{vmatrix} R_1 \\ R_2 \\ \frac{d}{dx}(R_3) \end{vmatrix}$$

$$\therefore \frac{d}{dx}(\Delta_1) = \begin{vmatrix} \frac{d}{dx}(x) & \frac{d}{dx}(b) & \frac{d}{dx}(b) \\ a & x & b \\ a & a & x \end{vmatrix}$$

$$+ \begin{vmatrix} x & b & b \\ \frac{d}{dx}(a) & \frac{d}{dx}(x) & \frac{d}{dx}(b) \\ a & a & x \end{vmatrix} + \begin{vmatrix} x & b & b \\ a & x & b \\ \frac{d}{dx}(a) & \frac{d}{dx}(a) & \frac{d}{dx}(x) \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ a & x & b \\ a & a & x \end{vmatrix} + \begin{vmatrix} x & b & b \\ 0 & 1 & 0 \\ a & a & x \end{vmatrix} + \begin{vmatrix} x & b & b \\ a & x & b \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix} = 3\begin{vmatrix} x & b \\ a & x \end{vmatrix} = 3\Delta_2.$$

</td><td>

Put a = 1 and b = 0

$\Delta_1 = x^3$ and $\Delta_2 = x^2$

$\frac{d}{dx}(\Delta_1) = 3x^2 = 3\Delta_2$

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Q.2 If $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Px - 12$ then-

(A) P = 24 (B) P = – 24 (C) P = 0 (D) P = 12

Sol. **[A]**

Proper Method	Short Trick
Applying - $R_2 \to R_2 - (R_1 + R_3)$ Determinant $\begin{vmatrix} x^2+x & x+1 & x-2 \\ -4 & 0 & 0 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$ Applying $= R_1 \to R_1 + \frac{1}{4}x^2R_2$ and $R_3 \to R_3 + \frac{1}{4}x^2R_2$ Determinant $= \begin{vmatrix} x & x+1 & x-2 \\ -4 & 0 & 0 \\ 2x+3 & 2x-1 & 2x-1 \end{vmatrix}$ Applying $= R_3 \to R_3 - 2R_1$ Determinant $= \begin{vmatrix} x+0 & x+1 & x-2 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}$ $= \begin{vmatrix} x & x & x \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 1 & -2 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}$ $= x\begin{vmatrix} 1 & 1 & 1 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 1 & -2 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}$ $= (24x - 12) \Rightarrow P = 24.$	Put $x = 1$ $\begin{vmatrix} 2 & 2 & -1 \\ 4 & 3 & 0 \\ 6 & 1 & 1 \end{vmatrix} = P - 12$ $\Rightarrow 2(3) - 2(4) - 1(4 - 18) = P - 12$ $\Rightarrow P = 24$

Q.3 If a, b, c, are p^{th}, q^{th} and r^{th} terms respectively of a H.P., then value of the determinant $\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ is -

(A) $p + q + r$ (B) 0 (C) $1/p + 1/q + 1/r$ (D) pqr

Sol. [B]

Proper Method	Short Trick
Let A and d be the first term and common difference of the corresponding A.P., then $\frac{1}{a} = A + (p-1)d$ $\frac{1}{b} = A + (q-1)d$ $\frac{1}{c} = A + (r-1)d$ Now det. $= abc\begin{vmatrix} 1/a & 1/b & 1/c \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix} = abc$	Let $p = 1, q = 2, r = 3$ then $a = 2, b = 3, c = 6$ $\begin{vmatrix} 18 & 12 & 6 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = 18(-1) - 12(-2) + 6(-1) = 0$

$\begin{vmatrix} A+(p-1)d & A+(q-1)d & A+(r-1)d \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ $= abc\begin{vmatrix} A & A & A \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix} + abcd\begin{vmatrix} p-1 & q-1 & r-1 \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$ $= 0 + 0 = 0$	

Q.4 If $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^2 & 3n^2-3n \end{vmatrix}$, then $\sum_{r=1}^{n} \Delta_r$ equals -

(A) 1 (B) – 1 (C) 0 (D) n

Sol. **[C]**

Proper Method	**Short Trick**
$\because \sum_{r=1}^{n}(r-1) = 1 + 2 + ...+ (n-1) = \frac{n(n-1)}{2}$ $\sum_{r=1}^{n}(r-1)^2 = 1^2 + 2^2 + ...+(n-1)^2 = \frac{n(n-1)(2n-1)}{6}$ $\sum_{r=1}^{n}(r-1)^3 = 1^3 + 2^3 + ...+ (n-1)^3 = \frac{n^2(n-1)^2}{4}$ $\therefore \sum_{r=1}^{n}\Delta_r = \begin{vmatrix} \frac{n(n-1)}{2} & n & 6 \\ \frac{1}{6}n(n-1)(2n-1) & 2n^2 & 2(2n-1) \\ \frac{1}{4}n^2(n-1)^2 & 3n^3 & 3n(n-1) \end{vmatrix}$ $= \frac{n(n-1)}{12}\begin{vmatrix} 6 & n & 6 \\ 2(2n-1) & 2n^2 & 2(2n-1) \\ 3n(n-1) & 3n^3 & 3n(n-1) \end{vmatrix} = 0$	Put r = 1, n = 1 $\Delta_1 = \begin{vmatrix} 0 & 1 & 6 \\ 0 & 2 & 0 \\ 0 & 3 & 0 \end{vmatrix} = 0$

Q.5 If $D_r = \begin{vmatrix} r & x & n(n+1)/2 \\ 2r-1 & y & n^2 \\ 3r-2 & z & n(3n-1)/2 \end{vmatrix}$, then $\sum_{r=1}^{n} D_r$ is equal to -

(A) $\frac{1}{6}n(n + 1)(2n + 1)$ (B) $\frac{1}{4}n^2(n + 1)^2$ (C) 0 (D) $\frac{n(n+1)}{2}$

Sol. **[C]**

Proper Method	**Short Trick**
$D_r = \begin{vmatrix} r & x & n(n+1)/2 \\ 2r-1 & y & n^2 \\ 3r-2 & z & n(3n-1)/2 \end{vmatrix}$ then $\Sigma D_r = \begin{vmatrix} \Sigma r & x & n(n+1)/2 \\ \Sigma(2r-1) & y & n^2 \\ \Sigma(3r-2) & z & n(3n-1)/2 \end{vmatrix}$	r = 1, n = 1 $D_r = \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix} = 0$

$\therefore \Sigma r = n(n+1)/2$

$\Sigma(2r-1) = 2\Sigma r - \Sigma 1 = \dfrac{2n(n+1)}{2} - n = n^2$

$\& \ \Sigma(3r-2) = 3\Sigma r - 2\Sigma 1 = \dfrac{3n(n+1)}{2} - 2n$

$= \dfrac{n(3n-1)}{2}$

$\therefore$ C_1 & C_3 are identical so, $\Sigma D_r = 0$

Q.6 If $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = k \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, then k is equal to-

(A) 1 (B) 2 (C) 4 (D) 0

Sol. **[C]**

Proper Method

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$$

$R_2 \to R_2 - R_3$

$$= \begin{vmatrix} a^2 & b^2 & c^2 \\ 4a & 4b & 4c \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$$

$$= 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$$

$R_3 \to R_3 - (R_1 - 2R_2)$

$$= 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

$\therefore k = 4$

Short Trick

Put a = 0, b = 1, c = 2

$$\begin{vmatrix} 0 & 1 & 4 \\ 1 & 4 & 9 \\ 1 & 0 & 1 \end{vmatrix} = k \begin{vmatrix} 0 & 1 & 4 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix}$$

$\Rightarrow -(-8) + 4(-4) = k(-2)$

$\Rightarrow k = 4$

Q.7 If x is real number such that $\begin{vmatrix} x+1 & x+2 & x+\alpha \\ x+2 & x+3 & x+\beta \\ x+3 & x+4 & x+\gamma \end{vmatrix} = 0$ then α, β, γ, are in

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Sol. **[A]**

Proper Method

Let, $R_2 = R_2 - \left(\dfrac{R_1+R_3}{2}\right)$

$$\therefore \Delta \Rightarrow \begin{vmatrix} x+1 & x+2 & x+\alpha \\ 0 & 0 & \beta-\left(\dfrac{\alpha+\gamma}{2}\right) \\ x+3 & x+4 & x+\gamma \end{vmatrix} = 0$$

Short Trick

Put x = 0

$$\begin{vmatrix} 1 & 2 & \alpha \\ 2 & 3 & \beta \\ 3 & 4 & \gamma \end{vmatrix} = 0$$

$\Rightarrow 1(3\gamma - 4\beta) - 2(2\gamma - 3\beta) + \alpha(-1) = 0$

$\Rightarrow 2\beta = \alpha + \gamma$

<table>
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<td>
Expanding along R_2, we get
$\beta - \left(\frac{\alpha+\gamma}{2}\right) = 0$
$\therefore\ \alpha + \gamma = 2\beta$
$\therefore\ \alpha, \beta, \gamma$ are in AP
</td>
<td>
α, β, γ are in A.P.
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Q.8 The value of the determinant $\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$ is-

(A) 0 (B) $8x^2$ (C) 8 (D) –8

Sol. [D]

<table>
<tr>
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Proper Method
$\begin{vmatrix} (x-2)^2 & (x-1)^2 & x^2 \\ (x-1)^2 & x^2 & (x+1)^2 \\ x^2 & (x+1)^2 & (x+2)^2 \end{vmatrix}$
$C_1 \to C_1 - C_2$
$= \begin{vmatrix} -2x+3 & (x-1)^2 & x^2 \\ -2x+1 & x^2 & (x+1)^2 \\ -2x-1 & (x+1)^2 & (x+2)^2 \end{vmatrix}$
$C_2 \to C_2 - C_3$
$= \begin{vmatrix} -2x+3 & -2x+1 & x^2 \\ -2x+1 & -2x-1 & (x+1)^2 \\ -2x-1 & -2x-3 & (x+2)^2 \end{vmatrix}$
$C_1 \to C_1 - C_2$
$\begin{vmatrix} 2 & -2x+1 & x^2 \\ 2 & -2x-1 & (x+1)^2 \\ 2 & -2x-3 & (x+2)^2 \end{vmatrix}$
$R_1 \to R_1 - R_2,\ R_2 \to R_2 - R_3$
$\begin{vmatrix} 0 & 2 & -2x-1 \\ 0 & 2 & -2x-3 \\ 2 & -2x-3 & (x+2)^2 \end{vmatrix}$
$= 2\{-4x - 6 + 4x + 2\} = -8$
</td>
<td>
Short Trick
Put $x = 2$
$\begin{vmatrix} 0 & 1 & 4 \\ 1 & 4 & 9 \\ 4 & 9 & 16 \end{vmatrix} = -1(-20) + 4(-7) = -8$
</td>
</tr>
</table>

Q.9 If $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = \lambda \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ then λ is equal to -

(A) 1 (B) 2 (C) 3 (D) 4

Sol. [B]

Proper Method	Short Trick
$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$ $C_1 \to C_1 - C_2,\ C_2 \to C_2 - C_3$ $= \begin{vmatrix} a-c & b-a & c+a \\ b-a & c-b & a+b \\ c-b & a-c & b+c \end{vmatrix}$ $C_1 \to C_1 + C_3$ $= \begin{vmatrix} 2a & b-a & c+a \\ 2b & c-b & a+b \\ 2c & a-c & b+c \end{vmatrix}$ $= 2\begin{vmatrix} a & b-a & c+a \\ b & c-b & a+b \\ c & a-c & b+c \end{vmatrix}$ $C_2 \to C_2 + C_1,\ C_3 \to C_3 - C_1$ $= 2\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ $\therefore \lambda = 2$	Put $a = 0, b = 1, c = 2$ $\Rightarrow \begin{vmatrix} 1 & 3 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \lambda \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{vmatrix}$ $\Rightarrow 1(5) - 3(7) + 2(-1)$ $= \lambda\{-1(1) + 2(-4)\}$ $\Rightarrow \lambda = 2$

Q.10 For any ΔABC, the value of determinant $\begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ is-

(A) 0
(B) 1
(C) $\sin A \sin B \sin C$
(D) $\sin A + \sin B + \sin C$

Sol. [A]

Proper Method	Short Trick
$\begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ $R_1 \to R_1 - R_2,\ R_2 \to R_2 - R_3$ $\Rightarrow \begin{vmatrix} \sin^2 A - \sin^2 B & \cot A - \cot B & 0 \\ \sin^2 B - \sin^2 C & \cot B - \cot C & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ $\Rightarrow \begin{vmatrix} \sin(A+B)\sin(A-B) & \dfrac{\sin(B-A)}{\sin A \cdot \sin B} & 0 \\ \sin(B+C)\sin(B-C) & \dfrac{\sin(C-B)}{\sin B \cdot \sin C} & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$	Let $A = B = C = 60°$ $\begin{vmatrix} 3/4 & 1/3 & 1 \\ 3/4 & 1/3 & 1 \\ 3/4 & 1/3 & 1 \end{vmatrix} = 0$

$$= \sin(A-B)\sin(B-C)\begin{vmatrix} \sin(A+B) & \dfrac{-1}{\sin A\cdot\sin B} & 0 \\ \sin(B+C) & \dfrac{-1}{\sin B\cdot\sin C} & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

$$= \sin(A-B)\sin(B-C)\left\{\frac{-\sin(A+B)}{\sin B\cdot\sin C} + \frac{\sin(B+C)}{\sin A\cdot\sin B}\right\}$$

$$= \sin(A-B)\cdot\sin(B-C)\left\{\frac{-\sin C}{\sin B\cdot\sin C} + \frac{\sin A}{\sin A\cdot\sin B}\right\}$$

$= \sin(A-B)\cdot\sin(B-C)\,(0) = 0$

Q.11 If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$ then $\sum_{r=1}^{n} S_r$ does not depends on

(A) x (B) y (C) n (D) all of these

Sol. [D]

Proper Method

$$\Sigma S_r = \begin{vmatrix} \Sigma 2r & x & n(n+1) \\ \Sigma(6r^2-1) & y & n^2(2n+3) \\ \Sigma(4r^3-2nr) & z & n^3(n+1) \end{vmatrix}$$

$\therefore\ \Sigma 2r = 2\Sigma r = n(n+1)$

& $\Sigma(6r^2-1) = 6\Sigma r^2 - \Sigma 1 = n(n+1)(2n+1) - n$

$= n[2n^2 + 3n + 1 - 1] = n^2(2n+3)$

& $\Sigma(4r^3 - 2nr) = 4\Sigma r^3 - 2n\Sigma r$

$$= 4\left[\frac{n(n+1)}{2}\right]^2 - 2n\,\frac{n(n+1)}{2}$$

$= [n(n+1)]^2 - n^2(n+1)$

$= n^2(n+1)\,[(n+1) - 1]$

$= n^3(n+1)$

$\because\ C_1$ & C_3 are identical so, $\Sigma S_r = 0$

Short Trick

Put r = 1, n = 1

$$S_1 = \begin{vmatrix} 2 & x & 2 \\ 5 & y & 5 \\ 2 & z & 2 \end{vmatrix} = 0$$

Q.12 In a ΔABC, if $\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$ then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to-

(A) 9/4 (B) 4/9 (C) 1 (D) $3\sqrt{3}$

Sol. [A]

Proper Method

$$\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$$

$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$

$$\Rightarrow \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2} = 0$$

$\therefore\ a = b = c$

$\therefore$ it is equilateral Δ so, $A = B = C = 60°$

$\therefore\ \sin^2 A + \sin^2 B + \sin^2 C = 9/4$

Short Trick

$\begin{vmatrix} 1 & a & b \\ 1 & c & a \\ 1 & b & c \end{vmatrix} = 0$ when $a = b = c$

i.e. triangle is equilaterals

$\therefore\ \sin^2 A + \sin^2 B + \sin^2 C = \dfrac{9}{4}$

Q.13 If $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots\ldots$ are in G.P., then the value of the determinant

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}, \text{ is-}$$ **[AIEEE-2004]**

(A) 0 (B) 1 (C) 2 (D) -2

Sol. **[A]**

Proper Method

$\because a_1, a_2, a_3 \ldots\ldots a_n$ are in G.P.

$\therefore\ a_{n+1}^2 = a_n\, a_{n+2} \Rightarrow 2\log a_{n+1} = \log a_n + \log a_{n+2}$

$a_{n+4}^2 = a_{n+3} \cdot a_{n+5} \Rightarrow 2\log a_{n+4} = \log a_{n+3} + \log a_{n+5}$

$a_{n+7}^2 = a_{n+6} \cdot a_{n+8} \Rightarrow 2\log a_{n+7} = \log a_{n+6} + \log a_{n+8}$

$$= \begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$$

$C_1 \to C_1 + C_3$

$$= \begin{vmatrix} 2\log a_{n+1} & \log a_{n+1} & \log a_{n+2} \\ 2\log a_{n+4} & \log a_{n+4} & \log a_{n+5} \\ 2\log a_{n+7} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$$

$$= 2\begin{vmatrix} \log a_{n+1} & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+4} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+7} & \log a_{n+7} & \log a_{n+8} \end{vmatrix} = 0\ (\because C_1 = C_2)$$

Short Trick

Let the G.P. be $2, 2^2, 2^3\ 2^4 \ldots$.

Let $n = 1$, then

$$\begin{vmatrix} \log 2 & \log 2^2 & \log 2^3 \\ \log 2^4 & \log 2^5 & \log 2^6 \\ \log 2^7 & \log 2^8 & \log 2^9 \end{vmatrix}$$

$$= (\log 2)^3 \begin{vmatrix} 2 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = (\log 2)^3 \times 0 = 0$$

Q.14 If p^{th}, q^{th}, r^{th} term of a GP are ℓ, m, n then the value of $\begin{vmatrix} \log \ell & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ is equal to- **[AIEEE-2002]**

(A) 0 (B) 1 (C) $\ell + m + n$ (D) $p + q + r$

Sol. **[A]**

Proper Method

If first term $= \alpha$, common ratio $= \beta$

$\ell = \alpha\beta^{p-1} \Rightarrow \log \ell = \log\alpha + (p-1)\log\beta$

$m = \alpha\beta^{q-1} \Rightarrow \log m = \log\alpha + (q-1)\log\beta$

$n = \alpha\beta^{r-1} \Rightarrow \log n = \log\alpha + (r-1)\log\beta$

$$\therefore \Delta = \begin{vmatrix} \log \ell & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$$

Let $R_3 = R_3 - R_2$; $R_2 = R_2 - R_1$

$$\therefore \Delta = \begin{vmatrix} \log \ell & p & 1 \\ (q-p)\log\beta & q-p & 0 \\ (r-q)\log\beta & r-q & 0 \end{vmatrix}$$

$\therefore \Delta = 0$

Short Trick

Let $p = 1, q = 2, r = 3$

and $\ell = 2, m = 2^2, n = 2^3$

$$\text{then} \begin{vmatrix} \log 2 & 1 & 1 \\ \log 2^2 & 2 & 1 \\ \log 2^2 & 3 & 1 \end{vmatrix} = \log 2 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = 0$$

Q.15 If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then f(x) is a polynomial of degree –

[AIEEE-2005]

(A) 1 (B) 0 (C) 3 (D) 2

Sol. **[D]**

Proper Method

Let, $C_1 = C_1 + C_2 + C_3$

$$f(x) = \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 1 & 1+b^2x & (1+c^2)x \\ 1 & (1+b^2)x & 1+c^2x \end{vmatrix}$$

Let $R_3 = R_3 - R_2$; $R_2 = R_2 - R_1$

$$\therefore f(x) = \begin{vmatrix} 1 & (1+b^2)x & (1+c^2)x \\ 0 & 1-x & 0 \\ 0 & x-1 & 1-x \end{vmatrix}$$

$\therefore f(x) = (1-x)^2$

$\therefore$ Hence, Degree = 2

Short Trick

$a^2 + b^2 + c^2 = -2$

let $a^2 = 0$, $b^2 = -1$, $c^2 = -1$

$$\therefore f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x & 1-x & 0 \\ x & 0 & 1-x \end{vmatrix}$$

$= 1 - 2x + x^2$

$\Rightarrow$ degree = 2

Q.16 Let a, b, c be such that $b(a + c) \neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$,

then the value of n is : **[AIEEE-2009]**

(A) any even integer (B) any odd integer (C) any integer (D) zero

Sol. **[B]**

Proper Method

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} = 0$$

n is odd integer then above relation is true

n is even integer then,

$$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} = 0$$

$$= \begin{vmatrix} a & 1 & -2 \\ -b & 2b+1 & -2 \\ c & -1 & 2 \end{vmatrix} = 0 \ (C_3 = C_3 - C_2,$$

$$= \begin{vmatrix} a+c & 0 & 0 \\ -b & 2b+1 & -2 \\ c & -1 & 2 \end{vmatrix} = 0 \ (R_1 \to R_1 + R_3)$$

$8b(a + c) = 0$ which is not true.

Short Trick

$b(a + c) \neq 0$

so let a = b = c = 1, then

$$\begin{vmatrix} 1 & 2 & 0 \\ -1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} + (-1)^n \begin{vmatrix} 2 & 2 & 0 \\ 0 & 0 & 2 \\ 1 & -1 & 1 \end{vmatrix} = 0$$

$\Rightarrow 1(4) - 2(-2) + (-1)^n \{2(2) - 2(-2)\} = 0$

$\Rightarrow 8 + (-1)^n 8 = 0$ is possible only if n is any odd integer

Q.17 If $\begin{vmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{vmatrix} = k\, a^2 b^2 c^2$, then k is equal to :

(A) 3 (B) 4 (C) 1 (D) 2

Sol. [B]

Proper Method	Short Trick
Let $\Delta = \begin{vmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{vmatrix}$ Multiply C_1, C_2 and C_3 by a, b, c respectively $\Delta = \frac{1}{abc}\begin{vmatrix} a(b^2+c^2) & ab^2 & ac^2 \\ a^2b & b(c^2+a^2) & bc^2 \\ a^2c & cb^2 & c(a^2+b^2) \end{vmatrix}$ $= \frac{abc}{abc}\begin{vmatrix} b^2+c^2 & b^2 & c^2 \\ a^2 & c^2+a^2 & c^2 \\ a^2 & b^2 & a^2+b^2 \end{vmatrix}$ Apply $C_1 \to C_1 - C_2 - C_3$ $\Delta = \begin{vmatrix} 0 & b^2 & c^2 \\ -2c^2 & c^2+a^2 & c^2 \\ -2b^2 & b^2 & a^2+b^2 \end{vmatrix} = 4a^2b^2c^2$ $\therefore k = 4$	Put a = b = c = 1, both sides then $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = k$ $\Rightarrow k = 2(4-1) - 1(1) + (-1) = 4$

Q.18 If $D_r = \begin{vmatrix} 2^{r-1} & 2.3^{r-1} & 4.5^{r-1} \\ \alpha & \beta & \gamma \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$, then the value of $\sum_{r=1}^{n} D_r$ is- [IIT-1993]

(A) 0 (B) $\alpha\beta\gamma$ (C) $\alpha+\beta+\gamma$ (D) $\alpha.2^n + \beta.3^n + \gamma.4^n$

Sol. [A]

Proper Method	Short Trick
$\Sigma D_r = \begin{vmatrix} \Sigma 2^{r-1} & 2\Sigma 3^{r-1} & 4\Sigma 5^{r-1} \\ \alpha & \beta & \gamma \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$ clearly $\Sigma 2^{r-1} = 1.\frac{(2^n-1)}{2-1} = 2^n - 1$ & $2.\Sigma 3^{r-1} = 2.\frac{(3^n-1)}{3-1} = 3^n - 1$ & $4.\Sigma 5^{r-1} = 4.\frac{(5^n-1)}{5-1} = 5^n - 1$ $\therefore R_1$ & R_3 are identical so, $\Sigma D_r = 0$	Let r = 1, n = 1 $D_1 = \begin{vmatrix} 1 & 2 & 4 \\ \alpha & \beta & \gamma \\ 1 & 2 & 4 \end{vmatrix} = 0$

Q.19 If $\begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & b+c & -2c \end{vmatrix} = \alpha(a+b)(b+c)(c+a) \neq 0$, then α is equal to **[AIEEE Online-2012]**

(A) 1 (B) a + b + c (C) abc (D) 4

Sol. **[D]**

Proper Method

Let $\Delta = \begin{vmatrix} -2a & a+b & a+c \\ b+a & -2b & b+c \\ c+a & b+c & -2c \end{vmatrix}$

Applying $C_1 + C_3$ and $C_2 + C_3$

$$\Delta = \begin{vmatrix} -a+c & 2a+b+c & a+c \\ 2b+a+c & -b+c & b+c \\ a-c & b-c & -2c \end{vmatrix}$$

Now, applying $R_1 + R_3$ and $R_2 + R_3$

$$\Delta = \begin{vmatrix} 0 & 2(a+b) & a-c \\ 2(a+b) & 0 & b-c \\ a-c & b-c & -2c \end{vmatrix}$$

On expanding, we get

$\Delta = -2(a+b)\{-2c[2(a+b)] - (a-c)(b-c)\}$
$\quad + (a-c)[2(a+b)(b-c)]$

$\Delta = 8c(a+b)(a+b) + 4(a+b)(a-c)(b-c)$

$= 4(a+b)[2ac + 2bc + ab - bc - ac + c^2]$

$= 4(a+b)[ac + bc + ab + c^2]$

$= 4(a+b)[c(a+c) + b(a+c)]$

$= 4(a+b)(b+c)(c+a)$

$= \alpha(a+b)(b+c)(c+a)$

Hence, $\alpha = 4$

Short Trick

Let a = 0, b = 1, c = 2

$\begin{vmatrix} 0 & 1 & 2 \\ 1 & -2 & 3 \\ 2 & 3 & -4 \end{vmatrix} = \alpha(1)(3)(2)$

$\Rightarrow -1(-10) + 2(7) = 6\alpha$

$\Rightarrow \alpha = 4$

Q.20 If a, b, c are sides of scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is : **[JEE Main Online-2013]**

(A) non-negative (B) negative (C) positive (D) non-positive

Sol. **[B]**

Proper Method

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$= (a+b+c)\begin{vmatrix} 0 & 0 & 1 \\ b-c & c-a & a \\ c-a & a-b & b \end{vmatrix}$$

Short Trick

$\because$ a, b, c are sides of scalene triangle

So let a = 1, b = 2, c = 3

$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = -18$ (negative)

$= (a + b + c)\,[ab + bc + ca - a^2 - b^2 - c^2]$
$= -(a + b + c)\,[(a - b)^2 + (b - c)^2 + (c - a)^2]$
Since a, b, c are sides of a scalene triangle, therefore at least two of the a, b, c will be unequal.
$\therefore\ (a - b)^2 + (b - c)^2 + (c - a)^2 > 0$
Also $a + b + c = 0$
$\therefore\ -(a + b + c)\,[(a - b)^2 + (b - c)^2 + (c - a)^2] < 0$

Q.21 If $\Delta_r = \begin{vmatrix} r & 2r-1 & 3r-2 \\ \frac{n}{2} & n-1 & a \\ \frac{1}{2}n(n-1) & (n-1)^2 & \frac{1}{2}(n-1)(3n+4) \end{vmatrix}$, then the value of $\sum_{r=1}^{n-1} \Delta_r$: **[JEE Main Online-2014]**

(A) Depends only on a
(B) Depends only on n
(C) Depends both on a and n
(D) Is independent of both a and n

Sol. [D]

Proper Method

$$\sum_{r=1}^{n-1} r = 1 + 2 + 3 + \dots + (n-1) = \frac{n(n-1)}{2}$$

$$\sum_{r=1}^{n-1} (2r-1) = 1 + 3 + 5 + \dots + [2(n-1) - 2] = (n-1)^2$$

$$\sum_{r=1}^{n-1} (3r-2) = 1 + 4 + 7 + \dots + (3n-3-2)$$

$$= \frac{(n-1)(3n-4)}{2}$$

$$\therefore \sum_{r=1}^{n-1} \Delta_r = \begin{vmatrix} \Sigma r & \Sigma(2r-1) & \Sigma(3r-2) \\ \frac{n}{2} & n-1 & a \\ \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \end{vmatrix}$$

$\sum_{r=1}^{n-1} \Delta_r$ consists of $(n-1)$ determinants in L.H.S. and in R.H.S. every constituent of first row consist of $(n-1)$ elements and hence it can be splitted into sum of $(n-1)$ determinants.

$$\therefore \sum_{r=1}^{n-1} \Delta_r = \begin{vmatrix} \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \\ \frac{n}{2} & n-1 & a \\ \frac{n(n-1)}{2} & (n-1)^2 & \frac{(n-1)(3n-4)}{2} \end{vmatrix}$$

$$= 0$$

($\because$ R_1 and R_3 are identical)

Hence, value of $\sum_{r=1}^{n-1} \Delta_r$ is independent of both 'a' and 'n'.

Short Trick

$r = 1,\ n = 2$

$$\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & a \\ 1 & 1 & 5 \end{vmatrix} = 0$$

Q.22 If a, b, c are in G.P., then the value of determinant $\Delta = \begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ ax+b & bx+c & 0 \end{vmatrix}$ is- **[IIT-1993]**

(A) 1 (B) 0 (C) – 1 (D) – 2

Sol. **[B]**

Proper Method	**Short Trick**
Let $R_3 = R_3 - (xR_1 + R_2)$ $\therefore \Delta = \begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ 0 & 0 & -[x(ax+b)+(bx+c)] \end{vmatrix}$ expanding along R_3 $\therefore = -[x(ax+b)+(bx+c)]\ (ac-b^2)$ but a,b,c are in GP so, $\Delta = 0$	Let a = 1, b = 2, c = 4 and x = 0 $\Delta = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 0 \end{vmatrix}$ $\Delta = 1(-16) - 2(-8) + 2(0) = 0$

Q.23 If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$ then f(100) is equal to - **[IIT-1999]**

(A) 0 (B) 1 (C) 100 (D) –100

Sol. **[A]**

Proper Method	**Short Trick**
$f(x) \begin{vmatrix} 1 & x & (x+1) \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix} \begin{matrix} \\ \to x \\ \to x(x-1) \end{matrix}$ $f(x) = x^2(x-1) \begin{vmatrix} 1 & x & (x+1) \\ 2 & (x-1) & (x+1) \\ 3 & (x-2) & (x+1) \end{vmatrix}$ Let $R_3 = R_3 - R_2$, $R_2 = R_2 - R_1$ $f(x) = x^2(x-1) \begin{vmatrix} 1 & x & x+1 \\ 1 & -1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = 0$ $\therefore f(100) = 0$	$\because$ here $f(0) = f(1) = f(-1) = ... = 0$ $\therefore$ The value of f(x) is independent of x so, it is an identity in x, so put x = 0 $f(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{vmatrix} = 0$ $\Rightarrow f(x) = 0$ $\Rightarrow f(100) = 0$

Q.24 If a, b, c, d, e and f are in G.P. then the value of $\begin{vmatrix} a^2 & d^2 & x \\ b^2 & e^2 & y \\ c^2 & f^2 & z \end{vmatrix}$ depends on

(A) x and y (B) x and z (C) y and z (D) Independent of x, y & z

Sol. **[D]**

Proper Method	**Short Trick**
$b = ar$, $c = ar^2$, $d = ar^3$, $e = ar^4$, $f = ar^5$ Determinant = $\begin{vmatrix} a^2 & a^2r^6 & x \\ a^2r^2 & a^2r^8 & y \\ a^2r^4 & a^2r^{10} & z \end{vmatrix}$ $= a^2a^2r^6 \begin{vmatrix} 1 & 1 & x \\ r^2 & r^2 & y \\ r^4 & r^4 & z \end{vmatrix} = 0$	Let a = 1, b = 2, $c = 2^2$, $d = 2^3$, $e = 2^4$, $f = 2^5$ $= \begin{vmatrix} 1 & 2^6 & x \\ 2^2 & 2^8 & y \\ 2^4 & 2^{10} & z \end{vmatrix}$ $= 2^6 \begin{vmatrix} 1 & 1 & x \\ 2^2 & 2^2 & y \\ 2^4 & 2^4 & z \end{vmatrix} = 0$

Q.25 $\begin{vmatrix} a & a+b & a+2b \\ a+2b & a & a+b \\ a+b & a+2b & a \end{vmatrix} = mb^n(a+b)$, then

(A) m = 9, n =1 (B) m = 1, n = 2 (C) m = 9, n = 2 (D) m = 9, n = 3

Sol. **[C]**

Proper Method	**Short Trick**
$\Delta = \begin{vmatrix} a & a+b & a+2b \\ a+2b & a & a+b \\ a+b & a+2b & a \end{vmatrix}$ $\Delta = 3(a+b)\begin{vmatrix} 1 & a+b & a+2b \\ 1 & a & a+b \\ 1 & a+2b & a \end{vmatrix}$ $C_1 \to C_1 + C_2 + C_3$ $= 3(a+b)\begin{vmatrix} 1 & a+b & a+2b \\ 0 & -b & -b \\ 0 & b & -2b \end{vmatrix}$ $= 9b^2(a+b) = mb^n(a+b)$ $\therefore$ m = 9, n = 2	Let a = 1, b = 2 $\begin{vmatrix} 1 & 3 & 5 \\ 5 & 1 & 3 \\ 3 & 5 & 1 \end{vmatrix} = m.2^n(3)$ $\Rightarrow 1(-14) - 3(-4) + 5(22) = m.2^n(3)$ $\Rightarrow 108 = m.2^n(3)$ $\Rightarrow 9 \times 2^2(3) = m.2^n(3)$ m= 9, n = 2

Q.26 Let $\Delta_r = \begin{vmatrix} 2^{r-1} & \dfrac{(r+1)!}{1+\frac{1}{r}} & 2r \\ a & b & c \\ 2^n-1 & (n+1)!-1 & n(n+1) \end{vmatrix}$, then value of $\sum_{r=1}^{n} \Delta_r$ is

(A) 0 (B) (n + 3)! (C) a (n)! + b (D) (n + 2)!

Sol. **[A]**

Proper Method	**Short Trick**
$\Delta_r = \begin{vmatrix} 2^{r-1} & r(r!) & 2r \\ a & b & c \\ 2^n-1 & (n+1)!-1 & n(n+1) \end{vmatrix}$ $\sum_{r=1}^{n}\Delta_r = \begin{vmatrix} \sum_{r=1}^{n} 2^{r-1} & \sum_{r=1}^{n} r(r!) & \sum_{r=1}^{n} 2r \\ a & b & c \\ 2^n-1 & (n+1)!-1 & n(n+1) \end{vmatrix} = 0$ $\because \sum_{r=1}^{n} 2^{r-1} = 2^n - 1$ $\sum_{r=1}^{n} r(r!) = (n+1)! - 1$	r = 1, n = 1 $\Delta_1 = \begin{vmatrix} 1 & 1 & 2 \\ a & b & c \\ 1 & 1 & 2 \end{vmatrix} = 0$

Q.27 Let $\Delta = \begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & a+b & c \end{vmatrix}$. Then $\Delta =$

(A) $(a+b+c)(a^2+b^2+c^2)$ (B) $a^3+b^3+c^3$
(C) $a^2+b^2+c^2$ (D) 0

Sol. **[A]**

Proper Method

$$\Delta = \begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & a+b & c \end{vmatrix}$$

$C_1 \to aC_1, C_2 \to bC_2, C_3 \to cC_3$ and then $C_1 \to C_1 + C_2 + C_3$, and taking $(a^2+b^2+c^2)$ common, we get

$$\Delta = \frac{(a^2+b^2+c^2)}{abc}\begin{vmatrix} 1 & b^2-bc & c^2+bc \\ 1 & b^2 & c^2-ac \\ 1 & b^2+ab & c^2 \end{vmatrix}$$

$$= \frac{a^2+b^2+c^2}{a}\begin{vmatrix} 1 & b-c & c+b \\ 1 & b & c-a \\ 1 & b+a & c \end{vmatrix}$$

$$= \frac{a^2+b^2+c^2}{a}\begin{vmatrix} 1 & b-c & c+b \\ 0 & c & -a-b \\ 0 & a+c & -b \end{vmatrix}$$

$[R_2 \to R_2 - R_1, R_3 \to R_3 - R_1]$

$$= \frac{a^2+b^2+c^2}{a}\{-bc + a^2 + ac + ab + bc\}$$

$= (a+b+c)(a^2+b^2+c^2)$

Short Trick

Let a = 0, b = 1, c = 2

$$\begin{vmatrix} 0 & -1 & 3 \\ 2 & 1 & 2 \\ -1 & 1 & 2 \end{vmatrix}$$

$= +1(6) + 3(3) = 15$

by option (A) $(a+b+c)(a^2+b^2+c^2)$ is correct option

Q.28 The value of the determinant $\begin{vmatrix} x & x+a & x+2a \\ x+1 & x+2a & x+4a \\ x+2 & x+3a & x+6a \end{vmatrix}$ is :

(A) 0 (B) $a^3 - x^3$ (C) $x^3 - a^3$ (D) $(x-a)^3$

Sol. **[A]**

Proper Method

$(R_2 \to R_2 - R_1; R_3 \to R_3 - R_2)$

$$\Delta = \begin{vmatrix} x & x+a & x+2a \\ 1 & a & 2a \\ 1 & a & 2a \end{vmatrix}$$

$= 0$

Short Trick

Let a = 1, x = 2

$$\begin{vmatrix} 2 & 3 & 4 \\ 3 & 4 & 6 \\ 4 & 5 & 8 \end{vmatrix} = 2(2) - 3(0) + 4(-1) = 0$$

Q.29 If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (y-z)(z-x)(x-y)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$, then :

(A) n = 2 (B) n = – 2 (C) n = – 1 (D) n = 1

Sol. **[C]**

<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
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$\Delta = x^n y^n z^n \begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix}$

$= x^n y^n z^n \begin{vmatrix} 1 & x^2 & x^3 \\ 0 & y^2 - x^2 & y^3 - x^3 \\ 0 & z^2 - x^2 & z^3 - x^3 \end{vmatrix}$

$= x^n y^n z^n (y - x)(z - x) \begin{vmatrix} 1 & x^2 & x^3 \\ 0 & y + x & y^2 + x^2 + yx \\ 0 & z + x & z^2 + x^2 + zx \end{vmatrix}$

$= x^n y^n z^n (y - x)(z - x) \begin{vmatrix} 1 & x^2 & x^3 \\ 0 & y + x & y^2 + x^2 + yx \\ 0 & z - y & z^2 - y^2 + x(z - y) \end{vmatrix}$

$= x^n y^n z^n (z - x)(y - x)(z - y) \begin{vmatrix} 1 & x^2 & x^3 \\ 0 & y + x & y^2 + x^2 + yx \\ 0 & 1 & z + y + x \end{vmatrix}$

$= x^n y^n z^n (x - y)(y - z)(z - x)(xy + yz + zx)$

$= x^n y^n z^n (xyz)(x - y)(y - z)(z - x)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

$= (x - y)(y - z)(z - x)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

$\Rightarrow n + 1 = 0 \Rightarrow n = -1$

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Sum of degree of element of principal diagonal = degree of function on R.H.S

$\Rightarrow n + (n + 2) + (n + 3) = (1 + 1 + 1 - 1)$

$\Rightarrow n = -1$

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Trick -2 Combination of method of substitution and balancing

Q.30 The value of $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$ is equal to -

(A) a^3 (B) b^3 (C) c^3 (D) $a^3 + b^3 + c^3$

Sol. [A]

<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
<tr><td>

$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$

$R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 3R_1$

$= \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix}$

$= a(7a^2 + 3ab - 6a^2 - 3ab)$

$= a(a^2)$

$= a^3$

</td><td>

Put a = 0, b = 1, c = 2

then value of determinant = 0 then option (A) a^3 is correct answer

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</table>

Q.31 The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is -

(A) k (a + b) (b + c) (c + a)
(B) k abc $(a^2 + b^2 + c^2)$
(C) k (a – b) (b – c) (c – a)
(D) k (a + b – c) (b + c – a) (c + a – b)

Sol. **[C]**

Proper Method

$$\Delta = \begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$$

$$\therefore \Delta = \begin{vmatrix} ka & k^2 & 1 \\ kb & k^2 & 1 \\ kc & k^2 & 1 \end{vmatrix} + \begin{vmatrix} ka & a^2 & 1 \\ kb & b^2 & 1 \\ kc & c^2 & 1 \end{vmatrix}$$

$$\Delta = k^2 \begin{vmatrix} ka & 1 & 1 \\ kb & 1 & 1 \\ kc & 1 & 1 \end{vmatrix} + k \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

$\Delta = 0 + k (a – b) (b –c) (c – a)$

Short Trick

Put a = 0, b = 1, c = 2

$$\begin{vmatrix} 0 & k^2 & 1 \\ k & k^2+1 & 1 \\ 2k & k^2+4 & 1 \end{vmatrix}$$

$= – k^2 (– k) + 1(k^3 + 4k – 2k^3 – 2k) = 2k$

by option (C) k(a – b) (b – c) (c – a) is correct option

Q.32 The value of the determinant $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is -

(A) abc (a – b) (b – c) (c – a)
(B) (a – b) (b – c) (c – a) (a + b + c)
(C) (a – b) (b – c) (c – a) (ab + bc + ca)
(D) 0

Sol. **[C]**

Proper Method

$$\Delta = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$$

$$= \frac{abc}{abc} \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$$

$$\Delta = \frac{1}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

$$\therefore \Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$\Delta = (a – b) (b – c) (c – a) (ab + bc + ca)$

Short Trick

Put a = 0, b = 1, c = 2

$$\begin{vmatrix} 0 & 1 & 2 \\ 0 & 1 & 4 \\ 2 & 0 & 0 \end{vmatrix} = 4$$

by option (C) (a – b) (b – c) (c – a) (ab + bc + ca) is correct option

Q.33 The value of $\begin{vmatrix} a & b+c & a^3 \\ b & c+a & b^3 \\ c & a+b & c^3 \end{vmatrix}$ is-

(A) $(a-b)(b-c)(c-a)$ (B) $abc(a-b)(b-c)(c-a)$
(C) $-(a+b+c)^2(a-b)(b-c)(c-a)$ (D) 0

Sol. **[C]**

Proper Method	Short Trick
$\Delta = \begin{vmatrix} a & b+c & a^3 \\ b & c+a & b^3 \\ c & a+b & c^3 \end{vmatrix}$ Let $C_2 = C_1 + C_2$ $\therefore \Delta = \begin{vmatrix} a & a+b+c & a^3 \\ b & a+b+c & b^3 \\ c & a+b+c & c^3 \end{vmatrix}$ $\therefore \Delta = (a+b+c)\begin{vmatrix} a & 1 & a^3 \\ b & 1 & b^3 \\ c & 1 & c^3 \end{vmatrix}$ $\therefore \Delta = -(a+b+c)\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$ $\Delta = -(a+b+c)[(a-b)(b-c)(c-a)(a+b+c)]$	$a = 0, b = 1, c = 2$ $\begin{vmatrix} 0 & 3 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 8 \end{vmatrix} = -18$ by option (c) $-(a+b+c)^2(a-b)(b-c)(c-a)$ is correct option

Q.34 $\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix}$ is equal to-

(A) abc (B) 2abc (C) 4abc (D) 0

Sol. **[C]**

Proper Method	Short Trick
$\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix}$ $= \frac{1}{abc}\begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ $R_1 \to R_1 - (R_2 + R_3)$ $= \frac{1}{abc}\begin{vmatrix} 0 & -2b^2 & -2a^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$	Put $a = -1, b = 1, c = 2$ $\begin{vmatrix} 1 & 2 & 2 \\ -1 & -5 & -1 \\ 1 & 1 & 5 \end{vmatrix}$ $= 1(-24) - 2(-4) + 2(4)$ $= -8$ by option (C) 4abc is correct option

$= \frac{-2}{abc}\begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ $R_2 \to R_2 - R_1$ $= \frac{-2}{abc}\begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & c^2 & 0 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ $R_3 \to R_3 - R_1$ $= \frac{-2}{abc}\begin{vmatrix} 0 & b^2 & a^2 \\ a^2 & c^2 & 0 \\ b^2 & 0 & c^2 \end{vmatrix}$ $= -\frac{-2}{abc}[0 - b^2(a^2c^2) + a^2(-b^2c^2)]$ $= \frac{4a^2b^2c^2}{abc}$ $= 4abc$	

Q.35 If a, b, c are non-zero real numbers, then $\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ is equal to

(A) abc (B) $a^2b^2c^2$ (C) ab + bc + ca (D) 0

Sol. [D]

Proper Method	**Short Trick**
$\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$ $= \begin{vmatrix} c^2(b^2-a^2) & c(b-a) & (b-a) \\ a^2(c^2-b^2) & a(c-b) & (c-b) \\ a^2b^2 & ab & a+b \end{vmatrix}$ $= (b-a)(c-b)\begin{vmatrix} c(b+a) & c & 1 \\ a(c+b) & a & 1 \\ a^2b^2 & ab & a+b \end{vmatrix}$ $= R_1 \to R_1 - R_2$ $(b-a)(c-b)\begin{vmatrix} b(c-a) & c-a & 0 \\ a(c+b) & a & 1 \\ a^2b^2 & ab & a+b \end{vmatrix}$ $= (b-a)(c-b)(c-a)\begin{vmatrix} b & 1 & 0 \\ a(c+b) & a & 1 \\ a^2b^2 & ab & a+b \end{vmatrix}$ $= (b-a)(c-b)(c-a)\{b(a^2+ab-ab)$ $-1(a^2c + a^2b + abc + ab^2 - a^2b^2)\}$ $= 0$	Put a = 1, b = – 1, c = 2 $\begin{vmatrix} 4 & -2 & 1 \\ 4 & 2 & 3 \\ 1 & -1 & 0 \end{vmatrix}$ $= 4(3) + 2(-3) + 1(-6) = 0$ by option (D) is correct option

Q.36 $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ is equal to -

(A) $(x + p)(x + q)(x - p - q)$ (B) $(x - p)(x - q)(x + p + q)$

(C) $(x - p)(x - q)(x - p - q)$ (D) $(x + p)(x + q)(x + p + q)$

Sol. **[B]**

Proper Method	**Short Trick**
$\Delta = \begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ Let $C_1 = C_1 + C_2 + C_3$ $\Delta = \begin{vmatrix} x+p+q & p & q \\ x+p+q & x & q \\ x+p+q & q & x \end{vmatrix}$ Let $R_3 = R_3 - R_2$; $R_2 = R_2 - R_1$ $\Delta = \begin{vmatrix} x+p+q & p & q \\ 0 & x-p & 0 \\ 0 & q-x & x-q \end{vmatrix}$ $\therefore \Delta = (x + p + q)(x - p)(x - q) \begin{vmatrix} 1 & p & q \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$ $\therefore \Delta = (x-p)(x - q)(x + p + q)$	Put $x = 0, p = 1, q = 2$ $\begin{vmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{vmatrix} = -1(-2) + 2(2) = 6$ by option (B) $(x - p)(x - q)(x + p + q)$ is correct option

Q.37 If $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+\lambda)^2 & (b+\lambda)^2 & (c+\lambda)^2 \\ (a-\lambda)^2 & (b-\lambda)^2 & (c-\lambda)^2 \end{vmatrix} = k\lambda \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, $\lambda \neq 0$. then k is equal to : **[JEE Main Online-2014]**

(A) $4\lambda abc$ (B) $-4\lambda abc$ (C) $4\lambda^2$ (D) $-4\lambda^2$

Sol. **[C]**

Proper Method	**Short Trick**
Let $\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+\lambda)^2 & (b+\lambda)^2 & (c+\lambda)^2 \\ (a-\lambda)^2 & (b-\lambda)^2 & (c-\lambda)^2 \end{vmatrix}$ Apply $R_2 \to R_2 - R_3$ $\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+\lambda)^2-(a-\lambda)^2 & (b+\lambda)^2-(b-\lambda)^2 & (c+\lambda)^2-(c-\lambda)^2 \\ (a-\lambda)^2 & (b-\lambda)^2 & (c-\lambda)^2 \end{vmatrix}$ $\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 4a\lambda & 4b\lambda & 4c\lambda \\ (a-\lambda)^2 & (b-\lambda)^2 & (c-\lambda)^2 \end{vmatrix}$ ($\because (x + y)^2 - (x - y)^2 = 4xy$) Taking out 4 common from R_2	Let $a = 1, b = 2, c = 3$ and $\lambda = 1$ $\therefore \begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 0 & 1 & 4 \end{vmatrix} = k \begin{vmatrix} 1 & 4 & 9 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$ $\Rightarrow -8 = -2k$ $\Rightarrow k = 4$ Put these value in options then (C) $4\lambda^2$ is correct option

$$= 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a\lambda & b\lambda & c\lambda \\ a^2+\lambda^2-2a\lambda & b^2+\lambda^2-2b\lambda & c^2+\lambda^2-2c\lambda \end{vmatrix}$$

Apply $R_3 \to [R_3 - (R_1 - 2R_2)]$

$$= 4\begin{vmatrix} a^2 & b^2 & c^2 \\ a\lambda & b\lambda & c\lambda \\ \lambda^2 & \lambda^2 & \lambda^2 \end{vmatrix}$$

Taking out λ common from R_2 and λ^2 from R_3.

$$= 4\lambda(\lambda^2)\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

$$= k\lambda\begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

$\Rightarrow k = 4\lambda^2$

Q.38 If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then K is equal to - **[JEE Main-2014]**

(A) -1 (B) $\alpha\beta$ (C) $\frac{1}{\alpha\beta}$ (D) 1

Sol. [D]

Proper Method

$f(n) = \alpha^n + \beta^n$

$$\begin{vmatrix} 3 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$= K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \cdot \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix}$$

$= K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$

$$\begin{vmatrix} 0 & 0 & 1 \\ 1-\alpha & \alpha-\beta & \beta \\ 1-\alpha^2 & \alpha^2-\beta^2 & \beta^2 \end{vmatrix}^2$$

$= K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$

$$(1-\alpha)^2(\alpha-\beta)^2\begin{vmatrix} 1 & 1 \\ 1+\alpha & \alpha+\beta \end{vmatrix}^2$$

$= K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$

$(\beta-1)^2 = K(1-\beta)^2$ $\Rightarrow K = 1$

Short Trick

Let $\alpha = 2$, $\beta = -1$, then $f(n) = 2^n + (-1)^n$

$f(1) = 2^1 + (-1)^1 = 1$,

$f(2) = 2^2 + (-1)^2 = 5$,

$f(3) = 2^3 + (-1)^3 = 7$,

$f(4) = 2^4 + (-1)^4 = 17$

$$\begin{vmatrix} 3 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{vmatrix} = k(1-2)^2(1+1)^2(2+1)^2 = 36k$$

$\Rightarrow 36 = 36k$

$\Rightarrow k = 1$

7

Trigonometric Ratio

KEY CONCEPTS

1. System of Measurement of Angles

There are three system for measuring angles :

(i) Sexagesimal system/Degree Measure (English System) :

In this system, a right angel is divided into 90 equal parts, called degrees. The symbol 1° is used to denote one degree. Each degree is divided into 60 equal parts, called minutes and one minute is divided into 60 equal parts, called seconds. Symbols $1'$ $1''$ are used to denote one minute and one second, respectively.

i.e. 1 right angle = 90°

$1^\circ = 60'$, $1' = 60''$

(ii) Centesimal System (French System) :

In this system, a right angle is divided into 100 equal parts, called grades. Each grade is subdivided into 100 min and each minute is divided into 100s.

i.e. 1 right angle = 100 grades = 100g

$1^g = 100'$, $1' = 100''$

(iii) Circular System (Radian System) :

In this system, angle is measured in radian. A radian is the angel subtended at the centre of a circle by an arc, whose length is equal to the radius of the circle. The number of radians in an angle subtended by an arc of circle at the centre equal to $\frac{\text{arc}}{\text{radius}}$.

2. Relation Between System of Measurement of Angles

(i) π radian = 180°

or 1 radian = $\left(\frac{180^\circ}{\pi}\right) = 57^\circ16'\,22'$ where, $\pi = \frac{22}{7} = 3.14159$

(ii) $1^\circ = \left(\frac{\pi}{180}\right)$ rad = 0.01746 rad

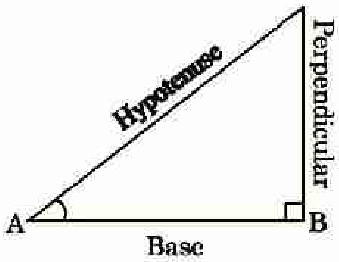
(iii) If D is the number of degrees, R is the number of radians and G is the number of grades in an angle θ, then $\frac{D}{90} = \frac{G}{100} = \frac{2R}{\pi}$

(iv) $\theta = \frac{1}{r}$, where θ = angle subtended by arc of length ℓ at the centre of the circle and r = radius of the circle.

3. Trigonometric Ratios

Relation between different sides and angles of a right angled triangle are called trigonometric ratios or T-ratios.

Trigonometric ratios can be represented is



$\sin\theta = \dfrac{\text{Perpendicular}}{\text{Hypotenuse}} = \dfrac{AB}{AC}$

$\cos\theta = \dfrac{\text{Base}}{\text{Hypotenuse}} = \dfrac{AB}{AC}$,

$\tan\theta = \dfrac{\text{Perpendicular}}{\text{Base}} = \dfrac{BC}{AB}$ $\text{cosec}\,\theta = \dfrac{1}{\sin\theta}$

$\sec\theta = \dfrac{1}{\cos\theta}$, $\cot\theta = \dfrac{\cos\theta}{\sin\theta} = \dfrac{1}{\tan\theta}$

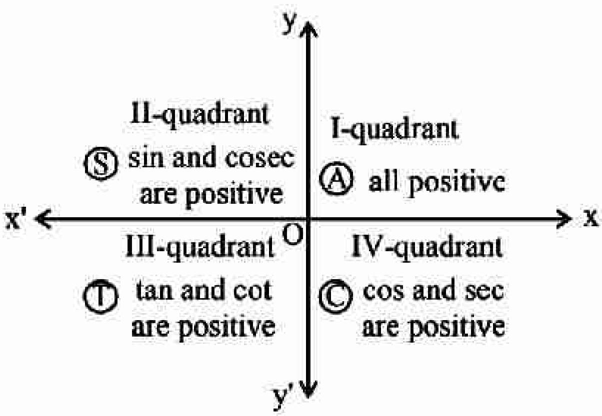
4. Fundamental Trigonometric Identities

(i) $\sin^2\theta + \cos^2\theta = 1$

(ii) $1 + \tan^2\theta = \sec^2\theta$

(iii) $1 + \cot^2\theta = \text{cosec}^2\theta$

5. Sign of Trigonometric Ratios or Functions



A crude aid to memorise the signs of trigonometrical ratio in different quadrant.

"*All* S*tudents* T*o* C*arreer point*"

6. Trigonometric Ratios of Some Standard Angles

Angle	0°	30°	45°	60°	90°	120°	135°	150°	80°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0
cot	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	$-\infty$
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	-2	$-\sqrt{2}$	$-\frac{2}{\sqrt{3}}$	-1
cosec	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞

7. Trigonometric Ratios of Some Special Angles

Angle	$7\frac{1°}{2}$	15°	18°	$22\frac{1°}{2}$	36°
sin θ	$\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\frac{\sqrt{5}-1}{4}$	$\frac{1}{2}\sqrt{2-\sqrt{2}}$	$\frac{1}{4}\sqrt{10-2\sqrt{5}}$
cos θ	$\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$\frac{1}{4}\sqrt{10+2\sqrt{5}}$	$\frac{1}{2}\sqrt{2+\sqrt{2}}$	$\frac{\sqrt{5}+1}{4}$
tan θ	$(\sqrt{3}-\sqrt{2})(\sqrt{2}-1)$	$2-\sqrt{3}$	$\frac{\sqrt{5}-1}{\sqrt{10+2\sqrt{5}}}$	$\sqrt{2}-1$	$\frac{\sqrt{5}+1}{\sqrt{10-2\sqrt{5}}}$

8. Trigonometric Ratios of Allied Angles

Allied angles → Trigo. Ratio ↓	$(-\theta)$	$(90°-\theta)$ OR $\left(\frac{\pi}{2}-\theta\right)$	$(90°+\theta)$ OR $\left(\frac{\pi}{2}+\theta\right)$	$(180°-\theta)$ OR $(\pi-\theta)$	$(180°+\theta)$ OR $(\pi+\theta)$	$(270°-\theta)$ OR $\left(\frac{3\pi}{2}-\theta\right)$	$(270°+\theta)$ OR $\left(\frac{3\pi}{2}+\theta\right)$	$(360°-\theta)$ OR $(2\pi-\theta)$
sin	$-\sin\theta$	$\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$
cos	$\cos\theta$	$\sin\theta$	$-\sin\theta$	$-\cos\theta$	$-\cos\theta$	$-\sin\theta$	$\sin\theta$	$\cos\theta$
tan	$-\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$	$\tan\theta$	$\cot\theta$	$-\cot\theta$	$-\tan\theta$

9. Trigonometric Ratios of Compound Angles

(i) $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

(ii) $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

(iii) $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

(iv) $\cot(A \pm B) = \dfrac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$

(v) $\sin(A + B).\sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

(vi) $\cos(A + B).\cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

(vii) $\tan(A + B + C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$

(viii) $\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$

(ix) $\cos(A + B + C) = \cos A \cos B \cos C - \cos A \sin B \sin C - \sin A \cos B \sin C - \sin A \sin B \cos C$

10. Transformation Formulae

(i) $2 \sin A \cos B = \sin(A + B) + \sin(A + B)$

(ii) $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$

(iii) $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$

(iv) $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

(v) $\sin C + \sin D = 2 \sin\left(\dfrac{C+D}{2}\right).\cos\left(\dfrac{C-D}{2}\right)$

(vi) $\sin C - \sin D = 2 \cos\left(\dfrac{C+D}{2}\right).\sin\left(\dfrac{C-D}{2}\right)$

(vii) $\cos C + \cos D = 2 \cos\left(\dfrac{C+D}{2}\right).\cos\left(\dfrac{C-D}{2}\right)$

(viii) $\cos C - \cos D = 2 \sin\left(\dfrac{C+D}{2}\right).\sin\left(\dfrac{D-C}{2}\right)$ (Note)

11. Trigonometrical Ratios of Multiple Angles

(i) $\sin 2\theta = 2\sin\theta\cos\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

(ii) $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

(iii) $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

(iv) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

(v) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

(vi) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$

(vii) $\sin(\theta/2) = \sqrt{\dfrac{1-\cos\theta}{2}}$

(viii) $\cos(\theta/2) = \sqrt{\dfrac{1+\cos\theta}{2}}$

(ix) $\tan(\theta/2) = \sqrt{\dfrac{1-\cos\theta}{1+\cos\theta}} = \left|\dfrac{1-\cos\theta}{\sin\theta}\right| = \left|\dfrac{\sin\theta}{1+\cos\theta}\right|$

(x) $\sqrt{1+\sin 2A} = |\sin A + \cos A|$

(xi) $\sqrt{1-\sin 2A} = |\sin A - \cos A|$

12. The Greatest & Least Value of the Expression [a sinθ+b cosθ]

Greatest value = $\sqrt{a^2+b^2}$ and least value = $-\sqrt{a^2+b^2}$

13. Some Useful Identities

(i) $\tan(A+B+C) = \dfrac{\sum \tan A - \tan A \tan B \tan C}{1-\sum \tan A.\tan B}$

(ii) $\tan\theta = \cot\theta - 2\cot 2\theta$

(iii) $\tan 3\theta = \tan\theta \cdot \tan(60° - \theta) \cdot \tan(60° + \theta)$

(iv) $\tan(A+B) - \tan A - \tan B = \tan A \cdot \tan B \cdot \tan(A+B)$

(v) $\sin\theta \sin(60° - \theta) \sin(60° + \theta) = \dfrac{1}{4}\sin 3\theta$

(vi) $\cos\theta \cos(60° - \theta) \cos(60° + \theta) = \dfrac{1}{4}\cos 3\theta$

14. Some Useful Series

(i) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots$ to n terms

$$= \frac{\sin\left[\alpha + \left(\frac{n-1}{2}\right)\beta\right]\left[\sin\left(\frac{n\beta}{2}\right)\right]}{\sin\left(\frac{\beta}{2}\right)} \;;\; \beta \neq 2n\pi$$

(ii) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots.$ to n terms

$$= \frac{\cos\left[\alpha + \left(\frac{n-1}{2}\right)\beta\right]\left[\sin\left(\frac{n\beta}{2}\right)\right]}{\sin\left(\frac{\beta}{2}\right)} \;;\; \beta \neq 2n\pi$$

(iii) $p = \cos\alpha \cdot \cos 2\alpha \cdot \cos 2^2\alpha \;\ldots\ldots\ldots\; \cos(2^{n-1}\alpha)$

$$= \begin{cases} \dfrac{\sin 2^n\alpha}{2^n \sin\alpha}, \text{if } \alpha \neq n\pi \\ 1, \text{if } \alpha = 2k\pi \\ -1, \text{if } \alpha = (2k+1)\pi \end{cases}$$

15. Conditional Trigonometrical Identities

If $A + B + C = \pi$ then

(i) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \cdot \sin B \cdot \sin C$

(ii) $\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cdot \cos B \cdot \cos C$

(iii) $\sin A + \sin B + \sin C = 4\cos\dfrac{A}{2}\cos\dfrac{B}{2}\cos\dfrac{C}{2}$

(iv) $\cos A + \cos B + \cos C = 1 + 4\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}$

(v) $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cdot \cos B \cos C$

(vi) $\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$

(vii) $\tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1$

(viii) $\cot B \cot C + \cot C \cot A + \cot A \cot B = 1$

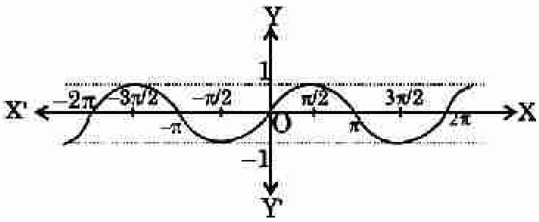
(ix) $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$

16. Domain, Range and Periodicity of Trigonometric Function

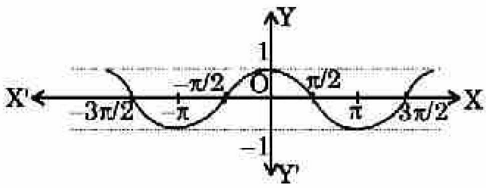
T-Ratio	Domain	Range	Period
sin x	R	[–1, 1]	2π
cos x	R	[–1, 1]	2π
tan x	$R-[(2n+1)\,\pi/2;\ n \in I]$	R	π
cot x	$R-[n\pi : n \in I]$	R	π
sec x	$R-[(2n+1)]\ \pi/2 : n \in I]$	$(-\infty, -1] \cup [1, \infty)$	2π
cosec x	$R-[n\pi : n \in I]$	$(-\infty, -1) \cup [1, \infty)$	2π

17. Graph of Different Trigonometric Functions

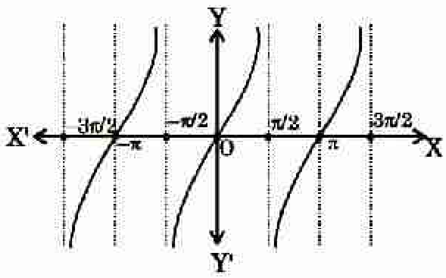
(i) $y = \sin x$



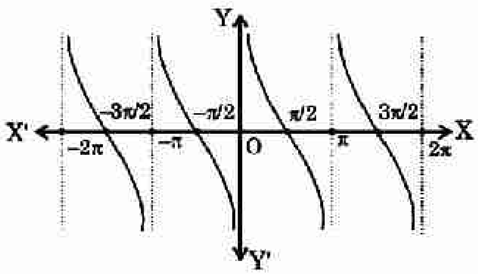
(ii) $y = \cos x$



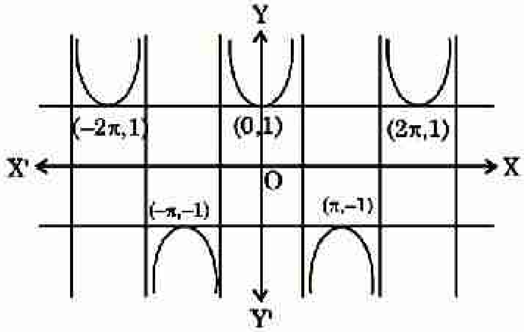
(iii) $y = \tan x$



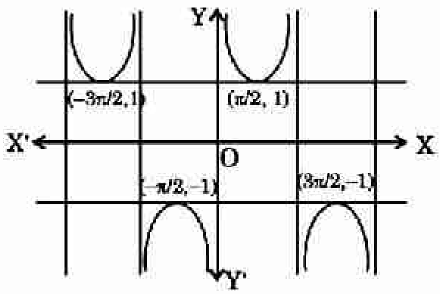
(iv) $y = \cot x$



(v) $y = \sec x$



(vi) $y = \operatorname{cosec} x$



SHORT TRICKS

Trick -1

$$\frac{\sin\theta_1+\sin\theta_2+\sin\theta_3+....+\sin\theta_n}{\cos\theta_1+\cos\theta_2+\cos\theta_3+....+\cos\theta_n}=\tan\left(\frac{\theta_1+\theta_2+....+\theta_n}{n}\right)$$

Q.1 $\dfrac{\sin 3\theta+\sin 5\theta+\sin 7\theta+\sin 9\theta}{\cos 30+\cos 50+\cos 70+\cos 90}=$

(A) $\tan 3\theta$ (B) $\cot 3\theta$ (C) $\tan 6\theta$ (D) $\cot 6\theta$

Sol. [D]

Proper Method

$$=\frac{(\sin 9\theta+\sin 3\theta)+(\sin 7\theta+\sin 5\theta)}{(\cos 9\theta+\cos 3\theta)+(\cos 7\theta+\cos 5\theta)}$$

$$=\frac{2\sin 6\theta.\cos 3\theta+2\sin 6\theta.\cos\theta}{2\cos 6\theta.\cos 3\theta+2\cos 6\theta.\cos\theta}$$

$$=\frac{2\sin 6\theta(\cos 3\theta+\cos\theta)}{2\cos 60(\cos 30+\cos 0)}$$

$= \tan 6\theta$

Short Trick

$$=\tan\left(\frac{3\theta+5\theta+7\theta+9\theta}{4}\right)=\tan 6\theta$$

Q.2 $\dfrac{\cos 3\theta+\cos 5\theta+\cos 7\theta}{\sin 3\theta+\sin 5\theta+\sin 7\theta}=$

(A) cot50 (B) tan150 (C) tan50 (D) cot30

Sol. [A]

Proper Method

$$=\frac{(\cos 7\theta+\cos 3\theta)+\cos 5\theta}{(\sin 70+\sin 30)+\sin 50}$$

$$=\frac{2\cos 5\theta.\cos\theta+\cos 5\theta}{2\sin 50\cos 0+\sin 50}$$

$$=\frac{\cos 5\theta(2\cos\theta+1)}{\sin 50(2\cos 0+1)}$$

$=\cot 50$

Short Trick

$$\cot\left(\frac{7\theta+5\theta+3\theta}{3}\right)=\cot 5\theta$$

Trick -2

Useful series

(i) $\sin\theta\,\sin(60°-\theta)\,\sin(60°+\theta)=\dfrac{1}{4}\sin 3\theta$

(ii) $\cos\theta.\,\cos(60°-\theta)\,\cos(60°+\theta)=\dfrac{1}{4}\cos 3\theta$

(iii) $\tan\theta\,\tan(60°-\theta)\,\tan(60°+\theta)=\tan 3\theta$

Q.3 The value of $\sin 20°\sin 40°\sin 60°\sin 80°$ is -

(A) $\dfrac{3}{8}$ (B) $\dfrac{1}{8}$ (C) $\dfrac{3}{16}$ (D) $\dfrac{3}{32}$

Sol. [C]

Proper Method

sin 20° sin 40° sin 60° sin 80°

$= \frac{\sqrt{3}}{2} \sin 20° \sin (60° - 20°) \sin (60° + 20°)$

$= \frac{\sqrt{3}}{2} \sin 20° (\sin^2 60° - \sin^2 20°)$

$= \frac{\sqrt{3}}{2} \sin 20° (\frac{3}{4} - \sin^2 20°)$

$= \frac{\sqrt{3}}{8} (3 \sin 20° - 4 \sin^3 20°)$

$= \frac{\sqrt{3}}{8} \sin 60°$

$= \frac{\sqrt{3}}{8} . \frac{\sqrt{3}}{2} = \frac{3}{16}$

Short Trick

$= (\sin 20° \sin 40° \sin 80°). \sin 60°$

$= \left(\frac{1}{4} \sin 60°\right) \sin 60°$

$= \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$

Q.4 $\frac{\cos 20° + 8 \sin 70° \sin 50° \sin 10°}{\sin^2 80°}$ is equal to -

(A) 1 (B) 2 (C) $\frac{3}{4}$ (D) $\frac{3}{8}$

Sol. [B]

Proper Method

$\frac{\cos 20 + 8 \sin 70 . \sin 50 \sin 10}{\sin^2 80}$

$= \frac{\cos 20 + 8 \sin 10 \sin (60 - 10) \sin (60 + 10)}{\sin^2 80}$

$\left\{ \because \sin A \sin (60 - A) \sin (60 + A) = \sin 3A / 4 \right.$

$= \frac{\cos 20 + \frac{8 \sin 30}{4}}{\sin^2 80}$

$= \frac{1 + \cos 20}{\frac{1 - \cos 160}{2}}$

$= \frac{2(1 + \cos 20)}{1 + \cos 20} = 2$

Short Trick

$\frac{\cos 20° + 8 . \frac{1}{4} \sin 30°}{\sin^2 80°} = \frac{\cos 20° + 1}{\sin^2 80°}$

$= \frac{2 \cos^2 10°}{\cos^2 10°} = 2$

Q.5 cos10° cos30° cos50° cos70° =

(A) $\frac{3}{16}$ (B) $\frac{4}{5}$ (C) $\frac{8}{15}$ (D) $\frac{9}{12}$

Sol. [A]

Proper Method

$= \frac{\sqrt{3}}{2} \cos 10° \cos 50° \cos 70°$

Short Trick

$(\cos 10° \cos 50° \cos 70°) \cos 30°$

$= \left(\frac{1}{4} \cos 30°\right) \cos 30° = \frac{3}{16}$

$= \frac{\sqrt{3}}{2} \cos 10° \cos(60° - 10°) \cos(60° + 10°)$ $= \frac{\sqrt{3}}{2} \cos 10° (\cos^2 10° - \sin^2 60°)$ $= \frac{\sqrt{3}}{2} \cos 10° \left(\cos^2 10 - \frac{3}{4}\right)$ $= \frac{\sqrt{3}}{8} (4\cos^3 10° - 3\cos 10°)$ $= \frac{\sqrt{3}}{8} \cos 30° = \frac{3}{16}$	

Trick -3 Method of Substitution ss
If an expression is independent of an unknown parameter, then we can put any suitable value for that parameter to minimize the calculation.

Q.6 The value of $6(\sin^6\theta + \cos^6\theta) - 9(\sin^4\theta + \cos^4\theta) + 4$ is :
(A) -3 (B) 0 (C) 1 (D) 3

Sol. **[C]**

Proper Method	**Short Trick**
$6(\sin^6\theta + \cos^6\theta) - 9(\sin^4\theta + \cos^4\theta) + 4$ $= 6\{(\sin^2\theta)^3 + (\cos^2\theta)^3\} - 9\{(\sin^2\theta)^2 + (\cos^2\theta)^2\} + 4$ $= 6\{(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)\}$ $- 9\{(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta.\cos^2\theta\} + 4$ $= 6 - 18\sin^2\theta.\cos^2\theta - 9 + 18\sin^2\theta.\cos^2\theta + 4$ $= 1$	Put $\theta = 0$, then expression $\Rightarrow 6(0 + 1) - 9(0 + 1) + 4 = 1$

Q.7 The value of $\sin^6\theta + \cos^6\theta + 3\sin^2\theta\cos^2\theta$ is equals to:
(A) 0 (B) -1 (C) 1 (D) 2

Sol. **[C]**

Proper Method	**Short Trick**
$\sin^6\theta + \cos^6\theta + 3\sin^2\theta.\cos^2\theta$ $= (\sin^2\theta)^3 + (\cos^2\theta)^3 + 3\sin^2\theta.\cos^2\theta$ $= \{(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta.\cos^2\theta\} + 3\sin^2\theta.\cos^2\theta$ $= 1$	Put $\theta = 0$, then expression $\Rightarrow 0 + 1 + 3 \times 0 = 1$

Q.8 The value of the expression $\cos\alpha\sin(\beta - \gamma) + \cos\beta\sin(\gamma - \alpha) + \cos\gamma\sin(\alpha - \beta)$ is ?
(A) 0 (B) -1 (C) 1 (D) 2

Sol. **[A]**

Proper Method	**Short Trick**
$\cos\alpha\sin(\beta - \gamma) + \cos\beta\sin(\gamma - \alpha) + \cos\gamma\sin(\alpha - \beta)$ $= \cos\alpha(\sin\beta\cos\gamma - \cos\beta\sin\gamma)$ $+ \cos\beta(\sin\gamma\cos\alpha - \cos\gamma\sin\alpha)$ $+ \cos\gamma(\sin\alpha\cos\beta - \cos\alpha.\sin\beta)$ $= \cos\alpha.\sin\beta\cos\gamma - \cos\alpha.\cos\beta.\sin\gamma$ $+ \cos\beta\sin\gamma.\cos\alpha - \cos\beta.\cos\gamma.\sin\alpha$ $+ \cos\gamma\sin\alpha.\cos\beta - \cos\gamma.\cos\alpha.\sin\beta$ $= 0$	$\therefore$ put $\alpha = \beta = \gamma = 0$ in the expression. Then, we will get $0 + 0 + 0 = 0$.

Q.9 The value of $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$ is equals to :

(A) 11 (B) 12 (C) 13 (D) 14

Sol. **[C]**

Proper Method	Short Trick
$3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$ $= 3\{{}^4C_0(\sin x)^4 - {}^4C_1 \sin^3 x.\cos x + {}^4C_2 \sin^2 x.\cos^2 x$ $- {}^4C_3 \sin x.\cos^3 x + {}^4C_4 \cos^4 x\} + 6(\sin^2 x + \cos^2 x$ $+ 2\sin x \cos x) + 4\{(\sin^2 x)^3 + (\cos^2 x)^3\}$ $= 3\sin^4 x - 12\sin^3 x.\cos x + 18\sin^2 x.\cos^2 x$ $- 12\sin x.\cos^3 x + 3\cos^4 x + 6 + 12\sin x\cos x + 4$ $- 12\sin^2 x\cos^2 x$ $= 3(\sin^4 x + \cos^4 x) - 12\sin x\cos x(\sin^2 x + \cos^2 x)$ $+ 6\sin^2 x\cos^2 x + 12\sin x.\cos x + 10$ $= 3[(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x\cos^2 x] + 6\sin^2 x\cos^2 x + 10$ $= 3 - 6\sin^2 x\cos^2 x + 6\sin^2 x\cos^2 x + 10$ $= 13$	Put $\theta = 0°$ in the expression. Then, we will get $3 + 6 + 4 = 13$

Q.10 The value of $\tan\alpha.\tan(\alpha + 60°) + \tan\alpha.\tan(\alpha - 60°) + \tan(\alpha + 60°)\tan(\alpha - 60°)$ is?

(A) 0 (B) 3 (C) – 3 (D) 1

Sol. **[C]**

Proper Method	Short Trick
$\tan\alpha\tan(\alpha + 60°) + \tan\alpha\tan(\alpha - 60°) + \tan(\alpha + 60°)\tan(\alpha - 60°)$ $= \tan\alpha\left(\frac{\tan\alpha + \sqrt{3}}{1 - \sqrt{3}\tan\alpha}\right) + \tan\alpha\left(\frac{\tan\alpha - \sqrt{3}}{1 + \sqrt{3}\tan\alpha}\right) + \left(\frac{\tan\alpha + \sqrt{3}}{1 - \sqrt{3}\tan\alpha}\right)\left(\frac{\tan\alpha - \sqrt{3}}{1 + \sqrt{3}\tan\alpha}\right)$ $= \frac{\left[(\tan^2\alpha + \sqrt{3}\tan\alpha)(1 + \sqrt{3}\tan\alpha) + (\tan^2\alpha - \sqrt{3}\tan\alpha)(1 - \sqrt{3}\tan\alpha) + \tan^2\alpha - 3\right]}{(1 - \sqrt{3}\tan\alpha)(1 + \sqrt{3}\tan\alpha)}$ $= \frac{\left[\tan^2\alpha + \sqrt{3}\tan^3\alpha + \sqrt{3}\tan\alpha + 3\tan^2\alpha + \tan^2\alpha - \sqrt{3}\tan^3\alpha - \sqrt{3}\tan\alpha + 3\tan^2\alpha + \tan^2\alpha - 3\right]}{1 - 3\tan^2\alpha}$ $= \frac{9\tan^2\alpha - 3}{1 - 3\tan^2\alpha} = -3$	Put $\alpha = 0°$ then, L.H.S. $= 0 + 0 + \tan 60° \tan(-60°)$ $= \sqrt{3} \times (-\sqrt{3}) = -3$

Q.11 The value of $3\left\{\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right\} - 2\left\{\sin^6\left(\frac{\pi}{2} + \theta\right) + \sin^6(5\pi + \theta)\right\}$ is :

(A) 5 (B) 0 (C) 1 (D) 3

Sol. **[C]**

Proper Method

$$3\left\{\sin^4\left(\frac{3\pi}{2}-\theta\right)+\sin^4(3\pi+\theta)\right\}-2\left\{\sin^6\left(\frac{\pi}{2}+\theta\right)+\sin^6(5\pi+\theta)\right\}$$

$= 3(\cos^4\theta + \sin^4\theta) - 2(\cos^6\theta + \sin^6\theta)$

$= 3\{(\cos^2\theta + \sin^2\theta)^2 - 2\sin^2\theta.\cos^2\theta\} - 2\{(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta.\cos^2\theta(\sin^2\theta + \cos^2\theta)\}$

$= 3(1 - 2\sin^2\theta.\cos^2\theta) - 2(1 - 3\sin^2\theta.\cos^2\theta)$

$= 1$

Short Trick

Put $\theta = 0°$ in the expression. Then,

$$3\left\{\sin^4\left(\frac{3\pi}{2}\right)+\sin^4(3\pi)\right\}-2\left\{\sin^6\left(\frac{\pi}{2}\right)+\sin^6(5\pi)\right\}$$

$= 3\{1 + 0\} - 2\{1 + 0\}$

$= 3 - 2 = 1$

Q.12 If $\tan\alpha = \dfrac{m}{m+1}$ and $\tan\beta = \dfrac{1}{2m+1}$, then $\alpha + \beta$ is equals to :

(A) $\dfrac{\pi}{3}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{6}$ (D) $\dfrac{\pi}{2}$

Sol. [B]

Proper Method

If $\tan\alpha = \dfrac{m}{m+1}$, $\tan\beta = \dfrac{1}{2m+1}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{\frac{m}{m+1}+\frac{1}{2m+1}}{1-\frac{m}{m+1}.\frac{1}{2m+1}}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{2m^2+m+m+1}{2m^2+m+2m+1-m}$

$= 1$

$\therefore \alpha+\beta = \dfrac{\pi}{4}$

Short Trick

Put m = 0 then, $\tan\alpha = 0$

$\Rightarrow \alpha = 0°$ and $\tan\beta = 1 \Rightarrow \beta = 45°$ therefore,

$\alpha+\beta = \dfrac{\pi}{4}$

Q.13 If $\tan\alpha = (1 + 2^{-x})^{-1}$ and $\tan\beta = (1 + 2^{1+x})^{-1}$, then $\alpha + \beta$ is equals to :

(A) $\dfrac{\pi}{3}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{6}$ (D) $\dfrac{\pi}{2}$

Sol. [B]

Proper Method

$\tan\alpha = (1 + 2^{-x})^{-1}$ and $\tan\beta = (1 + 2^{1+x})^{-1}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{\tan\alpha+\tan\beta}{1-\tan\alpha.\tan\beta}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{(1+2^{-x})^{-1}+(1+2^{1+x})^{-1}}{1-(1+2^{-x})^{-1}(1+2^{1+x})^{-1}}$

$\Rightarrow \tan(\alpha+\beta) = \dfrac{\frac{1}{1+2^{-x}}+\frac{1}{1+2^{1+x}}}{1-\frac{1}{(1+2^{-x})}\frac{1}{(1+2^{1+x})}}$

Short Trick

Put x = 0 $\therefore \tan\alpha = \dfrac{1}{2}$ & $\tan\beta = \dfrac{1}{3}$

$\Rightarrow \tan(\alpha+\beta) = 1 \Rightarrow \alpha+\beta = \dfrac{\pi}{4}$

$= \dfrac{1+2^{1+x}+1+2^{-x}}{1+2^{1+x}+2^{-x}+2-1}$ $= \dfrac{2+2^{-x}+2^{1+x}}{2+2^{-x}+2^{1+x}} = 1$ $\Rightarrow \alpha+\beta = \dfrac{\pi}{4}$	

Q.14 If $(\cos\theta + \sin\theta)^2 + k\cos\theta\sin\theta = 1,\ \forall\, \theta \in R - \{n\pi,\ n \in I\}$, then $k =$

(A) 1 (B) – 1 (C) 2 (D) – 2

Sol. [D]

Proper Method	**Short Trick**
$(\cos\theta + \sin\theta)^2 + k\cos\theta.\ \sin\theta = 1$ $\Rightarrow \cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta + k\cos\theta.\ \sin\theta = 1$ $\Rightarrow (k+2)\sin\theta\cos\theta = 0$ $\Rightarrow k+2 = 0$ $\quad (\because \sin\theta.\ \cos\theta \neq 0)$ $\Rightarrow k = -2$	$\because\ \theta \in R$, let $\theta = 45°$ (here $\theta = 0$ will vanish k) then $\left(\dfrac{1}{\sqrt{2}}+\dfrac{1}{\sqrt{2}}\right)^2 + k\,\dfrac{1}{\sqrt{2}}\dfrac{1}{\sqrt{2}} = 1 \Rightarrow k = -2$

Q.15 The value of $\cos^2(A - B) + \cos^2B - 2\cos(A - B)\cos A\cos B$ is :

(A) $\cos^2A$ (B) $\sin^2A$ (C) $\tan^2A$ (D) $\cot^2A$

Sol. [B]

Proper Method	**Short Trick**
$= \cos^2(A-B) + \cos^2B - 2\cos(A-B)\cos A.\ \cos B$ $= \cos^2(A-B) - 2\cos(A-B)\cos A\cos B + \cos^2B$ $= \cos(A-B)\{\cos(A-B) - 2\cos A\cos B\} + \cos^2B$ $= \cos(A-B)\{-\cos(A+B)\} + \cos^2B$ $= -\dfrac{1}{2}2\cos(A-B)\cos(A+B) + \cos^2B$ $= -\dfrac{1}{2}[\cos 2A + \cos 2B] + \cos^2B$ $= -\dfrac{1}{2}[2\cos^2B - 1 + 1 - 2\sin^2A] + \cos^2B$ $= \sin^2A$	$\because$ According to the options, the value of the expression is independent of B, so put $B = 0°$ and $A = 30°$ then L.H.S $= \dfrac{3}{4} + 1 - 2\dfrac{\sqrt{3}}{2}\dfrac{\sqrt{3}}{2} = \dfrac{1}{4}$

Q.16 The value of $\cos 2(\theta + \phi) + 4\cos(\theta + \phi) \times \sin\theta\sin\phi + 2\sin^2\phi$ is equal to:

(A) $\cos 2\theta$ (B) $\cos 3\theta$ (C) $\sin 2\theta$ (D) $\sin 3\theta$

Sol. [A]

Proper Method	**Short Trick**
$\cos 2(\theta + \phi) + 4\cos(\theta+\phi)\sin\theta\sin\phi + 2\sin^2\phi$ $= 2\cos^2(\theta+\phi) - 1 + 4\cos(\theta+\phi)\sin\theta\sin\phi + 2\sin^2\phi$ $= 2\cos(\theta+\phi)\{\cos(\theta+\phi) + 2\sin\theta.\ \sin\phi\} - 1 + 2\sin^2\phi$ $= 2\cos(\theta+\phi)\cos(\theta-\phi) - 1 + 2\sin^2\phi$ $= \cos 2\theta + \cos 2\phi - \cos 2\phi$ $= \cos 2\theta$	Put $\phi = 0°$. Then L.H.S. $= \cos 2(\theta + 0) - 4\cos(\theta+0)\sin\theta\sin 0 + 2\sin^2 0$ $= \cos 2\theta + 0 + 0 = \cos 2\theta$

Q.17 If $\text{cosec}\,\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$ then $(m^2n)^{2/3} + (n^2m)^{2/3}$ equals to -

(A) 0 (B) 1 (C) –1 (D) 2

Sol. [B]

Proper Method	Short Trick
$\text{cosec}\,\theta - \sin\theta = m$ $m = \dfrac{1}{\sin\theta} - \sin\theta = \dfrac{\cos^2\theta}{\sin\theta}$ ...(i) $n = \dfrac{1}{\cos\theta} - \cos\theta = \dfrac{\sin^2\theta}{\cos\theta}$ ...(ii) $m \times n = \dfrac{\cos^2\theta}{\sin\theta} \cdot \dfrac{\sin^2\theta}{\cos\theta} = \sin\theta\cos\theta$ from (i) and (ii) from (i) $\cos^2\theta = m \cdot \sin\theta$ or $\cos^3\theta = m\sin\theta\cos\theta$ $= m \cdot (mn) = m^2n$ Similarly $\sin^3\theta = n^2m$ since $\sin^2\theta + \cos^2\theta = 1$ $(n^2m)^{2/3} + (m^2n)^{2/3} = 1$	Let $\theta = \dfrac{\pi}{4} \Rightarrow m = \sqrt{2} - \dfrac{1}{\sqrt{2}}$ and $n = \sqrt{2} - \dfrac{1}{\sqrt{2}}$ so $m = n$ now $(m^2n)^{2/3} + (n^2m)^{2/3} = 2m^2 = 2\left(\sqrt{2} - \dfrac{1}{\sqrt{2}}\right)^2$ $= 1$

Q.18 $\tan^2\theta\sec^2\theta(\cot^2\theta - \cos^2\theta) =$

(A) 0 (B) – 1 (C) 1 (D) 2

Sol. [C]

Proper Method	Short Trick
$\tan^2\theta\sec^2\theta\,(\cot^2\theta - \cos^2\theta)$ $= \tan^2\theta\sec^2\theta\left(\dfrac{\cos^2\theta}{\sin^2\theta} - \cos^2\theta\right)$ $= \tan^2\theta\sec^2\theta\cos^2\theta\,(\text{cosec}^2\theta - 1)$ $= \tan^2\theta\cot^2\theta$ $= 1$	Put $\theta = \dfrac{\pi}{4}$ $\tan^2\theta\sec^2\theta(\cot^2\theta - \cos^2\theta) = 2\left(1 - \dfrac{1}{2}\right) = 1$

Q.19 If $\sin\theta + \text{cosec}\,\theta = 2$ then the value of $\sin^8\theta + \text{cosec}^8\theta$ is equal to-

(A) 2 (B) 2^8 (C) 2^4 (D) 2^{16}

Sol. [A]

Proper Method	Short Trick
$\sin\theta + \text{cosec}\,\theta = 1 + 1$ $\sin\theta + \dfrac{1}{\sin\theta} = 2 = 1 + 1$ $\left.\begin{array}{l}\sin\theta = 1\\ \dfrac{1}{\sin\theta} = 1\end{array}\right\} - \sin\theta = 1$ $\sin^8\theta + \text{cosec}^8\theta = 1 + 1 = 2$	Let $\theta = 90°$ Then $\sin^8\theta + \text{cosec}^8\theta = 2$

Q.20 If $A - B = \frac{\pi}{4}$, then $(1 + \tan A)(1 - \tan B) =$

(A) 1 (B) 2 (C) –1 (D) –2

Sol. [B]

Proper Method	Short Trick
$\tan(A - B) = \tan \pi/4$ $\frac{\tan A - \tan B}{1 + \tan A \tan B} = 1$ $\Rightarrow \tan A - \tan B = 1 + \tan A \tan B$ $\Rightarrow -1 + \tan A - \tan B - \tan A \tan B = 0$ Adding 2 we get $\Rightarrow (1 + \tan A) - \tan B (1 + \tan A) = 2$ $\Rightarrow (1 + \tan A)(1 - \tan B) = 2$	Let $A = \frac{\pi}{4}$, $B = 0$ $(1 + \tan A)(1 - \tan B) = (1 + 1)(1 - 0) = 2$

Q.21 If triangle ABC, $\angle C = \frac{2\pi}{3}$, then the value of $\cos^2 A + \cos^2 B - \cos A \cdot \cos B$ is equal to-

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Sol. [A]

Proper Method	Short Trick
$\cos^2 A + \cos^2 B - \cos A \cos B$ $= \frac{1+\cos 2A}{2} + \frac{1+\cos 2B}{2} - \cos A \cos B$ $= 1 + \frac{1}{2}[\cos 2A + \cos 2B] - \cos A \cos B$ $= 1 + \frac{1}{2} 2 \cos(A + B) \cos(A - B) - \frac{1}{2} 2 \cos A \cos B$ $\{A + B = \pi/3\}$ $= 1 + \cos \pi/3 \cos(A - B) - \frac{1}{2}[\cos(A + B) + \cos(A - B)]$ $= 1 + \frac{1}{2}\cos(A - B) - \frac{1}{2}\cos\frac{\pi}{3} - \frac{1}{2}\cos(A - B)$ $= 1 - \frac{1}{4} = 3/4$	Let $\angle A = 30°$ and $\angle B = 30°$ $\cos^2 A + \cos^2 B - \cos A . \cos B = \frac{3}{4} + \frac{3}{4} - \frac{3}{4} = \frac{3}{4}$

Q.22 If $A + B + C = \pi$, then $\Sigma \tan\frac{A}{2} \tan\frac{B}{2}$ is equal to -

(A) 0 (B) –1 (C) 1/2 (D) 1

Sol. [D]

Proper Method	Short Trick
$A + B + C = \pi$ $A/2 + B/2 + C/2 = \pi/2$ $\tan(A/2 + B/2 + C/2) = \tan \pi/2$ $\frac{\tan A/2 + \tan B/2 + \tan C/2 - \tan A/2 \tan B/2 \tan C/2}{1 - \tan A/2 \tan B/2 - \tan B/2 \tan C/2 - \tan C/2 \tan A/2} = \infty$ $1 - \tan A/2 \tan B/2 - \tan B/2 \tan C/2 - \tan C/2$ $\tan A/2 = 0$ $\Sigma \tan A/2 \tan B/2 = 1$	Let $A = B = C = \frac{\pi}{3}$ $\sum \tan\frac{A}{2}.\tan\frac{B}{2} = 3\left(\frac{1}{\sqrt{3}}.\frac{1}{\sqrt{3}}\right) = 1$

Trick -4 Combination of method of substitution and balancing

Q.23 If $A + B + C = \frac{3\pi}{2}$, then
$\cos 2A + \cos 2B + \cos 2C =$

(A) $1 - 4\cos A \cos B \cos C$
(B) $4 \sin A \sin B \sin C$
(C) $1 + 2\cos A \cos B \cos C$
(D) $1 - 4 \sin A \sin B \sin C$

Sol. [D]

Proper Method

$\cos 2A + \cos 2B + \cos 2C$
$= 2\cos (A + B) \cos (A - B) + \cos 2C$
$= 2\cos\left(\frac{3\pi}{2} - C\right) \cos (A - B) + \cos 2C$
$\because A + B + C = \frac{3\pi}{2}$
$= -2 \sin C \cos (A - B) + 1 - 2\sin^2 C$
$= 1 - 2\sin C [\cos (A - B) + \sin C)$
$= 1 - 2\sin C \left[\cos (A - B) + \sin\left(\frac{3\pi}{2} - (A + B)\right)\right]$
$= 1 - 2 \sin C [\cos (A - B) - \cos (A + B)]$
$= 1 - 4 \sin A \sin B \sin C$

Short Trick

Let $A = B = C = \frac{\pi}{2}$
then $\cos 2A + \cos 2B + \cos 2C = -3$
by option (D) is correct answer

Q.24 If $\alpha\cos^2 3\theta + \beta\cos^4\theta = 16\cos^6\theta + 9\cos^2\theta$ is an identity then-

(A) $\alpha = 1, \beta = 18$
(B) $\alpha = 1, \beta = 24$
(C) $\alpha = 3, \beta = 24$
(D) $\alpha = 4, \beta = 2$

Sol. [B]

Proper Method

$\alpha(4\cos^3\theta - 3\cos\theta)^2 + \beta\cos^4\theta = 16\cos^6\theta + 9\cos^2\theta$
$\alpha(16\cos^6\theta + 9\cos^2\theta - 24\cos^4\theta) + \beta\cos^4\theta$
$= 16\cos^6\theta + 9\cos^2\theta$
$\Rightarrow 16\alpha\cos^6\theta + \cos^4\theta(\beta - 24\alpha) + 9\alpha\cos^2\theta + 9\cos^2\theta$
$\alpha = 1$
$\beta - 24\alpha = 0$
$\beta = 24$

Short Trick

Put $\theta = 0$
$\alpha + \beta = 25$
option (B) $\alpha = 1, \beta = 24$ is correct answer

Q.25 The value of the expression $\frac{\sin 5\theta + \sin 2\theta - \sin\theta}{\cos 5\theta + 2\cos 3\theta + 2\cos^2\theta + \cos\theta}$ is equals to:

(A) $\tan\theta$
(B) $\cos\theta$
(C) $\cot\theta$
(D) $\sin\theta$

Sol. [A]

Proper Method

$$\frac{(\sin 5\theta - \sin\theta) + \sin 2\theta}{(\cos 5\theta + \cos\theta) + 2\cos 3\theta + 2\cos^2\theta}$$
$$= \frac{2\cos 3\theta \cdot \sin 2\theta + \sin 2\theta}{2\cos 3\theta \cdot \cos 2\theta + 2\cos 3\theta + 2\cos^2\theta}$$
$$= \frac{\sin 2\theta(2\cos 3\theta + 1)}{2\cos 3\theta(\cos 2\theta + 1) + 2\cos^2\theta}$$
$$= \frac{\sin 2\theta(2\cos 3\theta + 1)}{2\cos 3\theta(2\cos^2\theta) + 2\cos^2\theta}$$
$$= \frac{\sin 2\theta(2\cos 3\theta + 1)}{2\cos^2\theta(2\cos 3\theta + 1)}$$
$$= \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \tan\theta$$

Short Trick

Let $\theta = 30°$ then LHS
$$= \frac{\sin 150° + \sin 60° - \sin 30°}{\cos 150° + 2\cos 90° + 2\cos^2 30° + \cos 30°} = \frac{1}{\sqrt{3}}$$
Now put $\theta = 30°$ in options then (A) will match with L.H.S.

Q.26 The expression $\dfrac{1}{\tan 3\theta-\tan\theta}-\dfrac{1}{\cot 3\theta-\cot\theta}$ is equals to :

(A) $\tan\theta$ (B) $\tan 2\theta$ (C) $\cot\theta$ (D) $\cot 2\theta$

Sol. [D]

Proper Method

$$\frac{1}{\tan 3\theta-\tan\theta}-\frac{1}{\cot 3\theta-\cot\theta}$$

$$=\frac{1}{\left(\frac{\sin 3\theta}{\cos 3\theta}-\frac{\sin\theta}{\cos\theta}\right)}-\frac{1}{\left(\frac{\cos 3\theta}{\sin 3\theta}-\frac{\cos\theta}{\sin\theta}\right)}$$

$$=\frac{\cos 3\theta\cos\theta}{\sin 2\theta}-\frac{\sin 3\theta\sin\theta}{(-\sin 2\theta)}$$

$$=\frac{1}{\sin 2\theta}(\cos 3\theta\cdot\cos\theta+\sin 3\theta\cdot\sin\theta)$$

$$=\frac{1}{\sin 2\theta}\cos(3\theta-\theta)$$

$$=\cot 2\theta$$

Short Trick

Here suitable value for θ is $15°$ because for this value all four options are different.

$$\text{L.H.S.}=\frac{1}{\tan 45°-\tan 15°}-\frac{1}{\cot 45°-\cot 15°}$$

$$=\frac{1}{1-(2-\sqrt{3})}-\frac{1}{1-(2+\sqrt{3})}$$

$$=\frac{1}{\sqrt{3}+1}-\frac{1}{\sqrt{3}-1}=\sqrt{3}$$

If we put $\theta = 15°$ in the options then only (D) will match with L.H.S.

Q.27 The value of $\dfrac{\sec 8A-1}{\sec 4A-1}$ is equal to:

(A) $\dfrac{\tan 2A}{\tan 8A}$ (B) $\dfrac{\tan 8A}{\tan 2A}$ (C) $\dfrac{\cot 4A}{\cot 2A}$ (D) $\dfrac{\cot 16A}{\cot 8A}$

Sol. [B]

Proper Method

$$\frac{\sec 8A-1}{\sec 4A-1}=\frac{1-\cos 8A}{1-\cos 4A}\cdot\frac{\cos 4A}{\cos 8A}$$

$$=\frac{2\sin^2 4A}{2\sin^2 2A}\cdot\frac{\cos 4A}{\cos 8A}$$

$$=\frac{(2\sin 4A\cdot\cos 4A)\cdot\sin 4A}{2\sin^2 2A\cdot\cos 8A}$$

$$=\frac{\sin 8A}{\cos 8A}\cdot\frac{\sin 4A}{2\sin^2 2A}$$

$$=\tan 8A\cdot\frac{2\sin 2A\cdot\cos 2A}{2\sin^2 2A}$$

$$=\frac{\tan 8A}{\tan 2A}$$

Short Trick

Let $A = 15°$ then $\text{LHS}=\dfrac{\sec(120°)-1}{\sec(60°)-1}$

$$=\frac{-2-1}{2-1}=-3$$

After putting $A = 15°$ in the options, option (B) will balance the L.H.S.

Q.28 The value of $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha$ is:

(A) $\tan\alpha$ (B) $\tan 2\alpha$ (C) $\cot\alpha$ (D) $\cot 2\alpha$

Sol. [C]

Proper Method

$\tan\alpha + 2\tan 2\alpha + (4\tan 4\alpha + 8\cot 8\alpha)$

$$=\tan\alpha+2\tan 2\alpha+\left(4\tan 4\alpha+\frac{8}{\tan 8\alpha}\right)$$

Short Trick

Let $\alpha = 15°$, then LHS

$= \tan 15° + 2\tan 30° + 4\tan 60° + 8\cot 120°$

$$=2-\sqrt{3}+2\times\frac{1}{\sqrt{3}}+4\sqrt{3}+8\left(-\frac{1}{\sqrt{3}}\right)$$

$= \tan\alpha + 2\tan2\alpha + \left\{4\tan4\alpha + \dfrac{8(1-\tan^2 4\alpha)}{2\tan4\alpha}\right\}$

$= \tan\alpha + 2\tan2\alpha + \left\{\dfrac{4\tan^2 4\alpha + 4 - 4\tan^2 4\alpha}{\tan 4\alpha}\right\}$

$= \tan\alpha + \left(2\tan2\alpha + \dfrac{4}{\tan4\alpha}\right)$

$= \tan\alpha + 2\tan2\alpha + \dfrac{4(1-\tan^2 2\alpha)}{2\tan2\alpha}$

$= \tan\alpha + \left(\dfrac{2\tan^2 2\alpha + 2 - 2\tan^2 2\alpha}{\tan 2\alpha}\right)$

$= \tan\alpha + \dfrac{2}{\tan2\alpha}$

$= \tan\alpha + \dfrac{2}{2\tan\alpha}(1-\tan^2\alpha)$

$= \dfrac{\tan^2\alpha + 1 - \tan^2\alpha}{\tan\alpha}$

$= \cot\alpha$

$= 2+\sqrt{3} = \cot(15^\circ) = \cot\alpha$

Q.29 The value of $\sqrt{2+\sqrt{2+2\cos4\theta}}$ $(0^\circ \le \theta \le 15^\circ)$ is equals to : -

(A) $\sin\theta$ (B) $\cos\theta$ (C) $2\sin\theta$ (D) $2\cos\theta$

Sol. [D]

Proper Method

$\sqrt{2+\sqrt{2+2\cos4\theta}}$ $(0^\circ \le \theta \le 15^\circ)$

$= \sqrt{2+\sqrt{2(1+\cos4\theta)}}$

$= \sqrt{2+\sqrt{2(2\cos^2 2\theta)}}$

$= \sqrt{2+2|\cos2\theta|}$ $(\because\ 0^\circ \le 2\theta \le 30^\circ)$

$= \sqrt{2+2\cos2\theta}$

$= \sqrt{2(1+\cos2\theta)}$

$= \sqrt{2(2\cos^2\theta)}$

$= 2|\cos\theta|$ $(\because\ 0 \le \theta \le 15^\circ)$

$= 2\cos\theta$

Short Trick

Put $\theta = 0^\circ$ in L.H.S.

L.H.S. = $\sqrt{2+\sqrt{2+2\times1}} = 2$

After putting $\theta = 0^\circ$ in options only (D) option will match L.H.S.

Q.30 If $\cot^2\alpha \cot^2\beta = 3$, then the value of $(2-\cos2\alpha)(2-\cos2\beta)$ is :

(A) 2 (B) 1 (C) 4 (D) 3

Sol. [D]

Proper Method

$(2-\cos2\alpha)(2-\cos2\beta)$

$= \left(2-\dfrac{1-\tan^2\alpha}{1+\tan^2\alpha}\right)\left(2-\dfrac{1-\tan^2\beta}{1+\tan^2\beta}\right)$

Short Trick

$\because \cot^2\alpha\cot^2\beta = 3$ so let $\alpha = 30^\circ$ and $\beta = 45^\circ$

$\therefore (2-\cos60^\circ)(2-\cos90^\circ) = \left(2-\dfrac{1}{2}\right)(2-0) = 3$

Proper Method	Short Trick
$= \left(\frac{2+2\tan^2\alpha-1+\tan^2\alpha}{1+\tan^2\alpha}\right)\left(\frac{2+2\tan^2\beta-1+\tan^2\beta}{1+\tan^2\beta}\right)$ $= \frac{(1+3\tan^2\alpha)(1+3\tan^2\beta)}{(1+\tan^2\alpha)(1+\tan^2\beta)}$ $= \frac{1+3(\tan^2\beta+\tan^2\alpha)+9\tan^2\alpha\cdot\tan^2\beta}{1+(\tan^2\alpha+\tan^2\beta)+\tan^2\alpha\cdot\tan^2\beta}$ $\frac{1+3(\tan^2\alpha+\tan^2\beta)+3}{1+(\tan^2\alpha+\tan^2\beta)+\frac{1}{3}}$ $(\because \cot^2\alpha\cdot\cot^2\beta = 3)$ $= 3$	

Q.31 If $\cos x = \frac{1+2\cos y}{2+\cos y}$, then $\cot^2\left(\frac{x}{2}\right)\ \tan^2\left(\frac{y}{2}\right)$ is equals to:

(A) 3 (B) 1 (C) 4 (D) 6

Sol. [A]

Proper Method	**Short Trick**
$\cos x = \frac{1+2\cos y}{2+\cos y}$ $\Rightarrow \frac{1+\cos x}{1-\cos x} = \frac{2+\cos y+1+2\cos y}{2+\cos y-(1+2\cos y)}$ $\Rightarrow \frac{2\cos^2\frac{x}{2}}{2\sin^2\frac{x}{2}} = \frac{3(1+\cos y)}{1-\cos y}$ $\Rightarrow \cot^2\frac{x}{2} = \frac{3\left(2\cos^2\frac{y}{2}\right)}{2\sin^2\frac{y}{2}}$ $\Rightarrow \cot^2\frac{x}{2} = 3\cot^2\frac{y}{2}$ $\Rightarrow \cot^2\frac{x}{2}\ \tan^2\frac{y}{2}$ $= 3$	Here if we put $y = 90°$ in the given expression then we get $x = 60°$ $\therefore$ value of required expression $\cot^2\left(\frac{60°}{2}\right)\ \tan^2\left(\frac{90°}{2}\right) = \cot^2(30°)\ \tan^2(45°)$ $= 3$

Q.32 If $\tan x = \frac{b}{a}$, then $\sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}} =$

(A) $\frac{2\sin x}{\sqrt{\sin 2x}}$ (B) $\frac{2\cos x}{\sqrt{\cos 2x}}$ (C) $\frac{2\cos x}{\sqrt{\sin 2x}}$ (D) $\frac{2\sin x}{\sqrt{\cos 2x}}$

Sol. [B]

Proper Method	**Short Trick**
$\tan x = \frac{b}{a}$ by using componendo & dividendo $\Rightarrow \frac{1+\tan x}{1-\tan x} = \frac{a+b}{a-b}$	Put $b = 0$, then $\tan x = 0 \Rightarrow x = 0°$ and LHS $= 1 + 1 = 2$ again putting $x = 0°$ in the option then only (B) will balance the L.H.S.

$$\therefore \sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}} = \sqrt{\frac{1+\tan x}{1-\tan x}} + \sqrt{\frac{1-\tan x}{1+\tan x}}$$

$$= \frac{1+\tan x + 1 - \tan x}{\sqrt{1-\tan^2 x}}$$

$$= \frac{2}{\sqrt{1-\tan^2 x}}$$

$$= \frac{2\cos x}{\sqrt{\cos^2 x - \sin^2 x}}$$

$$= \frac{2\cos x}{\sqrt{\cos 2x}}$$

Q.33 If $\sin\theta + \sin\phi = a$ and $\cos\theta + \cos\phi = b$, then $\tan\left(\frac{\theta-\phi}{2}\right) =$

(A) $\sqrt{\frac{a^2+b^2}{4-a^2-b^2}}$ (B) $\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$ (C) $\sqrt{\frac{a^2+b^2}{4+a^2+b^2}}$ (D) $\sqrt{\frac{4+a^2+b^2}{a^2+b^2}}$

Sol. **[B]**

Proper Method

$\sin\theta + \sin\phi = \alpha$

$$\Rightarrow 2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a \quad \ldots(1)$$

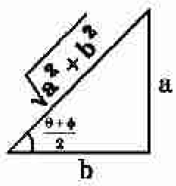
and $\cos\theta + \cos\phi = b$

$$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = b \quad \ldots(2)$$

divide (1) by (2), we get

$$\tan\left(\frac{\theta+\phi}{2}\right) = \frac{a}{b}$$

$$\text{so, } \sin\left(\frac{\theta+\phi}{2}\right) = \frac{a}{\sqrt{a^2+b^2}}$$



$$\text{From (1) } \frac{2a}{\sqrt{a^2+b^2}}\cos\left(\frac{\theta-\phi}{2}\right) = a$$

$$\Rightarrow \cos\left(\frac{\theta-\phi}{2}\right) = \frac{\sqrt{a^2+b^2}}{2}$$

$$\Rightarrow \tan\left(\frac{\theta-\phi}{2}\right) = \frac{\sqrt{4-(a^2+b^2)}}{\sqrt{a^2+b^2}}$$

$$= \sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$$

Short Trick

Put $\theta = \phi = 45°$, $a = b = \sqrt{2}$ therefore L.H.S. = $\tan 0° = 0$ after putting the values of a, b in the options (B) will balance the L.H.S.

Q.34 If $\tan\beta = \dfrac{n\sin\alpha\cos\alpha}{1-n\sin^2\alpha}$, then $\tan(\alpha-\beta) =$

(A) $n\tan\alpha$ (B) $(1-n)\tan\alpha$ (C) $(1+n)\tan\alpha$ (D) $(2+n)\tan\alpha$

Sol. **[B]**

Proper Method

$\tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1+\tan\alpha\cdot\tan\beta}$

$= \dfrac{\tan\alpha - \dfrac{n\sin\alpha\cdot\cos\alpha}{1-n\sin^2\alpha}}{1+\tan\alpha\cdot\dfrac{n\sin\alpha\cdot\cos\alpha}{1-n\sin^2\alpha}}$

$= \dfrac{\tan\alpha - n\tan\alpha\cdot\sin^2\alpha - n\sin\alpha\cdot\cos\alpha}{1-n\sin^2\alpha + n\tan\alpha\cdot\sin\alpha\cdot\cos\alpha}$

$= \dfrac{\tan\alpha - n\tan\alpha\cdot\sin^2\alpha - n\sin\alpha\cdot\cos\alpha}{1-n\sin^2\alpha + n\sin^2\alpha}$

$= \tan\alpha - n\sin\alpha(\tan\alpha\cdot\sin\alpha + \cos\alpha)$

$= \tan\alpha - n\sin\alpha\left(\dfrac{\sin^2\alpha}{\cos\alpha} + \cos\alpha\right)$

$= \tan\alpha - n\tan\alpha$

$= \tan\alpha(1-n)$

Short Trick

Let $\alpha = \beta = 45°$

$\therefore\ \tan 45° = \dfrac{n\sin(45°)\cos(45°)}{1-n\sin^2(45°)}$

$\Rightarrow n = 1$, then LHS $= \tan(45° - 45°) = 0$

after putting the values of n & α the option (B) will balance the L.H.S.

Q.35 If $\tan^2\alpha = 1 - e^2$, then $\sec\alpha + \tan^3\alpha\,\mathrm{cosec}\alpha =$

(A) $2 - e^2$ (B) $(2-e^2)^{1/2}$ (C) $(2-e^2)^{2/3}$ (D) $(2-e^2)^{3/2}$

Sol. **[D]**

Proper Method

$\tan^2\alpha = 1 - e^3$

$\Rightarrow \sec^2\alpha - 1 = 1 - e^2$

$\Rightarrow \sec\alpha = \sqrt{2-e^2}$ ......(1)

$\sec\alpha + \tan^3\alpha.\ \mathrm{cosec}\alpha$

$= \dfrac{1}{\cos\alpha} + \dfrac{\sin^3\alpha}{\cos^3\alpha}\cdot\dfrac{1}{\sin\alpha}$

$= \dfrac{1}{\cos\alpha} + \dfrac{\sin^2\alpha}{\cos^3\alpha}$

$= \dfrac{\cos^2 + \sin^2\alpha}{\cos^3\alpha} = \sec^3\alpha$

From (i) $\sec^3\alpha = (2-e^2)^{3/2}$

Short Trick

Let $\alpha = 45°$ then $e = 0$, and

L.H.S. $= \sec 45° + \tan^3 45°\ \mathrm{cosec}\ 45° = 2\sqrt{2}$ $= 2.2^{1/2} = 2^{3/2}$ after putting $e = 0$ in the options (D) will balance the L.H.S.

Q.36 If $\sin 2x = A - 1$ and $\cos 2x = B - 1$, then $\dfrac{\sec^2 x[(\cos^2 x - \sin^2 x) - 2\sin x\cos x]}{1+\sin 2x} =$

(A) $2(A-B)$ (B) $\dfrac{2}{A} - \dfrac{2}{B}$ (C) $\dfrac{2}{B} - \dfrac{2}{A}$ (D) $2(A+B)$

Sol. **[B]**

Proper Method	Short Trick
$= \dfrac{\sec^2 x[(\cos^2 x - \sin^2 x) - 2\sin x\cos x]}{1+\sin 2x}$ $= \dfrac{\sec^2 x(\cos 2x - \sin 2x)}{1+\sin 2x}$ $= \dfrac{(\cos 2x - \sin 2x)}{(1+\sin 2x)(\cos^2 x)}$ $= \dfrac{(\cos 2x - \sin 2x)}{(1+\sin 2x)\left(\dfrac{1+\cos 2x}{2}\right)}$ $= \dfrac{[(B-1)-(A-1)]}{(1+A-1)\left(\dfrac{1+B-1}{2}\right)}$ $= \dfrac{2(B-A)}{AB}$ $= \dfrac{2}{A} - \dfrac{2}{B}$	$\because \sin 2x = A - 1$ & $\cos 2x = B - 1$ $\therefore$ let $x = 0$, then $A = 1$ and $B = 2$, LHS $= \dfrac{1\{(1-0)-2\times 0\times 1\}}{1+0} = 1$, put values of A and B inoptions, only (B) will give R.H.S.

Q.37 If $\operatorname{cosec}\theta = \dfrac{p+q}{p-q}$, $(p \neq q \neq 0)$, then $\cot\left(\dfrac{\pi}{4}+\dfrac{\theta}{2}\right) =$

(A) $\sqrt{(p/q)}$ (B) $\sqrt{(q/p)}$ (C) $\sqrt{pq}$ (D) pq

Sol. **[B]**

Proper Method	Short Trick
$\operatorname{cosec}\theta = \dfrac{p+q}{p-q}$ $(p \neq q \neq 0)$ using componendo & dividendo $\Rightarrow \dfrac{\operatorname{cosec}\theta - 1}{\operatorname{cosec}\theta + 1} = \dfrac{2q}{2p}$ $\Rightarrow \dfrac{1-\sin\theta}{1+\sin\theta} = \dfrac{q}{p}$ $\Rightarrow \dfrac{\left(\cos\dfrac{\theta}{2} - \sin\dfrac{\theta}{2}\right)^2}{\left(\cos\dfrac{\theta}{2} + \sin\dfrac{\theta}{2}\right)^2} = \dfrac{q}{p}$ $\Rightarrow \dfrac{\cos\dfrac{\theta}{2} - \sin\dfrac{\theta}{2}}{\cos\dfrac{\theta}{2} + \sin\dfrac{\theta}{2}} = \sqrt{\dfrac{q}{p}}$ Divide by $\sin\dfrac{\theta}{2}$ $\Rightarrow \dfrac{\cot\dfrac{\theta}{2} - 1}{\cot\dfrac{\theta}{2} + 1} = \sqrt{\dfrac{q}{p}}$ $\Rightarrow \cot\left(\dfrac{\pi}{4}+\dfrac{\theta}{2}\right) = \sqrt{\dfrac{q}{p}}$	If we put $\theta = 30°$, then $2 = \dfrac{p+q}{p-q} \Rightarrow p = 3q$, $\therefore$ L.H.S $= \cot(45° + 15°) = \cot 60° = 1/\sqrt{3}$ Put the value of p in options then (B) will give $1/\sqrt{3}$ so it is right option.

Q.38 If $\alpha + \beta + \gamma = 2\pi$, then which of the followings is true?

(A) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = \tan\frac{\alpha}{2} . \tan\frac{\beta}{2} . \tan\frac{\gamma}{2}$

(B) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} + \tan\frac{\beta}{2}\tan\frac{\gamma}{2} + \tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 1$

(C) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = -\tan\frac{\alpha}{2} . \tan\frac{\beta}{2} . \tan\frac{\gamma}{2}$

(D) None of these

Sol. **[A]**

Proper Method

$\alpha + \beta + \gamma = 2\pi$

$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} + \frac{\gamma}{2} = \pi$

$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} = \pi - \frac{\gamma}{2}$

$\Rightarrow \tan\left(\frac{\alpha}{2} + \frac{\beta}{2}\right) = -\tan\frac{\gamma}{2}$

$= \dfrac{\tan\frac{\alpha}{2} + \tan\frac{\beta}{2}}{1 - \tan\frac{\alpha}{2} \cdot \tan\frac{\beta}{2}} = -\tan\frac{\gamma}{2}$

$\Rightarrow \tan\frac{\alpha}{2} + \tan\frac{\beta}{2} = -\tan\frac{\gamma}{2} + \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

$\Rightarrow \tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = \tan\frac{\gamma}{2} \cdot \tan\frac{\beta}{2} \cdot \tan\frac{\gamma}{2}$

Short Trick

Since $\alpha + \beta + \gamma = 360°$ therefore, let $\alpha = \beta = \gamma = 120°$ and put these values of α, β, γ in the options then (A) is satisfy.

Q.39 If $\cos^4\theta = a_0 + a_1\cos 2\theta + a_2\cos 4\theta$ is an identity then $\frac{a_0}{a_1 a_2}$ is equals to :

(A) 3 (B) 6 (C) 4 (D) 2

Sol. **[B]**

Proper Method

$\cos^4\theta = a_0 + a_1\cos 2\theta + a_2\cos 4\theta$

$\Rightarrow \cos^4\theta = a_0 + a_1(2\cos^2\theta - 1) + a_2(2\cos^2 2\theta - 1)$

$\Rightarrow \cos^4\theta = a_0 + 2a_1\cos^2\theta - a_1 + 2a_2\cos^2 2\theta - a_2$

$\Rightarrow \cos^4\theta = a_0 + 2a_1\cos^2\theta - a_1 + 2a_2(2\cos^2\theta - 1)^2 - a_2$

$\Rightarrow \cos^4\theta = (a_0 - a_1 - a_2) + 2a_1\cos^2\theta + 2a_2(4\cos^4\theta - 4\cos^2\theta + 1)$

$\Rightarrow \cos^4\theta = (a_0 - a_1 - a_2 + 2a_2) + 2a_1\cos^2\theta + 8a_2\cos^4\theta - 8a_2\cos^2\theta$

Compare, we get

$\Rightarrow 8a_2 = 1 \Rightarrow a_2 = \frac{1}{8}$

$\Rightarrow 2a_1 - 8a_2 = 0 \Rightarrow a_1 = \frac{1}{2}$

$\Rightarrow a_0 - a_1 + a_2 = 0 \Rightarrow a_0 = \frac{3}{8}$

So, $\dfrac{a_0}{a_1 a_2} = \dfrac{\frac{3}{8}}{\frac{1}{2} \times \frac{1}{8}} = 6$

Short Trick

$\therefore$ let $\theta = 0°$ then $1 = a_0 + a_1 + a_2$ .....(i),

again put $\theta = 45°$ then $\frac{1}{4} = a_0 - a_2$.....(ii)

and put $\theta = 90°$ then $0 = a_0 - a_1 + a_2$.....(iii),

By (i), (ii) and (iii) we get $a_0 = \frac{3}{8}$, $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{8}$

$\therefore \dfrac{a_0}{a_1 a_2} = \dfrac{3/8}{1/2 \times 1/8} = 6.$

Q.40 If $0 \le (\alpha_1, \alpha_2, \alpha_3, ... \alpha_n) \le \frac{\pi}{2}$ and if $\tan\alpha_1 \tan\alpha_2 \tan\alpha_3 .... \tan\alpha_n = 1$, then

$\cos\alpha_1 \cos\alpha_2 \cos\alpha_3 .... \cos\alpha_n =$

(A) $2^{n/2}$ (B) $2^{-n/2}$ (C) $2^{n/4}$ (D) $2^{-n/4}$

Sol. **[B]**

Proper Method	Short Trick
Given $\tan\alpha_1 \cdot \tan\alpha_2 \text{--------} \tan\alpha_n = 1$ $\Rightarrow \cos\alpha_1 \cdot \cos\alpha_2 \text{----} \cos\alpha_n = \sin\alpha_1 \cdot \sin\alpha_2 \text{----} \sin\alpha_n = y$(say) $\therefore y^2 = (\cos\alpha_1 \cos\alpha_2 \text{-------} \cos\alpha_n)^2$ $= (\cos\alpha_1 \cdot \cos\alpha_2 \text{-------} \cos\alpha_n)(\sin\alpha_1 \cdot \sin\alpha_2 \text{-------} \sin\alpha_n)$ $= \frac{1}{2^n} \sin2\alpha_1 \cdot \sin2\alpha_2 \text{--------} \sin2\alpha_n \le \left(\frac{1}{2}\right)^n$ $\Rightarrow y \le \left(\frac{1}{2}\right)^{n/2}$	Let $\alpha_1 = \alpha_2 = \alpha_3 = ... = \alpha_n = 45°$ (Satisfying given conditions) required value = $\cos45° \times \cos45° \times \cos45°$ .... n terms $= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$ .... n terms $= \frac{1}{(\sqrt{2})^n} = \frac{1}{2^{n/2}} = 2^{-n/2}$

8

Properties of Triangle

KEY CONCEPTS

1. **Basic Rules of Triangle**

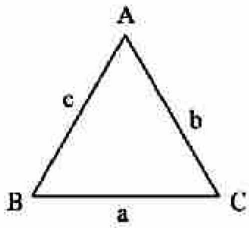
In a ΔABC, the angles are denoted by capital letters A,B and C and the lengths of the sides opposite to these angles are denoted by small letters a,b and c, respectively.

These six elements of a triangle are connected by the following relations

(i) $A + B + C = 180^\circ$ or π

(ii) $a + b > c,\ b + c > a,\ c + a > b$

(iii) $a > 0,\ b > 0,\ c > 0$



(i) **Sine Rule :** The sides of a triangle (any type of triangle) are proportional to the sines of the angle opposite to them.

In triangle ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The sine rule is very useful tool to express sides of a triangle in terms of sines of angle and vice-versa in the following manner.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \text{ (Let)} \Rightarrow a = k \sin A,\ b = k \sin B,\ c = k \sin C$$

(ii) **Cosine Rule :**

In triangle ABC

(a) $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

(b) $\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$

(c) $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

(iii) **Projection Formulae :**

In triangle ABC

(a) $a = b \cos C + c \cos B$

(b) $b = c \cos A + a \cos C$

(c) $c = a \cos B + b \cos A$

(iv) Napier's Analogy (Tangent rule) :

In any ΔABC,

(a) $\tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right)\cot\left(\frac{A}{2}\right)$

(b) $\tan\left(\frac{A-B}{2}\right) = \left(\frac{a-b}{a+b}\right)\cot\left(\frac{C}{2}\right)$

(c) $\tan\left(\frac{C-A}{2}\right) = \left(\frac{c-a}{c+a}\right)\cot\left(\frac{B}{2}\right)$

2. Trigonometrical Ratios of the Half Angles of a Triangle

If the perimeter of a triangle ABC is denoted by 2s then $2s = a + b + c$ and area denoted by Δ then.

(i) Formulae for $\sin\left(\frac{A}{2}\right)$, $\sin\left(\frac{B}{2}\right)$, $\sin\left(\frac{C}{2}\right)$

(a) $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{bc}}$

(b) $\sin\left(\frac{B}{2}\right) = \sqrt{\frac{(s-c)(s-a)}{ac}}$

(c) $\sin\left(\frac{C}{2}\right) = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(ii) Formulae for $\cos\left(\frac{A}{2}\right)$, $\cos\left(\frac{B}{2}\right)$, $\cos\left(\frac{C}{2}\right)$

(a) $\cos\left(\frac{A}{2}\right) = \sqrt{\frac{s(s-a)}{bc}}$

(b) $\cos\left(\frac{B}{2}\right) = \sqrt{\frac{s(s-b)}{ac}}$

(c) $\cos\left(\frac{C}{2}\right) = \sqrt{\frac{s(s-c)}{ab}}$

(iii) Formulae for $\tan\left(\frac{A}{2}\right)$, $\tan\left(\frac{B}{2}\right)$, $\tan\left(\frac{C}{2}\right)$

(a) $\tan\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

(b) $\tan\left(\frac{B}{2}\right) = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$

(c) $\tan\left(\frac{C}{2}\right) = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

3. Area of Triangle

The area of ΔABC is given by

(i) $\Delta = \frac{1}{2}bc\sin A$, $\Delta = \frac{1}{2}ca\sin B$, $\Delta = \frac{1}{2}ab\sin C$

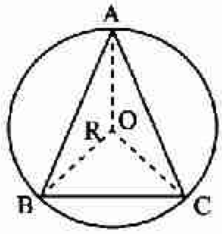
(ii) Hero's Formula :

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$

4. Circumcircle of a Triangle and its Radius (R)

Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'

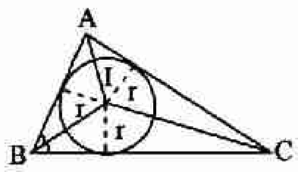
(i) $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$



(ii) $R = \frac{abc}{4\Delta}$

5. Inscribed Circle or Incircle of a Triangle and Its Radius (r)

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'



(i) $r = \frac{\Delta}{s}$

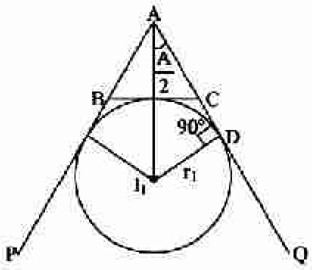
(ii) $r = (s-a)\tan\left(\frac{A}{2}\right)$, $r = (s-b)\tan\left(\frac{B}{2}\right)$ & $r = (s-c)\tan\left(\frac{C}{2}\right)$

(iii) $r = \frac{a\sin\left(\frac{B}{2}\right).\sin\left(\frac{C}{2}\right)}{\cos\left(\frac{A}{2}\right)}$, $r = \frac{b\sin\left(\frac{C}{2}\right).\sin\left(\frac{A}{2}\right)}{\cos\left(\frac{B}{2}\right)}$ & $r = \frac{c\sin\left(\frac{A}{2}\right).\sin\left(\frac{B}{2}\right)}{\cos\left(\frac{C}{2}\right)}$

(iv) $r = 4R\sin\left(\frac{A}{2}\right).\sin\left(\frac{B}{2}\right).\sin\left(\frac{C}{2}\right)$

6. Escribed Circles of a Triangle and Their Radii (r_1, r_2, r_3)

Point of intersection of two external angle and one internal angle bisectors is excentre and perpendicular of excentre from any side is called exradius. If r_1 is the radius of escribed circle opposite to angle A of Δ ABC and so on then :



(i) $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$, $r_3 = \frac{\Delta}{s-c}$

(ii) $r_1 = s\tan\frac{A}{2}$, $r_2 = s\tan\frac{B}{2}$, $r_3 = s\tan\frac{C}{2}$

(iii) $r_1 = \dfrac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$, $r_2 = \dfrac{b\cos\frac{C}{2}\cos\frac{A}{2}}{\cos\frac{B}{2}}$, $r_3 = \dfrac{c\cos\frac{A}{2}\cos\frac{B}{2}}{\cos\frac{C}{2}}$

(iv) $r_1 = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$,

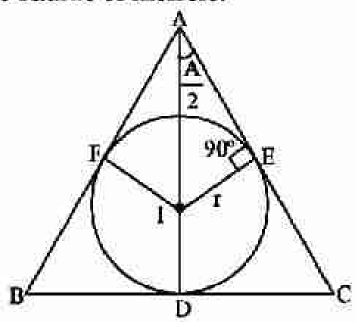
$r_2 = 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$,

$r_3 = 4R\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

7. Geometrical Distances

(i) Distance between the vertex of the triangle and the incentre:

I is centre of incircle, r is the radius of incircle.



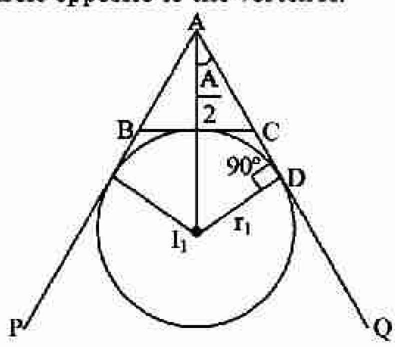
$AI = r\,\text{cosec}\left(\frac{A}{2}\right)$, $BI = r\,\text{cosec}\left(\frac{B}{2}\right)$ and

$CI = r\,\text{cosec}\left(\frac{C}{2}\right)$

(ii) Distance between the vertex of the triangle and the excentre opposite to the vertex A:

I_1 is the centre of ex-circle opposite to the vertex A.

r_1 is the radius of the ex-circle opposite to the vertex A.



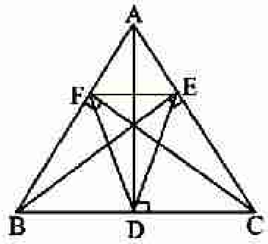
$AI_1 = r_1\,\text{cosec}\left(\frac{A}{2}\right)$

Also, $BI_2 = r_2\,\text{cosec}\left(\frac{B}{2}\right)$,

$CI_3 = r_3\,\text{cosec}\left(\frac{C}{2}\right)$

8. Orthocentre And Pedal Triangle

Point of intersection of altitudes is orthocenter & the triangle DEF which is formed by joining the feet of the altitudes is called the pedal triangle.



(i) The distances of the orthocenter from the angular points of the Δ ABC are 2R cosA, 2R cosB and 2R cosC.

(ii) The distance of orthocenter from sides are 2R cosB cosC, 2RcosC cosA and 2R cosA cosB

(iii) Area of pedal triangle = $2\Delta \cos A \cos B \cos C = \frac{1}{2}R^2 \sin 2A \sin 2B \sin 2C$

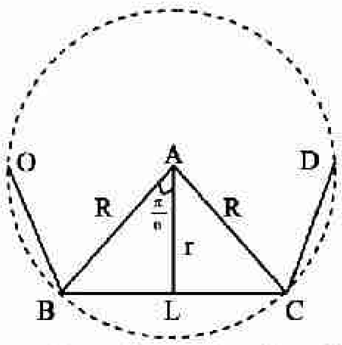
9. Length of Median, Altitude, Angle Bisector

(i) Length of Median through the vertex A = $\frac{1}{2}\sqrt{2b^2+2c^2-a^2}$

(ii) Length of Angle Bisector through the vertex A = $\dfrac{2bc\cos\frac{A}{2}}{b+c}$

(iii) Length of Altitude through the vertex A = $\dfrac{a}{\cot B+\cot C}$

10. Regular Polygon



If a be the length of every side of a regular polygon of n sides, then

(i) radius of incircle of this polygon is $r = \frac{a}{2}\cot\frac{\pi}{n}$

(ii) radius of circumcircle of this polygon is $R = \frac{a}{2}\operatorname{cosec}\frac{\pi}{n}$

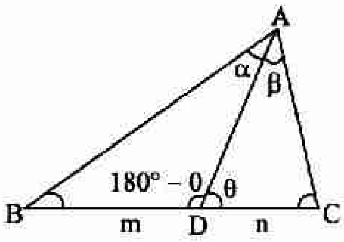
(iii) area of the polygon = $\frac{1}{4}na^2\cot\left(\frac{\pi}{n}\right) = nr^2\tan\left(\frac{\pi}{n}\right) = \frac{1}{2}nR^2\sin\left(\frac{2\pi}{n}\right)$

11. m-n Theorem

If D be a point on the side BC of a ΔABC such that $BD : DC = m : n$ and $\angle ADC = \theta$, $\angle BAD = \alpha$ and $\angle DAC = \beta$.

(i) $(m + n) \cot \theta = m \cot \alpha - n \cot \beta$

(ii) $(m + n) \cot \theta = n \cot B - m \cot C$



Important Points to be Remembered

(1) Distance between the circum-centre and the orthocentre is $R\sqrt{1 - 8\cos A \cos B \cos C}$

(2) Distance between the circum centre and the incentre $= \sqrt{R^2 - 2Rr}$

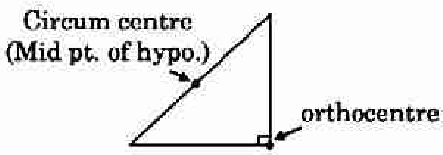
(3) Distance between incentree and orthocenter is $= \sqrt{2r^2 - 4R^2 \cos A \cos B \cos C}$

(4) Distance between circumcentre and centroid $= R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$

(5) In any triangle circum centre (O), orthocentre (H) and centroied (G) are colinear and GH : GO = 2 : 1

(6) In equilateral triangle, all centres coincides with centroid [all centre's mean ortho centre, incentre, circumcentre]

(7) In right angle triangle



(8) **Ambigious case** : Let a, b, $\angle A$ are given then following cases are possible

(a) $\angle A$ is acute, and $a < b \sin A$, no triangle is possible

(b) $\angle A$ is acute and $a = b \sin A$ only one Δ is possible which is Right angle $\Delta(\angle B = 90°)$ and if $\angle A$ is obtuse, No Δ is possible

(c) $\angle A$ is acute and $a > b \sin A$, in this case two values of $\angle B$ are possible hence two triangles are possible.

SHORT TRICKS

Trick -1 In an equilateral triangle $\frac{r}{R} = \frac{1}{2}$ (where r is inradius & R is circum radius)

Q.1 $\dfrac{a\cos A + b\cos B + c\cos C}{a+b+c} =$

(A) $1/r$ (B) r/R (C) R/r (D) $1/R$

Sol. [B]

Proper Method

$$\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$$

$$= \frac{k\,[\sin A\cos A + \sin B\cos B + \sin C\cos C]}{k\,(\sin A + \sin B + \sin C]}$$

$$= \frac{1}{2}\frac{(\sin 2A + \sin 2B + \sin 2C)}{(\sin A + \sin B + \sin C)}$$

$$= \frac{1}{2}\left[\frac{2\sin(A+B)\cos(A-B) + 2\sin C\cos C}{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2}}\right]$$

$$= \frac{1}{2}\left[\frac{\sin C\{\cos(A-B) - \cos(A+B)\}}{\cos\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) + \cos\left(\frac{A+B}{2}\right)\right\}}\right]$$

$$= \frac{1}{2}\left[\frac{\sin C(2\sin A\sin B)}{\cos\frac{C}{2}\left(2\cos\frac{A}{2}\cos\frac{B}{2}\right)}\right]$$

$$= \frac{1}{2}\left[\frac{2\sin\frac{A}{2}\cos\frac{A}{2}.2\sin\frac{B}{2}\cos\frac{B}{2}.2\sin\frac{C}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}\right]$$

$$= 4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2} = \frac{r}{R}, \left[\because r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right]$$

Short Trick

Let triangle be equilateral then value of expression $= \frac{1}{2}$ and in equilateral triangle $\frac{r}{R} = \frac{1}{2}$

Q.2 In triangle ABC, $\cos A + \cos B + \cos C$ is equal to

(A) $1 + R/r$ (B) $1 + r/R$ (C) $1 - R/r$ (D) $1 - r/R$

Sol. [B]

Proper Method

$\cos A + \cos B + \cos C$

$= 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ (identity relation)

$= 1 + \frac{r}{R}\ \left(\because r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$

Short Trick

Let the triangle be equilateral

$\cos A + \cos B + \cos C = \frac{3}{2}$

by option (B) is correct answer

Q.3 In a triangle ABC, $\frac{b-c}{r_1}+\frac{c-a}{r_2}+\frac{a-b}{r_3}$ is equal to

(A) 1 (B) 0 (C) abc (D) $r_1r_2r_3$

Sol. [B]

Proper Method	Short Trick
Exp. $=\frac{b-c}{\left(\frac{\Delta}{s-a}\right)}+\frac{c-a}{\left(\frac{\Delta}{s-b}\right)}+\frac{a-b}{\left(\frac{\Delta}{s-c}\right)}$ $=\frac{1}{\Delta}[(b-c)(s-a)+(c-a)(s-b)+(a-b)(s-c)]$ $=\frac{1}{\Delta}[s(b-c+c-a+a-b)-a(b-c)-b(c-a)-c(a-b)]$ $=0$	Let the triangle be equilateral then value of expression = 0

Q.4 In an equilateral triangle $r : R : r_1$ is equal to

(A) 1 : 2 : 3 (B) 2 : 3 : 5 (C) 1 : 3 : 7 (D) 3 : 7 : 9

Sol. [A]

Proper Method	Short Trick
If a be side of equilateral, then $s=\frac{3a}{2}, \Delta=\frac{\sqrt{3}}{4}a^2$ $\Rightarrow r=\frac{\Delta}{s}=\frac{a}{2\sqrt{3}}$ Also $R=\frac{a.a.a}{4\Delta}=\frac{a}{\sqrt{3}}$ $r_1=r_2=r_3=\frac{\Delta}{s-a}=\frac{\sqrt{3}a}{2}$ $\therefore r : R : r_1 = 1 : 2 : 3$	In an equilateral triangle $r : R = 1 : 2$ by option (A) is correct answer

Trick -2 Combination of method of substitution and balancing

Q.5 In triangle *ABC*, $(b+c)\cos A+(c+a)\cos B+(a+b)\cos C=$

(A) 0 (B) 1 (C) $a+b+c$ (D) $2(a+b+c)$

Sol. [C]

Proper Method	Short Trick
$(b+c)\cos A+(c+a)\cos B+(a+b)\cos C$ $= b\cos A + c.\cos A + c.\cos B + a\cos B + a\cos C + b.\cos C$ $= (b\cos A + a.\cos B) + (c\cos A + a\cos C) = (c\cos B + b\cos C)$ $= c + b + a$	Let triangle be equilateral $\therefore A = B = C = 60°$ and $a = b = c = x$ $(b + c)\cos A + (c + a)\cos B + (a + b)\cos C = 3x$ Hence (C) is correct option

Q.6 In ΔABC, $\left(\cot\frac{A}{2}+\cot\frac{B}{2}\right)\left(a\sin^2\frac{B}{2}+b\sin^2\frac{A}{2}\right)=$

(A) $\cot C$ (B) $c\cot C$ (C) $\cot\frac{C}{2}$ (D) $c\cot\frac{C}{2}$

Sol. [D]

Proper Method

$$\left\{\cot\frac{A}{2}+\cot\frac{B}{2}\right\}\left\{a\sin^2\frac{B}{2}+b\sin^2\frac{A}{2}\right\}$$

$$=\left\{\frac{\cos\frac{C}{2}}{\sin\frac{A}{2}\sin\frac{B}{2}}\right\}\left\{a\sin^2\frac{B}{2}+b\sin^2\frac{A}{2}\right\}$$

$$=\left\{\cos\frac{C}{2}\right\}\left\{a\frac{\sin\frac{B}{2}}{\sin\frac{A}{2}}+b\frac{\sin\frac{A}{2}}{\sin\frac{B}{2}}\right\}$$

$$=\sqrt{\frac{s(s-c)}{ab}}\left\{a\frac{\sqrt{\frac{(s-a)(s-c)}{ac}}}{\sqrt{\frac{(s-b)(s-c)}{bc}}}+b\frac{\sqrt{\frac{(s-b)(s-c)}{bc}}}{\sqrt{\frac{(s-a)(s-c)}{ac}}}\right\}$$

$$=\sqrt{\frac{s(s-c)}{ab}}\left\{\sqrt{\left(\frac{s-a}{s-b}\right)ab}+\sqrt{\left(\frac{s-b}{s-a}\right)ab}\right\}$$

$$=\sqrt{s(s-c)}\left\{\frac{s-a+s-b}{\sqrt{(s-a)(s-b)}}\right\}=\sqrt{s(s-c)}\left\{\frac{2s-a-b}{\sqrt{(s-a)(s-b)}}\right\}$$

$$=c\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}=c\cot\frac{C}{2}.$$

Short Trick

Let triangle be equilateral

$\therefore$ A = B = C = 60° and a = b = c = x, then value of expression

$(\sqrt{3}+\sqrt{3})\left(\frac{x}{4}+\frac{x}{4}\right)=\sqrt{3}x$ hence (D) is correct option

Q.7 In ΔABC, $1-\tan\frac{A}{2}\tan\frac{B}{2}=$

(A) $\frac{2c}{a+b+c}$ (B) $\frac{a}{a+b+c}$ (C) $\frac{2}{a+b+c}$ (D) $\frac{4a}{a+b+c}$

Sol. [A]

Proper Method

$$1-\tan\frac{A}{2}\tan\frac{B}{2}=\frac{\cos\frac{A}{2}\cos\frac{B}{2}-\sin\frac{A}{2}\sin\frac{B}{2}}{\cos\frac{A}{2}\cos\frac{B}{2}}$$

$$=\frac{\cos\left(\frac{A}{2}+\frac{B}{2}\right)}{\cos\frac{A}{2}\cos\frac{B}{2}}=\frac{\sin\frac{C}{2}}{\cos\frac{A}{2}\cos\frac{B}{2}}$$

$$=\left[\frac{(s-a)(s-b)bc.ac}{ab.s(s-a)s(s-b)}\right]^{1/2}=\frac{c}{s}=\frac{2c}{a+b+c}$$

Short Trick

Let triangle be equilateral

$1-\tan\frac{A}{2}\tan\frac{B}{2}=1-\frac{1}{3}=\frac{2}{3}$

Hence (A) is correct option

Q.8 In ΔABC, $a\sin(B-C)+b\sin(C-A)+c\sin(A-B)=$

(A) 0 (B) $a+b+c$ (C) $a^2+b^2+c^2$ (D) $2(a^2+b^2+c^2)$

Sol. [A]

Proper Method	**Short Trick**
$a\sin(B-C)+b\sin(C-A)+c\sin(A-B)$ $= k(\Sigma \sin A\sin(B-C) = k\{\Sigma\sin(B+C)\sin(B-C)\}$ $= k\left\{\Sigma\frac{1}{2}(\cos 2C-\cos 2B)\right\} = 0$ **Note:** Students should note here that most of the expressions containing the cyclic factor associating with '–' reduces to 0	Let triangle be equilateral then value of expression = 0 + 0 + 0 = 0

Q.9 In ΔABC, if $2(bc\cos A + ca\cos B + ab\cos C) =$

(A) 0 (B) $a+b+c$ (C) $a^2+b^2+c^2$ (D) $2(a^2+b^2+c^2)$

Sol. **[C]**

Proper Method	**Short Trick**
$2(bc\cos A + ca\cos B + ab\cos C)$ $= 2\left[bc\left(\frac{c^2+b^2-a^2}{2bc}\right)+ca\left(\frac{a^2+c^2-b^2}{2ac}\right)+ab\left(\frac{a^2+b^2-c^2}{2ab}\right)\right]$ $=\frac{2}{2}[(c^2+b^2-a^2)+(a^2+c^2-b^2)+(a^2+b^2-c^2)]$ $= a^2+b^2+c^2$	Let the triangle be equilateral then value of expression $= 2\left(\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}\right) = 3x^2$ Hence (C) is correct option

Q.10 In ΔABC, $\operatorname{cosec} A(\sin B\cos C + \cos B\sin C) =$

(A) c/a (b) a/c (c) 1 (d) c/ab

Sol. **[C]**

Proper Method	**Short Trick**
$\operatorname{cosec}A\ (\sin B\cos C + \cos B\sin C)$ $= \frac{1}{\sin A}\ (\sin B\cos C + \cos B\sin C)$ by sine rule ($\sin A = ak$, $\sin B = bk$, $\sin C = ck$) $= \frac{1}{ak}(bk.\cos C + \cos B.\ ck)$ $= \frac{1}{a}(b\cos C + c.\cos B)$ $= \frac{1}{a}(a)$ (by projection rule) $= 1$	Let the triangle be equilateral, then value of expression $\frac{2}{\sqrt{3}}\left(\frac{\sqrt{3}}{4}+\frac{\sqrt{3}}{4}\right) = 1$

Q.11 In a triangle ABC, $a^3\cos(B-C)+b^3\cos(C-A)+c^3\cos(A-B) =$

(A) abc (B) $3abc$ (C) $a+b+c$ (D) 0

Sol. **[B]**

Proper Method	**Short Trick**
$a^3\cos(B-C)+b^3\cos(C-A)+c^3\cos(A-B)$ $= k^3\sin^3 A\cos(B-C)+k^3\sin^3 B\cos(C-A)$ $+ k^3\sin^3 C\cos(A-B)$ $=\frac{1}{2}k^3[\sin^2 A(\sin 2B+\sin 2C)+\sin^2 B(\sin 2C+\sin 2A)$ $+\sin^2 C(\sin 2A+\sin 2B)]$ $= k^3[\sin A\sin B(\sin A\cos B+\cos A\sin B)$ $+\sin B\sin C(\sin B\cos C+\cos B\sin C)$ $= k^3[\sin A\sin B\sin C+\sin B\sin C\sin A+\sin C\sin A\sin B]$ $= 3k^3$ [$\sin A\sin B\sin C = 3abc$]	Let triangle be equilateral then value of expression = $x^3+x^3+x^3 = 3x^3$ Hence (B) is correct option

Q.12 In ΔABC, $a^2(\cos^2 B - \cos^2 C) + b^2(\cos^2 C - \cos^2 A) + c^2(\cos^2 A - \cos^2 B) =$

(A) 0 (B) 1 (C) $a^2 + b^2 + c^2$ (D) $2(a^2 + b^2 + c^2)$

Sol. [A]

Proper Method	**Short Trick**
$a^2(\cos^2B - \cos^2C) + b^2(\cos^2C - \cos^2A) + c^2(\cos^2A - \cos^2B)$ $= a^2(\sin^2C - \sin^2B) + b^2(\sin^2A - \sin^2C) + c^2(\sin^2B - \sin^2A)$ by sine rule $\{\sin A = ak, \sin B = bk, \sin C = ck\}$ $= a^2k^2(c^2 - b^2) + b^2k^2(a^2 - c^2) + c^2k^2(b^2 - a^2)$ $= k^2(a^2c^2 - a^2b^2 + b^2a^2 - b^2c^2 + c^2b^2 - c^2a^2)$ $= 0$	Let triangle be equilateral Hence (A) is correct option

Q.13 In ΔABC, $(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C =$

(A) 0 (B) $a^2 + b^2 + c^2$ (C) $2(a^2 + b^2 + c^2)$ (D) $\frac{1}{2abc}$

Sol. [A]

Proper Method	**Short Trick**
$(b^2 - c^2)\cot A = (b^2 - c^2)\frac{\cos A}{\sin A} = \frac{(b^2 - c^2)(b^2 + c^2 - a^2)}{2bc.ka}$ Hence L.H.S. $= \frac{1}{2k\ abc}$ $[(b^4 - c^4) + (c^4 - a^4) + (a^4 - b^4) - \{a^2(b^2 - c^2)$ $+ b^2(c^2 - a^2) + c^2(a^2 - b^2)\}] = 0$	Let triangle be equilateral Hence (A) is correct option

Q.14 In ΔABC, $(b - c)\cot\frac{A}{2} + (c - a)\cot\frac{B}{2} + (a - b)\cot\frac{C}{2}$ is equal to

(A) 0 (B) 1 (C) ±1 (D) 2

Sol. [A]

Proper Method	**Short Trick**
$(b - c)\cot\frac{A}{2} = k(\sin B - \sin C)\cot\frac{A}{2}$ $= 2k\cos\frac{B+C}{2}\sin\frac{B-C}{2}\cot\frac{A}{2}$ $= 2k\sin\frac{A}{2}.\sin\frac{B-C}{2}.\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}}$ $= 2k\sin\left(\frac{B-C}{2}\right)\sin\left(\frac{B+C}{2}\right) = 2k\left(\sin^2\frac{B}{2} - \sin^2\frac{C}{2}\right)$ or we get L.H.S. $= \Sigma 2k\left(\sin^2\frac{B}{2} - \sin^2\frac{C}{2}\right) = 0$	Let triangle be equilateral Hence (A) is correct option

Q.15 If α, β, γ are angles of a triangle, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma - 2\cos\alpha\cos\beta\cos\gamma$ is

(A) 2 (B) –1 (C) –2 (D) 0

Sol. **[A]**

Proper Method	**Short Trick**
We have, $\sin^2\alpha+\sin^2\beta+\sin^2\gamma-2\cos\alpha\cos\beta\cos\gamma$ $=3-[\cos^2\alpha+\cos^2\beta+\cos^2\gamma]-2\cos\alpha\cos\beta\cos\gamma$ $=3-\left[\frac{1+\cos 2\alpha}{2}+\frac{1+\cos 2\beta}{2}+\frac{1+\cos 2\gamma}{2}\right]-2\cos\alpha\cos\beta\cos\gamma$ $=3-\frac{1}{2}[3+\cos 2\alpha+\cos 2\beta+\cos 2\gamma]-2\cos\alpha\cos\beta\cos\gamma$ $=3-\frac{3}{2}-\frac{1}{2}(\cos 2\alpha+\cos 2\beta)-\frac{1}{2}\cos 2\gamma-2\cos\alpha\cos\beta\cos\gamma$ $=\frac{3}{2}-\frac{1}{2}[-2\cos\gamma\cos(\alpha-\beta)]-\frac{1}{2}[2\cos^2\gamma-1]-2\cos\alpha\cos\beta\cos\gamma$ $=\frac{3}{2}+\cos\gamma\cos(\alpha-\beta)-\cos^2\gamma+\frac{1}{2}-2\cos\alpha\cos\beta\cos\gamma$ $=2$	Let the triangle be equilateral then value of expression $=\frac{3}{4}+\frac{3}{4}+\frac{3}{4}-2\left(\frac{1}{8}\right)=2$

Q.16 If the area of a triangle ABC is Δ, then $a^2\sin 2B+b^2\sin 2A$ is equal to

(A) 3Δ (B) 2Δ (C) 4Δ (D) -4Δ

Sol. **[C]**

Proper Method	**Short Trick**
$\Delta=\frac{1}{2}bc\sin A\Rightarrow\frac{1}{2}k^2\sin B\sin C\sin A=\Delta$ $a^2\sin 2B+b^2\sin 2A=2(a^2\sin B\cos B+b^2\sin A\cos A)$ $=2k^2(\sin^2 A\sin B\cos B+\sin^2 B\sin A\cos A)$ $=2k^2(\sin A\sin B)(\sin C)$ $=2k^2(\sin A\sin B\sin C)=4\Delta$	Let triangle be equilateral then, value of expression $x^2\left(\frac{\sqrt{3}}{2}\right)+x^2\left(\frac{\sqrt{3}}{2}\right)=\sqrt{3}x^2$ Hence (C) is correct option

9

Inverse Trigonometric Function

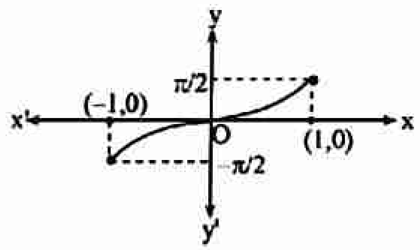
KEY CONCEPTS

1. Inverse Trigonometric Function

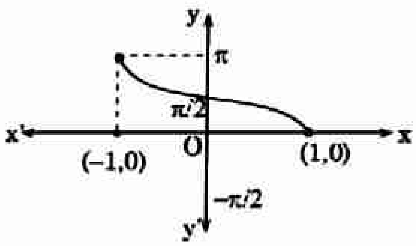
As we know that trigonometric functions are not one-one and onto in their natural domain and range, so their inverse do not exist but if we restrict their domain and range, then their inverse may exists.

2. Graph of Inverse Trigonometric Function

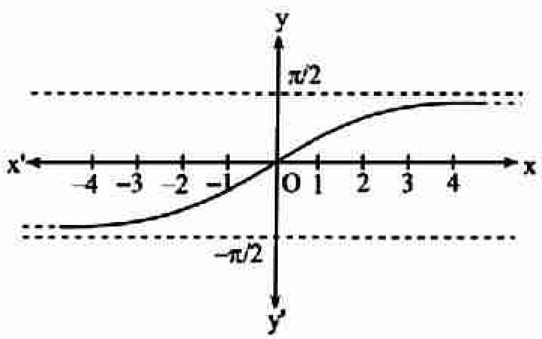
(i) Graph of $y = \sin^{-1}x$



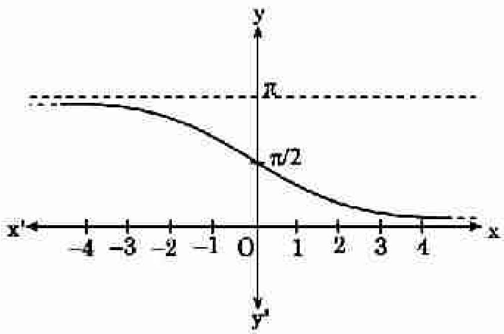
(ii) Graph of $y = \cos^{-1} x$



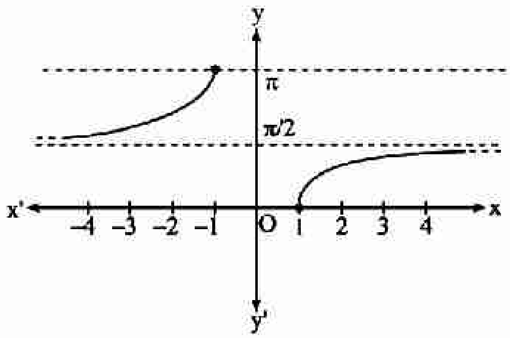
(iii) Graph of $y = \tan^{-1}x$



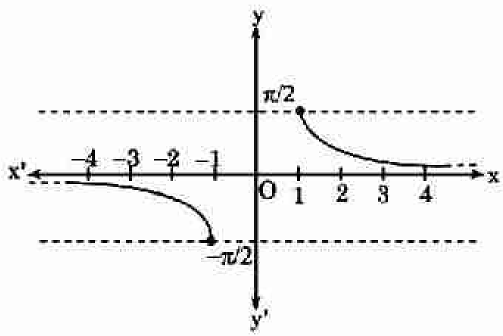
(iv) Graph of $y = \cot^{-1}x$



(v) Graph of $y = \sec^{-1}x$



(vi) Graph of $y = \text{cosec}^{-1}x$



3. Domain and Range of Inverse Trigonometric Functions

Function	**Domain**	**Range**
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
$\text{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

4. Properties of Inverse Trigonometric Function

Property-1 :

(i) $\sin^{-1}(\sin\theta) = \theta$, provided that $-\dfrac{\pi}{2} \le \theta \le \dfrac{\pi}{2}$

(ii) $\cos^{-1}(\cos\theta) = \theta$, provided that $0 \le \theta \le \pi$

(iii) $\tan^{-1}(\tan\theta) = \theta$, provided that $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$

(iv) $\cot^{-1}(\cot\theta) = \theta$, provided that $0 < \theta < \pi$

(v) $\sec^{-1}(\sec\theta) = \theta$, provided that $0 \le \theta < \dfrac{\pi}{2}$ or $\dfrac{\pi}{2} < \theta \le \pi$

(vi) $\text{cosec}^{-1}(\text{cosec}\,\theta) = \theta$, provided that $-\dfrac{\pi}{2} \le \theta < 0$ or $0 < \theta \le \dfrac{\pi}{2}$

Property-2 :

(i) $\sin(\sin^{-1}x) = x$, provided that $-1 \le x \le 1$

(ii) $\cos(\cos^{-1}x) = x$, provided that $-1 \le x \le 1$

(iii) $\tan(\tan^{-1}x) = x$, provided that $-\infty < x < \infty$

(iv) $\cot(\cot^{-1}x) = x$, provided that $-\infty < x < \infty$

(v) $\sec(\sec^{-1}x) = x$, provided that $-\infty < x \le -1$ or $1 \le x < \infty$

(vi) $\text{cosec}(\text{cosec}^{-1}x) = x$, provided that $-\infty < x \le -1$ or $1 \le x < \infty$

Property-3 :

(i) $\sin^{-1}(-x) = -\sin^{-1}x \;\; \forall\, x \in [-1, 1]$

(ii) $\cos^{-1}(-x) = \pi - \cos^{-1}x \;\forall\, x \in [-1, 1]$

(iii) $\tan^{-1}(-x) = -\tan^{-1}x \;\forall\, x \in R$

(iv) $\cot^{-1}(-x) = \pi - \cot^{-1}x \;\forall\, x \in R$

(v) $\sec^{-1}(-x) = \pi - \sec^{-1}x \;\forall\, x \in (-\infty, -1] \cup [1, \infty)$

(vi) $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1}x \;\forall\, x \in (-\infty, -1] \cup [1, \infty)$

Property-4 :

(i) $\sin^{-1}x + \cos^{-1}x = \dfrac{\pi}{2} \quad \forall\, x \in [-1, 1]$

(ii) $\tan^{-1}x + \cot^{-1}x = \dfrac{\pi}{2} \quad \forall\, x \in R$

(iii) $\sec^{-1}x + \text{cosec}^{-1}x = \dfrac{\pi}{2} \;\forall\, x \in (-\infty, -1] \cup [1, \infty)$

Property-5 :

(i) $\sin^{-1}x = \text{cosec}^{-1}\left(\dfrac{1}{x}\right), x \in [-1, 1] \sim \{0\}$ and $\text{cosec}^{-1}x = \sin^{-1}\left(\dfrac{1}{x}\right), x \in (-\infty, -1] \cup [1, \infty)$

(ii) $\cos^{-1}x = \sec^{-1}\left(\dfrac{1}{x}\right), x \in [-1, 1] \sim \{0\}$ and $\sec^{-1}x = \cos^{-1}\left(\dfrac{1}{x}\right),\; x \in (-\infty, -1] \cup [1, \infty)$

(iii) $\tan^{-1}x = \cot^{-1}\left(\dfrac{1}{x}\right), x \in (0, \infty) = -\pi + \cot^{-1}\left(\dfrac{1}{x}\right), \quad x \in (-\infty, 0)$ and

$\cot^{-1}x = \tan^{-1}\left(\dfrac{1}{x}\right), x \in (0, \infty) = \pi + \tan^{-1}\left(\dfrac{1}{x}\right), \quad x \in (-\infty, 0)$

Property-6 :

(i) $\sin^{-1}x \pm \sin^{-1}y = \sin^{-1}[x\sqrt{1-y^2} \pm y\sqrt{1-x^2}\,]$, if $x, y \ge 0$ and $x^2 + y^2 \le 1$

(ii) $\sin^{-1}x \pm \sin^{-1}y = \pi - \sin^{-1}[x\sqrt{1-y^2} \pm y\sqrt{1-x^2}\,]$, if $x, y \ge 0$ and $x^2 + y^2 > 1$

Property-7 :

(i) $\cos^{-1}x \pm \cos^{-1}y = \cos^{-1}[xy \mp \sqrt{1-x^2}\ \sqrt{1-y^2}\ ]$, if $x, y > 0$ and $x^2 + y^2 \le 1$

(ii) $\cos^{-1}x \pm \cos^{-1}y = \pi - \cos^{-1}[xy \mp \sqrt{1-x^2}\ \sqrt{1-y^2}\ ]$, if $x, y > 0$ and $x^2 + y^2 > 1$

Property-8 :

(i) $$\tan^{-1}x + \tan^{-1}y = \begin{cases} \tan^{-1}\left(\dfrac{x+y}{1-xy}\right); & \text{if } xy<1 \\ \pi + \tan^{-1}\left(\dfrac{x+y}{1-xy}\right); & \text{if } x>0, y>0 \text{ and } xy>1 \\ -\pi + \tan^{-1}\left(\dfrac{x+y}{1-xy}\right) & \text{if } x<0, y<0 \text{ and } xy>1 \end{cases}$$

(ii) $$\tan^{-1}x - \tan^{-1}y = \begin{cases} \tan^{-1}\left(\dfrac{x-y}{1+xy}\right); & \text{if } xy>-1 \\ \pi + \tan^{-1}\left(\dfrac{x-y}{1+xy}\right); & \text{if } x>0, y<0 \text{ and } xy<-1 \\ -\pi + \tan^{-1}\left(\dfrac{x-y}{1+xy}\right) & \text{if } x<0, y>0 \text{ and } xy<-1 \end{cases}$$

Property-9 :

(i) $$2\sin^{-1}x = \begin{cases} \sin^{-1}(2x\sqrt{1-x^2}\ ); & \text{if } -\dfrac{1}{\sqrt{2}} \le x \le \dfrac{1}{\sqrt{2}} \\ \pi - \sin^{-1}(2x\sqrt{1-x^2}\ ); & \text{if } \dfrac{1}{\sqrt{2}} \le x \le 1 \\ -\pi - \sin^{-1}(2x\sqrt{1-x^2}\ ); & \text{if } -1 \le x \le -\dfrac{1}{\sqrt{2}} \end{cases}$$

(ii) $$2\cos^{-1}x = \begin{cases} \cos^{-1}(2x^2-1); & \text{if } 0 \le x \le 1 \\ 2\pi - \cos^{-1}(2x^2-1); & \text{if } -1 \le x \le 0 \end{cases}$$

(iii) $$2\tan^{-1}x = \begin{cases} \tan^{-1}\left(\dfrac{2x}{1-x^2}\right); & \text{if } -1<x<1 \\ \pi + \tan^{-1}\left(\dfrac{2x}{1-x^2}\right); & \text{if } x>1 \\ -\pi + \tan^{-1}\left(\dfrac{2x}{1-x^2}\right); & \text{if } x<-1 \end{cases}$$

Property-10 :

(i) $$3\sin^{-1}x = \begin{cases} \sin^{-1}(3x-4x^3); & \text{if } -\dfrac{1}{2} \le x \le \dfrac{1}{2} \\ \pi - \sin^{-1}(3x-4x^3); & \text{if } \dfrac{1}{2} < x \le 1 \\ -\pi - \sin^{-1}(3x-4x^3); & \text{if } -1 \le x < -\dfrac{1}{2} \end{cases}$$

(ii) $$3\cos^{-1}x = \begin{cases} \cos^{-1}(4x^3-3x); & \text{if } \dfrac{1}{2} \le x \le 1 \\ 2\pi - \cos^{-1}(4x^3-3x); & \text{if } -\dfrac{1}{2} \le x \le \dfrac{1}{2} \\ 2\pi + \cos^{-1}(4x^3-3x) & \text{if } -1 \le x \le -\dfrac{1}{2} \end{cases}$$

(iii) $3\tan^{-1}x = \begin{cases} \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right); & \text{if } -\dfrac{1}{\sqrt{3}} < x < \dfrac{1}{\sqrt{3}} \\ \pi + \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right); & \text{if } x > \dfrac{1}{\sqrt{3}} \\ -\pi + \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right); & \text{if } x < -\dfrac{1}{\sqrt{3}} \end{cases}$

Property-11 :

(i) $2\tan^{-1}x = \begin{cases} \sin^{-1}\left(\dfrac{2x}{1+x^2}\right); & \text{if } -1 \le x \le 1 \\ \pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right); & \text{if } x > 1 \\ -\pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right); & \text{if } x < -1 \end{cases}$

(ii) $2\tan^{-1}x = \begin{cases} \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right); & \text{if } 0 \le x < \infty \\ -\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right); & \text{if } -\infty < x < 0 \end{cases}$

Property-12 :

(i) $\sin^{-1}x = \cos^{-1}\left(\sqrt{1-x^2}\right) = \tan^{-1}\left(\dfrac{x}{\sqrt{1-x^2}}\right) = \cot^{-1}\left(\dfrac{\sqrt{1-x^2}}{x}\right) = \sec^{-1}\left(\dfrac{1}{\sqrt{1-x^2}}\right) = \text{cosec}^{-1}\left(\dfrac{1}{x}\right)$

(ii) $\cos^{-1}x = \sin^{-1}\left(\sqrt{1-x^2}\right) = \tan^{-1}\left(\dfrac{\sqrt{1-x^2}}{x}\right) = \cot^{-1}\left(\dfrac{x}{\sqrt{1-x^2}}\right) = \sec^{-1}\left(\dfrac{1}{x}\right) = \text{cosec}^{-1}\left(\dfrac{1}{\sqrt{1-x^2}}\right)$

(iii) $\tan^{-1}x = \sin^{-1}\left(\dfrac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\dfrac{1}{\sqrt{1+x^2}}\right) = \cot^{-1}\left(\dfrac{1}{x}\right) = \sec^{-1}\left(\sqrt{1+x^2}\right) = \text{cosec}^{-1}\left(\dfrac{\sqrt{1+x^2}}{x}\right)$

5. Important Points to be remembered

(i) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left[\dfrac{x+y+z-xyz}{1-xy-yz-zx}\right]$ if $x > 0$, $y > 0$, $z > 0$ and $(xy + yz + zx) < 1$

(ii) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \dfrac{\pi}{2}$, then $xy + yz + zx = 1$

(iii) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, then $x + y + z = xyz$

(iv) $\tan^{-1}x_1 + \tan^{-1}x_2 + \ldots\ldots\ldots + \tan^{-1}x_n = \tan^{-1}\left[\dfrac{S_1 - S_3 + S_5 - \ldots\ldots\ldots\ldots}{1 - S_2 + S_4 - S_6 + \ldots\ldots}\right]$

Where S_k denotes the sum of the product $x_1, x_2, \ldots\ldots\ldots, x_n$ taken k at a time.

(v) $\tan^{-1}\left[\dfrac{x}{\sqrt{a^2-x^2}}\right] = \sin^{-1}\left(\dfrac{x}{a}\right)$

(vi) $\tan^{-1}\left[\dfrac{3a^2x - x^3}{a(a^2-3x^2)}\right] = 3\tan^{-1}\left(\dfrac{x}{a}\right)$

(vii) $\tan^{-1}\left[\dfrac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right] = \dfrac{\pi}{4} + \dfrac{1}{2}\cos^{-1}x^2$

SHORT TRICKS

Trick -1 Method of substitution

Q.1 If $0 \le x \le \frac{1}{2}$, then the value of $\tan\left(\sin^{-1}\left(\frac{x}{\sqrt{2}}+\frac{\sqrt{1-x^2}}{\sqrt{2}}\right)-\sin^{-1}x\right)$ is :

(A) 0 (B) – 1 (C) 1 (D) $\frac{\pi}{4}$

Sol. [C]

Proper Method	Short Trick
$\tan\left\{\sin^{-1}\left(\frac{x}{\sqrt{2}}+\frac{\sqrt{1-x^2}}{\sqrt{2}}\right)-\sin^{-1}x\right\}$ $=\tan\left\{\sin^{-1}\left(x\sqrt{1-\left(\frac{1}{\sqrt{2}}\right)^2}+\frac{1}{\sqrt{2}}\sqrt{1-x^2}\right)-\sin^{-1}x\right\}$ $=\tan\left\{\left(\sin^{-1}x+\sin^{-1}\frac{1}{\sqrt{2}}\right)-\sin^{-1}x\right\}$ $=\tan\left\{\sin^{-1}\frac{1}{\sqrt{2}}\right\}$ $=\tan\left(\frac{\pi}{4}\right)=1$	$\because 0 \le x \le 1/2$, so let $x = 0$ then $\tan\left(\sin^{-1}\left(0+\frac{1}{\sqrt{2}}\right)-\sin^{-1}0\right)=\tan 45° = 1$

Q.2 If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, then $x^2 + y^2 + z^2 + 2xyz =$

(A) 1 (B) – 1 (C) 0 (D) 2

Sol. [A]

Proper Method	Short Trick
$\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ $\Rightarrow \cos^{-1}x + \cos^{-1}y = \pi - \cos^{-1}z$ $\Rightarrow \cos^{-1}x + \cos^{-1}y = \cos^{-1}(-z)$ $\Rightarrow \cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right] = \cos^{-1}(-z)$ $\Rightarrow xy-\sqrt{1-x^2}\sqrt{1-y^2} = -z$ $\Rightarrow xy + z = \sqrt{1-x^2}\sqrt{1-y^2}$ $\Rightarrow$ square on both side $\Rightarrow (xy + z)^2 = (1 - x^2)(1 - y^2)$ $\Rightarrow x^2y^2 + z^2 + 2xyz = 1 - y^2 - x^2 + x^2y^2$ $\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1$	Let $\cos^{-1}x = \cos^{-1}y = \cos^{-1}z = 60°$ Then $x = y = z = \frac{1}{2}$ $\therefore x^2 + y^2 + z^2 + 2xyz$ $= \frac{1}{4}+\frac{1}{4}+\frac{1}{4}+2.\frac{1}{2}.\frac{1}{2}.\frac{1}{2} = 1$

Q.3 If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$, then which of the following is true?

(A) $x + y + z + xyz = 0$ (B) $x + y + z = xyz$

(C) $xy + yz + zx + 1 = 0$ (D) $xy + yz + zx = 1$

Sol. [D]

Proper Method	**Short Trick**
$\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$ $\Rightarrow \tan^{-1}\left\{\frac{x+y+z-xyz}{1-(xy+yz+zx)}\right\} = \frac{\pi}{2}$ $\Rightarrow \frac{x+y+z-xyz}{1-(xy+yz+zx)} = \infty$ $\Rightarrow 1 -(xy + yz + zx) = 0$ $\Rightarrow xy + yx + zx = 1$	$\therefore$ let $\tan^{-1} x = \tan^{-1} y = \tan^{-1} z = 30°$ $\Rightarrow x = y = z = \frac{1}{\sqrt{3}}$ after putting the values of x, y, z in the option (D) will give the correct answer

Q.4 If $\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$ and if $9x^2 - 12xy\cos\theta + 4y^2 = k^2\sin^2\theta$, then k =

(A) ± 3 (B) ± 1 (C) ± 6 (D) ± 2

Sol. [C]

Proper Method	**Short Trick**
$\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \theta$ $\Rightarrow \cos^{-1}\left(\frac{x}{2}.\frac{y}{3} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}}\right) = \theta$ $\Rightarrow \frac{xy}{6} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}} = \cos\theta$ $\Rightarrow \left(\frac{xy}{6} - \cos\theta\right)^2 = \left(1-\frac{x^2}{4}\right)\left(1-\frac{y^2}{9}\right)$ $\Rightarrow \frac{x^2y^2}{36} + \cos^2\theta - \frac{xy}{3}\cos\theta = 1 - \frac{y^2}{9} - \frac{x^2}{4} + \frac{x^2y^2}{36}$ $\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} - \frac{xy}{3}\cos\theta = 1 - \cos^2\theta$ $\Rightarrow 9x^2 + 4y^2 - 12xy\cos\theta = 36\sin^2\theta$ $\therefore k^2 = 36 \Rightarrow k = \pm 6$	Let x = 1 and y = 0, then $\theta = 150°$ $\Rightarrow 9x^2 - 12xy\cos\theta + 4y^2 = k^2\sin^2\theta$ $\Rightarrow 9 \times 1 - 0 + 0 = k^2\left(\frac{1}{2}\right)^2$ $\Rightarrow k^2 = 36 \Rightarrow k = \pm 6$

Q.5 If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$ and if $x^4 + y^4 + z^4 + 4x^2y^2z^2 = k(x^2y^2 + y^2z^2 + z^2x^2)$ then the value of k is :

(A) 2 (B) 3 (C) 1 (D) 4

Sol. [A]

Proper Method	**Short Trick**
$\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$ $\Rightarrow \sin^{-1}x + \sin^{-1}y = \pi - \sin^{-1}z$ $\Rightarrow \cos^{-1}\sqrt{1-x^2} + \cos^{-1}\sqrt{1-y^2} = \pi - \cos^{-1}\sqrt{1-z^2}$ $\Rightarrow \cos^{-1}\left[\sqrt{1-x^2}\sqrt{1-y^2} - xy\right] = \cos^{-1}(-\sqrt{1-z^2})$ $\Rightarrow \sqrt{1-x^2}\sqrt{1-y^2} - xy = -\sqrt{1-z^2}$	$\because \sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$ $\therefore$ Let $\sin^{-1}x = 0° \Rightarrow x = 0$, $\sin^{-1}y = 90°$ $\Rightarrow y = 1$, $\sin^{-1}z = 90° \Rightarrow z = 1$ Put the values of x, y, z in the given expression then we will get k = 2.

$\Rightarrow \sqrt{1-x^2}\sqrt{1-y^2} = xy - \sqrt{1-z^2}$

Square on both side

$\Rightarrow 1 - y^2 - x^2 + x^2y^2 = x^2y^2 + 1 - z^2 - 2xy\sqrt{1-z^2}$

$\Rightarrow x^2 + y^2 - z^2 = 2xy\sqrt{1-z^2}$

again square

$\Rightarrow x^4 + y^4 + z^4 + 2(x^2y^2 - y^2z^2 - z^2x^2) = 4x^2y^2(1 - z^2)$

$\Rightarrow x^4 + y^4 + z^4 + 4x^2y^2z^2 = 2(x^2y^2 + y^2z^2 + z^2x^2)$

$k = 2$

Q.6 Let $\sin^{-1}a + \sin^{-1}b + \sin^{-1}c = \pi$ then $a\sqrt{(1-a^2)} + b\sqrt{(1-b^2)} + c\sqrt{(1-c^2)} =$

(A) 2abc (B) abc (C) $\frac{1}{2}$abc (D) $\frac{abc}{3}$

Sol. [A]

Proper Method

Let $A = \sin^{-1} a$, $B = \sin^{-1} b$, $C = \sin^{-1} c$, then

$\sin A = a$, $\sin B = b$, $\sin C = c$

Also as given $A + B + C = \pi$

Further we know that if $A + B + C = \pi$, then

$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$ (identity result)

$\Rightarrow 2 \sin A \cos A + 2 \sin B \cos B + 2 \sin C \cos C = 4 \sin A \sin B \sin C$

$\Rightarrow a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2} = 2abc$

Short Trick

Let $\sin^{-1}a = \sin^{-1}b = \sin^{-1}c = 60°$

$\Rightarrow a = b = c = \frac{\sqrt{3}}{2}$

L.H.S. $= a\sqrt{1-a^2} + b\sqrt{1-b^2} + c\sqrt{1-c^2}$

$= \frac{\sqrt{3}}{2}\sqrt{1-\frac{3}{4}} + \frac{\sqrt{3}}{2}\sqrt{1-\frac{3}{4}} + \frac{\sqrt{3}}{2}\sqrt{1-\frac{3}{4}} = \frac{3\sqrt{3}}{4}$

after putting the values of a, b, c in the options (A) will give the R.H.S.

Q.7 If $x^2 + y^2 + z^2 = k^2$, then value of $\tan^{-1}\left(\frac{xy}{zk}\right) + \tan^{-1}\left(\frac{xz}{yk}\right) + \tan^{-1}\left(\frac{zy}{xk}\right)$ is equals to ?

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{2}$ (D) 0

Sol. [A]

Proper Method

$x^2 + y^2 + z^2 = k^2$

$= \tan^{-1}\left(\frac{xy}{zk}\right) + \tan^{-1}\left(\frac{xz}{yk}\right) + \tan^{-1}\left(\frac{zy}{xk}\right)$

$$= \tan^{-1}\left[\frac{\frac{xy}{zk} + \frac{xz}{yk} + \frac{zy}{xk} - \left(\frac{xy}{zk}\right)\left(\frac{xz}{yk}\right)\left(\frac{zy}{xk}\right)}{1 - \left\{\left(\frac{xy}{zk}\right)\left(\frac{xz}{yk}\right) + \left(\frac{xz}{yk}\right)\left(\frac{zy}{xk}\right) + \left(\frac{zy}{xk}\right)\left(\frac{xy}{zk}\right)\right\}}\right]$$

$$= \tan^{-1}\left[\frac{\left(\frac{xy}{zk}\right) + \left(\frac{xz}{yk}\right) + \left(\frac{zy}{xk}\right) - \left(\frac{xy}{zk}\right)\left(\frac{xz}{yk}\right)\left(\frac{zy}{xk}\right)}{1 - \left(\frac{x^2+y^2+z^2}{k^2}\right)}\right]$$

$= \tan^{-1}(\infty)$

$= \frac{\pi}{2}$

Short Trick

Let $x = y = z = 1$, then $k^2 = 3 \Rightarrow k = \sqrt{3}$

$\therefore \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$= 30° + 30° + 30° = 90° = \frac{\pi}{2}$

Q.8 The value of $\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{x}{y}\right)\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\left(\frac{x}{y}\right)\right\}$ is equals to ?

(A) $\frac{x}{y}$ (B) $\frac{y}{x}$ (C) $\frac{2x}{y}$ (D) $\frac{2y}{x}$

Sol. [D]

Proper Method	Short Trick
$\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{x}{y}\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{x}{y}\right\}$ Let $\frac{1}{2}\cos^{-1}\frac{x}{y}=\theta$ $\Rightarrow \cos^{-1}\frac{x}{y}=2\theta$ $\Rightarrow \cos 2\theta=\frac{x}{y}$ $=\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)$ $=\frac{1+\tan\theta}{1-\tan\theta}+\frac{1-\tan\theta}{1+\tan\theta}$ $=\frac{2(1+\tan^2\theta)}{1-\tan^2\theta}$ $=\frac{2}{\cos 2\theta}=\frac{2y}{x}$	Let x = 1 and y = 2 $\left(\because \cos^{-1}\left(\frac{1}{2}\right)=30°\right)$ $\therefore$ L.H.S $=\tan\left\{\frac{\pi}{4}+\frac{1}{2}.30°\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}.30°\right\}$ $=\tan 75°+\tan 15°=\cot 15°+\tan 15°$ $=2+\sqrt{3}+2-\sqrt{3}=4$ Now put x = 1 and y = 2 in options then (D) will give 4 and (D) is right choice.

Q.9 If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$, then $4x^2-4xy\cos\alpha+y^2$ is equal to :

(A) $4\sin^2\alpha$ (B) $-4\sin^2\alpha$ (C) $2\sin 2\alpha$ (D) 4

Sol. [A]

Proper Method	Short Trick
$\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$ $\Rightarrow \cos^{-1}\left\{\frac{xy}{2}-\sqrt{1-x^2}\sqrt{1-\frac{y^2}{4}}\right\}=\alpha$ $\Rightarrow \frac{xy}{2}-\sqrt{1-x^2}\sqrt{1-\frac{y^2}{4}}=\cos\alpha$ $\Rightarrow \left(\frac{xy}{2}-\cos\alpha\right)^2=(1-x^2)\left(1-\frac{y^2}{4}\right)$ $\Rightarrow \frac{x^2y^2}{4}+\cos^2\alpha-xy\cos\alpha=1-\frac{y^2}{4}-x^2+\frac{x^2y^2}{4}$ $\Rightarrow x^2+\frac{y^2}{4}-xy\cos\alpha=1-\cos^2\alpha$ $\Rightarrow 4x^2+y^2-4xy\cos\alpha=4\sin^2\alpha$	Let x = 0 and y = 1 Then $\alpha=\cos^{-1}0-\cos^{-1}\left(\frac{1}{2}\right)$ $=90°-60°=30°$ L.H.S. $=4x^2-4xy\cos\alpha+y^2=0+0+1=1$ Now put the value of α in options then only option (A) will give match with L.H.S.

Q.10 If a, b, c are positive real numbers and

$$\theta = \tan^{-1}\left(\sqrt{\frac{a(a+b+c)}{bc}}\right) + \tan^{-1}\left(\sqrt{\frac{b(a+b+c)}{ca}}\right) + \tan^{-1}\left(\sqrt{\frac{c(a+b+c)}{ab}}\right), \text{ then } \tan\theta =$$

(A) 1 (B) 0 (C) $\frac{a+b+c}{abc}$ (D) $\frac{abc}{a+b+c}$

Sol. [B]

Proper Method	Short Trick
$\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ Let $\frac{a+b+c}{abc} = x^2$ then $\theta = \tan^{-1} ax + \tan^{-1} bx + \tan^{-1} cx$ $\theta = \tan^{-1}\left\{\frac{(ax+bx+cx)-(abcx^3)}{1-(abx^2+bcx^2+cax^2)}\right\}$ $\Rightarrow \tan\theta = \left[\frac{x\left\{(a+b+c)-abc\left(\frac{a+b+c}{abc}\right)\right\}}{1-(abx^2+bcx^2+cax^2)}\right]$ $\Rightarrow \tan\theta = 0$	Put a = b = c = 1, then $\theta = \tan^{-1}(\sqrt{3}) + \tan^{-1}(\sqrt{3}) + \tan^{-1}(\sqrt{3})$ $\Rightarrow \theta = 3\tan^{-1}(\sqrt{3}) = 3.\frac{\pi}{3} = \pi$

Q.11 If $S_n = \tan^{-1}\left(\frac{x}{x^2+1.2}\right) + \tan^{-1}\left(\frac{x}{x^2+2.3}\right) + \tan^{-1}\left(\frac{x}{x^2+3.4}\right)$...... n terms, then S_n =

(A) $\tan^{-1}\left(\frac{n+1}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$ (B) $\tan^{-1}\left(\frac{n-1}{x}\right) + \tan^{-1}\left(\frac{1}{x}\right)$

(C) $\tan^{-1}\left(\frac{n+1}{x}\right) + \tan^{-1}\left(\frac{1}{x}\right)$ (D) $\tan^{-1}\left(\frac{n-1}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$

Sol. [A]

Proper Method	Short Trick
$f(x) = \tan^{-1}\left(\frac{\frac{2}{x}-\frac{1}{x}}{1+\frac{1}{x}.\frac{2}{x}}\right) + \tan^{-1}\left(\frac{\frac{3}{x}-\frac{2}{x}}{1+\frac{3}{x}.\frac{2}{x}}\right)$ $+ \tan^{-1}\left(\frac{\frac{4}{x}-\frac{3}{x}}{1+\frac{4}{x}.\frac{3}{x}}\right)$ ...... $\tan^{-1}\left(\frac{\frac{n+1}{x}-\frac{n}{x}}{1+\frac{n+1}{x}.\frac{n}{x}}\right)$ $f(x) = \tan^{-1}\left(\frac{2}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right) + \tan^{-1}\left(\frac{3}{x}\right)$ $- \tan^{-1}\left(\frac{2}{x}\right)$ ...... $+ \tan^{-1}\left(\frac{n+1}{x}\right) - \tan^{-1}\left(\frac{n}{x}\right)$ $f(x) = \tan^{-1}\left(\frac{n+1}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$	Put n = 1 $\Rightarrow S_1 = \tan^{-1}\left(\frac{x}{x^2+2}\right)$ by option (A) is correct answer

Q.12 The value of $\tan^{-1}\left(\frac{x}{1+2x^2}\right)+\tan^{-1}\left(\frac{x}{1+6x^2}\right)+\tan^{-1}\left(\frac{x}{1+12x^2}\right)+ \ldots\ldots$ n terms is :

(A) $\tan^{-1}(n+1)x - \tan^{-1}x$ (B) $\tan^{-1}(n+1)x + \tan^{-1}x$

(C) $\tan^{-1}(n-1)x - \tan^{-1}x$ (D) $\tan^{-1}(n-1)x + \tan^{-1}x$

Sol. **[A]**

Proper Method	**Short Trick**
$\because T_n = \tan^{-1}\left(\frac{x}{1+n(n+1)x^2}\right)$ $=\tan^{-1}\left(\frac{(n+1)x-nx}{1+(n+1)x.nx}\right) = \tan^{-1}((n+1)x) - \tan^{-1}(nx)$ $\therefore T_n = \tan^{-1}((n+1)x) - \tan^{-1}(nx)$ $T_1 = \tan^{-1}2x - \tan^{-1}x$ $T_2 = \tan^{-1}3x - \tan^{-1}2x$ $T_3 = \tan^{-1}4x - \tan^{-1}3x$ .... $T_n = \tan^{-1}(n+1)x - \tan^{-1}nx$ $\therefore S_n = T_1+T_2+T_3 \ldots\ldots + T_n = \tan^{-1}(n+1)x - \tan^{-1}x$	Put $n = 1 \Rightarrow \tan^{-1}\left(\frac{x}{1+2\times 2}\right)$ by option (A) is correct answer

Trick -2 Combination of method of substitution and balancing

Q.13 The value of $\cos^{-1}\left(\frac{2+3\cos x}{3+2\cos x}\right)$ is :

(A) $\tan^{-1}\left(\frac{1}{\sqrt{5}}\tan\left(\frac{x}{2}\right)\right)$ (B) $\tan^{-1}\left(\frac{1}{2\sqrt{5}}\tan\left(\frac{x}{2}\right)\right)$

(C) $2\tan^{-1}\left(\frac{1}{\sqrt{5}}\tan\left(\frac{x}{2}\right)\right)$ (D) $\tan^{-1}\left(\frac{\sqrt{5}}{2}\tan x\right)$

Sol. **[C]**

Proper Method	**Short Trick**
$y=\cos^{-1}\left(\frac{2+3\cos x}{3+2\cos x}\right)$ $y=\cos^{-1}\left[\frac{2+3\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right)}{3+2\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right)}\right]$ $y=\cos^{-1}\left[\frac{5-\tan^2 x/2}{5+\tan^2 x/2}\right]$ $y=\tan^{-1}\left[\frac{\sqrt{20}\tan x/2}{5-\tan^2 x/2}\right]$ $y=\tan^{-1}\left[\frac{2\left(\frac{1}{\sqrt{5}}\tan x/2\right)}{1-\left(\frac{1}{\sqrt{5}}\tan x/2\right)^2}\right]$ $y=2\tan^{-1}\left[\frac{1}{\sqrt{5}}\tan\frac{x}{2}\right]$	Put $x = 90°$, then L.H.S. $=\cos^{-1}\left(\frac{2+0}{3+0}\right)=\cos^{-1}\left(\frac{2}{3}\right)=\tan^{-1}\left(\frac{\sqrt{5}}{2}\right)$ after putting $x = 90°$ in the options, (C) will given required result.

Q.14 The value of $2\tan^{-1}\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{\theta}{2}\right)$ is :

(A) $\cos^{-1}\left(\frac{a\cos\theta+b}{a+b\cos\theta}\right)$ (B) $\cos^{-1}\left(\frac{a\cos\theta}{a\cos\theta+b}\right)$ (C) $\cos^{-1}\left(\frac{a\cos\theta}{a+b\cos\theta}\right)$ (D) $\cos^{-1}\left(\frac{b\cos\theta}{a\cos\theta+b}\right)$

Sol. [A]

Proper Method

$$2\tan^{-1}\left(\sqrt{\frac{a-b}{a+b}}\tan\frac{\theta}{2}\right)$$

$$=\cos^{-1}\left(\frac{1-\frac{a-b}{a+b}\tan^2\frac{\theta}{2}}{1+\frac{a-b}{a+b}\tan^2\frac{\theta}{2}}\right)$$

$$=\cos^{-1}\left[\frac{(a+b)-(a-b)\tan^2\frac{\theta}{2}}{(a+b)+(a-b)\tan^2\frac{\theta}{2}}\right]$$

$$=\cos^{-1}\left[\frac{b\left(1+\tan^2\frac{\theta}{2}\right)+a\left(1-\tan^2\frac{\theta}{2}\right)}{a\left(1+\tan^2\frac{\theta}{2}\right)+b\left(1-\tan^2\frac{\theta}{2}\right)}\right]$$

divide by $\left(1+\tan^2\frac{\theta}{2}\right)$

$$=\cos^{-1}\left[\frac{b+a\cos\theta}{a+b\cos\theta}\right]$$

Short Trick

Let $a = 2$, $b = 1$ and $\theta = 90°$

then L.H.S. $= 2\tan^{-1}\left(\sqrt{\frac{2-1}{2+1}}\tan\left(\frac{90°}{2}\right)\right)$

$$=2\tan^{-1}\left(\frac{1}{\sqrt{3}}\times 1\right)=2\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=2\times 30°=60°$$

put the values of a, b and θ in the options (A) will give the required result.

Q.15 If $a_1, a_2, a_3, \ldots\ldots a_n$ are in A.P. with common difference d, then

$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\ldots\ldots+\tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]$ is equal to -

(A) $\frac{nd}{1+a_1a_n}$ (B) $\frac{a_n-a_1}{a_n+a_1}$ (C) $\frac{(n-1)d}{a_1+a_n}$ (D) $\frac{(n-1)d}{1+a_1a_n}$

Sol. [D]

Proper Method

If exp. = tan(S), then

$$S=\tan^{-1}\left(\frac{a_2-a_1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{a_3-a_2}{1+a_2a_3}\right)+\ldots+\tan^{-1}\left(\frac{a_n-a_{n-1}}{1+a_{n-1}a_n}\right)$$

$= (\tan^{-1} a_2 - \tan^{-1} a_1) + (\tan^{-1} a_3 - \tan^{-1} a_2) + \ldots + (\tan^{-1} a_n - \tan^{-1} a_{n-1})$

$= -\tan^{-1} a_1 + \tan^{-1} a_n$

$$=\tan^{-1}\left(\frac{a_n-a_1}{1+a_1a_n}\right)=\tan^{-1}\frac{(n-1)d}{1+a_1a_n}$$

$$\therefore \text{Exp.}=\tan\left[\tan^{-1}\frac{(n-1)d}{1+a_1a_n}\right]=\frac{(n-1)d}{1+a_1a_n}$$

Short Trick

Let $n = 2$

$$\tan\left[\tan^{-1}\frac{d}{1+a_1a_2}\right]=\frac{d}{1+a_1a_2}=$$

Hence (D) is correct option

Q.16 If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi/2$, then $x^2 + y^2 + z^2 + 2xyz$ is equal to

(A) 0 (B) 1 (C) – 1 (D) $\frac{1}{2}$

Sol. **[B]**

Proper Method

From given relation

$\sin^{-1} x + \sin^{-1}y = \pi/2 - \sin^{-1}z$

$\Rightarrow \cos^{-1}\sqrt{1-x^2} + \cos^{-1}\sqrt{1-y^2} = \cos^{-1} z$

$\Rightarrow \cos^{-1}\left[\sqrt{1-x^2}\sqrt{1-y^2} - xy\right] = \cos^{-1}z$

$\Rightarrow \sqrt{1-x^2}\sqrt{1-y^2} - xy = z$

$\Rightarrow (1 - x^2)(1 - y^2) = (xy + z)^2$

$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1$

Short Trick

Let $\sin^{-1} x = 0$, $\sin^{-1}y = 30°$, $\sin^{-1}z = 60°$

Then value of expression $= \frac{1}{4} + \frac{3}{4} = 1$

Q.17 $\cot^{-1}\left(\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}\right)$ is equal to

(A) $x/2$ (B) $\pi - x/2$ (C) $\pi - x$ (D) $2\pi - x$

Sol. **[B]**

Proper Method

Exp. $= \cot^{-1}\left[\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}} \cdot \frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}+\sqrt{1+\sin x}}\right]$

$= \cot^{-1}\left(\frac{1+\cos x}{-\sin x}\right)$

$= \cot^{-1}\left(-\frac{2\cos^2 x/2}{2\sin x/2 \cos x/2}\right)$

$= \cot^{-1}(-\cot x/2)$

$= \cot^{-1}[\cot(\pi - x/2)]$

$= \pi - x/2$

Short Trick

Let $x = \frac{\pi}{2}$

$\Rightarrow \cot^{-1}\left(\frac{\sqrt{2}}{-\sqrt{2}}\right) = \cot^{-1}(-1)$

$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

Hence (B) is correct option

Q.18 If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, then $\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx}$ is equal to

(A) xyz (B) $1/xyz$ (C) 1 (D) 0

Sol. **[C]**

Proper Method

$\tan^{-1}x + \tan^{-1}y = \pi - \tan^{-1}z$

$\Rightarrow \tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(-z)$

$\Rightarrow \frac{x+y}{1-xy} = -z$

$\Rightarrow x + y + z = xyz$

$\Rightarrow \frac{1}{yz} + \frac{1}{zx} + \frac{1}{xy} = 1$

Short Trick

Let $\tan^{-1}x = \tan^{-1}y = \tan^{-1}z = \frac{\pi}{3}$

$\therefore x = y = z = \sqrt{3}$

So, $\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

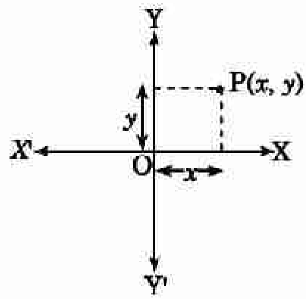
10

Point & Straight Line

KEY CONCEPTS

1. Cartesian Co-ordinates

Let XOX' (X-axis) and YOY' be (Y-axis) be two fixed straight lines, which meet at right angles at O (origin) (0, 0). Then



(i) The ordered pair of real number (x, y) is called cartesian co-ordinates.

(ii) The x-coordinates (or abscissa) is the distance of the point from the Y-axis in a direction parallel to the X-axis (horizontally)

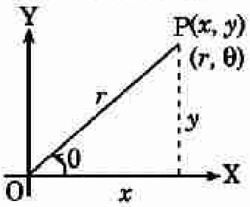
(iii) The y-coordinate (or ordinate) is the distance from the X-axis in a direction parallel to the Y-axis (vertically).

(iv) x-coordinate of a point on X-axis is zero.

(v) y-coordinate of a point on Y-axis is zero.

2. Polar Co-ordinates

The polar co-ordinates of the point are specified as (r, θ).



Where r = distance of point from origin (radius vector)
and θ = angle between the radius vector and x-axis (Vectorial angle)

3. Relation between Cartesian and Polar Coordinates

If (x, y) are the cartesian coordinates of a point P, then

$x = r\cos\theta$; $y = r\sin\theta \Rightarrow r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

4. Distance Formula

(i) The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $PQ = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

(ii) The distance between two points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ is $AB = \sqrt{r_1^2+r_2^2-2r_1r_2\cos(\theta_1-\theta_2)}$

(iii) Distance of a point (x, y) from origin is $\sqrt{x^2+y^2}$.

(iv) If the co-ordinates axes are inclined at an angle α then distance between (x_1, y_1) and (x_2, y_2) is $= \sqrt{(x_1-x_2)^2+(y_1-y_2)^2+2(x_1-x_2)(y_1-y_2)\cos\alpha}$

5. Application of distance formula

(i) Position of three points :

Three given points A, B and C are collinear, when sum of any two distances out of AB, BC, CA is equal to the remaining third otherwise the points will be the vertices of triangle.

(ii) Position of Four Points :

Four given points A, B, C and D are vertices of a

(a) Square if AB = BC = CD = DA and AC = BD

(b) Rhombus if AB = BC = CD = DA and AC $\neq$ BD

(c) Parallelogram if AB = DC; BC = AD; AC $\neq$ BD

(d) Rectangle if AB = CD; BC = DA; AC = BD

Note :

- Diagonal of square, rhombus, rectangle and parallelogram always bisect each other.
- Diagonal of rhombus and square bisect each other at right angle.

6. Section Formula

Coordinates of a point which divides the line segment joining two point $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m_1 : m_2$ are

(i) For internal division $= \left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

(ii) For external division $= \left(\frac{m_1x_2-m_2x_1}{m_1-m_2}, \frac{m_1y_2-m_2y_1}{m_1-m_2}\right)$

(iii) Co-ordinates of mid point of PQ are $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ put $m_1 = m_2$

(iv) Co-ordinates of any point on the line segment which divide the line joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio of $\lambda : 1$ is given by $\left(\frac{x_1+\lambda x_2}{\lambda+1}, \frac{y_1+\lambda y_2}{\lambda+1}\right)$, $(\lambda \neq -1)$

(v) A line $ax + by + c = 0$ divides PQ in the ratio $\lambda : 1$ then $\lambda = -\frac{ax_1+by_1+c}{ax_2+by_2+c}$

(vi) x-axis divides PQ the ratio $= -\frac{y_1}{y_2}$

(vii) y-axis divides PQ the ratio $= -\frac{x_1}{x_2}$

(viii) If P divides AB internally in the ratio $m : n$ & Q divides AB externally in the ratio $m : n$ then P & Q are said to be harmonic conjugate of each other w.r.t. AB.

Mathematically, $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$

i.e. AP, AB & AQ are in H.P.

Note :

➤ If ratio is positive then division will be internal and if ratio is negative then division will be external.

7. Co-ordinates of some particular points

If $A(x_1, y_1)$; $B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of a triangle ABC, then

(i) **Centroid :** The centroid is the point of intersection of medians of the triangle.

Co-ordinates of centroid is $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

Centroid devides the median in the ratio 2 : 1.

(ii) **Incentre :** The incentre is the point of intersection of internal bisectors of the angle of a triangle.

Co-ordinates of **incentre** $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$

(iii) **Circumcentre :** The circumcentre is the point of intersection of perpendicular bisectors of the sides of the triangle. The coordinates of circumcentre O(x, y) are obtained by solving the equations given by $OA^2 = OB^2 = OC^2$

(iv) **Orthocentre :** The orthocenter is the point of intersection of perpendicular drawn from vertices on opposite sides (called altitude) of a triangle.

It can be obtained by solving the equation of any two altitudes.

(v) **Ex-Centers :**

Co-ordiantes of excentre opposite to vertex A is

$$I_1 = \left(\frac{-ax_1 + bx_2 + cx_3}{-a+b+c}, \frac{-ay_1 + by_2 + cy_3}{-a+b+c}\right)$$

Co-ordiantes of excentre opposite to vertex B is

$$I_2 = \left(\frac{ax_1 - bx_2 + cx_3}{a-b+c}, \frac{ay_1 - by_2 + cy_3}{a-b+c}\right),$$

Co-ordiantes of excentre opposite to vertex C is

$$I_3 = \left(\frac{ax_1 + bx_2 - cx_3}{a+b-c}, \frac{ay_1 + by_2 - cy_3}{a+b-c}\right)$$

Important Points to be Remembered

(1) If a triangle is right angle , then its circumcentre is the mid point of hypotenuse.

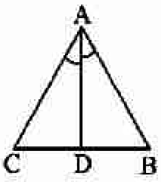
(2) If a triangle is right angle triangle, then ortho centre is the point where right angle is formed.

(3) If the triangle is equilateral, then centroid, incentre, orthocentre, circumcentre coincides.

(4) Orthocentre, centroid and circumcentre are always collinear and centroid divides the line joining orthocentre and circumcentre in the ratio 2 : 1.

(5) In an isosceles triangle centroid, orthocentre,incentre, circumcentre lies on the same line.

(6) Internal angle bisector of a triangle divides the opposite sides in the ratio of remaining sides.



e.g. $\dfrac{BD}{DC} = \dfrac{AB}{AC} = \dfrac{c}{b}$

(7) Incentre divides the angle bisector in the ratio
(b + c) : a, (c + a) : b and (a + b) : c.

8. Area of Triangle

(i) Area of ΔABC with vertices $A(x_1, y_1)$ $B(x_2, y_2)$ and $c(x_3, y_3)$ is.

(a) $\Delta = \left| \dfrac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$

(b) $\Delta = \dfrac{1}{2} \mid [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \mid$

(c) $\Delta = \dfrac{1}{2}\begin{bmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{bmatrix} = \dfrac{1}{2}\,[(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_1 - x_1y_3)] \mid$

(ii) Area of ΔABC with vertices $A(r_1, q_1)$ $B(r_2, q_2)$ and $C(r_3, q_3)$ is

$\Delta = \dfrac{1}{2}[r_1r_2 \sin(\theta_2 - \theta_1) + r_2r_3 \sin(\theta_3 - \theta_2) + r_3r_1 \sin(\theta_1 - \theta_3)]$

(iii) If a be the side of an equilateral triangle than area is. $= \left(\dfrac{a^2\sqrt{3}}{4}\right)$

(iv) If altitude of any equilateral triangle is p, then its area $= \dfrac{p^2}{\sqrt{3}}$.

(v) If the area of triangle joining three points is zero, then the points are collinear.

9. Area of Quadrilateral

If (x_1, y_1), (x_2, y_2), (x_3, y_3) and (x_4, x_4) are vertices of a quadrilateral then its area

$A = \dfrac{1}{2}\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{bmatrix}$

$= \dfrac{1}{2} \mid (x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_4 - x_4y_3) + (x_4y_1 - x_1y_4) \mid$

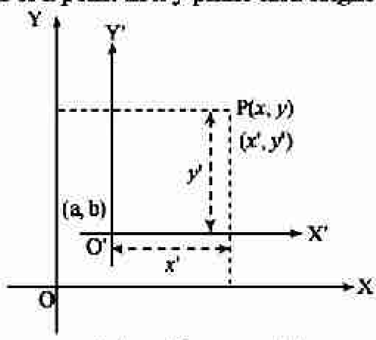
Note :

- If two opposite vertex of rectangle are (x_1, y_1) and (x_2, y_2) and sides are parallel to coordinate axes then its area is $= \mid (y_2 - y_1)(x_2 - x_1) \mid$
- If two opposite vertex of a square are $A(x_1, y_1)$ and $C(x_2, y_2)$ then its area is $= \dfrac{1}{2}AC^2 = \dfrac{1}{2}[(x_2 - x_1)^2 + (y_2 - y_1)^2]$

10. Transformation of Axes

(i) Shift of Origin without Rotation of Axis :

Let P(x, y) are the coordinates of a point in x-y plane and origin is shifted to (a, b) without rotating axis.

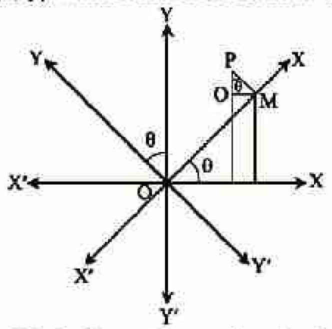


(a) Then the co-ordinates of same points with respect to new axis are $P(x', y') = P(x-a, y-b)$

(b) New equation of curve can be obtained by putting $x + a$ in place of x and $y + b$ in place of y.

(ii) Rotational Transformations without Shifting the Origin :

If coordinates of any point P(x, y) with reference to new axis will be (x', y') then



The relation between (x, y) and (x', y') can be easily obtained with the help of following table

Old / New	$x \downarrow$	$y \downarrow$
$x' \rightarrow$	$\cos\theta$	$\sin\theta$
$y' \rightarrow$	$-\sin\theta$	$\cos\theta$

(iii) Shitting of Origin and Rotation of Axes :

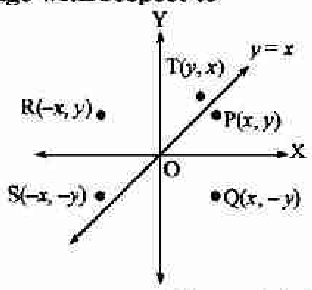
If origin is shifted to point (a, b) and system is also rotated by an angle θ in anti-clockwise, then coordinate of new point P′ (x′, y′) is obtained by replacing.

$x' = a + x\cos\theta + y\sin\theta$

and $y' = b - x\sin\theta + y\cos\theta$

11. Reflection of Point

Let $P(x, y)$ be any point, then its image with respect to



(a) x-axis is $Q(x, -y)$

(b) y-axis is $R(-x, y)$

(c) origin is $S(-x, -y)$

(d) line $y = x$ is $T(y, x)$

12. Locus

It is the path or the curve traced by a moving point satisfying the given condition.

Equation of locus :

The equation to the locus of a point which is satisfied by the coordinates of every point.

Step taken to find the equation of locus

(1) Assumes the coordinate of the point say (h, k) whose locus is to be find.

(2) Write the given condition involving (h, k)

(3) Eliminate the variable(s) if any

(4) Replace $h \longrightarrow x$ and $k \longrightarrow y$

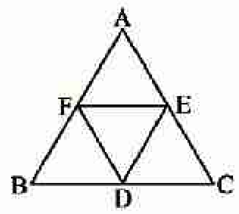
The equation so obtained is the locus of the point which moves under some definite condition.

Important Points to be Remembered

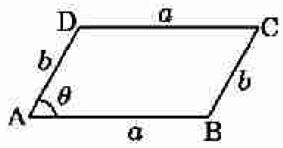
(1) A triangle having vertices $(at_1^2, 2at_1)$, $(at_2^2, 2at_2)$ and $(at_3^2, 2at_3)$, then area is $\Delta = a^2[(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)]$

(2) Area of triangle formed by Co-ordinate axis and the line $ax + by + c = 0$ is $\frac{c^2}{2ab}$

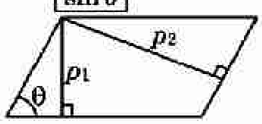
(3) In a triangle ABC, of D, E, F are midpoint of sides AB, BC and CA then $EF = \frac{1}{2}BC$ and $\Delta DEF = \frac{1}{4}(\Delta ABC)$



(4) Area of parallelogram whose sides are a and b and angle between them is θ is given by $\boxed{ab \sin\theta}$



(5) Area of parallelogram whose length of perpendicular from one vertices to the opposite sides are p_1 and p_2 and angle between sides is θ is given by $\boxed{\frac{p_1p_2}{\sin\theta}}$



(6) Area of Rhombus formed by $ax \pm by \pm c = 0$ is $\frac{2c^2}{|ab|}$

(7) To remove the term of xy in the equation $ax^2 + 2hxy + by^2 = 0$, the angle θ through which the axis must be turned (rotated) is given by $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$

(8) The area of the polygon whose vertices are $(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots, (x_n, y_n)$ is

$$A = \frac{1}{2}\begin{bmatrix} x_1 & x_2 & x_3 \cdots\cdots & x_n & x_1 \\ y_1 & y_2 & y_3 \cdots\cdots & y_n & y_1 \end{bmatrix}$$

$$A = \frac{1}{2}\left|(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \ldots + (x_ny_1 - x_1y_n)\right|.$$

13. Straight Line

A straight line is the locus of all those points which are collinear with two given points.
General equation of line is $ax + by + c = 0$

14. Equation of Straight Line Parallel to Axes

(i) Equation of x-axis $\Rightarrow y = 0$

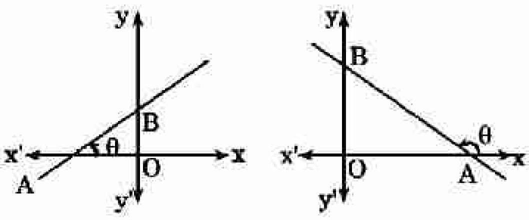
(ii) Equation of a line parallel to x-axes at a distance of b $\Rightarrow y = b$

(iii) Equation y-axis $\Rightarrow x = 0$

(iv) Equation of a line parallel to y-axis and at a distance of a $\Rightarrow x = a$

15. Slope (Gradient) of a Line

If θ is the angle made by a line with the positive direction of x axis in anticlockwise sense, then the slope of the line $\boxed{m = \tan\theta}$



Important Results on Slope of Line

(1) Slope of X-axis or a line parallel to X-axis is $m = 0$.

(2) Slope of Y-axis or a line parallel to Y-axis is $m = \infty$.

(3) Slope of a line equally inclined with axes is 1 or -1 as it makes an angle of 45° or 135°, with X-axis.

(4) The slope of a line joining two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

16. Different Forms of the Equation of Straight Line

(i) **Slope - Intercept form :** The equation of a line with slope m and making an intercept c on y-axis is

$\boxed{y = mx + c}$

(ii) **Slope point form :** The equation of a line with slope m and passing through a point (x_1, y_1) is

$\boxed{y - y_1 = m(x - x_1)}$

(iii) **Two point form :** The equation of a line passing through two given points (x_1, y_1) and (x_2, y_2) is

$$\boxed{y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)}$$

OR

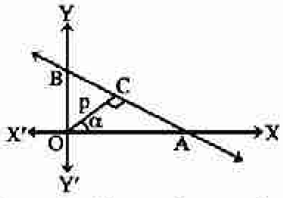
This can also be written in another form. $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

(iv) Intercept Form : The equation of a line which makes intercept a and b on the X-axis and Y-axis respectively is

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

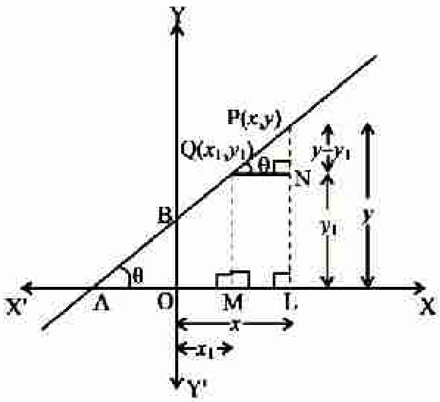
(v) Normal (perpendicular) form : If p is the length of perpendicular on a line from the origin and α is the inclination of perpendicular with x-axis then equation on this line is

$$\boxed{x \cos\alpha + y \sin\alpha = p}$$



(vi) Parametric form (distance form) : If θ is the angle made by straight line with x-axis which is passing through the point (x_1, y_2) and r be the distance of any point (x, y) on the line from the point (x_1, y_1) then its equation.

$$\boxed{\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r}$$



Thus the coordinates of any point on the line at a distance r from the given point (x_1, y_1) are $(x_1 \pm r\cos\theta, y_1 \pm r\sin\theta)$.

17. Position of Point(s) Relative to a Given Line

Let the equation of the given line be ax + by + c = 0 and let the coordinates of the two given points be $P(x_1, y_1)$ and $Q(x_2, y_2)$.

(i) The two points are on the same side of the straight line ax + by + c = 0, if $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have the same sign.

(ii) The two points are on the opposite side of the straight line ax + by + c = 0, if $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have the opposite sign.

(iii) A point (x_1, y_1) will lie on the side of the origin relative to a line ax + by + c = 0, if $ax_1 + by_1 + c$ and c have the same sign.

(iv) A point (x_1, y_1) will lie on the opposite side of the origin relative to a line ax + by + c = 0, if $ax_1 + by_1 + c$ and c have the opposite sign.

18. The angle between two Straight Lines

The angle between two straight line having slopes m_1 and m_2 is $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$

(i) Two lines are parallel if $m_1 = m_2$

(ii) Two lines are perpendicular if , $m_1 m_2 = -1$

(iii) If any one line is parallel to y axis then the angle between two straight line is given by $\tan\theta = \pm\frac{1}{m}$

Where m is the slope of other straight line

(iv) Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are coincident only and only if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

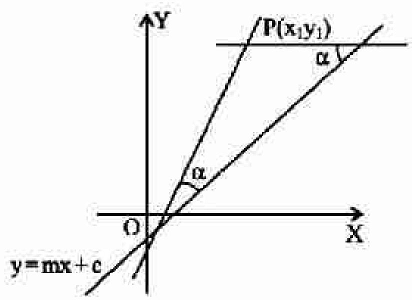
19. Line Parallel and Perpendicular to a Given Line

(i) Equation of a line which is parallel to $ax + by + c = 0$ is $ax + by + k = 0$

(ii) Equation of a line which is perpendicular to $ax + by + c = 0$ is $bx - ay + \lambda = 0$

The value of k and λ in both cases is obtained with the help of additional information given in the problem.

20. Equation of Straight Lines through (x_1, y_1) and making an angle α with $y = mx + c$



$$\boxed{y - y_1 = \frac{m \mp \tan\alpha}{1 \pm m\tan\alpha}(x - x_1)}$$

21. Length of Perpendicular

(i) Length of perpendicular From (x_1, y_1) to the line $ax + by + c = 0$ is given by $p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

(ii) Length of perpendicular from origin on the line $ax + by + c = 0$ is given by $\frac{c}{\sqrt{a^2 + b^2}}$

22. Distance between Two Parallel Lines

(i) The distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is given by $d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

(ii) Distance between two parallel lines $ax + by + c_1 = 0$ and $kax + kby + c_2 = 0$ is given by $\frac{\left|c_1 - \frac{c_2}{k}\right|}{\sqrt{a^2 + b^2}}$

23. Condition of Concurrency

Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are said to be concurrent, if they passes through a same point. The condition for their concurrency is

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

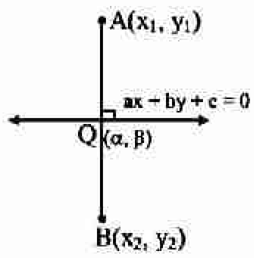
Note :

- If lines are concurrent then $\Delta = 0$, but if $\Delta = 0$ then the lines may or may not be concurrent (lines may be parallel)

24. Family of Lines

Equation of line passing through the point of intersection of the lines
$a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is
$(a_1x + b_1y + c_1) + \lambda (a_2x + b_2y + c_2) = 0$
Value of λ is obtained with the help of the additional information given in the problem.

25. Foot of Perpendicular and Image of a Point w.r.t to Line



(i) Let the foot of perpendicular drawn from the point (x_1, y_1) to the line $ax + by + c = 0$ is $Q(\alpha, \beta)$, then

$$\boxed{\frac{\alpha - x_1}{a} = \frac{\beta - y_1}{b} = -\frac{(ax_1 + by_1 + c)}{a^2 + b^2}}$$

(ii) Let the image of point (x_1, y_1) w.r.t. line $ax + by + c = 0$ is $B(x_2, y_2)$ then

$$\boxed{\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}}$$

26. Bisectors of Angles between Two Lines

Equation of the bisector of angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} \qquad ....(1)$$

(i) Equation of bisector of angle containing origin and not containing origin :
If the equation of the lines are written with constant terms c_1 and c_2 positive, then the equation of the bisectors of the angle containing the origin is obtained by taking positive sign in (1) and not containing origin is obtained by taking negative sign in (1).

(ii) Equation of bisector of acute / obtuse angles :

Rewrite the given equation such that constant terms are positive and the determine the sign of $a_1a_2 + b_2b_2$.

condition	Acuteangle bisector	obtuseangle bisector
$a_1a_2 + b_1b_2 > 0$	$-$	$+$
$a_1a_2 + b_1b_2 < 0$	$+$	$-$

27. General Equation of Second Degree

A general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

(i) represent a pair of two straight lines if $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$.

(ii) If θ is the angle between the pair of straight lines then $\tan\theta = \left|\dfrac{2\sqrt{h^2 - ab}}{a+b}\right|$

(iii) The general equation will represent two perpendicular liens if $a + b = 0$, i.e., coeff of x^2 +coeff of $y^2 = 0$.

(iv) The general equation will represent two parallel lines if $g^2 - ac > 0$ and $\dfrac{a}{h} = \dfrac{h}{b} = \dfrac{g}{f}$ and

Distance between them is $2\sqrt{\dfrac{g^2 - ac}{a(a+b)}}$ or $2\sqrt{\dfrac{f^2 - bc}{b(a+b)}}$

(v) Point of intersection of pair of lines is given by $\left(\dfrac{hf - bg}{ab - h^2}, \dfrac{gh - af}{ab - h^2}\right)$

(vi) The equation of bisector of the angles between the lines is

$$\frac{(x - x_1)^2 - (y - y_1)^2}{a - b} = \frac{(x - x_1)(y - y_1)}{h}$$

Where, (x_1, y_1) is the point of intersection of the lines represented by the given equation.

28. Homogeneous Equation of Second Degree

A homogeneous equation of degree two of the type $ax^2 + 2hxy + by^2 = 0$ always represents a pair of straight lines passing through the origin.

(i) If $y = m_1x$ and $y = m_2x$ be the two equation represented by $ax^2 + 2hxy + by^2 = 0$ then

$$m_1 = \frac{-h + \sqrt{h^2 - ab}}{b} \quad \text{and} \quad m_2 = \frac{-h - \sqrt{h^2 - ab}}{b}$$

$$\therefore \quad m_1 + m_2 = -\frac{2h}{b} \quad \text{and} \quad m_1m_2 = \frac{a}{b}$$

(a) If $h^2 > ab \Rightarrow$ lines are real & distinct.

(b) If $h^2 = ab \Rightarrow$ lines are coincident

(c) If $h^2 < ab \Rightarrow$ lines are imaginary with real point of intersection i.e. (0, 0)

(ii) If θ is the angle between the pair of straight lines then $\tan\theta = \left|\dfrac{2\sqrt{h^2 - ab}}{a+b}\right|$

The condition that these lines are :

(a) At right angles to each other is $a + b = 0$. i.e. coefficient of x^2 + coefficient of $y^2 = 0$.

(b) Coincident is $h^2 = ab$.

(c) Equally inclined to the axis of x is $h = 0$ i.e. coeff. of $xy = 0$.

(iii) The equation of bisector of the angles between the lines represented by the equation, $ax^2 + 2hxy + by^2 = 0$ is

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \quad (a \neq b, h \neq 0)$$

(iv) Pair of straight lines perpendicular to the lines $ax^2 + 2hxy + by^2 = 0$ and through origin are given by $bx^2 - 2hxy + ay^2 = 0$.

(v) Equation of the pair of lines through (α, β) and perpendicular to the pair of lines $ax^2 + 2hxy + by^2 = 0$ is $b(x - \alpha)^2 - 2h(x - \alpha)(y - \beta) + a(y - \beta)^2 = 0$.

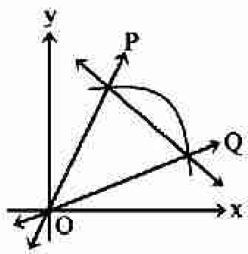
29. Equation of Lines Joining The Points Of Intersection of a Line and a Curve to The Origin

Let the equation of curve be :

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ....(i)

and straight line be

$\ell x + my + n = 0$ ....(ii)



Now joint equation of line OP and OQ joining the origin and points of intersection P and Q can be obtained by making the equation (i) homogeneous with the help of equation of the line. Thus, required equation is given by -

$$ax^2 + 2hxy + by^2 + 2(gx + fy)\left(\frac{\ell x + my}{-n}\right) + c\left(\frac{\ell x + my}{-n}\right)^2 = 0.$$

Important Points to be Remembered

(1) The area of the triangle formed by the lines $y = m_1x + c_1$, $y = m_2x + c_2$ and $y = m_3x + c_3$ is.

$$\Delta = \frac{1}{2}\left|\Sigma \frac{(c_1 - c_2)^2}{m_1 - m_2}\right|$$

(2) Area of the parallelogram formed by the lines

$a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_1x + b_1y + d_1 = 0$

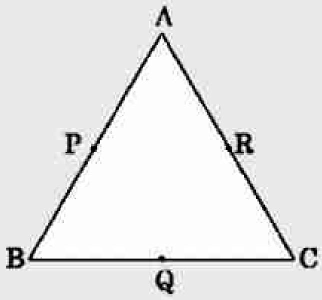
and $a_2x + b_2y + d_2 = 0$ is

$$A = \left|\frac{(d_1 - c_1)(d_2 - c_2)}{a_1b_2 - a_2b_1}\right|$$

(3) The image of the line $a_1x + b_1y + c_1 = 0$ about the line $ax + by + c = 0$ is $2(aa_1 + bb_1)(ax + by + c) = (a^2 + b^2)(a_1x + b_1y + c_1)$.

SHORT TRICKS

Trick -1 In a ΔABC, if P, Q, R are the mid points of sides of triangle, then



$A = P + R - Q$

$B = P + Q - R$

$C = R + Q - P$

Q.1 The mid-points of sides of a triangle are (2, 1), (–1, –3) and (4, 5). Then the coordinates of its vertices are

(A) (7, 9), (–3, –7), (1, 1) (B) (–3, –7), (1, 1), (7, 9)

(C) (1, 1), (2, 3), (–5, 8), (D) (1, 1) (– 7, 10) (– 3, – 7)

Sol. **[A]**

Proper Method

Let the vertices are

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$

then $\dfrac{x_1 + x_2}{2} = 2 \Rightarrow x_1 + x_2 = 4$ .... (1)

$\dfrac{x_2 + x_3}{2} = -1 \Rightarrow x_2 + x_3 = -2$ .... (2)

$\dfrac{x_3 + x_1}{2} = 4 \Rightarrow x_3 + x_1 = 8$ .... (3)

Solving (1) (2) & (3) we get

$x_1 = 7, x_2 = -3, x_3 = 1$

similarly

$\dfrac{y_1 + y_2}{2} = 1 \Rightarrow y_1 + y_2 = 2$ .... (4)

$\dfrac{y_2 + y_3}{2} = -3 \Rightarrow y_2 + y_3 = -6$ .... (5)

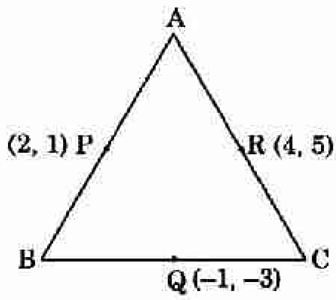
$\dfrac{y_3 + y_1}{2} = 5 \Rightarrow y_3 + y_1 = 10$ .... (6)

Solving (4) (5) & (6) we get

$y_1 = 9, y_2 = -7, y_3 = 1$

so vertices are (7, 9) (– 3, – 7) and (1, 1)

Short Trick



$A = P + R - Q = (7, 9)$

$B = P + Q - R = (-3, -7)$

$C = R + Q - P = (1, 1)$

Q.2 The mid points of the sides of a triangle are (5, 0), (5, 12) and (0, 12) the orthocentre of this triangle is -

(A) (0, 0) (B) (0, 24) (C) (10, 0) (D) $\left(\frac{13}{3}, 8\right)$

Sol. [A]

Proper Method

Let the vertices are (x_1, y_1) (x_2, y_2) (x_3, y_3)

$\therefore \frac{x_1 + x_2}{2} = 5 \Rightarrow x_1 + x_2 = 10$ .... (1)

$\frac{x_2 + x_3}{2} = 5 \Rightarrow x_2 + x_3 = 10$ .... (2)

$\frac{x_3 + x_1}{2} = 0 \Rightarrow x_3 + x_1 = 0$ .... (3)

Solving (1), (2), (3) we get

$x_1 = 0,\ x_2 = 10,\ x_3 = 0$

similarly $\frac{y_1 + y_2}{2} = 0 \Rightarrow y_1 + y_2 = 0$ .... (4)

$\frac{y_2 + y_3}{2} = 12 \Rightarrow y_2 + y_3 = 24$ .... (5)

$\frac{y_3 + y_1}{2} = 12 \Rightarrow y_3 + y_1 = 24$ .... (6)

Solving (4), (5) & (6) we get

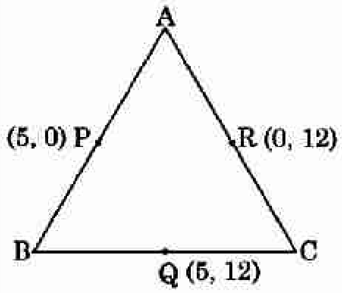
$y_1 = 0,\ y_2 = 0,\ y_3 = 24$

so vertices are (0, 0) (10, 0) (0, 24)

and it is a right angle triangle

$\therefore$ ortho centre is (0, 0)

Short Trick



A = P + R – Q = (0, 0)

B = P + Q – R = (10, 0)

C = Q + R – P = (0, 24)

Triangle is right angle at (0, 0)

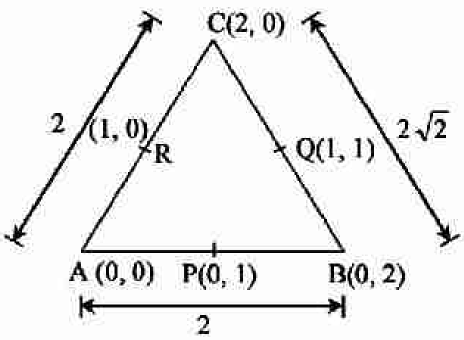
i.e. orthocentre (0, 0)

Q.3 The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1) (1, 1) and (1, 0) is -

(A) $1 + \sqrt{2}$ (B) $1 - \sqrt{2}$ (C) $2 + \sqrt{2}$ (D) $2 - \sqrt{2}$

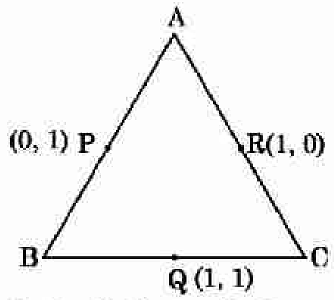
Sol. [D]

Proper Method



$$I \equiv \left(\frac{0(2\sqrt{2}) + 0(2) + 2(2)}{2 + 2 + 2\sqrt{2}}, \frac{0(2\sqrt{2}) + 2(2) + 0(2)}{2 + 2 + 2\sqrt{2}}\right)$$

Short Trick



A = (0, 0), B = (0, 2), C = (2, 0)

x co-ordinate of in centre

$$= \frac{2\sqrt{2}(0) + 2(0) + 2(2)}{2 + 2 + 2\sqrt{2}} = \frac{2}{2 + \sqrt{2}}$$

$$I \equiv \left(\frac{4}{4+2\sqrt{2}}, \frac{4}{4+2\sqrt{2}}\right)$$

$$I = \left(\frac{2}{2+\sqrt{2}}, \frac{2}{2+\sqrt{2}}\right)$$

x-coordinate $= \dfrac{2}{2+\sqrt{2}} \times \dfrac{2-\sqrt{2}}{2-\sqrt{2}} = 2-\sqrt{2}$.

$= 2-\sqrt{2}$

Trick -2 If (x_1, y_1) and (x_2, y_2) are the vertices of an equilateral triangle then, third vertex will be –

$$\left(\frac{x_1+x_2 \pm \sqrt{3}(y_1-y_2)}{2}, \frac{y_1+y_2 \mp \sqrt{3}(x_1-x_2)}{2}\right)$$

Q.4 If the points (4, –4), (–4, 4) and (x, y) form an equilateral triangle, then -

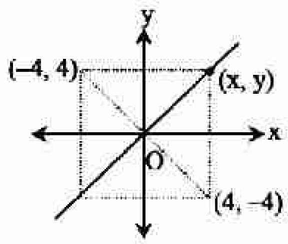
(A) $x = -4\sqrt{3}$, $y = 4\sqrt{3}$ (B) $x = 4\sqrt{3}$, $y = -4\sqrt{3}$

(C) $x = 4\sqrt{3}$, $y = 4\sqrt{3}$ (D) $x = 2\sqrt{3}$, $y = -2\sqrt{3}$

Sol. [C]

Proper Method

$x - y = 0$



Also, $(x-4)^2 + (y+4)^2 = (4+4)^2 + (-4-4)^2$

$\Rightarrow x^2 - 8x + 16 + y^2 + 8y + 16 = 64 + 64$

$\Rightarrow x^2 + y^2 - 8x + 8y = 64 + 32$

Now

$2x^2 = 96$

$x^2 = 48 \Rightarrow x = \pm 4\sqrt{3}$

$(x, y) \equiv (4\sqrt{3}, 4\sqrt{3})$

Short Trick

$$\left(\frac{4-4\pm\sqrt{3}(-4-4)}{2}, \frac{-4+4\mp\sqrt{3}(4+4)}{2}\right)$$

$(\mp 4\sqrt{3}, \mp 4\sqrt{3})$

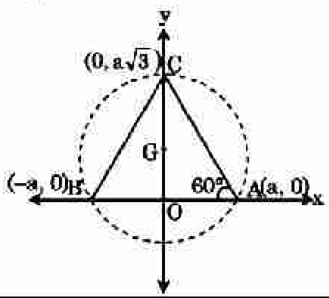
Q.5 If two vertices of an equilateral triangle are A (–a, 0) and B(a, 0), a > 0 and the third vertex C lies above x-axis then the equation of the circumcircle of ΔABC is - **[JEE Main Online-2013]**

(A) $3x^2 + 3y^2 - 2\sqrt{3}\,ay = 3a^2$ (B) $3x^2 + 3y^2 - 2ay = 3a^2$

(C) $x^2 + y^2 - 2ay = a^2$ (D) $x^2 + y^2 - \sqrt{3}\,ay = a^2$

Sol. [A]

Proper Method



Short Trick

Third vertex

$$\left(\frac{-a+a\pm\sqrt{3}(0)}{2}, \frac{0\mp\sqrt{3}(-a-a)}{2}\right)$$

$(0, \pm\sqrt{3}\,a)$

Now $(0, \sqrt{3}\,a)$ satisfy the option (A)

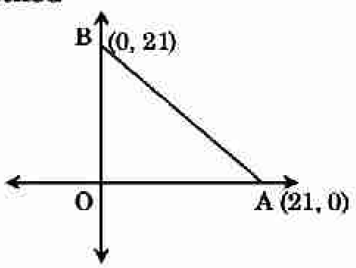
$OC = a \tan 60°$ $\Rightarrow OC = a\sqrt{3}$ triangle is equilateral $\therefore$ circumcentre = centroid $= \left(\frac{-a+a+0}{3}, \frac{0+0+a\sqrt{3}}{3}\right)$ $= \left(0, \frac{a}{\sqrt{3}}\right)$ Circum radius $= \sqrt{0 + \left(\frac{a}{\sqrt{3}} - a\sqrt{3}\right)^2}$ $= \frac{2a}{\sqrt{3}}$ $\therefore$ equation of circle $x^2 + \left(y - \frac{a}{\sqrt{3}}\right)^2 = \frac{4a^2}{3}$ $\Rightarrow x^2 + y^2 - \frac{2ay}{\sqrt{3}} + \frac{a^2}{3} = \frac{4a^2}{3}$ $\Rightarrow x^2 + y^2 - \frac{2ay}{\sqrt{3}} = a^2$ $\Rightarrow 3x^2 + 3y^2 - 2\sqrt{3}\,ay = 3a^2$	

Trick -8 Number of points having both co-ordinates as an integers that lies in the interior of a triangle with vertices (0, 0) (0, n) (n, 0), $n \in I^+$ is $\frac{(n-1)(n-2)}{2}$

Q.6 Number of integral points exactly in the interior of the triangle with vertices (0, 0) (0, 21) and (21, 0) is
(A) 133 (B) 190 (C) 233 (D) 105

Sol. [B]

Proper Method



Equation of line AB is $x + y = 21$
According to condition, points are in interior of triangle
$\therefore x + y < 21$
If we put x = 1 then y = 1, 2, 3 ..... 19 = 19 points
If we put x = 2 then y = 1, 2, 3 ..... 18 = 18 points
....
....
If we put x = 19 then y = 1 = 1 point
Adding all these points we get required no of point
$= 19 + 18 + 17 + ... + 1 = \frac{19}{2}(1 + 19) = 190$

Short Trick

Required no. of points $= \frac{(21-1)(21-2)}{2}$
$= 190$

Q.7 The number of points having both co-ordinates as an integers that lies in the interior of the triangle with vertices (0, 0) (0, 41) and (41, 0) **[JEE main 2015]**

(A) 901 (B) 861 (C) 820 (D) 780

Sol. [D]

Proper Method

Equation of line AB

$x + y = 41$

given that points are in interior of triangle

$\therefore x + y < 41$

If we put $x = 1$ then $y = 1, 2, 3 \ldots\ldots 39 = 39$ points

If we put $x = 2$ then $y = 1, 2, 3 \ldots\ldots 38 = 38$ points

....

....

If we put $x = 39$ then $y = 1$ $= 1$ point

Adding all these points we get required no. of points

$= 39 + 38 + 37 + \ldots\ldots\ldots + 1 = \frac{39}{2}(39 + 1) = 780$

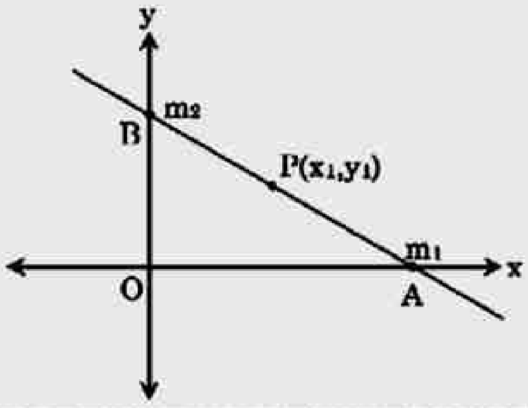
Short Trick

Trick required no. of points

$= \frac{(41-1)(41-2)}{2} = 780$

Trick -4 If the point (x_1, y_1) divides the line between the x-axis and y-axis in the ratio $m_1 : m_2$, then the equation of line is

$$\frac{m_2 x}{x_1} + \frac{m_1 y}{y_1} = m_1 + m_2$$

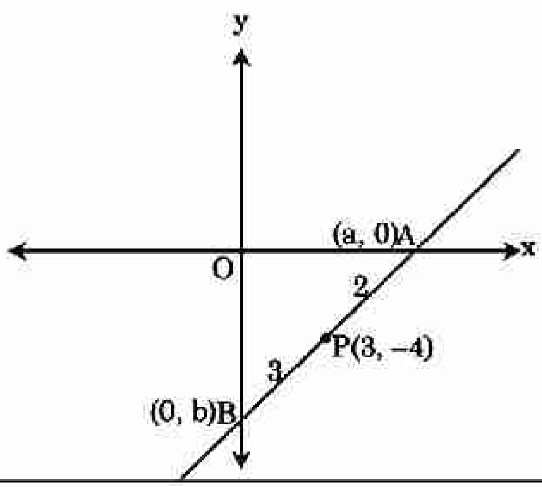


Q.8 If the point (3,–4) divides the line between the x-axis and y-axis in the ratio 2 : 3 then the equation of the line will be -

(A) $2x + y = 10$ (B) $2x - y = 10$ (C) $x + 2y = 10$ (D) $x - 2y = 10$

Sol. [B]

Proper Method



Short Trick

$m_1 : m_2 = 2 : 3$, $(x_1, y_1) = (3, -4)$

$\frac{3x}{3} + \frac{2y}{-4} = 2 + 3$

$\Rightarrow 2x - y = 10$

Let the equation of line $\frac{x}{a} + \frac{y}{b} = 1$

Given that P divides AB in the ratio 2 : 3

$\therefore \frac{3a}{5} = 3$ and $\frac{2b}{5} = -4$

$a = 5$ and $b = -10$

$\therefore$ equation of line $\frac{x}{5} - \frac{y}{10} = 1$

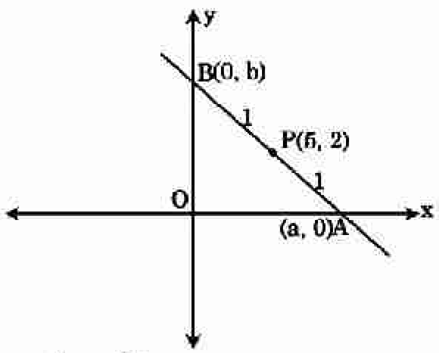
$2x - y = 10$

Q.9 If the point (5, 2) bisects the intercept of a line between the axes, then its equation is-

(A) $5x + 2y = 20$ (B) $2x + 5y = 20$ (C) $5x - 2y = 20$ (D) $2x - 5y = 20$

Sol. **[B]**

Proper Method



Let the equation of line

$\frac{x}{a} + \frac{y}{b} = 1$

given that P divides AB in the ratio 1 : 1

$\therefore \frac{a}{2} = 5, \frac{b}{2} = 2$

$\Rightarrow a = 10$ and $b = 4$

$\therefore$ equation of line the $\frac{x}{10} + \frac{y}{4} = 1$

$2x + 5y = 20$

Short Trick

$m_1 : m_2 = 1 : 1$, $(x_1, y_1) = (5, 2)$

equation $\Rightarrow \frac{x}{5} + \frac{y}{2} = 1 + 1$

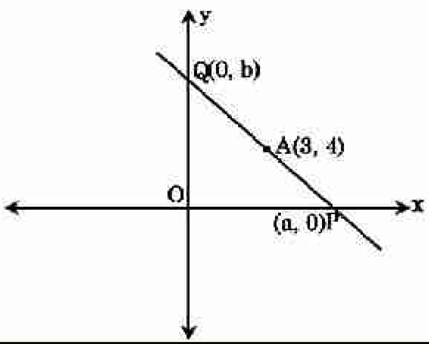
$\Rightarrow 2x + 5y = 20$

Q.10 A straight line through the point A(3, 4) is such that its intercept between the axes is bisected at A. Its equation is – **[AIEEE-2006]**

(A) $3x - 4y + 7 = 0$ (B) $4x + 3y = 24$ (C) $3x + 4y = 25$ (D) $x + y = 7$

Sol. **[B]**

Proper Method



Short Trick

$m_1 : m_2 = 1 : 1$, $(x_1, y_1) = (3, 4)$

$\therefore$ equation $\frac{x}{3} + \frac{y}{4} = 1 + 1$

$\Rightarrow 4x + 3y = 24$

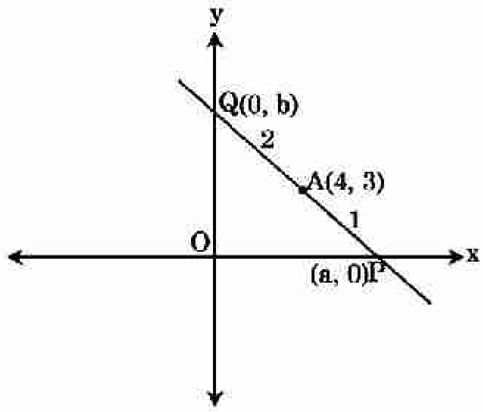
Let the equations of line $\frac{x}{a}+\frac{y}{b}=1$ Point A divides PQ in the ratio 1 : 1 $\therefore\ \frac{a}{2}=3$ and $\frac{b}{2}=4$ $\Rightarrow a = 6$ and $b = 8$ $\therefore$ equation of line $\frac{x}{6}+\frac{y}{8}=1$ $\Rightarrow 4x + 3y = 24$	

Q.11 If a line intercepted between the coordinate axes is trisected at a point A (4, 3), which is nearer to x-axis, then its equation is : **[JEE Main Online-2014]**

(A) $4x - 3y = 7$ (B) $3x + 2y = 18$ (C) $3x + 8y = 36$ (D) $x + 3y = 13$

Sol. **[B]**

Proper Method



Let the equation of line

$\frac{x}{a}+\frac{y}{b}=1$

Point A (4, 3) divides PQ in the ratio 1 : 2

($\because$ A is nearer to x-axis and trisect the line segment between axis)

So $\frac{2a}{3}=4$ and $\frac{b}{3}=3$

$\Rightarrow a = 6$ and $b = 9$

$\therefore$ equation of line $\frac{x}{6}+\frac{y}{9}=1$

$3x + 2y = 18$

Short Trick

$m_1 : m_2 = 1 : 2$, $(x_1, y_1) = (4, 3)$

$\therefore$ equation $\frac{2x}{4}+\frac{y}{3}=1+2$

$\Rightarrow 3x + 2y = 18$

Trick -5 condition of concurrency –

Make the constant term same in given condition and in equation of straight line and then compare

Q.12 The straight lines $ax + by + c = 0$ where $3a + 2b + 4c = 0$ are concurrent at the point

(A) (1/2, 3/4) (B) (3/4, 1/2) (C) (–3/4, –1/2) (D) (–3/4, 1/2)

Sol. [B]

Proper Method	Short Trick
Lin $ax + by + c = 0$ Condition : $3a + 2b + 4c = 0$ Eliminating c, we get $ax + by + \left(\frac{-3a-2b}{4}\right) = 0$ $\Rightarrow a(4x - 3) + b(4y - 2) = 0$ This passes through the intersection of line $4x - 3 = 0$ and $4y - 2 = 0$ i.e. $\left(\frac{3}{4}, \frac{1}{2}\right)$	line $ax + by + c = 0$ ....(1) condition $3a + 2b + 4c = 0$ or $\frac{3}{4}a + \frac{b}{2} + c = 0$ ....(2) comparing we get $\left(x = \frac{3}{4}, y = \frac{1}{2}\right)$

Q.13 A straight line cuts intercepts from the coordinate axes sum of whose reciprocals is 1/p. It passes through a fixed point -

(A) (1/p, p) (B) (p, 1/p) (C) (1/p, 1/p) (D) (p, p)

Sol. [D]

Proper Method	Short Trick
Let the equation $\frac{x}{a} + \frac{y}{b} = 1$ Condition $\frac{1}{a} + \frac{1}{b} = \frac{1}{p} \Rightarrow \frac{p}{a} + \frac{p}{b} = 1$ $\Rightarrow \frac{x}{a} + \frac{y}{b} = \frac{1}{a} - \frac{1}{b}$ $\frac{1}{a}(x - p) + \frac{1}{b}(y - p) = 0$ This passes through the intersecting of lines $x - p = 0$ and $y - p = 0$ $(x = p, y = p)$	line $\frac{x}{a} + \frac{y}{b} = 1$ ...... (1) condition $\frac{1}{a} + \frac{1}{b} = \frac{1}{p}$ or $\frac{p}{a} + \frac{p}{b} = 1$ ...... (2) comparing we get $(x = b, y = b)$

Q.14 If $16a^2 - 40\,ab + 25\,b^2 - c^2 = 0$, then the line $ax + by + c = 0$ passes through the points -

(A) (4, – 5) and (– 4, 5) (B) (5, – 4) and (– 5, 4)
(C) (1, – 1) and (– 1, 1) (D) (3, – 4) and (–3, 4)

Sol. [A]

Proper Method	Short Trick
line : $ax + by + c = 0$ ..... (1) condition : $16a^2 - 40ab + 25b^2 - c^2 = 0$ $\Rightarrow (4a - 5b)^2 - c^2 = 0$ $\Rightarrow (4a - 5b + c)(4a - 5b - c) = 0$ $\Rightarrow 4a - 5b + c = 0$ ..... (2) and $4a - 5b - c = 0$ ..... (3) eliminating c from (1) & (2) $ax + by - 4a + 5b = 0$ $\Rightarrow a(x - 4) + b(y + 5) = 0$ line passes through intersection of $x - 4 = 0$ and $y + 5 = 0$ $(x = 4, y = -5)$ Eliminating c from (1) & (3) $ax + by + 4a - 5b = 0$ $\Rightarrow a(x + 4) + b(y - 5) = 0$ Line passes through intersection of $x + 4 = 0$ & $y - 5 = 0$ $(x = -4, y = 5)$	line $ax + by + c = 0$ .... (1) condition $16a^2 - 40ab + 25\,b^2 - c^2 = 0$ $\Rightarrow (4a - 5b)^2 - c^2 = 0$ $\Rightarrow 4a - 5b + c = 0$ ...... (2) and $4a - 5b - c = 0$ or $-4a + 5b + c = 0$ ...... (3) compare (1) & (2) and (1) & (3) we get (4, – 5), (– 4, 5)

Trick -6 Without changing the direction of co-ordinate axes if origin is shifted to point (α, β) so that the equation $x^2 + y^2 + 2gx + 2fy + c = 0$ is changed to an equation which contains no term of first degree then (α, β) will be the centre of circle.

Q.15 Without changing the direction of coordinates axes, to which point origin should be transferred so that the equation $x^2 + y^2 - 4x + 6y - 7 = 0$ is changed to an equation which contains no term of first degree -

(A) (3, 2) (B) (2, –3) (C) (–2, 3) (D) (–2, –3)

Sol. **[B]**

Proper Method	**Short Trick**
Let the origin be shifted to (α, β) then transformed equation of curve is $(x + \alpha)^2 + (y + \beta)^2 - 4(x + \alpha) + 6(y + \beta) - 7 = 0$ $\Rightarrow x^2 + y^2 + x(2\alpha - 4) + y(2\beta + 6) + \alpha^2 + \beta^2 - 4\alpha + 6\beta - 7 = 0$ No term of first degree then $2\alpha - 4 = 0$ and $2\beta + 6 = 0$ So $(\alpha, \beta) = (2, -3)$	$(\alpha, \beta) = (2, -3)$

Q.16 Without changing the direction of co-ordinate axes, to which point origin should be transferred so that the equation $3x^2 + 3y^2 - 6x + 9y - 8 = 0$ is changed to an equation which contains no term of first degree.

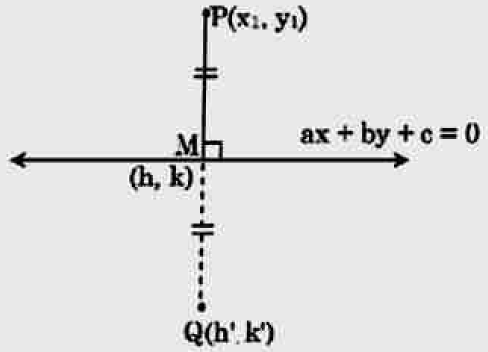
(A) $\left(\frac{3}{2}, 1\right)$ (B) $\left(\frac{-3}{2}, -1\right)$ (C) $\left(1, -\frac{3}{2}\right)$ (D) $\left(-1, -\frac{3}{2}\right)$

Sol. **[C]**

Proper Method	**Short Trick**
Let origin be shifted to (a, b) then transformed equation of curve is $3(x + a)^2 + 3(y + b)^2 - 6(x + a) + 9(y + b) - 8 = 0$ $\Rightarrow 3(x^2 + 2ax + a^2) - 3(y^2 + 2by + b^2) - 6x - 6a - 9y - 9b - 8 = 0$ $\Rightarrow 3(x^2 + y^2) + x(6a - 6) + y(-6b - 9) + 3a^2 - 3b^2 - 6a - 9b - 8 = 0$ Given that equation contains no term of first degree $\therefore 6a - 6 = 0 \Rightarrow a = 1$ and $-6b - 9 = 0 \Rightarrow b = -\frac{3}{2}$ so origin should be shifted to $\left(1, -\frac{3}{2}\right)$	Equation of circle $x^2 + y^2 - 2x + 3y - \frac{8}{3} = 0$ (a, b) ≡ centre of circle) $\equiv \left(1, -\frac{3}{2}\right)$

Trick -7 Foot of perpendicular and reflection of a point with respect to a line

Foot of perpendicular (M) and reflection of P w.r.t. line (Q) can be determined using following formula.



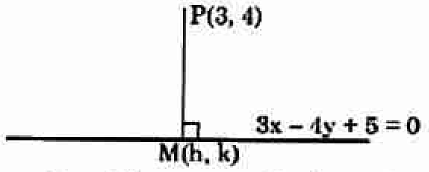
(i) $\dfrac{h-x_1}{a} = \dfrac{k-y_1}{b} = \dfrac{-(ax_1+by_1+c)}{a^2+b^2}$ (ii) $\dfrac{h'-x_1}{a} = \dfrac{k'-y_1}{b} = \dfrac{-2(ax_1+by_1+c)}{a^2+b^2}$

Q.17 The foot of perpendicular from the point (3, 4) on the line $3x - 4y + 5 = 0$ is –

(A) $\left(\dfrac{81}{25}, \dfrac{92}{25}\right)$ (B) $\left(\dfrac{92}{25}, \dfrac{81}{25}\right)$ (C) $\left(\dfrac{46}{25}, \dfrac{54}{25}\right)$ (D) $\left(\dfrac{-81}{25}, \dfrac{92}{25}\right)$

Sol. [A]

Proper Method



Let the equation of line perpendicular to line

$3x - 4y + 5 = 0$ .... (1)

is $4x + 3y + k = 0$

it passes through (3, 4)

$\therefore 12 + 12 + k = 0 \Rightarrow k = -24$

So, equation of line $4x + 3y - 24 = 0$ .... (2)

Solving equation (1) & (2) we get co-ordinates of foot of perpendicular

$M\left(\dfrac{81}{25}, \dfrac{92}{25}\right)$

Short Trick

$\dfrac{h-3}{3} = \dfrac{k-4}{-4} = \dfrac{-(9-16+5)}{9+16}$

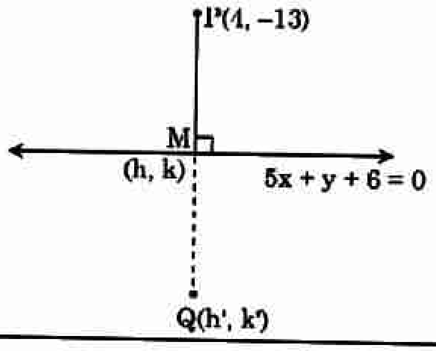
$\left(h = \dfrac{81}{25}, k = \dfrac{92}{25}\right)$

Q.18 The reflection of the point (4, –13) in the line $5x + y + 6 = 0$ is –

(A) (–1, –14) (B) (3, 4) (C) (1, 2) (D) (–4, 13)

Sol. [A]

Proper Method



Short Trick

$\dfrac{h'-x_1}{a} = \dfrac{k'-y_1}{b} = \dfrac{-2(ax_1+by_1+c)}{a^2+b^2}$

$\Rightarrow \dfrac{h^1-4}{5} = \dfrac{k^1+13}{1} = \dfrac{-2(20-13+6)}{25+1}$

$\Rightarrow (h' = -1, k' = -14)$

Let the equation of line perpendicular to line $5x + y + 6 = 0$ .... (1) is $x - 5y + k = 0$ it passes through $(4, -13)$ $\therefore 4 + 65 + k = 0 \Rightarrow k = -69$ So, equation of line $x - 5y - 69 = 0$ .... (2) solving (1) & (2) we get $M(h, k) = \left(\frac{3}{2}, -\frac{27}{2}\right)$ let the reflection of P is $Q \Rightarrow PM = MQ$ M is the mid point of PQ So $Q(-1, -14)$	

Trick -8 Questions based on locus

Q.19 If the equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of c is - **[AIEEE-2003]**

(A) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$ (B) $a_1^2 - a_2^2 + b_1^2 - b_2^2$

(C) $\frac{1}{2}(a_1^2 + a_2^2 + b_1^2 + b_2^2)$ (D) $\frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$

Sol. **[D]**

Proper Method	**Short Trick**
Let the point P be (h, k) and $A(a_1, b_1)$ $B(a_2, b_2)$ given that PA = PB $\Rightarrow (h - a_1)^2 + (k - b_1)^2 = (h - a_2)^2 + (k - b_2)^2$ $\Rightarrow h^2 + k^2 - 2a_1h - 2b_1k + a_1^2 + b_1^2$ $= h^2 + k^2 - 2a_2h - 2b_2k + a_2^2 + b_2^2$ $\Rightarrow (a_1 - a_2)h + (b_1 - b_2)k + \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2) = 0$ $\therefore$ locus $(a_1 - a_2)x + (b_1 - b_2)y + \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2) = 0$ $\therefore c = \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$	Let the point be mid point $\therefore x = \frac{a_1 + a_2}{2}$ and $y = \frac{b_1 + b_2}{2}$ Put these value in locus $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$ We get $c = \frac{1}{2}(a_2^2 + b_2^2 - a_1^2 - b_1^2)$

Q.20 Locus of centroid of the triangle whose vertices are (a cos t, a sin t), (b sin t, – b cos t) and (1, 0), where t is a parameter, is- **[AIEEE 2003]**

(A) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$ (B) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$ (D) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

Sol. **[C]**

Proper Method	**Short Trick**
Let the centroid be (h, k) $\therefore h = \frac{a\cos t + b\sin t + 1}{3}$ and $k = \frac{a\sin t - b\cos t}{3}$ $\Rightarrow a\cos t + b\sin t = 3h - 1$ and $a\sin t - b\cos t = 3k$ Squaring and adding we get $\Rightarrow a^2(\cos^2 t + \sin^2 t) + b^2(\sin^2 t + \cos^2 t) = (3h - 1)^2 + (3k)^2$ $\Rightarrow (3h - 1)^2 + (3k)^2 = a^2 + b^2$ $\therefore$ locus $(3x - 1)^2 + (3y)^2 = a^2 + b^2$	Put t = 0 Now vertices are (a, 0) (0, – b) and (1, 0) so centroid $x = \frac{a+1}{3}$ and $y = \frac{-b}{3}$ put these value of x and y in option so option (C) is correct answer

Q.21 If A(cosα, sinα), B(sinα, –cosα), C(1, 2) are the vertices of a ΔABC, then as α varies, the locus of its centroid is -

(A) $x^2 + y^2 - 2x - 4y + 3 = 0$ (B) $x^2 + y^2 - 2x - 4y + 1 = 0$
(C) $3(x^2 + y^2) - 2x - 4y + 1 = 0$ (D) None of these

Sol. [C]

<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
<tr><td>$h = \dfrac{\cos\alpha + \sin\alpha + 1}{3}$
$\Rightarrow \cos\alpha + \sin\alpha = 3h - 1$
$k = \dfrac{\sin\alpha - \cos\alpha + 2}{3} \Rightarrow \sin\alpha - \cos\alpha = 3k - 2$
Squaring and adding, we get
$2 = (3h-1)^2 + (3k-2)^2$
$\Rightarrow 9h^2 + 9k^2 - 6h - 12k + 3 = 0$
$\Rightarrow 3(x^2 + y^2) - 2x - 4y + 1 = 0$</td><td>Put α = 0
vertices are (1, 0), (0, –1), (1, 2)
So centroid $x = \dfrac{2}{3}$, $y = \dfrac{1}{3}$
Put these value of x and y in option so option (C) is correct answer</td></tr>
</table>

Trick -9 Method of substitution

Q.22 The equation of the straight line passing through the point (4, 3) and making intercepts on the coordinate axes whose sum is – 1 is - **[AIEEE 2004]**

(A) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$ (B) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$ (D) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

Sol. [D]

<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
<tr><td>Let the equation of line $\dfrac{x}{a} + \dfrac{y}{b} = 1$ given that it passes through (4, 3)
$\dfrac{4}{a} + \dfrac{3}{b} = 1$
and $a + b = -1$
$\Rightarrow \dfrac{4}{a} - \dfrac{3}{1+a} = 1$
$\Rightarrow 4 + 4a - 3a = a + a^2$
$\Rightarrow a = \pm 2$
So the equation of lines are
$\dfrac{x}{2} - \dfrac{y}{3} = 1$ and $\dfrac{x}{-2} + \dfrac{y}{1} = 1$</td><td>Line passes through (4, 3) and sum of intercept is –1
So cheking the option only (D) satisfy these conditions</td></tr>
</table>

Q.23 The equation to a line passing through the point (2, –3) and sum of whose intercept on the axes is equal to –2 is -

(A) $x + y + 2 = 0$ or $3x + 3y = 7$ (B) $x + y + 1 = 0$ or $3x - 2y = 12$
(C) $x + y + 3 = 0$ or $3x - 3y = 5$ (D) $x - y + 2 = 0$ or $3x + 2y = 12$

Sol. [B]

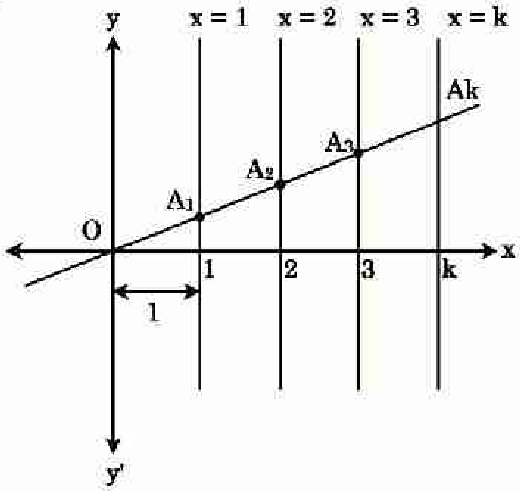
<table>
<tr><th>Proper Method</th><th>Short Trick</th></tr>
<tr><td>$\dfrac{x}{a} + \dfrac{y}{b} = 1$
$a + b = -2$</td><td>Line passes through (2, –3) so cheking the option only (B) satisfy the condition.</td></tr>
</table>

<table>
<tr><td>
Also $\frac{2}{a}-\frac{3}{b}=1$
$\Rightarrow \frac{2}{a}=\frac{3}{b}+1 \Rightarrow \frac{a}{2}=\frac{b}{3+b}$
$\Rightarrow (-b-2)(3+b)=2b$
$\Rightarrow -b^2-3b-2b-6=2b$
$\Rightarrow b^2+7b+6=0$
$b=-1,-6$
then $a=-1, 4$
Equations are $\frac{x}{-1}+\frac{y}{-1}=1 \Rightarrow x+y+1=0$
$\frac{x}{4}-\frac{y}{6}=1 \Rightarrow 3x-2y=12$
</td><td></td></tr>
</table>

Q.24 If a line $x=\frac{y}{\sqrt{5}}$ meets the lines $x=1, x=2 \ldots\ldots x=k$ at points $A_1, A_2 \ldots. A_k$ respectively then $(OA_1)^2+(OA_2)^2+(OA_3)^2 \ldots..+(OA_k)^2=$

(A) $3(k^2+k)$ (B) $2k^3+3k^2+k$ (C) $3(k^3+k^2)+2$ (D) $k^4+2k^3+k^2$

Sol. **[B]**

<table>
<tr><td>
Proper Method


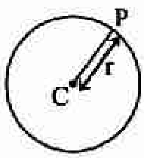
$A_1 \equiv (1, \sqrt{5})$
$A_2 \equiv (2, 2\sqrt{5})$
$A_3 \equiv (3, 3\sqrt{5})$
— — — — —
— — — — —
$A_k \equiv (k, k\sqrt{5})$
Now $(OA_1)^2+(OA_2)^2+\ldots..+(OA_k)^2$
$=6+24+54+\ldots\ldots+6k^2$
$=6(1+4+9+\ldots\ldots+k^2)$
$=6(1^2+2^2+3^2+\ldots..+k^2)$
$=6\,\frac{k(k+1)(2k+1)}{6}$
$=k(k+1)(2k+1)$
$=2k^3+3k^2+k$
</td><td>
Short Trick
Let $k=2$
Here $A_1=(1, \sqrt{5})$
$A_2=(2, 2\sqrt{5})$
So, $(OA_1)^2+(OA_2)^2=6+24=30$
Hence option (B) is correct answer
</td></tr>
</table>

11

Circle

KEY CONCEPTS

A circle is the locus of a point which moves in a plane in such a way that its distance from a fixed point remains constant.



1. General Equation of Circle

The general equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Centre of circle is $(-g, -f)$

i.e. $\left(-\frac{1}{2}\right.$ coefficient of x, $-\frac{1}{2}$ coefficient of y) and radius of circle is $r = \sqrt{g^2 + f^2 - c}$

(i) If $(g^2 + f^2 - c) > 0$, then the circle is a real circle.

(ii) If $(g^2 + f^2 - c) = 0$, then the circle is a point circle .

(iii) If $(g^2 + f^2 - c) < 0$, then the circle is an imaginary circle with real centre.

(iv) The general equation of second degree $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents a circle if

(a) $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0.$

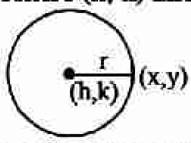
(b) coefficient of x^2 = coefficient of y^2 or $a = b \neq 0$

(c) coefficient of $xy = 0$ or $h = 0$

2. Standard Forms of Circle

(i) Central Form

(a) The equation of a circle having centre (h, k) and radius r, is $(x - h)^2 + (y - k)^2 = r^2$



(b) If the centre is origin then the equation of a circle is $x^2 + y^2 = r^2$

(ii) Diameter Form :

If (x_1, y_1) and (x_2, y_2) are the end points of a diameter of the circle, then the equation of circle is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

Noet :

➢ This will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2)

(iii) The Parametric Equations of a Circle :

(a) The parametric equations of a circle $x^2 + y^2 = r^2$ are $x = r\cos\theta$, $y = r\sin\theta$.

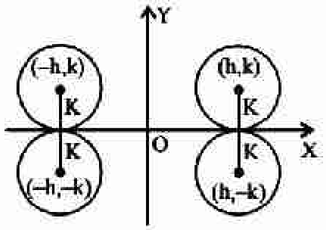
(b) The parametric equations of the circle $(x - h)^2 + (y - k)^2 = r^2$ are $x = h + r\cos\theta$, $y = k + r\sin\theta$.

(c) Parametric equations of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ are $x = -g + \sqrt{g^2 + f^2 - c}\ \cos\theta$, $y = -f + \sqrt{g^2 + f^2 - c}\ \sin\theta$ Where θ is the parameter and $\theta \in [0, 2\pi)$

3. Equation of Circle in Some Special Cases

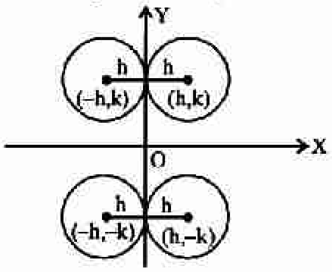
Centre of circle is (h, k)

(i) If the circle touches x axis then its equation is (Four cases)



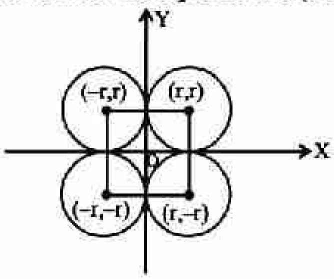
$(x \pm h)^2 + (y \pm k)^2 = k^2$

(ii) If the circle touches y axis then its equation is (Four cases)



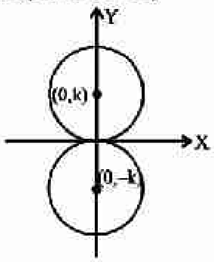
$(x \pm h)^2 + (y \pm k)^2 = h^2$

(iii) If the circle touches both the axis then its equation is (Four cases)



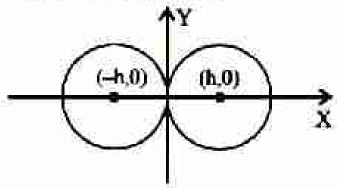
$(x \pm r)^2 + (y \pm r)^2 = r^2$

(iv) If the circle touches x axis at origin (Two cases)



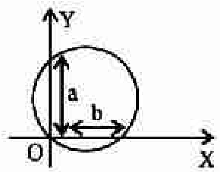
$x^2 + (y \pm k)^2 = k^2 \Rightarrow x^2 + y^2 \pm 2ky = 0$

(v) If the circle touches y axis at origin (Two cases)



$(x \pm h)^2 + y^2 = h^2 \quad \Rightarrow \quad x^2 + y^2 \pm 2xh = 0$

(vi) If the circle passes through origin and cut intercept of a and b on axes, the equation of circle is (Four cases)



$x^2 + y^2 - ax - by = 0$

4. Position of a Point with Respect to a Circle

A point (x_1, y_1) lies outside, on or inside a circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ according as

$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ is positive, zero or negative i.e.

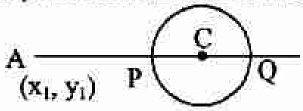
$S_1 > 0 \Rightarrow$ Point is outside the circle.

$S_1 = 0 \Rightarrow$ Point is on the circle.

$S_1 < 0 \Rightarrow$ Point is inside the circle.

5. The Least and Greatest Distance of a Point From a Circle

The greatest & the least distance of a point A from a circle with centre C & radius r is.



$AQ = AC + r$ (greatest distance)

$AP = AC - r$ (least distance)

6. Intercepts made on Coordinate axes by the circle

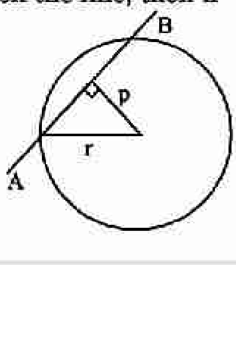
The intercept made by the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ on -

(i) x axis $= 2\sqrt{g^2 - c}$ (ii) y axis $= 2\sqrt{f^2 - c}$

7. Line and Circle

Let L = 0 be a line and S = 0 be a circle, if 'r' be the radius of a circle and p be the length of perpendicular from the centre of circle on the line, then if



$p > r \Rightarrow$ Line is outside the circle
$p = r \Rightarrow$ Line touches the circle (tangent)
$p < r \Rightarrow$ Line is the chord of circle (secant)
$p = 0 \Rightarrow$ Line is diameter of circle

Note :

- Length of the intercept made by the circle on the line is $= 2\sqrt{r^2 - p^2}$
- The length of the intercept made by line $y = mx + c$ with the circle $x^2 + y^2 = a^2$ is $2\sqrt{\dfrac{a^2(1+m^2) - c^2}{1+m^2}}$

8. Condition of Tangency

A line L = 0 touches the circle S = 0, if length of perpendicular drawn from the centre of the circle to the line is equal to radius of the circle i.e. $p = r$.

Circle $x^2 + y^2 = a^2$ will touch the line $y = mx + c$ if $\boxed{c = \pm a\sqrt{1+m^2}}$

9. Equation of Tangent

(i) Point Form (T = 0) :

(a) The equation of tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at a point (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

(b) The equation of tangent to circle $x^2 + y^2 = a^2$ at point (x_1, y_1) is $xx_1 + yy_1 = a^2$

Note :

- To get the equation of tangent at the point (x_1y_1) on any curve we replace xx_1 in place of x^2, yy_1 in place of y^2, $\dfrac{x+x_1}{2}$ in place of x, $\dfrac{y+y_1}{2}$ in place of y, $\dfrac{xy_1+yx_1}{2}$ in place of xy.

(ii) Slope Form :

(a) The equation of tangent of slope m to the circle $x^2 + y^2 = a^2$ are $y = mx \pm a\sqrt{1+m^2}$ and its point of contact is $\left(\dfrac{\mp am}{\sqrt{1+m^2}}, \dfrac{\pm a}{\sqrt{1+m^2}}\right)$.

(b) The equation of tangent of slope m to the circle $(x - h)^2 + (y - k)^2 = a^2$ are $y - k = m(x - h) \pm a\sqrt{1+m^2}$ and its point of contact is $\left(h \pm \dfrac{ma}{\sqrt{1+m^2}}, a \mp \dfrac{a}{\sqrt{1+m^2}}\right)$.

(iii) Parameter Form :

The equation of tangent to the circle $(x - h)^2 + (y - k)^2 = a^2$ at the point $(h + r\cos\theta, k + r\sin\theta)$ is $(x - h)\cos\theta + (y - k)\sin\theta = a$

10. Equation of Normal

(i) Point form :

(a) The equation of normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at any point (x_1, y_1) is $y - y_1 = \dfrac{y_1+f}{x_1+g}(x - x_1)$

(b) The equation of normal to the circle $x^2 + y^2 = a^2$ at any point (x_1, y_1) is $xy_1 - x_1y = 0$

(ii) Slope form :

The equation of a normal of slope m to the circle $x^2+y^2=a^2$ is $my=-x\pm a\sqrt{1+m^2}$.

(iii) Parametric Form :

The equation of normal to the circle $x^2+y^2=a^2$ at the point $(a\cos\theta, a\sin\theta)$ is $\boxed{y = x\tan\theta}$.

Note :

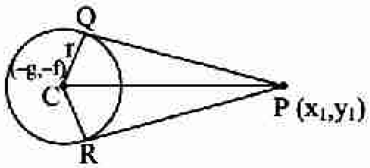
- Normal always passes through the centre of the circle.

11. Length of Tangent

The length of the tangent drawn from a point $P(x_1,y_1)$ to the circle
$S = x^2 + y^2 + 2gx + 2fy + c = 0$ is given by

$PQ = PR = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$

Length of tangent = $\sqrt{S_1}$

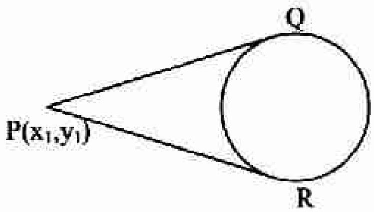


(i) If PQ is a length of the tangent from a point P to a given circle. Then PQ^2 is called the power of the point with respect to a given circle.

(ii) Area of quadrilateral PQCR = $r\sqrt{S_1}$ and angle between tangents PQ and PR is $\theta = 2\tan^{-1}\frac{r}{\sqrt{S_1}}$

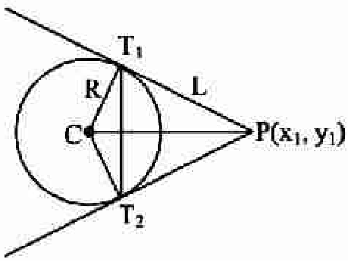
12. Pair of Tangents

From a given point $P(x_1, y_1)$, two tangents PQ and PR can be drawn to the circle
$S = x^2 + y^2 + 2gx + 2fy + c = 0$.
Their combined equation is $SS_1 = T^2$



13. Chord of Contact

A line joining the two points of contacts of two tangents drawn from a point out side the circle, is called chord of contact of that point.
The equation of the chord of contact is T = 0
i.e. $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$



(i) Length of chord of contact

$$T_1T_2 = \frac{2LR}{\sqrt{R^2 + L^2}}$$

(ii) Area of the triangle formed by the pair of the tangents & its chord of contact = $\frac{RL^3}{R^2 + L^2}$, where R is the radius of the circle & L is the length of the tangent from (x_1, y_1) on $S = 0$.

14. Equation of a chord whose middle point is given

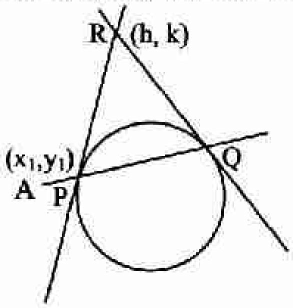
The equation of code of circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ bisected at the point (x_1, y_1) is given by $T = S_1$.
i.e. $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

15. Director Circle

The locus of the point of intersection of two perpendicular tangents to a circle is called the Director circle. Let the circle be $x^2 + y^2 = a^2$, the equation of director circle is $x^2 + y^2 = 2a^2$, director circle is a concentric circle whose radius is $\sqrt{2}$ times the radius of the given circle.

16. Pole and Polar

Let any straight line through the given point $A(x_1, y_1)$ intersect the given circle $S = 0$ in two points P and Q and if the tangent of the circle at P and Q meet at the point R then locus of point R is called polar of the point A and point A is called the pole, with respect to the given circle.



(i) The equation of the polar is the $T = 0$, so the polar of point (x_1, y_1) w.r.t. circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

(ii) Pole of polar $Ax + By + C = 0$ with respect to circle $x^2 + y^2 = a^2$ is $\left(-\frac{Aa^2}{C}, -\frac{Ba^2}{C}\right)$

(iii) Conjugate Points : Two points A and B are conjugate points with respect to given circle, if each lies on the polar of the other with respect to the circle.

(iv) Conjugate Lines : If two lines be such that the pole of one lies on the other, then they are called conjugate lines with respect to the given circle.

17. Family of Circles

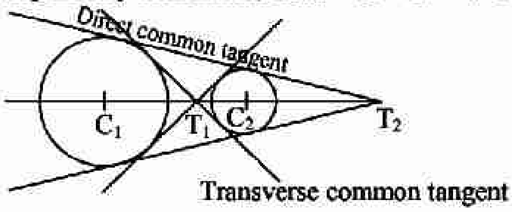
(i) The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is $S_1 + KS_2 = 0$ $(K \neq -1)$

(ii) The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + KL = 0$.

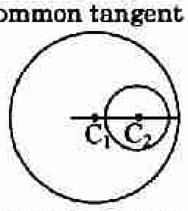
18. Relative position of two circles and No. of common Tangents

Let C_1 (h_1, k_1) and C_2 (h_2, k_2) be the centre of two circle and r_1, r_2 be their radius and C_1C_2 is the distance between their centres then

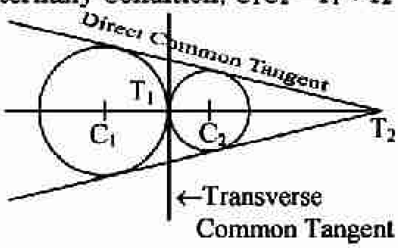
(i) When two circles are separately Condition, $C_1C_2 > r_1 + r_2$ $\Rightarrow$ 4 common tangents



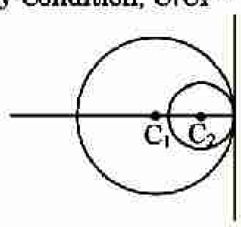
(ii) When one circle completely lie inside the other circle
Condition, $C_1C_2 < |r_1 - r_2|$ $\Rightarrow$ 0 common tangent



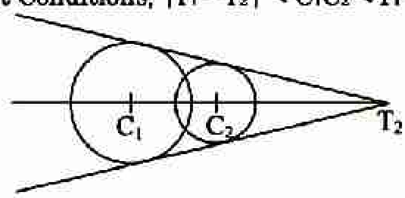
(iii) When two circles touch externally Condition, $C_1C_2 = r_1 + r_2$ $\Rightarrow$ 3 common tangents



(iv) When two circles touch internally Condition, $C_1C_2 = |r_1 - r_2|$ $\Rightarrow$ 1 common tangent



(v) When two circles intersect Conditions, $|r_1 - r_2| < C_1C_2 < r_1 + r_2$ $\Rightarrow$ 2 common tangents



(vi) **Points of intersection of common tangents:** The points T_1 and T_2 (points of intersection of transverse and direct common tangents) divides C_1C_2 internally and externally in the ratio $r_1 : r_2$.

(vii) **Equation of the common tangents at point of contact :** $S_1 - S_2 = 0$.

(viii) **Point of contact :** The point of contact divides C_1C_2 in the ratio $r_1 : r_2$ internally or externally as the case may be.

(ix) **Length of direct common tangent**

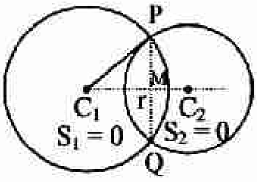
$$= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$$

(x) **Length of transverse common tangent**

$$= \sqrt{(C_1C_2)^2 - (r_1 + r_2)^2}$$

19. Equation of common chord

The chord joining the points of intersection of two given circles is known as **Common chord**.
The equation of the common chord of two circles $S_1 = 0$ and $S_2 = 0$ is $S_1 - S_2 = 0$

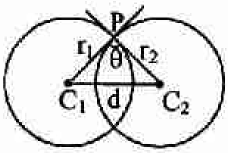


and the length of the common chord $PQ = 2\ PM = 2\ \sqrt{C_1P^2 - C_1M^2}$

Where C_1P = radius of the circle $S_1 = 0$ and C_1M = length of the $\perp$ from centre C_1 to the common chord PQ.

20. The Angle of Intersection of two Circles :

The angle between the tangents of two circles at the point of intersection of the two circles is called angle of intersection of two circles



$$\cos\theta = \left(\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right)$$

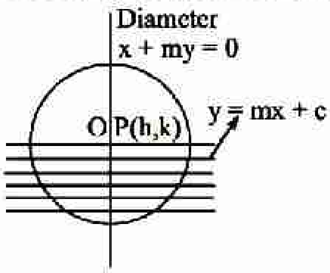
Here r_1 and r_2 are the radii of the circles and d is the distance between their centres.

20. Orthogonal Circles

Two circles are to be intersect orthogonally, if their angle of intersection is a right angle, if two circles
$S_1 = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and
$S_2 = x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are orthogonal, then
$2g_1g_2 + 2f_1f_2 = c_1 + c_2$

21. Diameter of a Circle

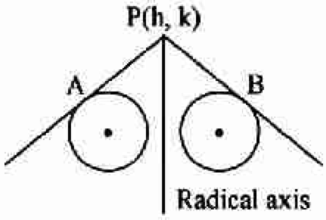
The locus of middle points of a system of parallel chords of a circle is called the diameter of a circle.



The diameter of the circle $x^2 + y^2 = r^2$ corresponding to the system of parallel chord $y = mx + c$ is $x + my = 0$

22. Radical Axis

(i) **Definition :** The locus of a point, which moves in such a way that the length of tangents drawn from it to the circles are equal and is called the radical axis.



(ii) $S_1 = 0$ and $S_2 = 0$ are two circles then equation of radical axes is $S_1 - S_2 = 0$

(iii) If circles touch each other then Radical axis is the common tangent to both the circles.

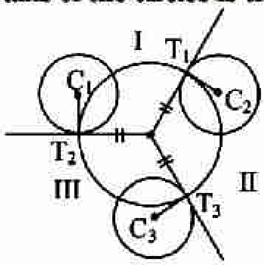
(iv) When the two circles intersect on real point then common chord is the Radical axis of the two circles.

(v) The Radical axis of the two circles is perpendicular to the line joining the centre of two circles but not always pass through mid point of it.

(vi) If two circles are orthogonal to the third circle then Radical axis of both circle passes through the centre of the third circle.

23. Radical centre

The radical centre of three circles is the point from which length of tangents on three circles are equal i.e. the point of intersection of radical axis of the circles is the radical centre of the circles.



(i) If $S_1 = 0$, $S_2 = 0$ and $S_3 = 0$ be any three given circles then the radical centre can be obtained by solving any two of the following equations
$S_1 - S_2 = 0$, $S_2 - S_3 = 0$, $S_3 - S_1 = 0$.

(ii) The circle with centre as radical centre and radius equal to the length of tangent from radical centre to any of the circle, will cut the three circles orthogonally.

24. Co-axial System of Circles

A system of circle is said to be coaxial system of circles, if every pair of the circles in the system has same radical axis.

(i) The equation of a system of coaxial circles, when the equation of the radical axis $P \equiv \ell x + my + n = 0$ and one of the circle of the system $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, is $S + \lambda P = 0$. where λ is an arbitrary constant.

(ii) Since, the lines joining the centres of two circles is perpendicular to their radical axis. Therefore, the centres of all circles of a coaxial system lie on a straight line, which is perpendicular to the common radical axis.

25. Limiting Points

Limiting points of a system of coaxial circles are the centres of the point circles belonging to the family.

Let equation of circle b $x^2 + y^2 + 2gx + c = 0$

$\therefore$ Radius of circle = $\sqrt{g^2 - c}$

For limiting point, r = 0 $\quad \therefore \sqrt{g^2 - c} = 0 \Rightarrow g = \pm\sqrt{c}$

Thus, limiting points, of the given coaxial system as $(\sqrt{c}, 0)$ and $(-\sqrt{c}, 0)$.

SHORT TRICKS

Trick -1 The x co-ordinates of two points A and B are roots of equation $px^2 + qx + r = 0$ and y co-ordinates are roots of equation $ay^2 + by + c = 0$, then to find the equation of circle with AB as a diameter in both the quadratic equation first makes the coefficient of x^2 and y^2 equal to one and then add.

$$\left(x^2 + \frac{q}{p}x + \frac{r}{p}\right) + \left(y^2 + \frac{b}{a}y + \frac{c}{a}\right) = 0$$

$$\Rightarrow x^2 + y^2 + \frac{q}{p}x + \frac{b}{a}y + \frac{r}{p} + \frac{c}{a} = 0$$

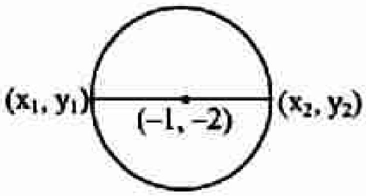
Q.1 The x coordinates of two points A and B are roots of equation $x^2 + 2x - a^2 = 0$ and y coordinate are roots of equation $y^2 + 4y - b^2 = 0$ then equation of the circle which has diameter AB is-

(A) $(x - 1)^2 + (y - 2)^2 = 5 + a^2 + b^2$ (B) $(x + 1)^2 + (y + 2)^2 = \sqrt{(5 + a^2 + b^2)}$

(C) $(x + 1)^2 + (y + 2)^2 = (a^2 + b^2)$ (D) $(x + 1)^2 + (y + 2)^2 = 5 + a^2 + b^2$

Sol. [D]

Proper Method



$x_1 + x_2 = -2$

$x_1x_2 = -a^2$

$y_1 + y_2 = -4$

$y_1y_2 = -b^2$

Equation of circle

$(x + 1)^2 + (y + 2)^2 = [(x_1 - x_2)^2 + (y_1 - y_2)^2]/4$

$= [(x_1 + x_2)^2 - 4x_1x_2 + (y_1 + y_2)^2 - 4y_1y_2]/4$

$= [4 + 4a^2 + 16 + 4b^2]/4$

$= 4[5 + a^2 + b^2]/4$

$= 5 + a^2 + b^2$

Short Trick

Equation of circle

$(x^2 + 2x - a^2) + (y^2 + 4y - b^2) = 0$

$(x + 1)^2 + (y + 2)^2 = 5 + a^2 + b^2$

Q.2 The abscissa of two points A and B are the roots of the equation $x^2 + 2ax - b^2 = 0$ and their ordinates are the roots of the equation $y^2 + 2py - q^2 = 0$. The radius of the circle with AB as a diameter will be -

(A) $\sqrt{a^2 + b^2 + p^2 + q^2}$ (B) $\sqrt{b^2 + q^2}$ (C) $\sqrt{a^2 + b^2 - p^2 - q^2}$ (D) $\sqrt{a^2 + p^2}$

Sol. [A]

Proper Method

Let $A \equiv (\alpha, \beta)$; $B \equiv (\gamma, \delta)$. Then

$\alpha + \gamma = -2a$, $\alpha\gamma = -b^2$

and $\beta + \delta = -2p$, $\beta\delta = -q^2$

Now equation of the required circle is

$(x - \alpha)(x - \gamma) + (y - \beta)(y - \delta) = 0$

$\Rightarrow x^2 + y^2 - (\alpha + \gamma)x - (\beta + \delta) + \alpha\gamma + \beta\delta = 0$

$\Rightarrow x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$

Its radius $= \sqrt{a^2 + b^2 + p^2 + q^2}$

Short Trick

Equation of circle

$x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$

$C(-a, -p)$, $r = \sqrt{a^2 + p^2 + b^2 + q^2}$

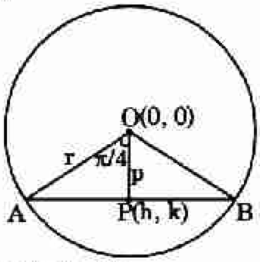
Trick -2 The locus of mid point of chord of circle $(x - h)^2 + (y - k)^2 = r^2$ which subtends an angle θ at centre of circle is $(x - h)^2 + (y - k)^2 = r^2 \cos^2\dfrac{\theta}{2}$

Q.3 Find the locus of mid point of chords of circle $x^2 + y^2 = 25$ which subtends right angle at origin-

(A) $x^2 + y^2 = 25/4$ (B) $x^2 + y^2 = 5$ (C) $x^2 + y^2 = 25/2$ (D) $x^2 + y^2 = 5/2$

Sol. [C]

Proper Method



Let the mid point be (α, β)

In ΔOAP

$\cos 45° = \dfrac{p}{r}$

$\Rightarrow \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{h^2 + k^2}}{5}$

$\Rightarrow h^2 + k^2 = \dfrac{25}{2}$

$\therefore$ locus $x^2 + y^2 = \dfrac{25}{2}$

Short Trick

$x^2 + y^2 = 25 \cos^2 45°$

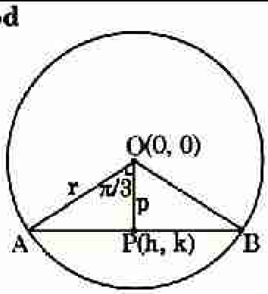
$x^2 + y^2 = \dfrac{25}{2}$

Q.4 Let C be the circle with centre (0, 0) and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\dfrac{2\pi}{3}$ at its centre is – **[AIEEE-2006]**

(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = \dfrac{27}{4}$ (C) $x^2 + y^2 = \dfrac{9}{4}$ (D) $x^2 + y^2 = \dfrac{3}{2}$

Sol. [C]

Proper Method



In ΔOAP

$\cos\dfrac{\pi}{3} = \dfrac{p}{r}$

$\dfrac{1}{2} = \dfrac{\sqrt{h^2 + k^2}}{3}$

$\Rightarrow h^2 + k^2 = \dfrac{9}{4}$

$\therefore$ locus $x^2 + y^2 = \dfrac{9}{4}$

Short Trick

$x^2 + y^2 = 9 \cos^2\dfrac{\pi}{3}$

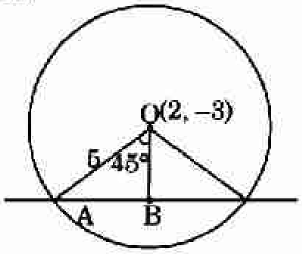
$x^2 + y^2 = \dfrac{9}{4}$

Q.5 Find the locus of mid point of chord of circle $x^2 + y^2 - 4x + 6y - 12$ which subtends a right angle at centre of the circle.

(A) $(x-2)^2 + (y+3)^2 = \frac{25}{2}$
(B) $(x+2)^2 + (y-3)^2 = \frac{25}{2}$
(C) $(x-2)^2 + (y-3)^2 = \frac{25}{2}$
(D) $(x+2)^2 + (y+3)^2 = \frac{25}{2}$

Sol. [A]

Proper Method



Let the mid point of chord is (h, k).

In ΔOAB $\cos 45° = \frac{OB}{5}$

$\Rightarrow OB = 5\cos 45°$

$\Rightarrow \sqrt{(h-2)^2 + (k+3)^2} = 5\frac{1}{\sqrt{2}}$

$\Rightarrow (h-2)^2 + (k+3)^2 = \frac{25}{2}$

$\therefore$ locus $(x-2)^2 + (y+3)^2 = \frac{25}{2}$

Short Trick

$(x-h)^2 + (y-k)^2 = r^2\cos^2\frac{\theta}{2}$

$\Rightarrow (x-2)^2 + (y+3)^2 = \frac{25}{2}$

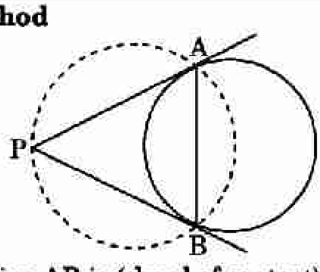
Trick -3 Method of substitution

Q.6 Tangents drawn from the point P(1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is - **[IIT-2009]**

(A) $x^2 + y^2 + 4x - 6y + 19 = 0$
(B) $x^2 + y^2 - 4x - 10y + 19 = 0$
(C) $x^2 + y^2 - 2x + 6y - 29 = 0$
(D) $x^2 + y^2 - 6x - 4y + 19 = 0$

Sol. [B]

Proper Method



Equation of line AB is (chord of contact)

$x + 8y - 3(x+1) - 2(y+8) - 11 = 0$

$\Rightarrow -2x + 6y - 30 = 0$

$\Rightarrow x - 3y + 15 = 0$

Equation of circle passes through A & B is given by

$(x^2 + y^2 - 6x - 4y - 11) + \lambda(x - 3y + 15) = 0$

As, (3, 2) will lie on it $-24 + 12\lambda = 0$

$\lambda = 2$

Hence equation is $x^2 + y^2 - 4x - 10y + 19 = 0$

Short Trick

As the circumcircle passes through the point (1, 8), so this point will satisfy equation of circle. Hence (B) is correct option.

Q.7 The equations of the tangents drawn from the point (0, 1) to the circle $x^2 + y^2 - 2x + 4y = 0$ are-

(A) $2x - y + 1 = 0,\ x + 2y - 2 = 0$
(B) $2x - y - 1 = 0,\ x + 2y - 2 = 0$
(C) $2x - y + 1 = 0,\ x + 2y + 2 = 0$
(D) $2x - y - 1 = 0,\ x + 2y + 2 = 0$

Sol. [A]

Proper Method	**Short Trick**
$T^2 = SS_1$ $(y - x + 2(y + 1))^2 = (x^2 + y^2 - 2x + 4y)\ (5)$ $\Rightarrow (3y - x + 2)^2 = 5x^2 + 5y^2 - 10x + 20y$ $\Rightarrow 9y^2 + x^2 + 4 - 6xy - 4x + 12y$ $= 5x^2 + 5y^2 - 10x + 20y$ $\Rightarrow 4x^2 - 4y^2 + 6xy - 6x + 8y - 4 = 0$ $\Rightarrow 2x^2 - 2y^2 + 3xy - 3x + 4y - 2 = 0$ $\Rightarrow (2x - y + 1)\ (x + 2y - 2) = 0$	(0, 1) satisfy both the tangents Hence (A) is correct option.

Q.8 The equation of the normal to the circle $x^2 + y^2 - 8x - 2y + 12 = 0$ at the points whose ordinate is -1, will be-

(A) $2x - y - 7 = 0,\ 2x + y - 9 = 0$
(B) $2x + y - 7 = 0,\ 2x + y + 9 = 0$
(C) $2x + y + 7 = 0,\ 2x + y + 9 = 0$
(D) $2x - y + 7 = 0,\ 2x - y + 9 = 0$

Sol. [A]

Proper Method	**Short Trick**
As ordinate is -1 $x^2 - 8x + 15 = 0 \Rightarrow x = 3, 5$ that is points are $(3, -1)$ & $(5, -1)$ At (3, 1), equation of normal $y + 1 = \left(\frac{-2}{-1}\right)(x - 3)$ $\Rightarrow 2x - y - 7 = 0$ At $(5, -1)$ $\quad y + 1 = \left(\frac{-2}{1}\right)(x - 5)$ $\Rightarrow 2x + y = 9$	Normal of the circle passes through centre of circle Hence, centre (4, 1) satisfy the option (A)

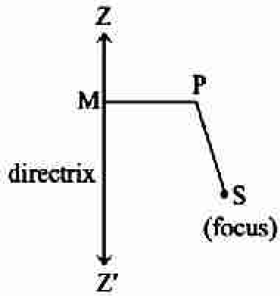
12

Conic Sections

KEY CONCEPTS

Parabola

A conic is the locus of a point whose distance from a fixed point bears a constant ratio to its distance from a fixed line. The fixed point is the focus S and the fixed line is directrix.



The constant ratio is called "eccentricity" and denoted by "e".

$$\frac{PS}{PM} = e$$

(i) If e = 1, conic section is parabola

(ii) e > 1, conic section is Hyperbola

(iii) e < 1, conic section is Ellipse

A_1A_2 = Latus rectum = 4 (OS) = 2(SS')

1. General Equation of Conic

A second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents.

(i) Pair of straight lines, if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

(ii) Circle, if $a = b$, $h = 0$

(iii) Parabola, if $h^2 = ab$ and $\Delta \neq 0$

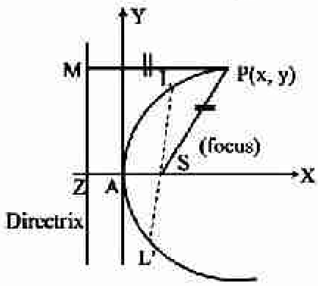
(iv) Ellipse, if $h^2 < ab$ and $\Delta \neq 0$

(v) Hyperbola, if $h^2 > ab$ and $\Delta \neq 0$

(vi) Rectangular hyperbola, if $a + b = 0$ and $\Delta \neq 0$

2. Parabola

A parabola is the locus of a point which moves in such a way that its distance from a fixed point (focus) is equal to its perpendicular distance from a fixed straight line (directrix).



3. Terms Related to Parabola

(i) Axis : A straight line passes through the focus and perpendicular to the directrix is called the axis of parabola.

(ii) Vertex : The point of intersection of a parabola and its axis is called the vertex of the Parabola.

Note :

➤ The vertex is the middle point of the focus and the point of intersection of axis and directrix.

(iii) Focal Length (Focal distance) :

The distance of any point P (x, y) on the parabola from the focus is called the focal length.

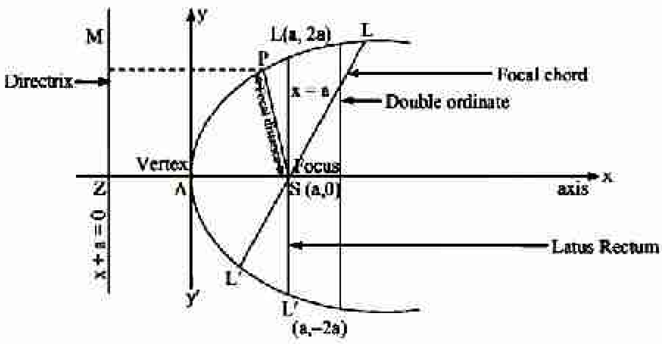
(iv) Double ordinate : The chord which is perpendicular to the axis of Parabola or parallel to Directrix is called double ordinate of the Parabola.

(v) Focal chord : Any chord of the parabola passing through the focus is called Focal chord.

(vi) Latus Rectum : If a double ordinate passes through the focus of parabola then it is called as latus rectum.

4. Standard Equation of Parabola

If we take vertex as the origin, axis as x-axis and distance between vertex and focus as 'a' then equation of the parabola in the simplest form will be. $y^2 = 4ax$



(i) Parameters of the Parabola $y^2 = 4ax$

(a) Vertex A $\Rightarrow$ (0, 0)

(b) Focus S $\Rightarrow$ (a, 0)

(c) Directrix $\Rightarrow x + a = 0$

(d) Axis $\Rightarrow y = 0$ or x– axis

(e) Equation of Latus Rectum $\Rightarrow x = a$

(f) Length of L.R. $\Rightarrow 4a$

(g) Ends of L.R. $\Rightarrow (a, 2a), (a, -2a)$

(h) The focal distance $\Rightarrow$ sum of abscissa of the point and distance between vertex and L.R. $= |x_1 + a|$

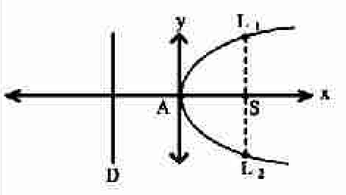
(i) If length of any double ordinate of parabola $y^2 = 4ax$ is 2ℓ then coordinates of end points of this

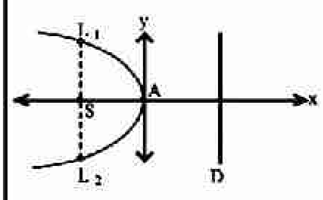
Double ordinate are $\left(\frac{\ell^2}{4a}, \ell\right)$ and $\left(\frac{\ell^2}{4a}, -\ell\right)$.

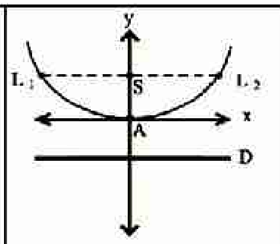
Note :

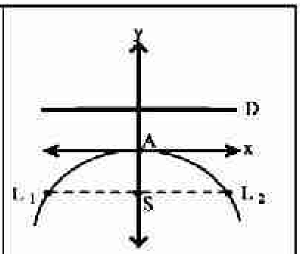
➤ The length of the Latus Rectum = 2 × perpendicular distance of focus from the directrix.

(ii) Other Standard Parabola :









Equation	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Coordinates of Vertex (A)	(0, 0)	(0, 0)	(0, 0)	(0, 0)
Axis of Parabola	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Coordinates of focus (S)	(a, 0)	(–a, 0)	(0, a)	(0, –a)
Equation of directrix (D)	$x = -a$	$x = a$	$y = -a$	$y = a$
Length of latus rectum L_1L_2	4a	4a	4a	4a
Coordinates of of L_1, L_2	(a, 2a), (a, –2a)	(–a, 2a), (–a, –2a)	(–2a, a), (2a, a)	(–2a, –a), (2a, – a)

5. Parametric equation of Parabola

The parameter equation parabola $y^2 = 4ax$ are $x = at^2, y = 2at$ Parametric Equation for parabola

(i) $y^2 = -4ax$ is $(-at^2, 2at)$

(ii) $x^2 = 4ay$ is $(2at, at^2)$

(iii) $x^2 = -4ay$ is $(2at, -at^2)$

6. Equation of Chord Joining any two Point of a Parabola

Let $P(at_1^2, 2at_1)$, $Q(at_2^2, 2at_2)$ be any two point on the parabola $y^2 = 4ax$. Then the equation of the chord joining these points is given by $y(t_1 + t_2) = 2x + 2at_1t_2$

Important Points to be Remembered

(1) If t_1 & t_2 are the ends of a focal chord of the parabola $y^2 = 4ax$ then $t_1t_2 = -1$, Hence the co-ordinates at the extremities of a focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$.

(2) Length of the chord $= a(t_1 - t_2)\sqrt{(t_1 + t_2)^2 + 4}$.

(3) The length of the focal chord joining parameter t_1 and t_2 on the parabola $y^2 = 4ax$ is $a(t_2 - t_1)^2$

(4) If ℓ_1 & ℓ_2 are the lengths of segments of a focal chord of a parabola. Then length of its latus rectum is $\frac{4\ell_1\ell_2}{\ell_1 + \ell_2}$.

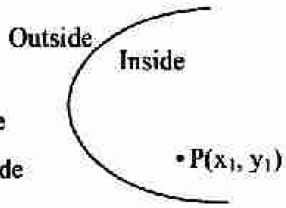
7. Position of a point relative to a parabola

The point (x_1, y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.

$S_1 : y_1^2 - 4ax_1$

$S_1 < 0 \rightarrow$ Inside

$S_1 > 0 \rightarrow$ Outside



8. Line & a Parabola

(i) The line $y = mx + c$ meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $c <, = \text{ or } > a/m$

$\Rightarrow$ condition of tangency is $\boxed{c = \frac{a}{m}}$

(ii) Line $y = mx + c$ will be tangent to the parabola $x^2 = 4ay$ if $\boxed{c = -am^2}$

(iii) Then the length of the chord intercepted by the parabola $y^2 = 4ax$ on the line $y = mx + c$ is given by

$\boxed{\frac{4}{m^2}\sqrt{a(1+m^2)(a - mc)}}$

(iv) Length of the focal chord making an angle α with the x-axis is $= 4a\,\text{cosec}^2\alpha$.

9. Equation of Tangent

(i) Point Form : The equation of tangent to the parabola $y^2 = 4ax$ at the point (x_1, y_1) is $yy_1 = 2a(x + x_1)$ or $T = 0$

(ii) Parametric Form : The equation of the tangent to the parabola at t.

i.e. $(at^2, 2at)$ is $ty = x + at^2$

Note :

➢ The point of intersection of tangents at t_1 and t_2 on the parabola is $[at_1t_2, a(t_1 + t_2)]$ i.e. (GM of abscissa, AM of ordinate).

(iii) Slope Form :

(a) The equation of the tangent of the parabola $y^2 = 4ax$ is

$y = mx + \frac{a}{m}$ and the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

(b) $y = mx - am^2$ is a tangent to the parabola $x^2 = 4ay$ for all value of m and its point of contact is $(2am, am^2)$

(c) The equation of the tangent of slope m to the parabola

$(y - k)^2 = 4a(x - h)$ is given by $(y - k) = m(x - h) + \frac{a}{m}$ and the coordinates of point of contact are $\left(h + \frac{a}{m^2}, k + \frac{2a}{m}\right)$

10. Equation of Normal

(i) Point Form : The equation to the normal at the point (x_1, y_1) of the parabola $y^2 = 4ax$ is given by $y - y_1 = \frac{-y_1}{2a}(x - x_1)$

(ii) Slope Form : The equation of normal to the parabola $y^2 = 4ax$ having slope m is $y = mx - 2am - am^3$ and its point of contact is given by $(am^2, -2am)$

Note :

➤ The equation of normal of slope m to $y^2 = -4ax$, $x^2 = 4ay$ and $x^2 = -4ay$ are $y = mx + 2am + am^3$, $x = \frac{y}{m} - \frac{2a}{m} - \frac{a^3}{m}$ and $x = \frac{y}{m} + \frac{2a}{m} + \frac{a^3}{m}$ respectively.

(iii) Parametric Form : The equation to the normal at the point $(at^2, 2at)$ is $y + tx = 2at + at^3$

Important Points to be Remembered

(1) If the normals to the parabola $y^2 = 4ax$ at the point t_1 meets the parabola again at the point t_2 then $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

(2) If the normals to the parabola $y^2 = 4ax$ at the point t_1 and t_2 intersect again on the parabola at point t_3 then $t_1t_2 = 2$ & $t_3 = -(t_1 + t_2)$

(3) The point of intersection of normals at t_1 & t_2 are $(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))$

(4) The length of normal chord $= a(t_1 - t_2)\sqrt{(t_1 + t_2)^2 + 4} = \frac{4a(t_1^2 + 1)^{3/2}}{t_1^2}$

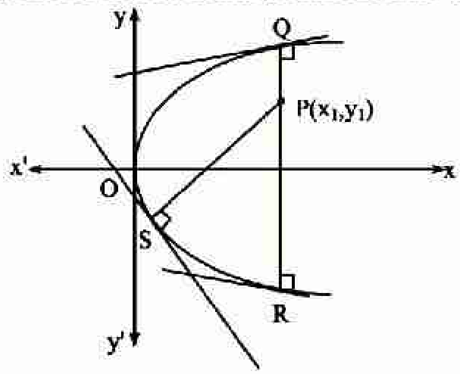
(5) The tangent at one extremity of the focal chord of parabola is parallel to the normal of other extremity.

(6) The normal at any point of a parabola is equally inclined to the focal distance of the point and the axis of the parabola.

(7) Three normals can be drawn from a point to a parabola.

11. Co-Normal Point

The points on the curve at which the normals pass through a common point are called co-normal points. Q, R, S are co-normal points. The co-normal points are also called the feet of the normal.



Note :

- The sum of the slopes of the normals at co-normals points is zero.
  $m_1 + m_2 + m_3 = 0$
- The sum of the ordinate of the co-normal points is zero.
- The centroid of the triangle formed by the co-normal points lies on the axis of the parabola.

12. Pair of Tangents

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the parabola $y^2 = 4ax$ is given by $SS_1 = T^2$ where $S = y^2 - 4ax$; $S_1 = y_1^2 - 4ax_1$; $T = yy_1 - 2a(x + x_1)$

13. Chord of Contact

(i) The equation of chord of contact of tangents drawn from a point (x_1, y_1) to the parabola $y^2 = 4ax$ is $T = 0$ or $yy_1 = 2a(x + x_1)$.

(ii) Lengths of the chord of contact is $\frac{1}{a}\sqrt{(y_1^2 - 4ax)(y_1^2 + 4a^2)}$

(iii) Area of triangle formed by tangents drawn from (x_1, y_1) and their chord of contact is $\frac{1}{2a}(y_1^2 - 4ax_1)^{3/2}$

14. Director Circle

The locus of the point of intersection of the perpendicular tangents to the parabola $y^2 = 4ax$ is called the director circle. Its equation is $x + a = 0$ which is parabola own directrix.

Note :

- In case of parabola director circle is a line.

15. Equation of the chord with a given middle point

Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) will be given by $(y - y_1) = \frac{2a}{y_1}(x - x_1)$

This can be reduced to $T = S_1$

where $T = yy_1 - 2a(x + x_1)$ & $S_1 = y_1^2 - 4ax_1$

16. Pole and Polar

(i) Equation of polar of the point (x_1, y_1) with respect to parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$ or $T = 0$

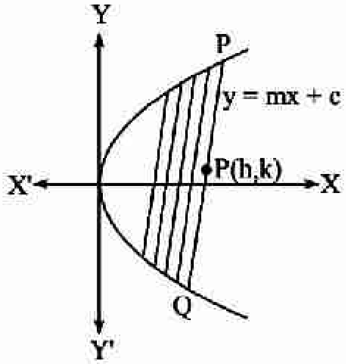
(ii) The pole of the line $\ell x + my + n = 0$ with respect to the parabola $y^2 = 4ax$ is $\left(\frac{n}{\ell}, \frac{-2am}{\ell}\right)$

(iii) If the polar of a point P passes through the point Q, then the polar of Q goes through P, then P and Q are called **conjugate points**.

(iv) Two straight lines are said to be conjugated to each other w.r.t. a parabola when the pole of one lies on the other.

17. Diameter of the Parabola

The locus of the mid points of a system of parallel chords of a parabola is known as diameter of a parabola.



The equation of the diameter bisecting chords of slope m of the parabola $y^2 = 4ax$ is $y = 2a/m$.

18. Geometrical properties of the Parabola

(i) The semi latus rectum of a parabola is the H.M. between the segments of any focal chord of a parabola i.e. if PQR is a focal chord, then $2a = \dfrac{2PQ.QR}{PQ + QR}$

(ii) The tangents at the extremities of any focal chord of a parabola intersect at right angles and their point of intersection lies on directrix i.e. the locus of the point of intersection of perpendicular tangents is directrix.

(iii) If the tangent and normal at any point P of parabola meet the axes in T and G respectively, then
(a) ST = SG = SP
(b) $\angle$ PSK is a right angle, where K is the point where the tangent at P meets the directrix.
(c) The tangent at P is equally inclined to the axis and the focal distance.

(iv) The area of triangle formed inside the parabola $y^2 = 4ax$ is
$\dfrac{1}{8a}(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)$ where y_1, y_2, y_3 are ordinate of vertices of the triangle.

(v) The abscissa of point of intersection R of tangents at $P(x_1, y_1)$ and $Q(x_2, y_2)$ on the parabola is G.M. of abscissa of P and Q and ordinate of R is A.M. of ordinate of P and Q thus $R\left(\sqrt{x_1x_2}, \dfrac{x_1 + x_2}{2}\right)$

(vi) The area of triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

Ellipse

It is a locus of a point which moves in such a way that the ratio of its distance from a fixed point and a fixed line (not passes through fixed point and all points and line lies in same plane) is constant (e) which is less than one.

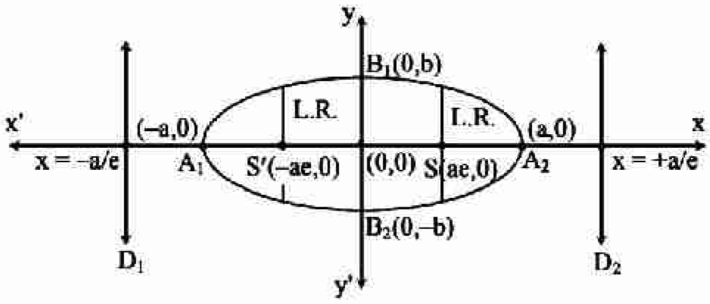
OR

In other word, an ellipse is the locus of a point which moves in a plane so that the sum of it distances from two fixed points is constant.

1. Standard Form of the Equation of Ellipse

(i) Horizontal Ellipse :

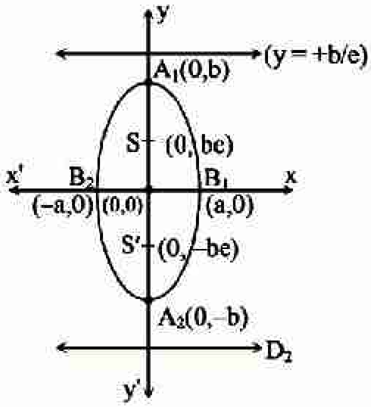
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (when $a > b > 0$)



(a) Centre (0, 0)
(b) Length of major axis $A_1A_2 = 2a$
(c) Length of minor axis $B_1B_2 = 2b$
(d) Vertices (a, 0) and (–a, 0)
(e) Directrices : $x = \frac{a}{e}$ and $x = -\frac{a}{e}$
(f) Foci : S (ae, 0) & S' (– ae, 0)
(g) Length of Latus Rectum $= \frac{2b^2}{a}$.
(h) Ends of Latus Rectum : $\left(ae, \pm\frac{b^2}{a}\right)$ and $\left(-ae, \pm\frac{b^2}{a}\right)$
(i) Eccentricity $b^2 = a^2(1 - e^2) \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}}$
(j) Focal distance :
$SP = a - ex_1$ or $S'P = a + ex_1 \Rightarrow SP + S'P = 2a$ (major axis)
(k) Distance between foci = 2ae
(l) Disatnce between directrices $= \frac{2a}{e}$

(ii) Vertical Ellipse :

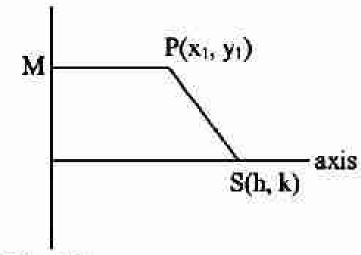
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (when $0 < b < a$)



(a) Centre (0, 0)
(b) Length of Major axis $A_1A_2 = 2b$
(c) Length of Minor axis $B_1B_2 = 2a$
(d) Vertices (0, b) and (0, −b)
(e) Directrices, $y = \pm \dfrac{b}{e}$,
(f) Foci $(0, \pm be)$
(g) Length of latus rectum = $\dfrac{2a^2}{b}$
(h) Eccentricity $a^2 = b^2(1 - e^2) \Rightarrow e = \sqrt{1 - \dfrac{a^2}{b^2}}$
(i) Focal distance :
$SP = b - ey_1$ or $S'P = b + ey_1 \Rightarrow SP + S'P = 2b$ (major axis)
(j) Distance between foci = 2be
(k) Distance between directrices = $\dfrac{2b}{e}$

2. General Equation of an Ellipse

The general equation of an ellipse whose focus is $S(h, k)$ and the directrix is the line $ax + by + c = 0$ and the eccentricity will be e is.



$SP = e\,PM$

$$\Rightarrow (x_1 - h)^2 + (y_1 - k)^2 = \frac{e^2(ax_1 + by_1 + c)^2}{a^2 + b^2}$$

Hence the locus of (x_1, y_1) will be given by
$(a^2 + b^2)[(x - h)^2 + (y - k)^2] = e^2(ax + by + c)^2$

Note :

- *Condition for second degree in X & Y to represent an ellipse is that if $h^2 - ab < 0$ &*
  $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$.

3. Special form of Ellipse

If centre of the ellipse is (h, k) and the direction of the axes are parallel to the coordinate axes, then its equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

4. Parametric Form of the Ellipse

Let the equation of ellipse in standard form will be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Then the equation of ellipse in the parametric form will be given by $x = a\cos\phi$, $y = b\sin\phi$ where ϕ is the eccentric angle whose value vary from $0 \le \phi < 2\pi$.

5. Position of a point with respect to an Ellipse

The position of a point $P(x_1, y_1)$ with respect to ellipse

$S = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$ can be given in following manner if

$S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$ then

$S_1 < 0$ Point is inside

$S_1 = 0$ Point lies on ellipse

$S_1 > 0$ Point lies out side the ellipse

6. Line and Ellipse

The line $y = mx + c$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in two real points, coincident or imaginary according as c^2 is $<$ $=$ or $>$ $a^2m^2 + b^2$

Hence $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

if $c = \pm\sqrt{a^2m^2 + b^2}$

7. Equation of the Tangent

(i) Point form : The equation of the tangent at any point (x_1, y_1) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$. $(T = 0)$

(ii) Parametric form : The equation of tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at any point $(a\cos\phi, b\sin\phi)$ is $\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1$.

(iii) Slope Form : The equation of the tangent of slope of m to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are $y = mx \pm \sqrt{a^2m^2+b^2}$ and Its point of contact is $\left(\frac{\mp a^2m}{\sqrt{a^2m^2+b^2}}, \frac{\pm b^2}{\sqrt{a^2m^2+b^2}}\right)$

8. Equation of the Normal

(i) Point form : The equation of the normal at any point (x_1, y_1) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2$$

(ii) Parametric form : The equation of the normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at any point $(a\cos\phi, b\sin\phi)$ is

$$ax\sec\phi - by\,\mathrm{cosec}\,\phi = a^2-b^2$$

(iii) Slope Form : The equation of the normal of slope m to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are given by $y = mx \pm \frac{m(a^2-b^2)}{\sqrt{a^2+b^2m^2}}$ and the coordinate of the foot of normal are $\left(\pm\frac{a^2}{\sqrt{a^2+b^2m^2}}, \pm\frac{b^2m}{\sqrt{a^2+b^2m^2}}\right)$

9. Equation of chord whose mid point is given

The equation of the chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose mid point be (x_1, y_1) is

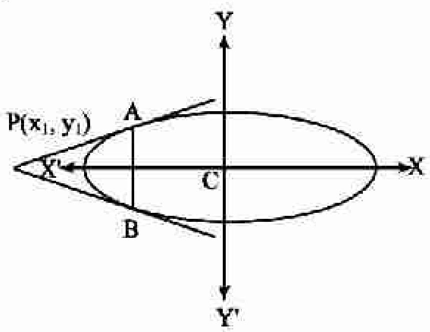
$T = S_1$

where $T = \frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1=0$

$S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$

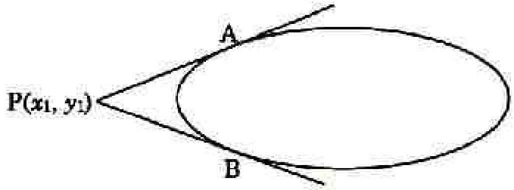
10. Chord of contact

If PA and PB be the tangents through point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then the equation of the chord of contact AB is $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ or $T = 0$



11. Pair of Tangents

Let P (x_1, y_1) be any point lies outside the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and let a pair of tangents PA, PB can be drawn to it from P.s



then the equation of pair of tangents of PA & PB is $SS_1 = T^2$ where $S \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$

$$S_1 \equiv \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$$

$$T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = 0.$$

12. Pole and Polar

(i) Equation of polar of $P(x_1, y_1)$ w.r.t. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \ (T = 0)$$

(ii) The pole of the line $\ell x + my + n = 0$ with respect to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(-\frac{a^2\ell}{n}, -\frac{b^2m}{n}\right)$

13. Director Circle

Locus of the point of intersection of the tangents which meet at right angles is called the **Director Circle.**

The equation of the director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 + b^2$.

14. Diameter

The locus of the middle points of a system of parallel chords is called a diameter.

If $y = mx + c$ represent a system of parallel chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then the equation of the diameter is $y = -\frac{b^2}{a^2m}x$

15. Conjugate Diameters

Two diameters are said to be conjugate when each bisects all chords parallel to the other.

If m_1 and m_2 be the slope of the conjugate diameters of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $\boxed{m_1m_2 = \frac{-b^2}{a^2}}$

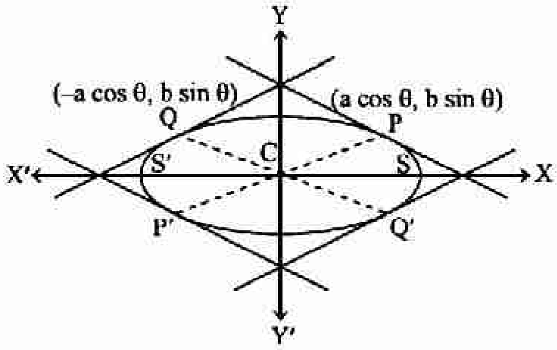
Important Points to be Remembered

(1) The eccentric angles of the extremities of a pair of conjugate diameters differ by a right angle.

(2) The sum of the squares of two conjugate semi diameters of an ellipse is constant and equal to sum of the squares of the semi- axes of the ellipse. i.e. $CP^2 + CD^2 = a^2 + b^2$.

(3) The tangent at the extremities of a pair of conjugate diameters form a parallelogram whose area is constant and equal to product of the axes.

(4) If CP, CQ are two conjugate semi-diameters of an ellipse and S, S′ be two foci of an ellipse, then $SP \times S'P = CQ^2$



16. Auxiliary Circle

A circle described on major axis of ellipse as diameter is called the **auxiliary circle**.

The equation of Auxiliary circle of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$x^2 + y^2 = a^2$ (a > b) or $x^2 + y^2 = b^2$ (b > a)

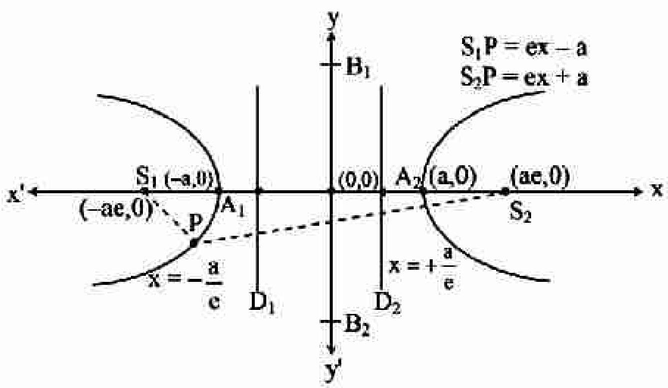
Hyperbola

1. Definition

A hyperbola is defined as the locus of a point moving in a plane in such a way that the ratio of its distance from a fixed point to that from a fixed line (the point does not lie on the line) is a fixed constant greater than 1.

2. Standard Equation of Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



(i) Centre (0, 0)

(ii) Length of transverse axis = 2a

(iii) Length of conjugate axis = 2b

(iv) Vertices (a, 0) and (–a, 0)

(v) Directrices : $x = a/e$ and $x = -a/e$

(vi) Foci : S (ae, 0) and S' (– ae, 0)

(vii) Distance between foci = 2ae

(viii) Distance between directrices = $\frac{2a}{e}$

(ix) Length of Latus Rectum = $\frac{2b^2}{a}$.

(x) End of Latus Rectum $\left(ae, \pm\frac{b^2}{a}\right)$ and $\left(-ae, \pm\frac{b^2}{a}\right)$

(xi) Eccentricity : $b^2 = a^2(e^2 - 1)$ and $e = \sqrt{1 + \frac{b^2}{a^2}}$

(xii) Focal distance : $PS = ex_1 - a$ and $PS' = ex_1 + a$

Note :

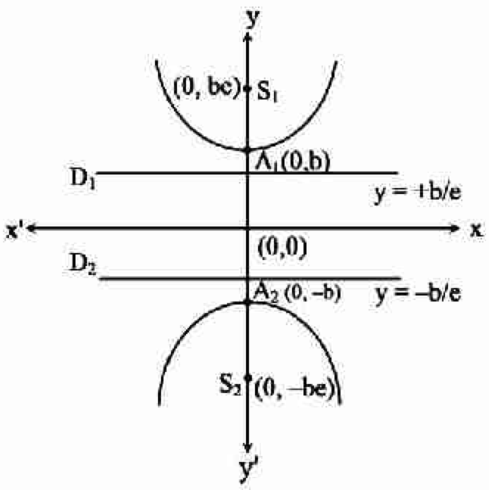
- Focal Property : The difference of the focal distances of any point on the hyperbola is constant and equal to transverse axis i.e. $|PS| - |PS'| = 2a$. The distance SS' - focal length.

3. Conjugate Hyperbola

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate & the transverse axes of the other are called Conjugate Hyperbola of each other.

The equation of the conjugate hyperbola of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ or } \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$



(i) Centre (0, 0)
(ii) Length of transverse axis = 2b
(iii) Length of conjugate axis = 2a
(iv) Vertices (0, b) and (0, –b)
(v) Directrices : y = b/e and y = – b/e
(vi) Foci : S (0, be) and S' (0, – be)
(vii) Distance between foci = 2be
(viii) Distance between directrices = $\frac{2b}{e}$
(ix) Length of Latus Rectum = $\frac{2a^2}{b}$.
(x) Eccentricity : $b^2 = a^2 (e^2 - 1)$ and $e = \sqrt{1 + \frac{b^2}{a^2}}$

Important Points to be Remembered

(1) If e_1 & e_2 are the eccentricities of the hyperbola & its conjugate hyperbola then $\boxed{\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1}$.
(2) The foci of a hyperbola and its conjugate hyperbola are concyclic.
(3) Two hyperbolas are said to be similar if they have the same eccentricity.
(4) Two similar hyperbolas are said to be equal if they have same latus rectum.
(5) The equation $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ will represent an hyperbola if $h^2 - ab > 0$ & $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$.

4. Special Form of Hyperbola

If centre of the hyperbola is (h, k) and the direction of the axes are parallel to the coordinate axes, then its equation is

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

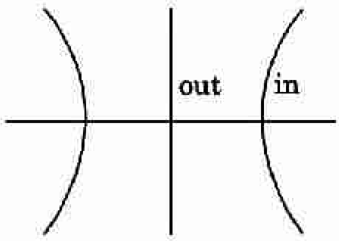
5. Parametric Equation

(i) The parametric equation of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ will be given by $x = a\sec\phi$, $y = b\tan\phi$. (where ϕ is eccentric angle) $(0 \le \phi < 2\pi)$

(ii) The equation $x = a\left(\frac{e^{\theta}+e^{-\theta}}{2}\right)$, $y = b\left(\frac{e^{\theta}-e^{-\theta}}{2}\right)$ are also the parametric equations of the hyperbola.

6. Position of a point with respect to Hyperbola

Let S = 0 be the hyperbola and P (x_1, y_1) be the point and $S_1 \equiv S(x_1, y_1)$. Then



$S_1 < 0 \Rightarrow$ P is outside the hyperbola (same side of centre)

$S_1 > 0 \Rightarrow$ P is inside the hyperbola (Opposite side of centre)

$S_1 = 0 \Rightarrow$ P lies on the hyperbola

7. Line and Hyperbola

The straight line $y = mx + c$ is a secant, a tangent or passes outside the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ according as : $c^2 > = < a^2m^2 - b^2$.

Hence the line $y = mx + c$ touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, if

$$\boxed{c = \pm\sqrt{a^2m^2-b^2}}$$

8. Equation of Tangent

(i) **Point form :** The equation of the tangent at any point (x_1, y_1) on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1.\ (T=0)$$

(ii) Parametric Form : Equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $(a \sec\theta, b\tan\theta)$ is $\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$.

(iii) Slope form : The equation of tangent of slope m to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are given by $y = mx \pm \sqrt{a^2m^2 - b^2}$ and the point of contacts is $\left(\pm\frac{a^2m}{\sqrt{a^2m^2-b^2}}, \pm\frac{b^2}{\sqrt{a^2m^2-b^2}}\right)$

9. Equation of the Normal

(i) Point form : The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at (x_1, y_1) is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 = a^2e^2$.

(ii) Parametric Form : The equation of normal at $(a \sec\theta, b\tan\theta)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $ax\cos\theta + by\cot\theta = a^2 + b^2$.

(iii) Slope form : The equation of normal of slope m to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $y = mx - \frac{m(a^2+b^2)}{\sqrt{a^2-b^2m^2}}$ and foot of normal are $\left(\pm\frac{a^2}{\sqrt{a^2-b^2m^2}}, \mp\frac{b^2m}{\sqrt{a^2-b^2m^2}}\right)$

10. Pair of Tangents

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by : $SS_1 = T^2$ where $S = \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

11. Chord of Contact

Chord of contact of tangents drawn from a point outside to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$T = 0$ i.e. $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

12. Equation of the chord whose middle point is given

Equation of the chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $T = S_1$, where $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

13. Director Circle

The locus of the point of intersection of perpendicular to tangent to the hyperbola is called a director circle.

The equation of the director circle of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2$ $(a^2 > b^2)$

Note :

- If $b^2 > a^2$, then the radius of the director circle is imaginary, so that there is no such circle and so no pair of tangents at right angle can be drawn to the curve.

14. Diameter

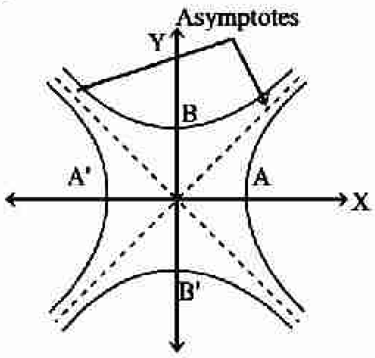
If y = mx + c represent a system of parallel chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then the equation of the diameter is $y = \frac{b^2}{a^2 m}x$.

15. Conjugate Diameter

The diameters of a hyperbola are said to be conjugate diameter, if each bisect the chords parallel to the other. The diameters $y = m_1x$ and m_2x are conjugate, if $m_1m_2 = \frac{b^2}{a^2}$

16. Asymptotes

If the length of perpendicular let fall from a point on a hyperbola straight line ends to zero as the point on the hyperbola moves to the Hyperbola.



Equation of the asymptotes of hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x$.

Important Points to be Remembered

(1) A hyperbola and its conjugate hyperbola have the same asymptotes.

(2) The angle between the asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $2\tan^{-1}\left(\frac{b}{a}\right)$

(3) The equation of the pair of Asymptotes differ the hyperbola and the conjugate hyperbola same by the constant only i.e. Hyperbola – Asymptotes = Asymptotes – Conjugate hyperbola.

$$H\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right), C\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1\right) \& \quad A\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = 0$$

Clearly → $\boxed{C + H = 2A}$

(4) The Asymptotes pass through the center of the hyperbola.

17. Rectangular Hyperbola

A hyperbola whose asymptotes include a right angle is said to be rectangular hyperbola or we can say that, if the lengths of transverse and conjugate axes of any hyperbola be equal, then it is said to be a rectangular hyperbola.

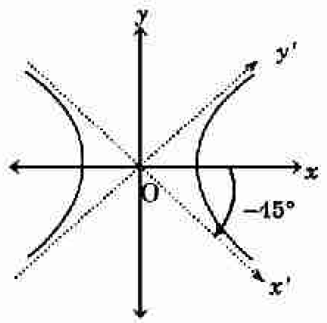
i.e. In a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if b = a, then it said to be rectangular hyperbola. The eccentricity of a rectangular hyperbola is always $\sqrt{2}$.

(i) Rectangular Hyperbola of the Form $x^2 - y^2 = a^2$

(a) Asymptotes are perpendicular lines i.e. $x \pm y = 0$

(b) Eccentricity $e = \sqrt{2}$

(c) Centre (0, 0)



(d) Foci : $(\pm\sqrt{2}\,a, 0)$

(e) Vertices : A(a, 0) and $A_1(-a, 0)$

(f) Directrices : $x = \pm \frac{a}{\sqrt{2}}$

(g) Latusrectum $= 2\alpha$

(h) Parametric form : $x = a\sec\theta$, $y = a\tan\theta$

(i) Equation of tangent : $x\sec\theta - y\tan\theta = \alpha$

(j) Equation of normal : $\frac{x}{\sec\theta} + \frac{y}{\tan\theta} = 2a$

(ii) Rectangular Hyperbola of the form $xy = c^2$

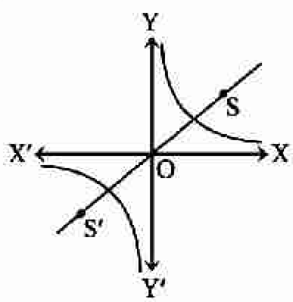
(a) Asymptotes are perpendicular liens i.e. x = 0 and y = 0

(b) Eccentricity : $e = \sqrt{2}$

(c) Centre : (0, 0)

(d) Foci : $S(\sqrt{2}\,c, \sqrt{2}\,c)$, $S_1(-\sqrt{2}\,c, -\sqrt{2}\,c)$

(e) Vertices : A(c, c), $A_1(-c, -c)$



(f) Directrices : $x + y = \pm \sqrt{2}$ e

(g) Latusrectum = $2\sqrt{2}$ c

(h) Parametric form : $x = ct,\ y = \frac{c}{t}$.

18. Equation of Tangent of Rectangular Hyperbola $xy=c^2$

(i) Point Form : The equation of tangent at x_1, y_1 to the rectangular hyperbola is $xy_1 + yx_1 = 2c^2$ or $\frac{x}{x_1} + \frac{y}{y_1} = 2$

(ii) Parametric Form : The equation of tangent at $\left(ct, \frac{c}{t}\right)$ to the rectangular hyperbola is $x + yt^2 - 2ct = 0$

19. Equation of Normal of Rectangular Hyperbola $xy=c^2$

(i) Point Form : The equation of the normal at (x_1, y_1) to the rectangular hyperbola is $xx_1 - yy_1 = x_1^2 - y_1^2$

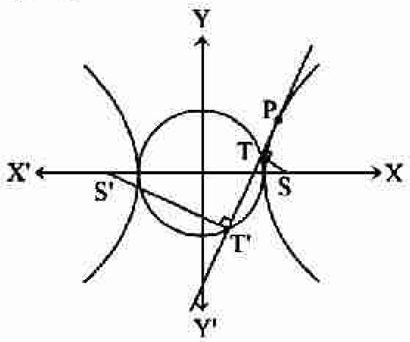
(ii) Parametric Form : The equation of Normal at $\left(ct, \frac{c}{t}\right)$ to the rectangular hyperbola is $xt^3 - yt - ct^4 + c = 0$

(iii) Four normals can be drawn from a point to the hyperbola $xy = c^2$.

20. Equation of Auxiliary Circle

Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ on any tangent is its auxiliary circle

Equation of auxilliary circle is $x^2 + y^2 = a^2$



SHORT TRICKS

Trick -1 Length of latus rectum of parabola

(i) Length of latus rectum for the parabola $ax^2 + by + cx + d = 0$ is $= \left|\dfrac{\text{coeff. of } y}{\text{coeff. of } x^2}\right|$

(ii) Length of latus rectum for the parabola $py^2 + qy + rx + s = 0$ is $= \left|\dfrac{\text{coeff. of } x}{\text{coeff. of } y^2}\right|$

Q.1 The length of latus rectum of the parabola $y = x\tan\alpha - \dfrac{gx^2}{2u^2\cos^2\alpha}$ is-

(A) $\dfrac{2u^2\cos^2\alpha}{g}$ (B) $\dfrac{u^2\sin^2 2\alpha}{g}$ (C) $\dfrac{u^2\cos^2 2\alpha}{g}$ (D) $\dfrac{u^2\cos^2\alpha}{g}$

Sol. [A]

Proper Method	**Short Trick**
$y = x\tan\alpha - \dfrac{gx^2}{2u^2\cos^2\alpha}$ $\dfrac{gx^2}{2u^2\cos^2\alpha} - x\tan\alpha = -y$ $\Rightarrow \left(x^2 - x\dfrac{2u^2\cos^2\alpha\tan\alpha}{g}\right) = -y\dfrac{2u^2\cos^2\alpha}{g}$ $\Rightarrow \left(x - \dfrac{u^2\cos^2\alpha\tan\alpha}{g}\right)^2 = \dfrac{-2u^2\cos^2\alpha}{g}y + \dfrac{u^4\cos^4\alpha\tan^2\alpha}{g^2}$ $\Rightarrow \left(x - \dfrac{u^2\cos^2\alpha\tan\alpha}{g}\right)^2 = -\dfrac{2u^2\cos^2\alpha}{g}\left(y - \dfrac{u^2\cos^2\alpha\tan^2\alpha}{2g}\right)$ $\therefore$ length of L.R $= \dfrac{2u^2\cos^2\alpha}{g}$	Length of L.R $= \left\|\dfrac{1}{\frac{-g}{2u^2\cos^2\alpha}}\right\|$ $= \dfrac{2u^2\cos^2\alpha}{g}$

Q.2 The length of the latus rectum of the parabola $x = ay^2 + by + c$ is-

(A) $\dfrac{a}{4}$ (B) $\dfrac{a}{3}$ (C) $\dfrac{1}{a}$ (D) $\dfrac{1}{4a}$

Sol. [C]

Proper Method	**Short Trick**
$x = ay^2 + by + c$ $\Rightarrow x = a\left[y^2 + \dfrac{b}{a}y + \dfrac{c}{a}\right]$ $\Rightarrow x = a\left[y^2 + 2.\dfrac{b}{2a}y + \dfrac{b^2}{4a^2}\right] + c - \dfrac{b^2}{4a}$ $\Rightarrow \left(y + \dfrac{b}{2a}\right)^2 = 4\left(\dfrac{1}{4a}\right)\left(x + \dfrac{b^2 - 4ac}{4a}\right)$ $\Rightarrow Y^2 = 4AX$ Length of LR $= 4 \times \dfrac{1}{4a} = \dfrac{1}{a}$	Length $= \left\|\dfrac{\text{coeff. of } x}{\text{coeff. of } y^2}\right\| = \left\|\dfrac{1}{a}\right\|$

Q.3 The length of the latus-rectum of the parabola $x^2 - 4x - 8y + 12 = 0$ is- **[AIEEE-2002]**

(A) 4 (B) 6 (C) 8 (D) 10

Sol. **[C]**

Proper Method

$x^2 - 4x - 8y + 12 = 0$

$\Rightarrow x^2 - 4x = 8y - 12$

$\Rightarrow (x-2)^2 = 8y - 12 + 4$

$\Rightarrow (x-2)^2 = 8y - 8$

$\Rightarrow (x-2)^2 = 8(y-1)$

$\therefore$ length of L.R = 8

Short Trick

$$\text{Length} = \left|\frac{\text{coeff. of } y}{\text{coeff. of } x^2}\right| = \left|\frac{-8}{1}\right| = 8$$

Q.4 Find the latus-rectum of the parabola $3x^2 - 6x - y + 6 = 0$ is-

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) $\frac{2}{3}$ (D) $\frac{2}{7}$

Sol. **[A]**

Proper Method

$3x^2 - 6x - y + 6 = 0$

$\Rightarrow 3x^2 - 6x = y - 6$

$\Rightarrow x^2 - 2x = \frac{y}{3} - \frac{6}{3}$

$\Rightarrow (x-1)^2 = \frac{y}{3} - 2 + 1$

$\Rightarrow (x-1)^2 = \frac{1}{3}(y-3)$

comparing with $X^2 = 4aY$

$(X = x - 1, Y = y - 3, a = \frac{1}{12})$

length of L.R $= 4a = \frac{1}{3}$

Short Trick

$$\text{Length} = \left|\frac{\text{coeff. of } y}{\text{coeff. of } x^2}\right| = \left|\frac{-1}{3}\right| = \frac{1}{3}$$

Trick -2 If parabola is in the form

$ax^2 + bx + cy + d = 0$ ...... (1)

or $py^2 + qy + rx + s = 0$ ...... (2)

then to find equation of axis

For parabola (1) $\Rightarrow$ differentiate given equation w.r.t. x keeping y as constant

For parabola (2) $\Rightarrow$ differentiate given equation w.r.t. y keeping x as constant

Q.5 The equation of the axis of the parabola $x^2 - 4x - 3y + 10 = 0$ is

(A) $y + 2 = 0$ (B) $x + 2 = 0$ (C) $x - 2 = 0$ (D) $y - 2 = 0$

Sol. **[C]**

Proper Method

$x^2 - 4x - 3y + 10 = 0$

$\Rightarrow x^2 - 4x + 4 = 3y - 6$

$\Rightarrow (x-2)^2 = 4.\frac{3}{4}(y-2)$

$(x-2)^2 = 4.\frac{3}{4}(y-2)$

comparing with $X^2 = 4aY$

equation of axis $x - 2 = 0$

Short Trick

Differentiating given equation w.r.t. x keeping y as constant $2x - 4 = 0$

$\Rightarrow x - 2 = 0$

Q.6 The axis of the parabola $4y^2 - 6x - 4y = 5$ is-

(A) $y - 1 = 0$ (B) $2y + 1 = 0$ (C) $2y - 1 = 0$ (D) $y - 2 = 0$

Sol. **[C]**

Proper Method	**Short Trick**
Here $4y^2 - 4y = 6x + 5$ $\Rightarrow 4(y^2 - y) = 6x + 5$ $\Rightarrow 4\left(y-\frac{1}{2}\right)^2 = 6x + 5 + 1$ $\Rightarrow \left(y-\frac{1}{2}\right)^2 = \frac{1}{4}(6x + 6)$ $\Rightarrow \left(y-\frac{1}{2}\right)^2 = \frac{6}{4}(x + 1)$ Comparing with $Y^2 = 4aX$, then equation of axis is $y-\frac{1}{2} = 0$ $2y - 1 = 0$	Differentiating w.r.t y keeping x as constant $8y - 4 = 0$ $\Rightarrow 2y - 1 = 0$

Q.7 Find the axis of the parabola $3x^2 - 6x - y + 6 = 0$ is-

(A) $x - 2 = 0$ (B) $x - 1 = 0$ (C) $x + 1 = 0$ (D) $2x - 1 = 0$

Sol. **[B]**

Proper Method	**Short Trick**
$3x^2 - 6x - y + 6 = 0$ $\Rightarrow 3x^2 - 6x = y - 6$ $\Rightarrow x^2 - 2x = \frac{y}{3} - \frac{6}{3}$ $\Rightarrow (x - 1)^2 = \frac{y}{3} - 2 + 1$ $\Rightarrow (x - 1)^2 = \frac{1}{3}(y - 3)$ comparing with $X^2 = 4aY$ $(X = x - 1, Y = y - 3, a = \frac{1}{12})$ axis : $X = 0 \Rightarrow x - 1 = 0$	Differentiating w.r.t x keeping y as constant $6x - 6 = 0$ $\Rightarrow x - 1 = 0$

Trick -8 Tangents drawn from point (x_1, y_1) to the parabola $y^2 = 4ax$ makes an angle θ, then

$$\tan\theta = \left|\frac{\sqrt{S_1}}{x_1 + a}\right| \text{ (where } S_1 = y_1^2 - 4ax_1)$$

Q.8 Tangents are drawn from the point $(-2, -1)$ to the parabola $y^2 = 4x$. If α is the angle between these tangents then $\tan\alpha$ equals - **JEE Main Online 2014**

(A) 3 (B) 2 (C) 1/3 (D) 1/2

Sol. [A]

Proper Method	Short Trick
Any tangent to $y^2 = 4x$ is $y = mx + 1/m$ If it is drawn from (–2, –1), then $-1 = -2m + 1/m$ $\Rightarrow 2m^2 - m - 1 = 0$ If $m = m_1$, m_2 then $m_1 + m_2 = 1/2$, $m_1m_2 = -1/2$ $\therefore \tan a \ \dfrac{m_1 - m_2}{1+m_1m_2} = \dfrac{\sqrt{(m_1+m_2)^2 - 4m_1m_2}}{1+m_1m_2}$ $\dfrac{\sqrt{1/4+2}}{1-1/2} = 3$	$\tan\alpha = \left\|\dfrac{\sqrt{1+8}}{-2+1}\right\| = 3$

Trick -4 Eccentricity of an ellipse

Eccentricity of an ellipse $ax^2 + by^2 + cx + dy + e = 0$ is

If $a > b$, then $e = \sqrt{1 - \dfrac{\text{coeff. of } y^2}{\text{coeff. of } x^2}}$

If $b > a$, then $e = \sqrt{1 - \dfrac{\text{coeff. of } x^2}{\text{coeff. of } y^2}}$

Q.9 The equation $x^2 + 4y^2 + 2x + 16y + 13 = 0$ represents a ellipse then eccentricity is-

(A) $\dfrac{1}{\sqrt{3}}$ (B) $\dfrac{1}{2}$ (C) $\dfrac{\sqrt{3}}{2}$ (D) $\dfrac{2}{\sqrt{3}}$

Sol. [C]

Proper Method	Short Trick
We have $x^2 + 4y^2 + 2x + 16y + 13 = 0$ $(x^2 + 2x + 1) + 4(y^2 + 4y + 4) = 4$ $(x + 1)^2 + 4(y + 2)^2 = 4$ $\dfrac{(x+1)^2}{2^2} + \dfrac{(y+2)^2}{1^2} = 1$ Comparing with $\dfrac{X^2}{a^2} + \dfrac{Y^2}{b^2} = 1$ where $X = x + 1$, $Y = y + 2$ and $a = 2$, $b = 1$ eccentricity of the ellipse $e = \sqrt{1 - \dfrac{b^2}{a^2}} = \sqrt{1 - \dfrac{1}{4}} = \dfrac{\sqrt{3}}{2}$	$e = \sqrt{1 - \dfrac{1}{4}} = \dfrac{\sqrt{3}}{2}$

Q.10 The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is-

(A) 1/3 (B) 2/3 (C) 3/4 (D) 5/9

Sol. [B]

Proper Method	Short Trick
$9x^2 + 5(y^2 - 6y + 9) = 45$ $\Rightarrow \frac{x^2}{5} + \frac{(y-3)^2}{9} = 1$ $e = \sqrt{1 - \frac{a^2}{b^2}}$ $(\because b > a)$ $e = \sqrt{1 - \frac{5}{9}} = \frac{2}{3}$	$e = \sqrt{1 - \frac{5}{9}} = \frac{2}{3}$

Q.11 Eccentricity of the ellipse $4x^2 + y^2 - 8x + 2y + 1 = 0$ is-

(A) $1/\sqrt{3}$ (B) $\sqrt{3}/2$ (C) 1/2 (D) 1/4

Sol. [B]

Proper Method	Short Trick
$4x^2 + y^2 - 8x + 2y + 1 = 0$ $\Rightarrow 4x^2 - 8x + y^2 + 2y = -1$ $\Rightarrow 4(x^2 - 2x) + y^2 + 2y = -1$ $\Rightarrow 4(x-1)^2 + (y+1)^2 = -1 + 4 + 1$ $\Rightarrow \frac{(x-1)^2}{1} + \frac{(y+1)^2}{4} = 1$ $e = \sqrt{1 - \frac{a^2}{b^2}}$ $(b > a)$ $e = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$	$e = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

Q.12 The eccentricity of the ellipse represented by the equation $25x^2 + 16y^2 - 150x - 175 = 0$ is -

(A) 2/5 (B) 3/5 (C) 4/5 (D) 7/5

Sol. [B]

Proper Method	Short Trick
$25x^2 + 16y^2 - 150x - 175 = 0$ $\Rightarrow 25x^2 - 150x + 16y^2 = 175$ $\Rightarrow 25(x^2 - 6x) + 16y^2 = 175$ $\Rightarrow 25(x-3)^2 + 16y^2 = 175 + 225$ $\Rightarrow \frac{(x-3)^2}{16} + \frac{y^2}{25} = 1$ $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$	$e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

Trick -5 **Length of latus rectum of an ellipse**

Let the equation of ellipse $ax^2 + by^2 + cx + dx + e = 0$

$$\text{Length of latus rectum} = \frac{2(\text{coff. of } y^2)}{\sqrt{\text{coff. of } x^2}} \quad (a > b)$$

$$\text{Length of latus rectum} = \frac{2(\text{coeff. of } x^2)}{\sqrt{\text{coeff. of } y^2}} \quad (b > a)$$

Q.13 Latus rectum of ellipse $4x^2 + 9y^2 - 8x - 36y + 4 = 0$ is-

(A) 8/3 (B) 4/3 (C) $\frac{\sqrt{5}}{3}$ (D) 16/3

Sol. [A]

Proper Method	Short Trick
$4(x^2 - 2x + 1) + 9(y^2 - 4y + 4) = 36$ $\Rightarrow \frac{4(x-1)^2}{36} + \frac{9(y-2)^2}{36} = 1$ $\Rightarrow \frac{(x-1)^2}{9} + \frac{(y-2)^2}{4} = 1$ Latus rectum $= \frac{2\times4}{3} = \frac{8}{3}$	Length of latus rectum $= \frac{2(4)}{\sqrt{9}} = \frac{8}{3}$

Q.14 The length of the latus rectum of ellipse $9x^2 + 16y^2 - 36x + 96y + 36 = 0$ is.

(A) $\frac{9}{2}$ (B) $\frac{7}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

Sol. [A]

Proper Method	Short Trick
Given: $9x^2 + 16y^2 - 36x + 96y + 36 = 0$ ...(1) $\Rightarrow [9x^2 - 36x] + [16y^2 + 96y] = -36$ $\Rightarrow 9[x^2 - 4x + 4] + 16[y^2 + 6y + 9] = -36 + 36 + 144$ $\Rightarrow 9[x-2]^2 + 16[y+3]^2 = 144$ Dividing throughout by 144, we get $\frac{[x-2]^2}{16} + \frac{[y+3]^2}{9} = 1$ ...(2) Length of LR $= \frac{2b^2}{a} = \frac{2(9)}{4} = \frac{9}{2}$ units.	length of the latus rectum $= \frac{2(9)}{\sqrt{16}} = \frac{9}{2}$

Trick -6 Eccentricity of hyperbola

Let the equation of hyperbola is $ax^2 - by^2 + cx + dy + c = 0$, then

$$e = \sqrt{1 + \frac{\text{coeff. of } x^2}{|\text{coeff. of } y^2|}} = \sqrt{1 + \frac{a}{b}}$$

Q.15 The equation $16x^2 - 3y^2 - 32x + 12y - 44 = 0$ represents a hyperbola, then its eccentricity is

(A) $\frac{16}{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) $\sqrt{\frac{19}{3}}$

Sol. [D]

Proper Method	Short Trick
$16(x^2 - 2x + 1) - 3(y^2 - 4y + 4) = 48$ $\Rightarrow \frac{(x-1)^2}{3} - \frac{(y-2)^2}{16} = 1$ $e = \frac{\sqrt{19}}{\sqrt{3}}$	$e = \sqrt{1 + \frac{16}{3}} = \sqrt{\frac{19}{3}}$

Q.16 The eccentricity of the hyperbola $9x^2 - 4y^2 + 18x - 8y - 31 = 0$ is -

(A) $\sqrt{\frac{11}{2}}$ (B) $\sqrt{\frac{13}{4}}$ (C) $\sqrt{\frac{7}{4}}$ (D) $\sqrt{\frac{9}{2}}$

Sol. **[B]**

Proper Method

Given: $9x^2 - 4y^2 + 18x - 8y - 31 = 0$ ...(1)

$\Rightarrow 9(x^2 + 2x + 1) - 4(y^2 + 2y + 1) = 31 + 9 - 4$

$\Rightarrow 9(x + 1)^2 - 4(y + 1)^2 = 36$

Dividing throughout by 36, we get

$$\frac{(x+1)^2}{4} - \frac{(y+1)^2}{9} = 1 \quad ...(2)$$

Eccentricity

$$e = \frac{\sqrt{a^2 + b^2}}{a} = \frac{\sqrt{13}}{2}$$

Short Trick

$$\sqrt{1 + \frac{9}{4}} = \frac{\sqrt{13}}{2}$$

Q.17 The eccentricity of $x^2 - 3y^2 - 4x - 6y - 11 = 0$ is -

(A) $\frac{2}{\sqrt{5}}$ (B) $\frac{5}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{2}}$

Sol. **[C]**

Proper Method

Given : $x^2 - 3y^2 - 4x - 6y - 11 = 0$ ...(1)

$\Rightarrow (x^2 - 4x) - 3(y^2 + 2y) = 11$

$\Rightarrow (x^2 - 4x + 4) - 3(y^2 + 2y + 1) = 11 + 4 - 3$

$\Rightarrow [x - 2]^2 - 3[y + 1]^2 = 12$

$$\Rightarrow \frac{[x-2]^2}{12} - \frac{[y+1]^2}{4} = 1 \quad ...(2)$$

Eccentricity

$$e = \frac{\sqrt{a^2 + b^2}}{a} = \frac{\sqrt{16}}{2\sqrt{3}} = \frac{4}{2\sqrt{3}} = \frac{2}{\sqrt{3}}$$

Short Trick

$$e = \sqrt{1 + \frac{1}{3}} = \frac{2}{\sqrt{3}}$$

Trick -7 Question solved with the help of diagram

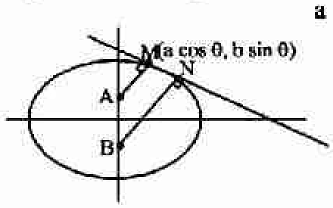
Q.18 The sum of the squares of the perpendicular on any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from two points on the minor axis each distance $\sqrt{a^2 - b^2}$ from the centre is -

(A) a^2 (B) b^2 (C) $2a^2$ (D) $2b^2$

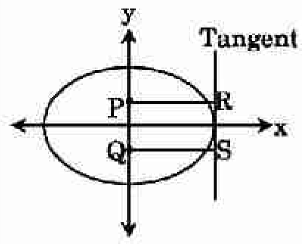
Sol. **[C]**

Proper Method

Equation of tangent is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$



Short Trick



$(PR)^2 + (QS)^2 = a^2 + a^2 = 2a^2$

$$AM = \frac{0+\frac{\sqrt{a^2-b^2}}{b}\sin\theta-1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

$$BN = \frac{0-\frac{\sqrt{a^2-b^2}}{b}\sin\theta-1}{\sqrt{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}}$$

$$\text{Now } AM^2+BN^2 = \frac{2\left(\frac{(a^2-b^2)}{b^2}\sin^2\theta+1\right)}{\frac{\cos^2\theta}{a^2}+\frac{\sin^2\theta}{b^2}}$$

$$= \frac{2(a^2\sin^2\theta-b^2\sin^2\theta+b^2)}{b^2}\times\frac{a^2b^2}{(b^2\cos^2\theta+a^2\sin^2\theta)}$$

$$= \frac{2a^2(a^2\sin^2\theta+b^2\cos^2\theta)}{(b^2\cos^2\theta+a^2\sin^2\theta)}$$

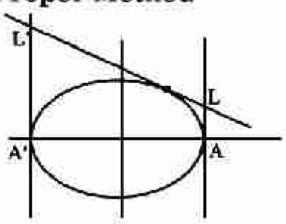
$= 2a^2$

Q.19 The tangent at any point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the tangents at the vertices A, A′ in L and L′. Then AL. A′L′ =

(A) a + b (B) $a^2 + b^2$ (C) a^2 (D) b^2

Sol. [D]

Proper Method



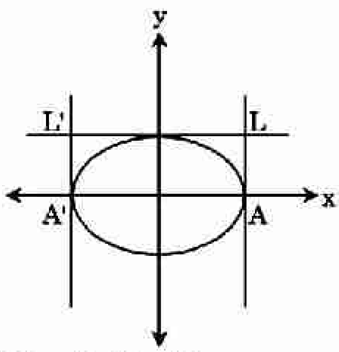
$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$$

$$L \equiv \left(a, \frac{b(1-\cos\theta)}{\sin\theta}\right)$$

$$L' \equiv \left(-a, \frac{b(1+\cos\theta)}{\sin\theta}\right)$$

$$\text{Now AL . A′L′} = \frac{b^2(1-\cos^2\theta)}{\sin^2\theta} = b^2$$

Short Trick



(AL) (A′L′) = (b) (b) = b^2

Q.20 The product of perpendiculars on a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ drawn its foci is

(A) a^2 (B) b^2 (C) $a^2 + b^2$ (D) $\sqrt{a^2+b^2}$

Sol. [B]

Proper Method

Equation of any tangent to the given ellipse is

$\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1$

$\Rightarrow bx\cos\phi + ay\sin\phi = ab$ .... (1)

Product of perpendiculars drawn from foci (ae, 0), (–ae, 0) on (1)

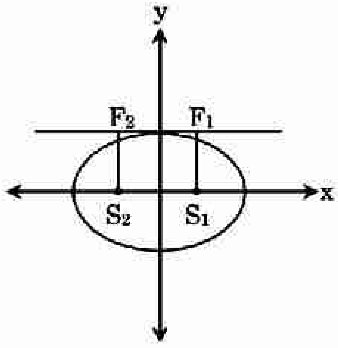
$= \frac{|(abe\cos\phi - ab)(-abe\cos\phi - ab)|}{b^2\cos^2\phi + a^2\sin^2\phi}$

$= \frac{b^2(a^2 - a^2e^2\cos^2\phi)}{b^2\cos^2\phi + a^2\sin^2\phi}$

$= \frac{b^2[a^2 - (a^2 - b^2)\cos^2\phi]}{b^2\cos^2\phi + a^2\sin^2\phi}$ $(\because a^2e^2 = a^2 - b^2)$

$= \frac{b^2(b^2\cos^2\phi + a^2\sin^2\phi)}{b^2\cos^2\phi + a^2\sin^2\phi} = b^2$

Short Trick



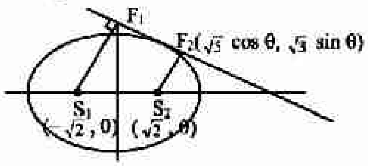
$(S_1F_1)(S_2F_2) = b^2$

Q.21 If F_1 and F_2 are the feet of the perpendiculars from the foci S_1 & S_2 of an ellipse $\frac{x^2}{5} + \frac{y^2}{3} = 1$ on the tangent at any point P on the ellipse, then (S_1F_1). (S_2F_2) is equal to-

(A) 2 (B) 3 (C) 4 (D) 5

Sol. [B]

Proper Method



$e = \frac{\sqrt{5-3}}{\sqrt{5}} = \sqrt{\frac{2}{5}}$

$ae = \sqrt{2}$

$\frac{x\cos\theta}{\sqrt{5}} + \frac{y\sin\theta}{\sqrt{3}} = 1$

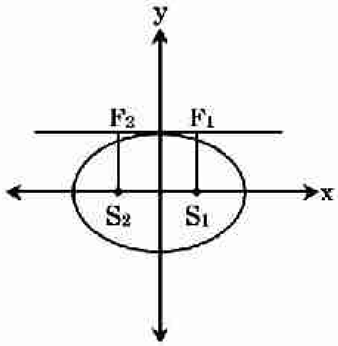
(S_2F_2) (S_1F_1)

$= \frac{\left(\frac{-\sqrt{2}\cos\theta}{\sqrt{5}} - 1\right)}{\sqrt{\frac{\cos^2\theta}{5} + \frac{\sin^2\theta}{3}}} \times \frac{\left(\frac{\sqrt{2}\cos\theta}{\sqrt{5}} - 1\right)}{\sqrt{\frac{\cos^2\theta}{5} + \frac{\sin^2\theta}{3}}}$

$= -\frac{\left(\frac{2}{5}\cos^2\theta - 1\right)}{\frac{\cos^2\theta}{5} + \frac{1-\cos^2\theta}{3}}$

$= \frac{5 - 2\cos^2\theta}{5} \times \frac{15}{3\cos^2\theta + 5 - 5\cos^2\theta} = 3$

Short Trick



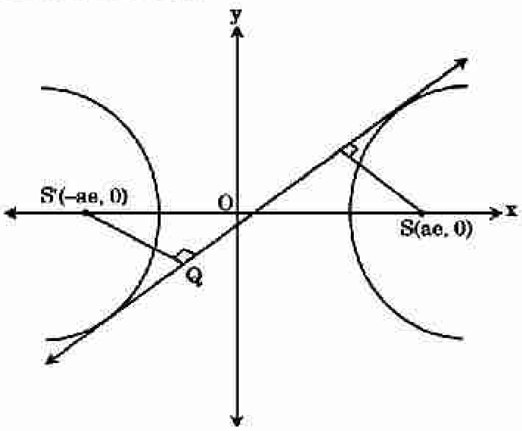
$(S_1F_1)(S_2F_2) = b^2 = 3$

Q.22 The product of the lengths of the perpendiculars drawn from foci on any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is -

(A) a^2 (B) b^2 (C) a^2b^2 (D) a^2 / b^2

Sol. **[B]**

Proper Method



Let the equation of tangent $\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$

$$SP.S'Q = \left|\frac{\frac{ae}{a}\sec\theta - 1}{\sqrt{\frac{\sec^2\theta}{a^2} + \frac{\tan^2\theta}{b^2}}}\right| \left|\frac{-\frac{ae}{a}\sec\theta - 1}{\sqrt{\frac{\sec^2\theta}{a^2} + \frac{\tan^2\theta}{b^2}}}\right|$$

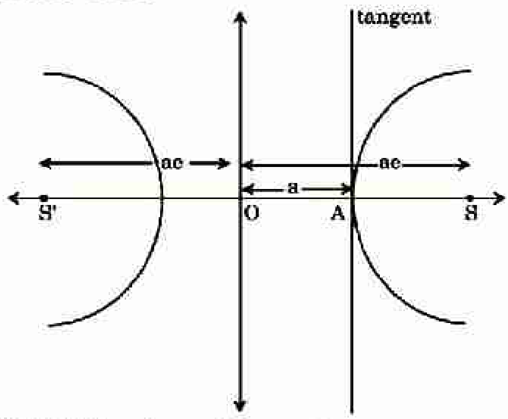
$$SP.S'Q = \frac{e^2\sec^2\theta - 1}{\frac{\sec^2\theta}{a^2} + \frac{\tan^2\theta}{b^2}} = \frac{a^2b^2(e^2\sec^2\theta - 1)}{b^2\sec^2\theta + a^2\tan^2\theta}$$

$$= \frac{a^2b^2(e^2\sec^2\theta - 1)}{a^2(e^2-1)\sec^2\theta + a^2(\sec^2\theta - 1)}$$

$$= \frac{a^2b^2(e^2\sec^2\theta - 1)}{a^2(e^2\sec^2\theta - 1)}$$

$= b^2$

Short Trick



$(SA)(S'A) = (ae - a)(ae + a)$

$= a^2(e^2 - 1)$

$= b^2$

13

Vector

KEY CONCEPTS

1. Scalar & Vector Quantity

(i) **Scalar :** A physical quantity which is completely specified by its magnitude only is called **scalar**. It is represented by a real number along with suitable unit.
Example, Distance, Mass, Length, Time, Volume, Speed.

(ii) **Vector :** On the other hand, a physical quantity which has magnitude as well as direction is called a **vector.**
Example, Displacement, velocity, acceleration, force etc.

2. Representation of Vector :

Vectors are represented by directed straight line segment
magnitude of $\vec{a} = |\vec{a}| =$ length AB
direction of $\vec{a} =$ A to B.
"AB" is magnitude of vector and its direction is from A to B.

3. Types of Vectors

(i) **Zero or Null Vector :** A vector whose magnitude is zero is called zero or null vector & it is denoted by **0** or $\vec{0}$.

(ii) **Unit Vector :** A vector of unit magnitude is called a unit vector. A unit vector in the direction of $\vec{a}$ is denoted by $\hat{a}$. Thus $\hat{a} = \dfrac{\vec{a}}{|a|}$

(iii) **Equal vector :** Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal, if $|\vec{a}| = |\vec{b}|$ & they have the same direction.

(iv) **Negative Vector :** A vector having the same magnitude as that of a given vector $\vec{a}$ and the direction opposite to that of $\vec{a}$ is called the negative vector $\vec{a}$ and it is denoted by $-\vec{a}$.

(v) **Collinear or Parallel Vectors :** Two vectors are said to be collinear if their directed lines segments are parallel disregards to their direction. Collinear vectors are also called parallel vectors. Two non zero vectors $\vec{a}$ and $\vec{b}$ are collinear or parallel iff $\boxed{\vec{a} = \lambda\vec{b}}$. where $\lambda \in R_0$.

(vi) **Like and Unlike Vectors :** Vectors are said to be like when they have the same direction and unlike when they have opposite directing.

(vii) **Free Vector :** If the initial point of a vector is not specified, then it is said to be a free vector.

(viii) **Localised Vectors :** A vector which is drawn parallel to a given vector through a specified point in space is called localised vector.

(ix) **Coterminus Vectors :** Vectors having the same terminal point are called coterminus vectors.

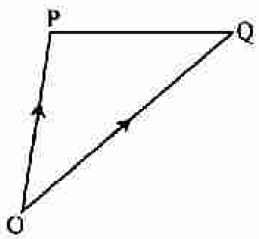
(x) **Coplanar Vectors :** A given number of vectors are called coplanar if their supports are parallel to the same plane.

(xi) **Position Vector of a point :** The position vector of a point P with respect to a fixed point say O, is the vector $\overrightarrow{OP}$. The fixed point is called the origin.

Let $\overrightarrow{PQ}$ be any vector. We have,

$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ} = -\overrightarrow{OP} + \overrightarrow{OQ} = \overrightarrow{OQ} - \overrightarrow{OP}$

= Position vector of $\vec{O}$ – Position vector of $\vec{P}$.



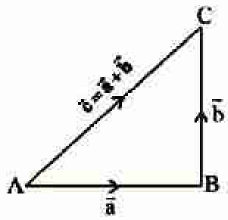
i.e. $\overrightarrow{PQ}$ = PV or $\vec{Q}$ – PV of $\vec{P}$

4. Addition of Vectors

(i) **Triangle Law of Addition :** If two vectors are represented by two consecutive sides of a triangle then their sum is represented by the third side of the triangle but in opposite direction. This is known as the triangle law of addition of vectors. Thus,

if $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{BC} = \vec{b}$, and $\overrightarrow{AC} = \vec{c}$

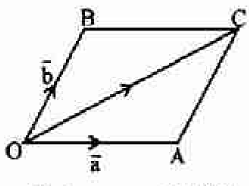
then $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ i.e. $\vec{a} + \vec{b} = \vec{c}$



(ii) **Parallelogram Law of Addition :** If two vectors are represented by two adjacent sides of a parallelogram, then their sum is represented by the diagonal of the parallelogram.

Thus if $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, and $\overrightarrow{OC} = \vec{c}$

then $\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ i.e. $\vec{a} + \vec{b} = \vec{c}$



Where OC is a diagonal of the parallelogram OABC.

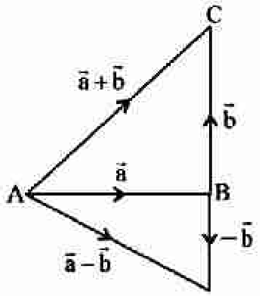
(iii) Addition in Component Form :

If $\vec{a} = a_1\mathbf{i} + a_2\,\mathbf{j} + a_3\mathbf{k}$ and $\vec{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$

then their sum is defined as $\vec{a} + \vec{b} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k}$.

5. Subtraction of Vectors

If $\vec{a}$ and $\vec{b}$ are two vectors, then their subtraction $\vec{a} - \vec{b}$ is defined as $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$



where $-\vec{b}$ is the negative of $\vec{b}$ having magnitude equal to that of $\vec{b}$ and direction opposite to $\vec{b}$.

6. Multiplication of a Vector by a Scalar

If $\vec{a}$ is a vector and m is a scalar (i.e. a real number) then $m\vec{a}$ is a vector whose magnitude is m times that of $\vec{a}$ and whose direction is the same as that of $\vec{a}$.

If $\vec{a} = a_1\mathbf{i} + a_2\,\mathbf{j} + a_3\mathbf{k}$ then $m\vec{a} = (ma_1)\mathbf{i} + (ma_2)\mathbf{j} + (ma_3)\mathbf{k}$

7. Distance between Two Points

Let A and B be two given points whose coordinate are respectively (x_1, y_1, z_1) and (x_2, y_2, z_2)

If $\vec{a}$ and $\vec{b}$ are p.v. of A and B relative to point O, then

$\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$; $\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

Now $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \vec{b} - \vec{a} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

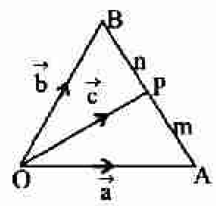
Distance between the points and = magnitude of $\overrightarrow{AB}$

$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

8. Section Formula

(i) Internal Division : If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A and B, then the position vector c of a point P dividing AB in the ratio m : n is given by

$$\vec{c} = \frac{m\vec{b} + n\vec{a}}{m+n}$$



(ii) **External Division :** If the point P divides AB in the ratio m : n externally, then p.v. $\bar{c}$ of P is given by $\bar{c} = \dfrac{m\bar{b} - n\bar{a}}{m-n}$

(iii) **Angle bisector :** Any vector along the internal bisector of $\angle AOB$ is given by $\lambda(\hat{a}+\hat{b})$ or $\lambda\left(\dfrac{\bar{a}}{|\bar{a}|}+\dfrac{\bar{b}}{|\bar{b}|}\right)$

(iv) **Centroid of Triangle :** If $\vec{a}, \vec{b}, \vec{c}$ are position vectors of vertices of a triangle, then p.v. of its centroid is $\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

(v) **Centroid of Tetrahedron :** If a,b,c,d are position vectors of vertices of a tetrahedron, then p.v. of its centroid is $\dfrac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$

9. Collinearity of Three Points

(i) If $\bar{a}$, $\bar{b}$, $\bar{c}$ be position vectors of three points A,B and C respectively and x, y, z be three scalars so that all are not zero, then the necessary and sufficient conditions for three points to be collinear is that $x\bar{a} + y\bar{b} + z\bar{c} = 0$ and $x + y + z = 0$

(ii) Three points A, B and C are collinear, if any two vectors $\overline{AB}$, $\overline{BC}$ and $\overline{CA}$ are parallel i.e. one of them is scalar multiple of any one of the remaining vectors.

If $\overline{AB} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\overline{BC} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ then from the property of parallel vector, we have

$\overline{AB} \parallel \overline{BC} \Rightarrow \dfrac{a_1}{b_1} = \dfrac{a_2}{b_2} = \dfrac{a_3}{b_3}$

10. Coplanar & Non-Coplanar Vector

(i) If $\vec{a}, \vec{b}, \vec{c}$ be three coplanar vectors, then a vector c can be expressed uniquely as linear combination of remaining two vectors i.e.

$\vec{c} = \lambda\vec{a} + \mu\vec{b}$, Where λ and μ are suitable scalars.

Again $\vec{c} = \lambda\vec{a} + \mu\vec{b} \Rightarrow$ vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.

If $\vec{a}, \vec{b}, \vec{c}$ be three coplanar vectors, then there exist three non zero scalars x, y, z

so that, $x\vec{a} + y\vec{b} + z\vec{c} = 0$

(ii) If $\vec{a}, \vec{b}, \vec{c}$ be three non coplanar non zero vector then

$x\vec{a} + y\vec{b} + z\vec{c} = 0 \Rightarrow x = 0, y = 0, z = 0$

(iii) Any vector r can be expressed uniquely as the linear combination of three non coplanar and non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$

i.e. $r = x\vec{a} + y\vec{b} + z\vec{c}$ where x, y and z are scalars.

11. Linear Combination of Vectors

Given a finite set of vector $\vec{a}, \vec{b}, \vec{c}$ .... then the vector $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + ...$ is called a linear combination of $\vec{a}$, $\vec{b}$, $\vec{c}$ .... for any x, y, z, ... $\in$ R.

(i) Linear Independent :

If $\bar{x}_1, \bar{x}_2, ..., \bar{x}_n$ are n non zero vectors, and $k_1, k_2, ... k_n$ are n scalars and if the linear combination $k_1\bar{x}_1 + k_2\bar{x}_2 + ...+ k_n\bar{x}_n = 0$,

$\Rightarrow$ $k_1 = 0, k_2 = 0, ... k_n = 0$ then, we say that vectors $\bar{x}_1, \bar{x}_2, ..., \bar{x}_n$ are Linearly Independent vectors.

(ii) Linear dependent :

If $k_1\vec{x}_1 + k_2\vec{x}_2$ .... $k_n\vec{x}_n = 0$ and if there exists at least one $k_r \neq 0$ then $\vec{x}_1, \vec{x}_2$ .... are said to be Linearly Dependent.

Note :

- Two non-collinear vectors $\vec{a}$ and $\vec{b}$ are linearly independent.
- Three non-coplanar vectors, $\vec{a}$, $\vec{b}$ and $\vec{c}$ are linearly independent.
- More than three vectors are always linearly dependent.

12. Product of Vectors

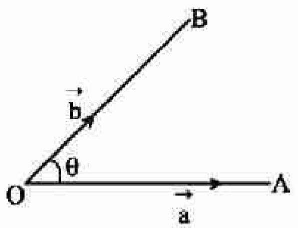
(i) Dot Product : Product of two vectors is done by two methods when the product of two vectors results in a scalar quantity then it is called scalar product. It is also called as dot product because this product is represented by putting a dot (.).

(ii) Vector Product : When the product of two vectors results in a vector quantity then this product is called Vector Product. This product is represented by (×) sign so that it is also called as Cross Product.

13. Scalar or Dot Product of Two Vectors

If $\vec{a}$ and $\vec{b}$ are two non zero vectors and θ be the angle between them, then their scalar product (or dot product) is defined as

$\vec{a} . \vec{b} = |\vec{a}| \, |\vec{b}| \cos\theta = a\,b\,\cos\theta$



Note :

(i) If $\vec{a}$ and $\vec{b}$ are perpendicular to each other then $\theta = \pi/2$, so

$\vec{a} . \vec{b} = 0$

i.e. the scalar product of two perpendicular vectors is always zero.

But its converse may not be true i.e.

$\vec{a} . \vec{b} = 0 \not\Rightarrow \vec{a} \perp \vec{b}$

But if a and b are non zero vectors, then

$\vec{a} . \vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$

Thus $\vec{a} \neq 0$, $\vec{b} \neq 0$, $\vec{a} . \vec{b} = 0, \Rightarrow \vec{a} \perp \vec{b}$

(ii) If $\vec{a}$ and $\vec{b}$ are like vectors, then $\theta = 0$ so $\vec{a}.\vec{b} = |\vec{a}|\,|\vec{b}| = \vec{a}\,\vec{b}$

In particular $\vec{a}.\vec{a} = |\vec{a}|^2$

Let $\vec{a} = \vec{a}_1\,\hat{i} + \vec{a}_2\,\hat{j} + \vec{a}_3\,\hat{k}$, then $\vec{a}.\vec{a} = |\vec{a}|^2 = \vec{a}_1{}^2 + \vec{a}_2{}^2 + \vec{a}_3{}^2$

(iii) If $\vec{a}$ and $\vec{b}$ are unlike vectors then $\theta = \pi$ so

$\vec{a}.\vec{b} = -\vec{a}\,\vec{b}$.

i.e. the scalar product of two perpendicular vectors is always zero.
But its converse may not be true i.e.

$\vec{a}.\vec{b} = 0 \not\Rightarrow \vec{a} \perp \vec{b}$

But if a and b are non zero vectors, then

$\vec{a}.\vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$

Thus $\vec{a} \neq 0$, $\vec{b} \neq 0$, $\vec{a}.\vec{b} = 0, \Rightarrow \vec{a} \perp \vec{b}$

(iv) With the help of the above cases, we get the following important results:

(a) $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$

(b) $\hat{i}.\hat{j} = \hat{j}.\hat{k}.\hat{k}.\hat{j} = 0$

14. Properties of Scalar Product

(i) (a . b) . b is not defined

(ii) $(a + b)^2 = a^2 + 2\,a.b + b^2$

(iii) $(a - b)^2 = a^2 - 2a.b + b^2$

(iv) $(a + b).(a - b) = a^2 - b^2 = a^2 - b^2$

(v) $|a + b| = |a| + |b| \Rightarrow a \,||\, b$

(vi) $|a + b|^2 = |a|^2 + |b|^2 \Rightarrow a \perp b$

(vii) $|a + b| = |a - b| \Rightarrow a \perp b$

15. Scalar product in terms of Components :

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\vec{a} = \vec{a}_1\,\hat{i} + \vec{a}_2\,\hat{j} + \vec{a}_3\,\hat{k}$

and $\vec{b} = \vec{b}_1\,\hat{i} + \vec{b}_2\,\hat{j} + \vec{b}_3\,\hat{k}$

Then $\vec{a}.\vec{b} = \vec{a}_1\,\vec{b}_1 + \vec{a}_2\,\vec{b}_2 + \vec{a}_3\,\vec{b}_3$

16. Angle between Two Vectors

(i) If $\vec{a}$ and $\vec{b}$ be two vectors and θ be the angle between them, then

$$\cos\theta = \frac{\vec{a}.\vec{b}}{|\vec{a}|\,|\vec{b}|}$$

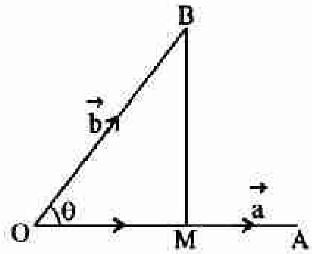
(ii) If $\vec{a} = \vec{a}_1\,\hat{i} + \vec{a}_2\,\hat{j} + \vec{a}_3\,\hat{k}$ and

$\vec{b} = \vec{b}_1\,\hat{i} + \vec{b}_2\,\hat{j} + \vec{b}_3\,\hat{k}$ then

$$\cos\theta = \frac{\vec{a}_1\vec{b}_1 + \vec{a}_2\vec{b}_2 + \vec{a}_3\vec{b}_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}$$

If $\vec{a}$ and $\vec{b}$ are perpendicular to each other then $\vec{a}\cdot\vec{b} = 0$ or $a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$

17. Projection & Components of a Vector



(i) Similarly projection of $\vec{a}$ on $\vec{b}$ = $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

(ii) Projection of $\vec{b}$ on $\vec{a}$ = $\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$

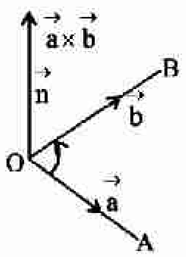
(iii) Component of $\vec{b}$ along $\vec{a}$ $(\overline{OM}) = \dfrac{(\vec{a}\cdot\vec{b})}{|\vec{a}|^2}.\vec{a}$

(iv) Component of $\vec{b}$ perpendicular to $\vec{a}$ $(\overline{MB}) = \vec{b} - \dfrac{(\vec{a}.\vec{b})}{a^2}.\vec{a}$

18. Work Done by the Force

If a constant force F acting on a particle displaces it from point A to B, then work done by the force $W = f.d$ (where $d = \overline{AB}$)

19. Vector or Cross Product of Two Vectors



If $\vec{a}$ and $\vec{b}$ be two vectors and θ $(0 \le \theta \le \pi)$ be the angle between them, then their vector (or cross) product is defined as

$\therefore\ \vec{a}\times\vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n}$

$\qquad\quad = \vec{a}\,\vec{b}\,\sin\theta\,\hat{n}$

Where $\hat{n}$ is a unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$ such that $\vec{a}$, $\vec{b}$ and $\hat{n}$ form a right handed system.

(i) if vectors a and b are parallel, then $\vec{a} \times \vec{b} = 0$

In particular $\vec{a} \times \vec{b} = 0$

(ii) If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, then $\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}|\,\hat{n}$

(iii) If $\hat{i}, \hat{j}, \hat{k}$ be three mutually perpendicular unit vectors, then

(a) $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$

(b) $\hat{i} \times \hat{j} = \hat{k}$, $\hat{i} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{i}$

(c) $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{j} = -\hat{i}$, $\hat{i} \times \hat{k} = -\hat{j}$

20. Vector Product in Terms of Components

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\,\hat{i} + (a_3b_1 - a_1b_3)\,\hat{j} + (a_1b_2 - a_2b_1)\,\hat{k}$$

21. Angle between Two Vectors

If θ is the angle between $\vec{a}$ and $\vec{b}$, then

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$$

If $\hat{n}$ is the unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$, then

$$\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

22. Properties of Vector Product

If a, b, c are any vectors and m, n any scalars then

(i) $a \times b \neq b \times a$ (Non-commutativity) but $a \times b = -(b \times a)$ and $|a \times b| = |b \times a|$

(ii) $(ma) \times b = a \times (mb) = m(a \times b)$

(iii) $(ma) \times (nb) = (mn)(a \times b)$

(iv) $a \times (b \times c) \neq (a \times b) \times c$

(v) $a \times (b + c) = (a \times b) + (a \times c)$ (Distributivity)

(vi) $a \times b = a \times c \not\Rightarrow b = c$

23. Area of Triangle

(i) Area of triangle ABC $= \frac{1}{2}|\overline{AB} \times \overline{AC}|$

(ii) If $\vec{a}, \vec{b}, \vec{c}$ are position vectors of vertices of a ΔABC then its

Area $= \frac{1}{2}\,|(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a})|$

(iii) Three points with position vectors $\vec{a}, \vec{b}, \vec{c}$ are collinear if

$(\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) = 0$

24. Area of Parallelogram

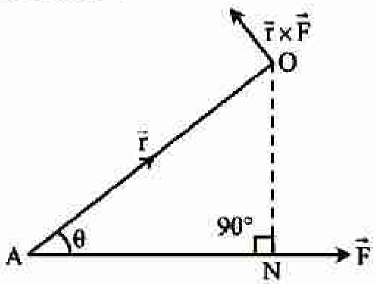
(i) If $\vec{a}$ and $\vec{b}$ are two adjacent sides of a parallelogram then the area = $|\vec{a} \times \vec{b}|$

(ii) If $\vec{a}$ and $\vec{b}$ represent two diagonals of a parallelogram then the area = $\frac{1}{2}\,|\vec{a} \times \vec{b}|$

(iii) Area of a quadrilateral ABCD = $\frac{1}{2}\,|\overrightarrow{AC} \times \overrightarrow{BD}|$

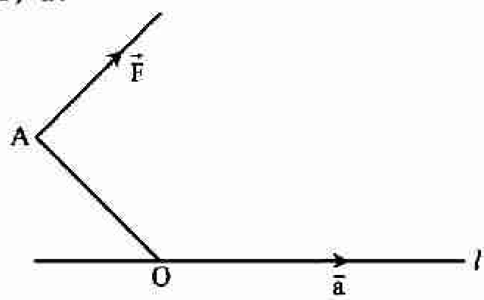
25. Moment of Force

(i) **Moment of force about a point :** The vector moment of torque $\vec{M}$ of a force $\vec{F}$ about the point O is the vector whose magnitude is equal to the product of $\vec{F}$ and the perpendicular distance of the point O from the line of action $\vec{F}$.



$\overrightarrow{M} = \overrightarrow{OA} \times \vec{F} = \vec{r} \times \vec{F}$

(ii) **The moment of a force about a line :** Let $\vec{F}$ be a force acting at a point A, O be any point on the given line l and $\vec{a}$ be the unit vector along the line, then moment of $\vec{F}$ about the line l is a scalar given by $(\overrightarrow{OA} \times \vec{F}) \cdot \vec{a}$.



26. Formula for Scalar Triple Product

(i) If **a** = $a_1\ell + a_2m + a_3n$, **b** = $b_1\ell + b_2m + b_3n$ and **c** = $c_1\ell + c_2m + c_3n$, then scalar triple product is

$$[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(ii) For any three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$

(a) $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}] = 2\,[\vec{a}\ \vec{b}\ \vec{c}]$

(b) $[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}] = 0$

(c) $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = 2\,[\vec{a}\ \vec{b}\ \vec{c}]^2$

27. Properties of Scalar Triple Product

(i)) The position of (.) and (×) can be interchanged i.e.

$\vec{a}.(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b}).\vec{c}$

but $(\vec{a}\times\vec{b}).\vec{c} = \vec{c}.(\vec{a}\times\vec{b})$

So $[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}] = [\vec{c}\ \vec{a}\ \vec{b}]$

Therefore if we don't change the cyclic order of a, b and c then the value of scalar triple product is not changed by interchanging dot and cross.

(ii) If the cyclic order of vectors is changed, then sign of scalar triple product is changed i.e.

$\vec{a}.[\vec{b}\times\vec{c}] = -\vec{a}.(\vec{c}\times\vec{b})$

or $[\vec{a}\ \vec{b}\ \vec{c}] = -[\vec{a}\ \vec{c}\ \vec{b}]$

from (i) and (ii) we have

$[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}] = [\vec{c}\ \vec{a}\ \vec{b}]$

$= -[\vec{a}\ \vec{c}\ \vec{b}] = -[\vec{b}\ \vec{a}\ \vec{c}] = -[\vec{c}\ \vec{b}\ \vec{a}]$

(iii) The scalar triple product of three vectors when two of them are equal or parallel, is zero i.e.

$[\vec{a}\ \vec{b}\ \vec{b}] = [\vec{a}\ \vec{b}\ \vec{a}] = 0$

(iv) The scalar triple product of three mutually perpendicular unit vectors is ±1 Thus

$[\hat{i}\ \hat{j}\ \hat{k}] = 1, [\hat{i}\ \hat{k}\ \hat{j}] = -1$

(v)) Three non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, if and only if $[\vec{a}\ \vec{b}\ \vec{c}] = 0$.

(vi) Four points A, B, C, D with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively are coplanar, if and only if

$[\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0$

i.e. if and only if $[\vec{b}-\vec{a}\ \ \vec{c}-\vec{a}\ \ \vec{d}-\vec{a}] = 0$

(vii) For any vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$

$[\vec{a}+\vec{b}\ \ \vec{c}\ \ \vec{d}] = [\vec{a}\ \vec{c}\ \vec{d}] + [\vec{b}\ \vec{c}\ \vec{d}]$

(viii) Scalar product of four vector's : $(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d}) = \begin{vmatrix} \vec{a}.\vec{c} & \vec{b}.\vec{c} \\ \vec{a}.\vec{d} & \vec{b}.\vec{d} \end{vmatrix}$

(ix) $[\vec{a}\ \vec{b}\ \vec{c}][\vec{u}\ \vec{v}\ \vec{w}] = \begin{vmatrix} \vec{a}.\vec{u} & \vec{b}.\vec{u} & \vec{c}.\vec{u} \\ \vec{a}.\vec{v} & \vec{b}.\vec{v} & \vec{c}.\vec{v} \\ \vec{a}.\vec{w} & \vec{b}.\vec{w} & \vec{c}.\vec{w} \end{vmatrix}$

28. Volume of Parallelopiped

If coterminous edges of a parallelopiped are **a**, **b** and **c** then volume = **[a b c]**

29. Volume of Tetrahedron

(i) If **a**, **b**, **c** are position vectors of vertices A, B and C with respect to O, then volume of tetrahedron OABC $= \frac{1}{6}$ [**a** **b** **c**]

(ii) If **a**, **b**, **c**, **d** are position vectors of vertices A, B, C, D of a tetrahedron ABCD, then its volume

$$= \begin{cases} \frac{1}{6}\left[\overline{AB} \quad \overline{AC} \quad \overline{AD}\right] \\ \text{or} \\ \frac{1}{6}\left[\mathbf{b}-\mathbf{a} \quad \mathbf{c}-\mathbf{a} \quad \mathbf{d}-\mathbf{a}\right] \end{cases}$$

30. Vector Triple Product

The vector triple product of three vectors $\vec{a}, \vec{b}, \vec{c}$ is defined as the vector product of two vectors $\vec{a}$ and $\vec{b} \times \vec{c}$. It is denoted by $\vec{a} \times (\vec{b} \times \vec{c})$.

$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c}$

$(\vec{b} \times \vec{c}) \times \vec{a} = (\vec{b}.\vec{a})\vec{c} - (\vec{c}.\vec{a})\vec{b}$.

The direction of $\vec{a} \times (\vec{b} \times \vec{c})$ is perpendicular to $\vec{a}$ and parallel to and $\vec{b}$ and $\vec{c}$.

Note :

- $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$

31. Reciprocal System of Vectors

If $\vec{a}, \vec{b}, \vec{c}$ be any three non coplanar vectors so that $[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$ then the three vectors $\vec{a}', \vec{b}', \vec{c}'$ define equation

$$\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]},\ \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]},\ \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

are called reciprocal system of vectors to the $\vec{a}, \vec{b}, \vec{c}$

32. Properties of reciprocal system of vectors

(a) $\vec{a}.\vec{a}' = \vec{b}.\vec{b}' = \vec{c}.\vec{c}' = 1$

(b) $\vec{a}.\vec{b}' = \vec{a} \cdot \vec{c}' = 0,\ \vec{b} \cdot \vec{a}' = \vec{b} \cdot \vec{c}' = 0,\ \vec{c} \cdot \vec{a}' = \vec{c} \cdot \vec{b}' = 0$

(c) $[\vec{a}\ \vec{b}\ \vec{c}]\,[\vec{a}'\ \vec{b}'\ \vec{c}'] = 1$

(d) $\vec{a} = \frac{\vec{b}' \times \vec{c}'}{[\vec{a}'\ \vec{b}'\ \vec{c}']}, \vec{b} = \frac{\vec{c}' \times \vec{a}'}{[\vec{a}'\ \vec{b}'\ \vec{c}']}, \vec{c} = \frac{\vec{a}' \times \vec{b}'}{[\vec{a}'\ \vec{b}'\ \vec{c}']}$

Note :

- System of $\hat{i}, \hat{j}, \hat{k}$ are its own Reciprocal System of vector's.

SHORT TRICKS

Trick -1 Method of substitution

Q.1 If $\vec{p}$, $\vec{q}$, $\vec{r}$ be three mutually perpendicular vectors of equal magnitude, then the angle between $\vec{p}$ and $\vec{p}+\vec{q}+\vec{r}$ is-

(A) $\cos^{-1}(1/\sqrt{3})$ (B) $\sin^{-1}(1/\sqrt{3})$ (C) $\cos^{-1}(1/3)$ (D) $\sin^{-1}(1/3)$

Sol. **[A]**

Proper Method	Short Trick
$\vec{p}.\vec{q} = \vec{q}.\vec{r} = \vec{r}.\vec{p} = 0$ $\lvert\vec{p}\rvert = \lvert\vec{q}\rvert = \lvert\vec{r}\rvert = x$ $\cos\theta = \dfrac{\vec{p}.(\vec{p}+\vec{q}+\vec{r})}{\lvert\vec{p}\rvert\lvert\vec{p}+\vec{q}+\vec{r}\rvert} = \dfrac{\lvert\vec{p}\rvert^2}{\lvert\vec{p}\rvert\lvert\vec{p}+\vec{q}+\vec{r}\rvert}$ $\lvert\vec{p}+\vec{q}+\vec{r}\rvert^2 = \lvert\vec{p}\rvert^2+\lvert\vec{q}\rvert^2+\lvert\vec{r}\rvert^2 + 2(\vec{p}.\vec{q}+\vec{q}.\vec{r}+\vec{r}.\vec{p})$ $\Rightarrow \lvert\vec{p}+\vec{q}+\vec{r}\rvert^2 = 3x^2$ $\Rightarrow \lvert\vec{p}+\vec{q}+\vec{r}\rvert = \sqrt{3}x$ then $\cos\theta = \dfrac{x^2}{(x)(\sqrt{3}x)}$ $\cos\theta = \dfrac{1}{\sqrt{3}}$ $\theta = \cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$	Let $\vec{p} = \hat{i}$, $\vec{q} = \hat{j}$, $\vec{r} = \hat{k}$ $\cos\theta = \dfrac{\hat{i}.(\hat{i}+\hat{j}+\hat{k})}{\lvert\hat{i}\rvert\lvert\hat{i}+\hat{j}+\hat{k}\rvert} = \left(\dfrac{1}{\sqrt{3}}\right)$ $\Rightarrow \theta = \cos^{-1}\left(\dfrac{1}{\sqrt{3}}\right)$

Q.2 Let $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and $\vec{a}'$, $\vec{b}'$, $\vec{c}'$ are reciprocal system of vectors $\vec{a}, \vec{b}, \vec{c}$. Then value of the expression $\vec{a}\times\vec{a}' + \vec{b}\times\vec{b}' + \vec{c}\times\vec{c}'$ is

(A) a unit vector (B) null vector

(C) $\dfrac{\lvert\vec{a}\rvert^2+\lvert\vec{b}\rvert^2+\lvert\vec{c}\rvert^2}{\lvert\vec{a}'\rvert^2+\lvert\vec{b}'\rvert^2+\lvert\vec{c}'\rvert^2}$ (D) arbitrary vector

Sol. **[B]**

Proper Method	Short Trick
$\vec{a}\times\vec{a}' + \vec{b}\times\vec{b}' + \vec{c}\times\vec{c}'$ $= \vec{a}\times\dfrac{(\vec{b}\times\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]} + \vec{b}\times\dfrac{(\vec{c}\times\vec{a})}{[\vec{a}\,\vec{b}\,\vec{c}]} + \vec{c}\times\dfrac{(\vec{a}\times\vec{b})}{[\vec{a}\,\vec{b}\,\vec{c}]}$ $= \dfrac{(\vec{a}.\vec{c})\vec{b}-(\vec{a}.\vec{b})\vec{c}+(\vec{b}.\vec{a})\vec{c}-(\vec{b}.\vec{c})\vec{a}+(\vec{c}.\vec{b})\vec{a}-(\vec{c}.\vec{a})\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ $= \vec{0}$	Let $\vec{a} = \hat{i}$, $\vec{b} = \hat{j}$, $\vec{c} = \hat{k}$ then $\vec{a}' = \hat{i}$, $\vec{b}' = \hat{j}$, $\vec{c}' = \hat{k}$ $\vec{a}\times\vec{a}' + \vec{b}\times\vec{b}' + \vec{c}\times\vec{c}' = 0$

Q.3 Let $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors $\vec{a}, \vec{b}, \vec{c}$. Then value of the expression $\vec{a}' \times \vec{b}' + \vec{b}' \times \vec{c}' + \vec{c}' \times \vec{a}'$ is-

(A) $\frac{\vec{a}+\vec{b}-\vec{c}}{[\vec{a}\ \vec{b}\vec{c}]}$ (B) $\frac{\vec{a}-\vec{b}+\vec{c}}{[\vec{a}\ \vec{b}\vec{c}]}$ (C) $\frac{-\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\ \vec{b}\vec{c}]}$ (D) $\frac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\ \vec{b}\vec{c}]}$

Sol. **[D]**

Proper Method	**Short Trick**
$\vec{a}' \times \vec{b}' + \vec{b}' \times \vec{c}' + \vec{c}' \times \vec{a}'$ $= \frac{\vec{b}\times\vec{c}}{[\vec{a}\vec{b}\vec{c}]} \times \frac{\vec{c}\times\vec{a}}{[\vec{a}\vec{b}\vec{c}]} + \frac{\vec{c}\times\vec{a}}{[\vec{a}\vec{b}\vec{c}]} \times \frac{\vec{a}\times\vec{b}}{[\vec{a}\vec{b}\vec{c}]} + \frac{\vec{a}\times\vec{b}}{[\vec{a}\vec{b}\vec{c}]} \times \frac{\vec{b}\times\vec{c}}{[\vec{a}\vec{b}\vec{c}]}$ $= \frac{\{(\vec{b}\times\vec{c}).\vec{a}\}\vec{c} - 0 + \{(\vec{c}\times\vec{a}).\vec{b}\}\vec{a} - 0 + \{(\vec{a}\times\vec{b}).\vec{c}\}\vec{b} - 0}{[\vec{a}\vec{b}\vec{c}]^2}$ $= \frac{[\vec{b}\ \vec{c}\ \vec{a}]\vec{c} + [\vec{c}\ \vec{a}\ .\vec{b}]\vec{a} + [\vec{a}\ \vec{b}\ \vec{c}]\vec{b}}{[\vec{a}\vec{b}\vec{c}]^2} = \frac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\vec{b}\vec{c}]}$	Let $\vec{a} = \hat{i}, \vec{b} = \hat{j}, \vec{c} = \hat{k}$ then $\vec{a}' = \hat{i}$, $\vec{b}' = \hat{j}$, $\vec{c}' = \hat{k}$ $\vec{a}' \times \vec{b}' + \vec{b}' \times \vec{c}' + \vec{c}' \times \vec{a}' = \hat{i}\times\hat{j} + \hat{j}\times\hat{k} + \hat{k}\times\hat{i}$ $= \hat{k} + \hat{i} + \hat{j}$ Hence option (D) is correct answer

Q.4 Let $\vec{u} = \hat{i}+\hat{j}$, $\vec{v} = \hat{i} - \hat{j}$ and $\vec{w} = \hat{i} + 2\hat{j} + 3\hat{k}$. If $\hat{n}$ is a unit vector such that $\vec{u}\ .\ \hat{n} = 0$ and $\vec{v}\ .\ \hat{n} = 0$, then $|\ \vec{w}\ .\hat{n}\ |$ is equal to- **[AIEEE- 2003]**

(A) 3 (B) 0 (C) 1 (D) 2

Sol. **[A]**

Proper Method	**Short Trick**
Let $\vec{n} = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$ given $\|\vec{n}\| = 1 \Rightarrow n_1^2 + n_2^2 + n_3^2 = 1$ and $\vec{u}.\hat{n} = 0 \Rightarrow n_1 + n_2 = 0$ ..... (1) $\vec{v}.\hat{n} = 0 \Rightarrow n_1 - n_2 = 0$ ..... (2) Solving (1) & (2) we get $n_1 = 0$, $n_2 = 0$ and $n_3 = \pm 1$ $\therefore \vec{n} = \pm \hat{k}$ $\|\ \vec{w}\ .\hat{n}\ \| = \|\pm 3\| = 3$	Let $\hat{n} = \hat{i}$ satisfy all the given conditions $\therefore \|w.\ \hat{n}\ \| = 3$

Q.5 For any vector $\vec{a}$, $|\ \vec{a} \times \hat{i}\ |^2 + |\ \vec{a} \times \hat{j}\ |^2 + |\ \vec{a} \times \hat{k}\ |^2$ is equal to - **[AIEEE- 2005]**

(A) $|\ \vec{a}\ |^2$ (B) $2|\ \vec{a}\ |^2$ (C) $3|\ \vec{a}\ |^2$ (D) $4|\ \vec{a}\ |^2$

Sol. **[B]**

Proper Method	**Short Trick**
Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $\|\vec{a}\times\hat{i}\| = \|-a_2\hat{k} + a_3\hat{j}\| = \sqrt{a_2^2 + a_3^2}$ $\|\vec{a}\times\hat{i}\|^2 = a_2^2 + a_3^2$ Similarly $\|\vec{a}\times\hat{j}\|^2 = a_1^2 + a_3^2$ and $\|\vec{a}\times\hat{k}\|^2 = a_1^2 + a_2^2$ $\therefore \|\vec{a}\times\hat{i}\|^2 + \|\vec{a}\times\hat{j}\|^2 + \|\vec{a}\times\hat{k}\|^2 = 2(a_1^2 + a_2^2 + a_3^2)$ $= 2\|\ \vec{a}\ \|^2$	Let $\vec{a} = \hat{i}$ $\|\vec{a}\times\hat{i}\|^2 + \|\vec{a}\times\hat{j}\|^2 + \|\vec{a}\times\hat{k}\|^2 = 2\|\ \vec{a}\ \|^2$ Hence, option (B) is correct answer

Q.6 If $\vec{a}$ and $\vec{b}$ are two unit vectors and θ is the angle between them, then the unit vector along the angular bisector of $\vec{a}$ and $\vec{b}$ will be given by -

(A) $\dfrac{\vec{a}-\vec{b}}{2\cos\theta/2}$ (B) $\dfrac{\vec{a}+\vec{b}}{2\cos\theta/2}$ (C) $\dfrac{\vec{a}-\vec{b}}{\cos\theta/2}$ (D) $\dfrac{\vec{a}+\vec{b}}{\cos\theta/2}$

Sol. [B]

Proper Method

$$|\vec{a}+\vec{b}| = \sqrt{|\vec{a}|^2+|\vec{b}|^2+2|\vec{a}||\vec{b}|\cos\theta}$$

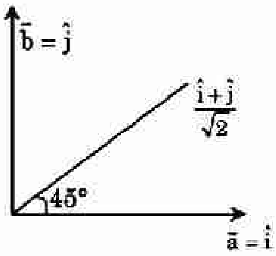
$$= \sqrt{2(1+\cos\theta)}$$

$$= \sqrt{4\cos^2(\theta/2)} = 2\cos(\theta/2)$$

$$\text{Angle bisector} = \frac{\vec{a}+\vec{b}}{|\vec{a}+\vec{b}|}$$

$$= \frac{\vec{a}+\vec{b}}{2\cos\theta/2}$$

Short Trick



Let a = $\hat{i}$, b = $\hat{j}$, $\theta = 90°$

Hence, options (B) is correct answer

Q.7 'P' is any arbitrary point on the circumcircle of the equilateral triangle of side length l units, then $|\overrightarrow{PA}| + |\overrightarrow{PB}|^2 + |\overrightarrow{PC}|^2$ is always equal to -

(A) $2l^2$ (B) $2\sqrt{3}\,l^2$ (C) l^2 (D) $3l^2$

Sol. [A]

Proper Method

Let position vector of P, A, B, and C are $\vec{p}$, $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively

O ≡ circumcentre of ΔABC

$$|\vec{p}| = |\vec{a}| = |\vec{b}| = |\vec{c}| = \frac{l}{\sqrt{3}}$$

$$|\overrightarrow{PA}|^2 = |\vec{a}-\vec{p}|^2$$

$$|\overrightarrow{PB}|^2 = |\vec{b}-\vec{p}|^2$$

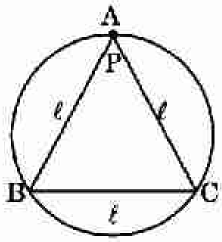
$$|\overrightarrow{PC}|^2 = |c-p|^2$$

$$|\overrightarrow{PA}|^2 + |\overrightarrow{PB}|^2 + |\overrightarrow{PC}|^2$$

$$= \frac{6l^3}{3} - 2\vec{p}.(\vec{a}+\vec{b}+\vec{c})$$

$$= 2l^2 \quad \text{as} \quad \frac{\vec{a}+\vec{b}+\vec{c}}{3} = 0$$

Short Trick



Let P lies on A then

$|PA|^2 + |PB|^2 + |PC|^2$

$= \ell^2 + \ell^2 = 2\ell^2$

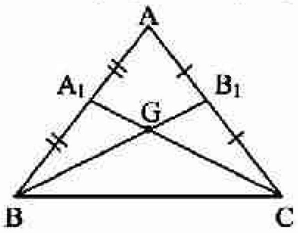
Q.8 G is the centroid of triangle ABC and A_1 and B_1 are the midpoints of sides AB and AC, respectively. If Δ_1 be the area of quadrilateral GA_1AB_1 and Δ be the area of triangle ABC, then Δ/Δ_1 is equal to -

(A) $\dfrac{3}{2}$ (B) 3 (C) $\dfrac{1}{3}$ (D) 5

Sol. [B]

Proper Method

Let position vector of A, B, C are $\vec{0}$, $\vec{b}$, $\vec{c}$ respectively



position vector of G = $\dfrac{\vec{b}+\vec{c}}{3}$

position vector of A_1 = $\dfrac{\vec{b}}{2}$

position vector of B_1 = $\dfrac{\vec{c}}{2}$

area of quadrilateral

$AA_1GB_1 = \dfrac{1}{2}\ |\overrightarrow{AG} \times \overrightarrow{A_1B_1}|$

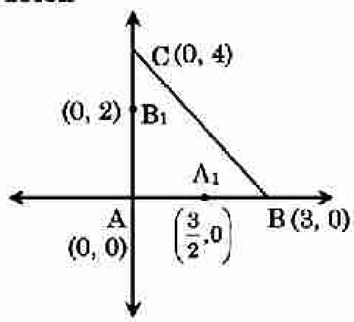
$= \dfrac{1}{2}\left|\left(\dfrac{\vec{b}+\vec{c}}{3}\right)\times\left(\dfrac{\vec{c}}{2}-\dfrac{\vec{b}}{2}\right)\right|$

$= \dfrac{1}{12}\ |2(\vec{b}\times\vec{c})|$

$= \dfrac{1}{6}\ |\vec{b}\times\vec{c}|$

area of $\Delta ABC = \dfrac{1}{2}\ |\vec{b}\times\vec{c}|$

Short Trick



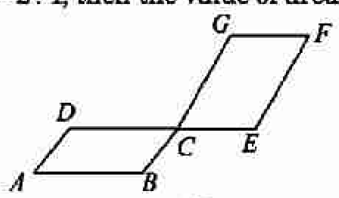
Centroid of ΔABC $G\left(1, \dfrac{4}{3}\right)$

Area of triangle $\Delta = 6$

Area of quadrilateral $\Delta_1 = \dfrac{1}{2}\begin{vmatrix} 1 & 4/3 \\ 3/2 & 0 \\ 0 & 0 \\ 0 & 2 \\ 1 & 4/3 \end{vmatrix} = 2$

$\dfrac{\Delta}{\Delta_1} = \dfrac{6}{2} = 3$

Q.9 In the following figure, AB, DE and GF are parallel to each other and AD, BG and EF are parallel to each other. If CD : CE = CG : CB = 2 : 1, then the value of area (ΔAEG) : area (ΔABD) is equal to –



(A) 7/2 (B) 3 (C) 4 (D) 9/2

Sol. [A]

Proper Method

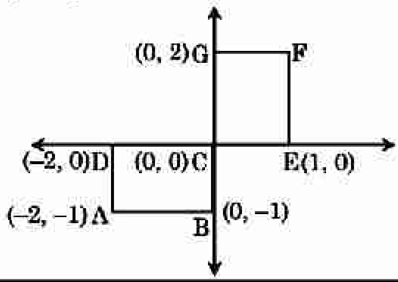
Let A be origin

$\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AB} = \vec{b}$

$\overrightarrow{AE} = \vec{b} + \dfrac{3}{2}\vec{a}$

$\overrightarrow{AC} = \vec{a} + 3\vec{b}$

Short Trick



Required ratio = $\dfrac{\frac{1}{2}\lvert\overrightarrow{AE}\times\overrightarrow{AG}\rvert}{\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AD}\rvert}$	$\dfrac{\text{Area of }\Delta\text{ AEG}}{\text{Area of }\Delta\text{ABD}} = \dfrac{\frac{1}{2}[-2(-2)+1(3)+0]}{\frac{1}{2}[-2(-1)+0-2(0)]} = \dfrac{7}{2}$

Q.10 $[(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\ \ (\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})\ \ (\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b})]$; (where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non zero non-coplanar vector) is equal to

(A) $[\vec{a}\ \vec{b}\ \vec{c}]^2$ (B) $[\vec{a}\ \vec{b}\ \vec{c}]^3$ (C) $[\vec{a}\ \vec{b}\ \vec{c}]^4$ (D) $[\vec{a}\ \vec{b}\ \vec{c}]$

Sol. [A]

Proper Method	**Short Trick**
$[(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c})\ \ (\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})\ \ (\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b})]$ $[\{(\vec{a}\times\vec{b}).\vec{c}\}\vec{b}-0\ \ \ \{(\vec{b}\times\vec{c}).\vec{a}\}\vec{c}-0\ \ \ \{(\vec{c}\times\vec{a}).\vec{b}\}\vec{a}-0]$ $[[\vec{a}\ \vec{b}\ \vec{c}]\vec{b}\ \ \ [\vec{a}\ \vec{b}\ \vec{c}]\vec{c}\ \ \ [\vec{a}\ \vec{b}\ \vec{c}]\vec{a}]$ $=[\vec{a}\ \vec{b}\ \vec{c}]\ \ \ [\vec{b}\ \vec{c}\ \vec{a}]$ $=[\vec{a}\ \vec{b}\ \vec{c}]^2$	Let $\vec{a}=\hat{i}$, $\vec{b}=\hat{j}$, $\vec{c}=\hat{k}$ then given expression $[\hat{k}\times\hat{i}\ \ \hat{i}\times\hat{j}\ \ \hat{j}\times\hat{k}]=[\hat{i}\ \hat{j}\ \hat{k}]^2$ Hence option (A) is correct answer

Q.11 Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar unit vectors equally inclined to one another at an acute angle θ. Then $[\vec{a}, \vec{b}, \vec{c}\,]$ in terms of θ is equal to :

(A) $(1+\cos\theta)\sqrt{\cos 2\theta}$ (B) $(1+\cos\theta)\sqrt{(1-2\cos 2\theta)}$

(C) $(1-\cos\theta)\sqrt{(1+2\cos\theta)}$ (D) $(2+2\cos\theta)\sqrt{(1-2\cos 2\theta)}$

Sol. [C]

Proper Method	**Short Trick**
$\lvert\vec{a}\rvert=\lvert\vec{b}\rvert=\lvert\vec{c}\rvert=1$ $\Rightarrow \vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=\cos\theta$, θ is acute $[\vec{a}\ \vec{b}\ \vec{c}]^2=\begin{vmatrix}\vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c}\\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c}\\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c}\end{vmatrix}$ $=\begin{vmatrix}1 & \cos\theta & \cos\theta\\ \cos\theta & 1 & \cos\theta\\ \cos\theta & \cos\theta & 1\end{vmatrix}$ $=(1+2\cos\theta)\begin{vmatrix}1 & \cos\theta & \cos\theta\\ 1 & 1 & \cos\theta\\ 1 & \cos\theta & 1\end{vmatrix}$ $=(1+2\cos\theta)(1-\cos\theta)^2$ $\therefore\ [\vec{a}\ \vec{b}\ \vec{c}]=(1-\cos\theta)\sqrt{1+2\cos\theta}$.	Let $\vec{a}=\hat{i}$, $\vec{b}=\hat{j}$, $\vec{c}=\hat{k}$ $[\vec{a}\ \vec{b}\ \vec{c}]=1$ Putting $\theta=90°$ in option then option (C) is correct answer

Q.12 If $\vec{b}$ and $\vec{c}$ are any two orthogonal unit vectors and $\vec{a}$ is any vector,

then $(\vec{a}.\vec{b})\vec{b} + (\vec{a}.\vec{c})\vec{c} + \frac{\vec{a}.(\vec{b}\times\vec{c})}{|\vec{b}\times\vec{c}|^2}(\vec{b}\times\vec{c}) =$

(A) $\vec{0}$ (B) $\vec{a}$ (C) $\vec{a}\cdot(\vec{b}\times\vec{c})$ (D) $\vec{b}$

Sol. [B]

Proper Method	**Short Trick**
$\lvert\vec{b}\rvert = \lvert\vec{c}\rvert = 1$ and $\vec{b}.\vec{c} = 0$ Therefore $\vec{b}$, $\vec{c}$ and $\vec{b}\times\vec{c}$ are non-coplanar. Let $\vec{a} = x\vec{b} + y\vec{c} + z(\vec{b}\times\vec{c})$ Then $\vec{a}.\vec{b} = x$, $\vec{a}.\vec{c} = y$ and $\vec{a}.(\vec{b}\times\vec{c}) = z(\vec{b}\times\vec{c})^2 \Rightarrow z = \frac{\vec{a}.(\vec{b}\times\vec{c})}{\lvert\vec{b}\times\vec{c}\rvert^2}$.	Let $\vec{b} = \hat{i}$, $c = \hat{j}$, $\vec{a} = \hat{k}$ $\Rightarrow (\vec{a}.\vec{b})\vec{b} + (\vec{a}.\vec{c})\vec{c} + \frac{\vec{a}.(\vec{b}\times\vec{c})}{\lvert\vec{b}\times\vec{c}\rvert^2}(\vec{b}\times\vec{c})$ $= \hat{k}$ $= \vec{a}$

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Function

KEY CONCEPTS

1. Numbers and their Sets

(i) Natural Numbers : N = {1, 2, 3, 4,...}

(ii) Whole Numbers : W = {0, 1, 2, 3, 4,....}

(iii) Integer Numbers : I or Z = {...–3, –2, –1, 0, 1, 2, 3,...}

$Z^+ = \{1, 2, 3,....\}$, $Z^- = \{-1, -2, -3,....\}$

$Z_0 = \{\pm 1, \pm 2, \pm 3,....\}$

(iv) Rational Numbers : $Q = \{p/q \,;\, p, q \in Z, q \neq 0\}$

(v) Irrational numbers : The numbers which are not rational or which can not be written in the form of p/q, called irrational numbers

eg. $\{\sqrt{2}, \sqrt{3}, 2^{1/3}, 5^{1/4}, \pi, e,....\}$

2. Interval

(i) Close interval $[a, b] = \{x, a \leq x \leq b\}$

(ii) Open interval (a, b) or $]a, b[= \{x, a < x < b\}$

(iii) Semi open or semi close interval

$[a, b[$ or $[a, b) = \{x; a \leq x < b\}$

$]a, b]$ or $(a, b] = \{x; a < x \leq b\}$

3. Function

Let A and B be two given sets and if each element $a \in A$ is associated with a unique element $b \in B$ under a rule f , then this relation is called Function.

Here, b is called the image of a and a is called the pre-image of b under f.

Note :

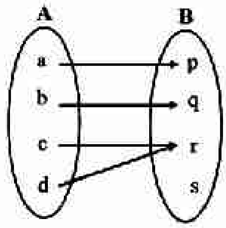
- Every element of A should be associated with B but vice-versa is not essential.
- Every element of A should be associated with a unique (one and only one) element of but any element of B can have two or more relations in A.

4. Domain, Codomain and Range of Function

If a function f is defined from a set of A to set B then for f: $A \to B$ set A is called the **domain** of function f and set B is called the **co-domain** of function f. The set of the f- images of the elements of A is called the **range** of function f.

Domain = All possible values of x for which f(x) exists.

Range = For all values of x, all possible values of f(x).



Domain $= \{a, b, c, d\} = A$
Co-domain $= \{p, q, r, s\} = B$
Range $= \{p, q, r\}$

5. Identification of a function with its graph

If we are given a graph of the relation then we can check whether the given relation is function or not . If it is possible to draw a vertical line which cuts the given curve at more than one point then given relation is not a function and when this vertical line means line parallel to Y - axis cuts the curve at only one point then it is a function.

(i) (ii)

fig. (i) does not represents a function but (ii) represents a function.

6. Classification of Real Function

Real functions are generally classified under two categories algebric functions and transcendental functions.

(i) Algebric Function :

Algebric functions further can be classified as-

(a) Polynomial or integral Function :

The function of the form

$f(x) = a_0 x^n + a_1 x^{n-1} + ... + a_{n-1}x + a_n$, $a_0 \neq 0$

Where $a_0, a_1, a_2, ...a_n$ are constants and $n \in N$

(b) Rational Function :

If a function y = f(x) is given by $f(x) = \dfrac{\Phi(x)}{\Psi(x)}$

Where, $\Phi(x)$ and $\psi(x)$ are polynomial functions, then f(x) is called rational function in x

(c) Irrational Function :

The algebric function containing one or more terms having non-integral rational power of x is called irrational functions.

(ii) Transcendental Functions :

A function which is not algebraic is called a transcendental function. Trigonometric, Inverse trigonometric, Exponential, Logarithmic, etc are transcendental functions.

7. Explicit and Implicit Function

(i) Explicit Function :

A function is said to be explicit if it can be expressed directly in terms of the independent variable.

$y = f(x)$ or $x = \phi(y)$

Eg. $f(x) = x^2 \sin x$

(ii) Implicit Function :

A function is said to be implicit if it can not be expressed directly in terms of the independent variable.

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

8. Increasing and Decreasing Function

(i) Increasing Function

A function f(x) is called increasing function in the domain D if the value of the function does not decrease by increasing the value of x.

If $x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$

or $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$

or $f'(x) > 0$ for increasing and $f'(x) \geq 0$ for not decreasing.

(ii) Decreasing Function :

A function f(x) is said to be decreasing function in the domain D if the value of the function does not increase by increasing the value of x (variable).

If $x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$

or $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$

or $f'(x) < 0$ for decreasing and $f'(x) \leq 0$ for not increasing.

9. Even and Odd Function

(i) Even Function :

If we put $(-x)$ in place of x in the given function and if $f(-x) = f(x)$, $\forall x \in$ domain then function f(x) is called even function.

(ii) Odd Function :

If we put $(-x)$ in place of x in the given function and if $f(-x) = -f(x)$, $\forall x \in$ domain then f(x) is called odd function.

(iii) Properties of Even and Odd Function :

(i) The product of two even functions is even function.

(ii) The sum and difference of two even functions is even function.

(iii) The sum and difference of two odd functions is odd function.

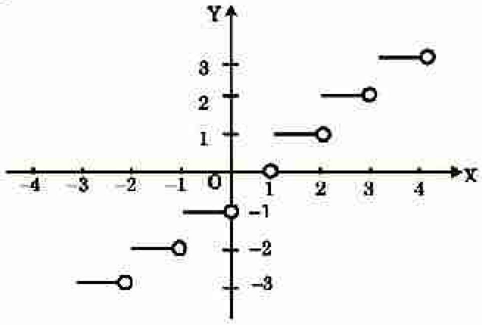
(iv) The product of two odd functions is even function.

(v) The product of an even and an odd function is odd function.

(vi) The sum of even and odd function is neither even nor odd function.

10. Greatest Integer Function

A function is said to be greatest integer function if it is of the form of f(x) = [x] where [x] = integer equal or less than x. i.e. [4.2] = 4, [– 4. 4] = – 5



Properties of G.I.F. :

(i) [x] = x if x is integer

(ii) [x + I] = [x] + I, if I is an integer

(iii) $[x + y] \geq [x] + [y]$

(iv) If $[\phi(x)] \geq I$ then $\phi(x) \geq I$

(v) If $[\phi(x)] \leq I$ then $\phi(x) < I + 1$

(vi) If $[x] > n \Rightarrow x \geq n + 1$

(vii) If $[x] < n \Rightarrow x < n, n \in I$

(viii) $[-x] = -[x]$ if $\forall x \in I$

(ix) $[-x] = -[x] - 1$ if $x \notin$ Integer

(x) $[x + y] = [x] + [y + x - [x]] \ \forall x, y \in R$

(xi) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + ... + \left[x + \frac{n-1}{n}\right] = [nx]$

11. Periodic Function

A function is said to be periodic function if its each value is repeated after a definite interval. So a function f(x) will be periodic if a positive real number T exist such that, f(x + T) = f(x) , $\forall$ x $\in$ Domain. Here the least positive value of T is called the period of the function.

Note :

- Constant function is periodic with no fundamental period.
- If function f (x) has period T then

  (a) f(nx) has period T/n

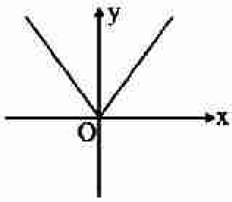
  (b) f(x/n) has period nT

  (c) f(ax + b) has period $\frac{T}{|a|}$
- If the period of f(x) and g(x) are same (T) then the period of af(x) + bg(x) will also be T.
- If the period of f(x) is T_1 and g(x) has T_2, then the period of f(x) $\pm$ g(x) will be LCM of T_1 and T_2 provided it satisfies the definition of periodic function.
- sinx, cosx, secx and cosec x are periodic functions with period 2π
- tan x and cotx are periodic functions with period π .
- $|\sin x|$, $|\cos x|$, $|\tan x|$, $|\cot x|$, $|\sec x|$ and $|\text{cosec}x|$ are peiodic functions with period π

- $\sin^n x$, $\cos^n x$, $\sec^n x$ and $\text{cosec}^n x$ are periodic functions with period 2π, when n is odd, or π when n is even.
- $\tan^n x$ and $\cot^n x$ are periodic functions with period π
- $|\sin x| + |\cos x|$, $|\tan x| + |\cot x|$ and $|\sec x| + |\text{cosec} x|$ are periodic with period $\pi/2$

12. Modulus Function

It is given $n \in N$ by $y = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$



Properties of Modules Function :

(i) $|x| \leq a \Rightarrow -a \leq x \leq a$

(ii) $|x| \geq a \Rightarrow x \leq -a$ or $x \geq a$

(iii) $|x + y| = |x| + |y| \Rightarrow x, y \geq 0$ or $x \leq 0, y \leq 0$

(iv) $|x - y| = |x| - |y| \Rightarrow x \geq 0$ and $|x| \geq |y|$ or $x \leq 0$ and $y \leq 0$ and $|x| \geq |y|$

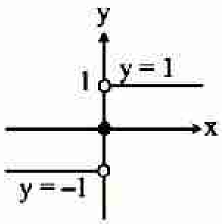
(v) $|x \pm y| \leq |x| + |y|$

(vi) $|x \pm y| \geq |x| - |y|$

13. Signum Function

The signum function f is defined as

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$



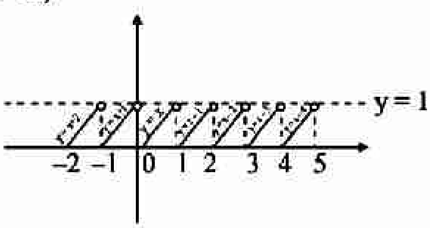
14. Fractional Part Function

It is denoted as $f(x) = \{x\}$ and defined as

(i) $\{x\} = f$ if $x = n + f$ where $n \in I$ and $0 \leq f < 1$

(ii) $\{x\} = x - [x]$

(For proper fraction $0 < f < 1$)



15. Even & Odd Extension

(i) Even Extension : If a function f(x) is defined on the interval [0, a], $0 \le x \le a \Rightarrow -a \le -x \le 0$ we define f(x) in the [– a, 0] such that f(x) = f(–x). Let $I(x) = \begin{cases} f(x) & : x \in [0,a] \\ f(-x) & : x \in [-a,0] \end{cases}$

(ii) Odd Extension : If a function f(x) is defined on the interval [0, a], $0 \le x \le a \Rightarrow -a \le -x \le 0$

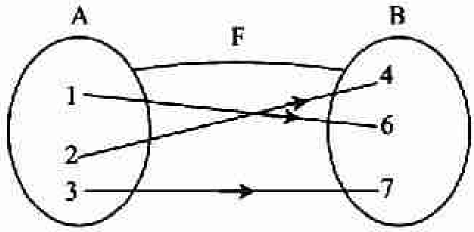
$\therefore$ $x \in [-a, 0]$, we define f(x) = – f(–x). Let be the odd extension then

$I(x) = \begin{cases} f(x), x \in [0,a] \\ -f(-x), x \in [-a,0] \end{cases}$

16. Kinds of Mapping

(i) One-One Function : The mapping $f : A \to B$ is called one-one function, if different elements in A have different images in B. Such a mapping is known as injective function or an injection.

Methods to Test One-One :



(a) Analytically : If $x_1, x_2 \in A$, then $f(x_2) = (x_2) \Rightarrow x_1 = x_2$
or equivalently $x_1 \ne x_2 \Rightarrow f(x_1) \ne f(x_2)$

(b) Graphically : If any line parallel to X-axis cuts the graph of the function atmost at one point, then the function is one-one.

(c) Monotonically : Any function which is entirely increasing or decreasing in whole domain, then f (x) is one-one.

(d) Number of One-One Functions : Let $f : A \to B$ be a function, such that A and B are finite sets having m and n elements respectively, (where, n > m).

The number of one-one functions

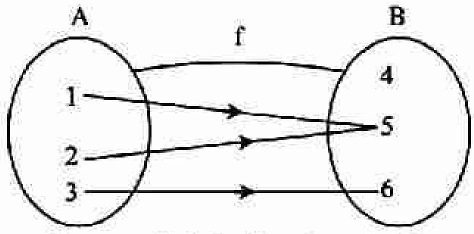
$$n(n-1)(n-2)...(n-m+1) = \begin{cases} {}^nP_m n \ge m \\ 0, n < m \end{cases}$$

(ii) Many-One Function :

The function $f: A \longrightarrow B$ is called many-one function, if two or more than two different elements in A have the same image in B.

Method to Test Many-one :

(a) Analytically If $x_1, x_2 \in A$, then
$x_1 \ne x_2 \Rightarrow f(x_1) = f(x_2)$



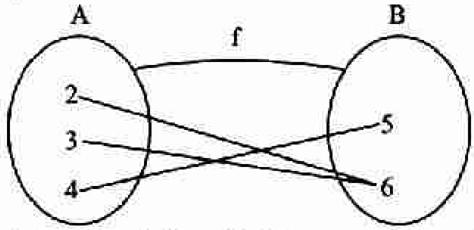
(b) Graphically If any line parallel to X-axis cuts the graphs of the function atleast two points, then f(x) is many-one

(c) **Monotonically** Many-one function is neither strictly increasing nor strictly decreasing. So, f(x) is not monotonic function

(d) **Number of Many-one Function** Let f: $A \rightarrow B$ be a function such that A and B are finite sets having m and n elements respectively.
The number of many-one function = n^m

(iii) Onto (Surjective) **Function :**

If the function f:$A \rightarrow B$ is such that each element in B (co-domain) is the image of atleast one element of A then we say that f is function of A onto B. Thus. f: $A \rightarrow B$, such that f(A) = B,



Note : Every polynomial function f : $R \rightarrow R$ of degree odd is onto

Number of Onto (surjective) **Functions** Let A and B are finite sets having m and n elements respectively, such that $1 \leq n \geq m$, then number of onto (surjective) functions from A to B is

$\sum_{r=1}^{n}(-1)^{2-m}C_r r^m$ Coefficient of x^2 in $n!(e^x - 1)^r$

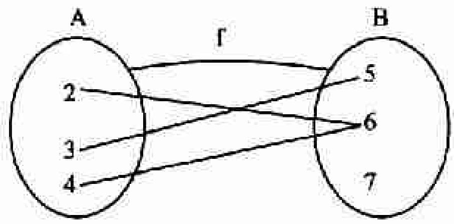
Note A one-one and onto function is called bijective function.

(iv) Into Function :

If f : $A \rightarrow B$ is such that there exists atleast one element in co-domain which is not the image of any element in domain, then f(x) is into.

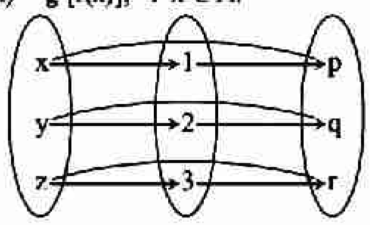
Thus, f : $A \rightarrow B$, such that $f(x) \subset B$

i.e. Range $\subset$ Co-domain



17. Composite Function

If f : $A \rightarrow B$ and g : $B \rightarrow C$ are two function then the composite function of f and g, gof $A \rightarrow C$ will be defined as gof (x) = g [f(x)], $\forall x \in A$.



Properties of Composite Function :

(i) If f and g are two functions then for composite of two functions fog $\neq$ gof.

(ii) Composite functions obeys the property of associativity i.e. fo(goh) = (fog)oh.

(iii) Composite function of two one-one onto functions if exist, will also be a one-one onto function.

18. Inverse Function

If $f : A \to B$ be a one-one onto (bijection) function, then the mapping $f^{-1} : B \to A$ which associates each element $b \in B$ with element $a \in A$, such that $f(a) = b$, is called the inverse function of the function $f : A \to B$

$f^{-1} : B \to A$, $f^{-1}(b) = a \Rightarrow f(a) = b$

(i) **Properties:** For the existence of inverse function, it should be one-one and onto.

(a) Inverse of a bijection is also a bijection function.

(b) Inverse of a bijection is unique.

(c) $(f^{-1})^{-1} = f$

(d) If f and g are two bijections such that (gof) exists then $(gof)^{-1} = f^{-1}og^{-1}$

(e) If $f : A \to B$ is a bijection then $f^{-1}: B \to A$ is an inverse function of f.

$f^{-1}of = I_A$ and $fof^{-1} = I_B$.

(ii) **Algebra of Function :**

(a) $(fog)(x) = f[g(x)]$

(b) $(fof)(x) = f[f(x)]$

(c) $(gog)(x) = g[g(x)]$

(d) $(fg)(x) = f(x) . g(x)$

(e) $(f \pm g)(x) = f(x) \pm g(x)$

(f) $(f/g)(x) = \dfrac{f(x)}{g(x)}$, $g(x) \neq 0$

* Composite functions is not commutative
* Let f and g are two functions then if f & g are injective or surjective or bijective then "gof " also injective or surjective or bijective.

19. Important points to be remembered

If x, y are Independent Variables then :

(i) If $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \log x$

(ii) If $f(xy) = f(x) . f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) If $f(x + y) = f(x).f(y) \Rightarrow f(x) = a^{kx}$

(iv) If $f(x + y) = f(x) + f(y) \Rightarrow f(x) = x$

(v) If $f(x + y) = f(x) = f(y) \Rightarrow f(x) = k$, here k is constant

(vi) By considering a general nth degree polynomial and writing the expression

$$f(x) . f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \Rightarrow f(x) = \pm x^n + 1$$

20. Some Functions & Their Graphs

Classification of Function :

Function

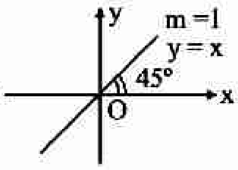
- Algebric
  - Polynomial
  - Rational
  - Irrational
  - Piecewise
    - Modulus
    - Greatest integer function
    - Fractional part
    - Signum
- Transcendental
  - Trigonometric
  - Exponential
  - Logarithmic
  - Inverse Trigonometric

Graphs of Function :

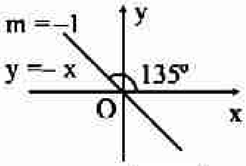
(i) Polynomials

(a) $y = f(x) = mx$, where m is a constant($m \neq 0$)

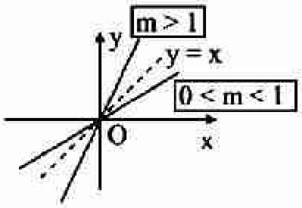
(1)



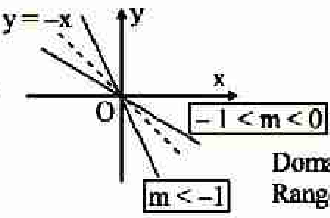
(2)



(3)



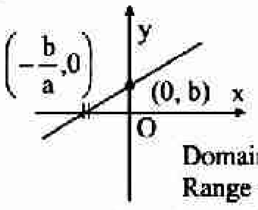
(4)



Domain = R
Range = R

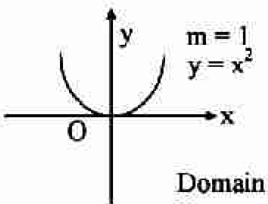
$y = mx$ is continuous function & one-one function ($m \neq 0$)

(5) $y = ax + b$, where a & b are constant

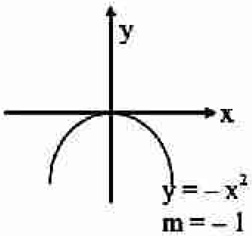


Domain = R
Range = R

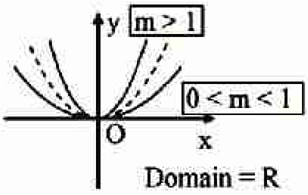
(b) $y = mx^2$, ($m \neq 0$), $x \in R$



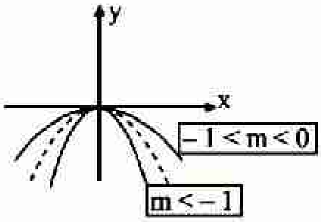
Domain = R
Range = $[0, \infty]$



Domain = R
Range = $(\infty, 0]$



Domain = R
Range = [0, ∞)

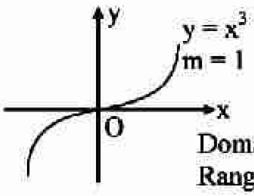


Domain = R
Range = [0, ∞)

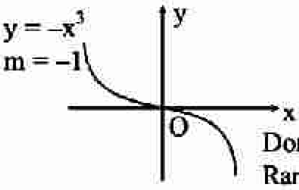
$y = mx^2$ is continuous, even function and many-one function.

Graph is symmetric about y-axis

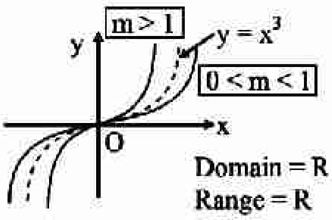
(c) $y = mx^3$, $(m \neq 0)$, $x \in R$



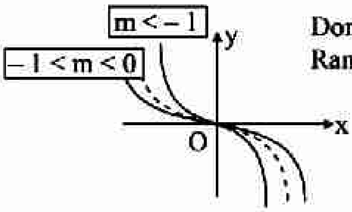
Domain = R
Range = R



Domain = R
Range = R



Domain = R
Range = R

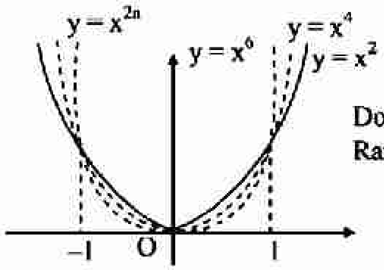


Domain = R
Range = R

$y = mx^3$ is continuous, odd one-one & increasing function.

Graph is symmetric about origin.

(d) $y = x^{2n}, n \in N$



Domain = R
Range = [0, ∞)

$f(x) = x^{2n}$ is an even, continuous function. It is many one function.

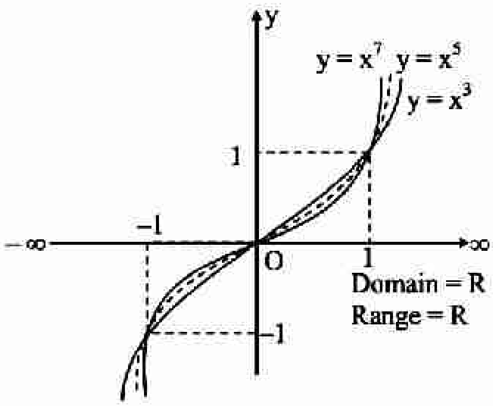
From the Graph :

For $|x| \le 1 \Rightarrow x^6 \le x^4 \le x^2$

For $|x| > 1 \Rightarrow x^6 > x^4 > x^2$

Graph is symmetric about y-axis

(e) $y = x^{2n+1}, n \in N$



Domain = R
Range = R

$f(x) = x^{2n+1}$, $n \in N$ is an odd continuous function. It is always one-one function.

From the Graph :

For $x \in (-\infty, -1] \cup [0, 1]$

$\Rightarrow \quad x^3 \ge x^5 \ge x^7$

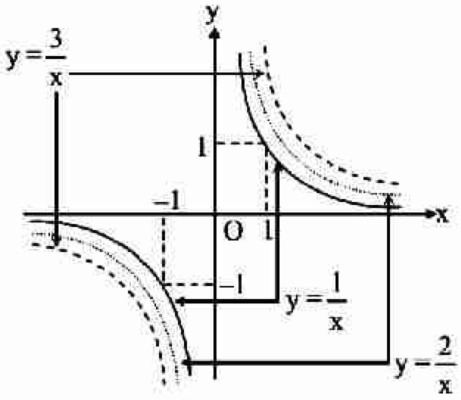
For $x \in (-1, 0) \cup (1, \infty)$

$\Rightarrow \quad x^3 < x^5 < x^7$

Graph is symmetric about origin.

(ii) Rational Function :

(a) $y = \dfrac{m}{x}, x \ne 0, m \ne 0$



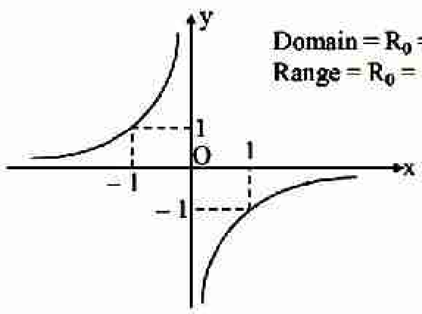
$y = \frac{m}{x}$ is odd function discontinuous at x = 0.

Domain = R_0 = R – {0}

Range = R_0 = R – {0}

This type of graph xy = m is called rectangular hyperbola.

(b) $y = -\frac{1}{x}, x \neq 0$



Domain = R_0 = R – {0}
Range = R_0 = R – {0}

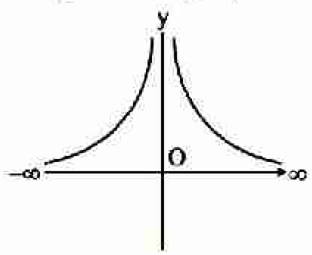
$y = -\frac{1}{x}$ is an odd & one-one function discontinuous at x = 0

(c) $y = \frac{1}{x^2}, x \neq 0$

$y = \frac{1}{x^2}$ is even function, many one and discontinuous at x = 0

Domain = R_0

Range = $R^+ = (0, \infty)$



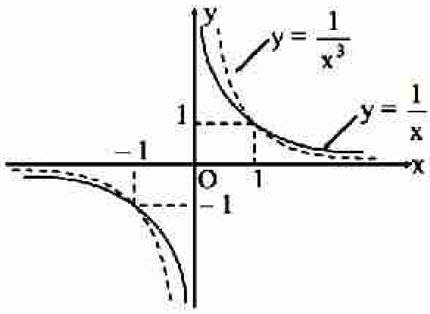
(d) $y = \frac{1}{x^{2n-1}}, n \in N$

An odd one-one function discontinuous at x = 0

From Graph :

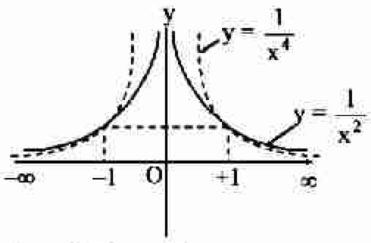
$0 < x < 1 \Rightarrow \frac{1}{x^3} > \frac{1}{x}$

$x > 1 \Rightarrow \frac{1}{x^3} < \frac{1}{x}$.



(e) $y = \frac{1}{x^{2n}}$, $n \in N$

$y = \frac{1}{x^{2n}}$ is an even function & many one, discontinuous at $x = 0$.

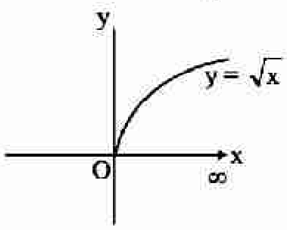


Domain = R_0
Range = R^+

(iii) Irrational Function

(a) $y = \sqrt{x} = x^{1/2}$

This function is neither even nor odd, one-one and continuous function in it's domain. It is an increasing function.

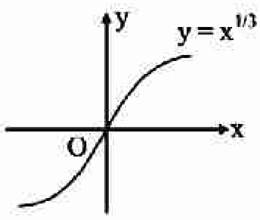


Domain = $[0, \infty)$

Range = $[0, \infty)$

(b) $y = \sqrt[3]{x} = x^{1/3}$

This function is an odd, continuous and one-one function. It is an increasing function.



Domain = R
Range = R

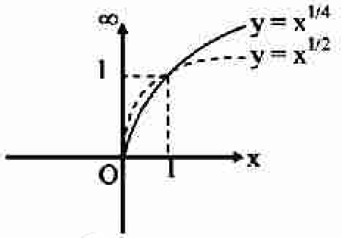
(c) $y = x^{1/2n}$, $n \in N$

This function is neither even nor odd, continuous and increasing function it is also one-one function

From graph

$0 < x < 1 \Rightarrow > \sqrt[2]{x} > \sqrt[4]{x}$

$x > 1 \Rightarrow x^{1/2} < x^{1/4}$



Domain = $R^+ = [0, \infty)$
Range = $R^+ = [0, \infty)$

(d) $y = x^{\frac{1}{2n+1}}$, $n \in N$

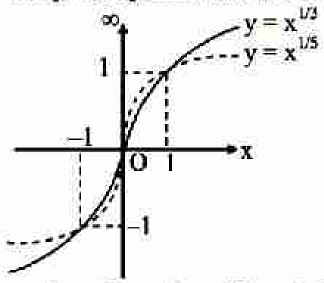
This function is an odd, continuous, one-one and increasing function

From Graph :

$x \in (-\infty, -1) \cup (0, 1)$, $x^{1/3} > x^{1/5}$

$x \in (-1, 0) \cup (1, \infty)$, $x^{1/3} < x^{1/5}$

Graph is symmetric about origin.



Domain = R
Range = R

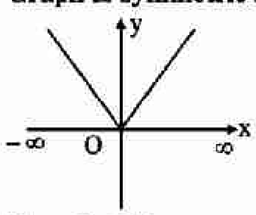
(iv) Piecewise Function (Special Function)

(a) Modulus Function

$$y = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

This is an even, continuous, many one function.

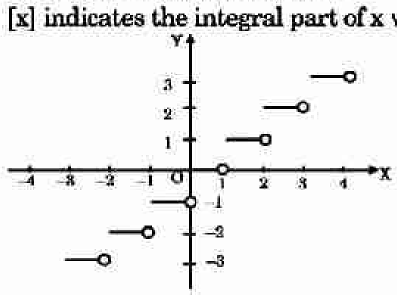
Graph is symmetric about y-axis



Domain = R
Range = $[0, \infty)$

(b) Greatest Integer Function

[x] indicates the integral part of x which is nearest and smaller to x. It is also called step function.



$$y = [x] = \begin{cases} -1, & -1 \le x < 0 \\ 0, & 0 \le x < 1 \\ +1, & 1 \le x < 2 \\ +2, & 2 \le x < 3 \\ +3, & 3 \le x < 4 \end{cases}$$

$\Rightarrow \quad y = [x] = k,\ k \le x < k + 1,\ k \in I$

Important Result from graph

$x - 1 < [x] \le x$

(c) Fractional Part of x

$y = \{x\}$

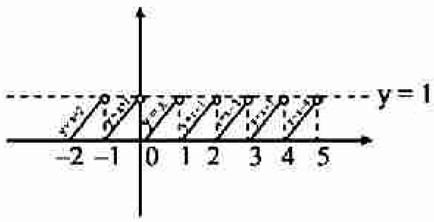
Any number is sum of integral & fractional part i.e. $x = [x] + \{x\}$

$\therefore\ y = \{x\} = x - [x] = x - k,\ k \le x < k + 1$

$$y = \{x\} = \begin{cases} x+1, & \ldots\ldots \\ x, & 0 \le x < 1 \\ x-1, & 1 \le x < 2 \\ x-2, & 2 \le x < 3 \\ \ldots\ldots, & \ldots\ldots \end{cases}$$

Domain = R

Range = [0, 1)

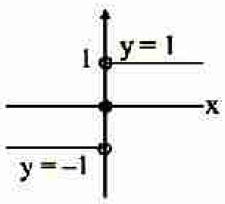


From graph $0 \le \{x\} < 1$

This function is a periodic function with period 1. This is also many-one function discontinuous at $x \in I$.

(d) $y = \text{sgn}(x) = \begin{cases} \dfrac{|x|}{x}, & x \ne 0 \\ 0, & x = 0 \end{cases} = \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ -x, & x < 0 \end{cases}$

This is odd, many-one function discontinuous at $x = 0$

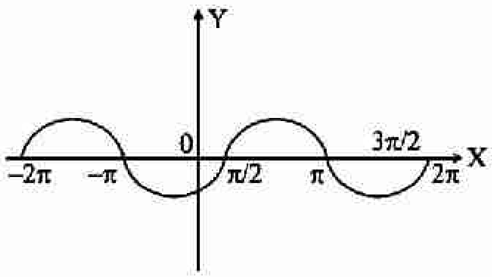


Domain = R

Range = {–1, 0, 1}

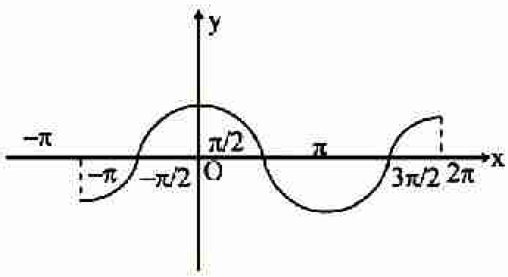
(v) Trigonometrical Function :

(a) $f(x) = \sin x$.



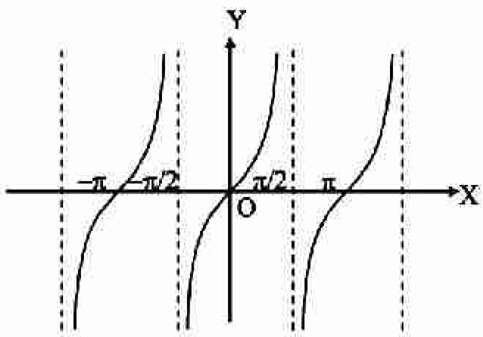
It is a continuous and many one function with period 2π.

(b) $f(x) = \cos x$



It is a continuous and many-one function having a period $2.\pi$

(c) $f(x) = \tan x$



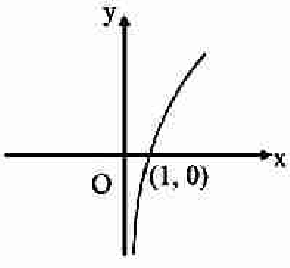
Domain = $R - (2n+1)\,\pi/2,\ n \in I$

Range = R

It is a discontinuous function at $x = (2n+1)\pi/2,\ n \in I$ and periodic with period π.

(vi) Logarithmic Function :

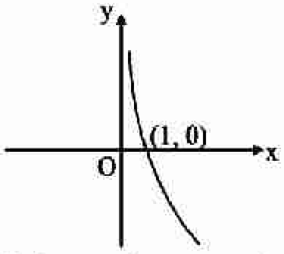
(a) $f(x) = \log_a x \quad (a > 1)$



Domain = R^+
Range = R

It is a continuous and one-one function.

(b) $f(x) = \log_a x$ $(a<1)$

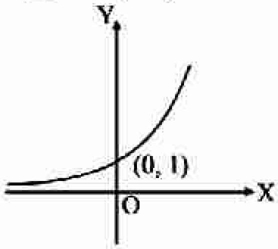


Domain = R^+
Range = R

It is a continuous and one-one function.

(vii) Exponential Function :

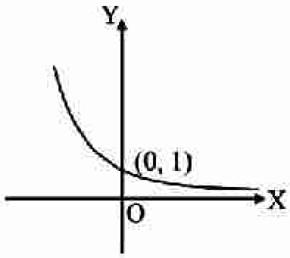
(a) $f(x) = a^x$ $(a>1)$



Domain = R
Range = R^+

It is a continuous and one-one function.

(b) $f(x) = a^x$ $(a<1)$

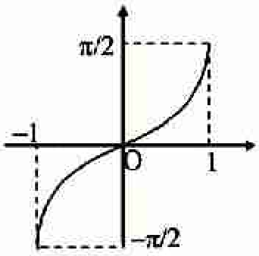


Domain = R
Range = R^+

It is a continuous and one-one function.

(viii) Inverse Trigonometric Function :

$y = \sin^{-1} x$

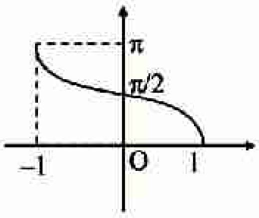


Domain = [–1, 1]

Range = $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

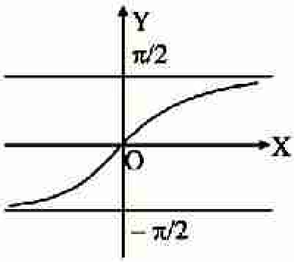
Odd function

$y = \cos^{-1}x$



Domain = [–1, 1]
Range = $[0, \pi]$
Neither even nor odd

$y = \tan^{-1}x$

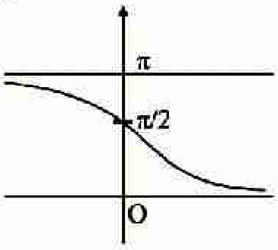


Domain = R

Range = $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

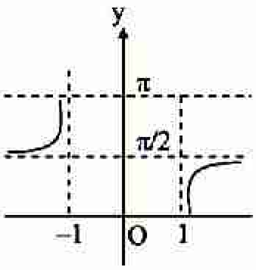
Odd function

y = cot^{-1} x



Domain = R
Range = (0, π)
Neither even nor odd

y = sec^{-1}x

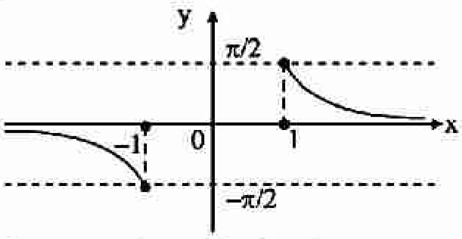


Domain = (–∞, –1] ∪ [1, ∞)

Range = [0, π] – $\left\{\frac{\pi}{2}\right\}$

Neither even nor odd

y = cosec^{-1}x



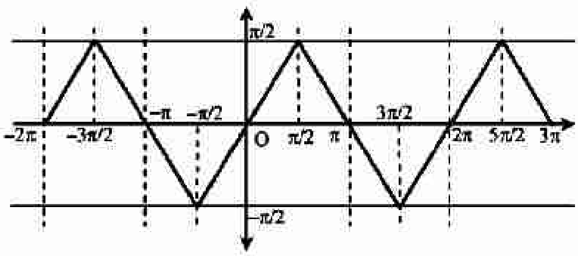
Domain = (–∞, –1] ∪ [1, ∞)

Range = $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

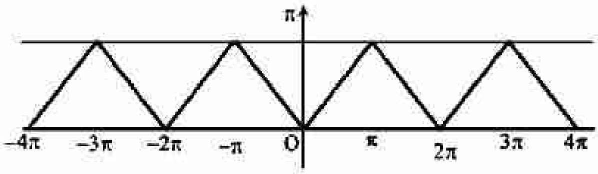
Odd function

All inverse trigonometric functions are monotonic.

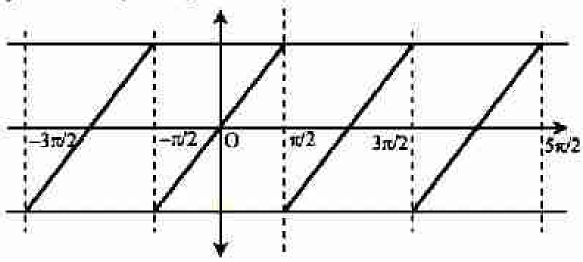
(ix) (a) **y = sin^{-1}(sin x) :**



(b) **y = cos^{-1}(cos x) :**



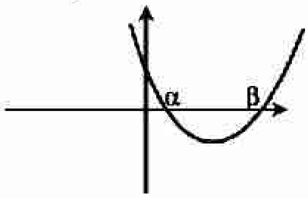
(c) $y = \tan^{-1}(\tan x)$



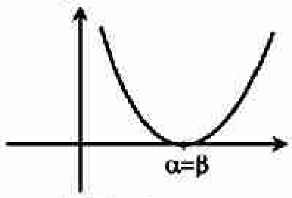
(x) Graph of Quadratic Function

$f(x) = ax^2 + bx + c$, $D = b^2 - 4ac$

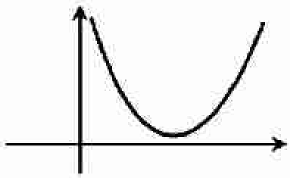
(a) $a > 0$; $D > 0$



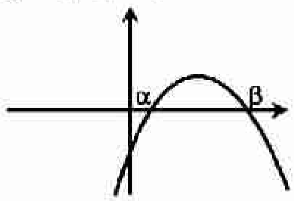
(b) $a > 0$; $D = 0$



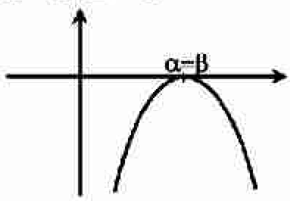
(c) $a > 0$; $D < 0$



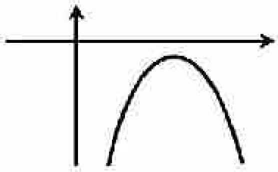
(d) $a < 0$; $D > 0$



(e) $a < 0$; $D = 0$



(f) $a < 0$; $D < 0$



(xi) Plotting of Algebraic Curve

From : $f(x) = (x - \alpha)^a (x - \beta)^b$ ...., $a, b \in N$.

Method :

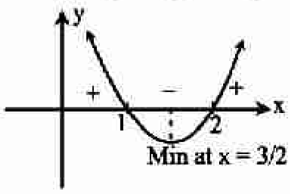
(a) Put $f(x) = 0$ to get change points.

(b) Write these points on the number line.

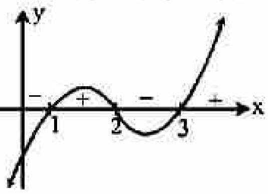
(c) Write sign scheme of wavy curve method.

(d) Wherever positive sign is occurring, the graph is above x-axis and wherever the sign is negative the graph is below x-axis.

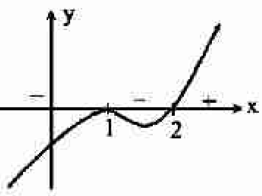
Ex. $y = (x - 1)(x - 2)$



Ex. $y = (x - 1)(x - 2)(x - 3)$



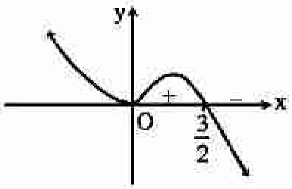
Ex. $y = (x - 1)^2 (x - 2)$



Ex. $y = 3x^2 - 2x^3$

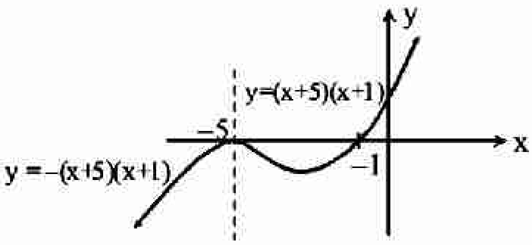
$\Rightarrow$ $y = x^2(3 - 2x)$

$\Rightarrow$ $y = -x^2(2x - 3)$



Ex. $y = |x + 5|\ (x + 1)$

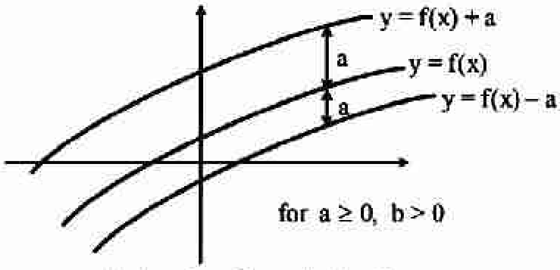
$$\Rightarrow \quad y = \begin{cases} (x+5)(x+1) & ; \quad x \ge -5 \\ -(x+5)(x+1) & ; \quad x < -5 \end{cases}$$



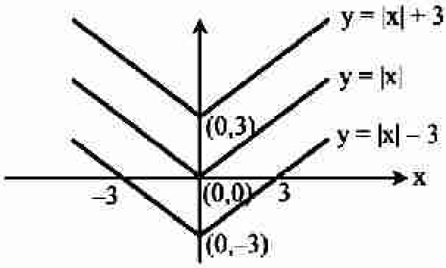
(xii) To draw $y = f(x) \pm a$ from $y = f(x)$

For $f(x) + a$, shift $f(x)$ upward by 'a' units.

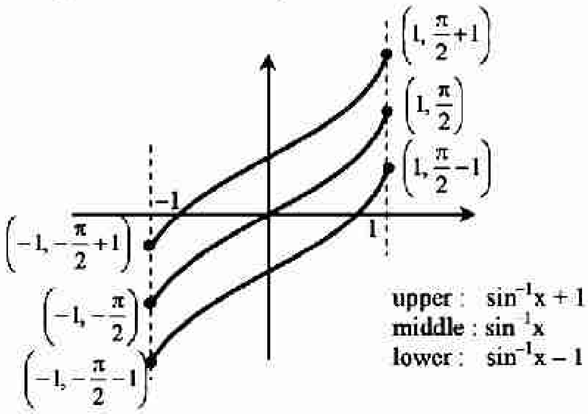
For $f(x) - a$, shift $f(x)$ downward by 'a' units.



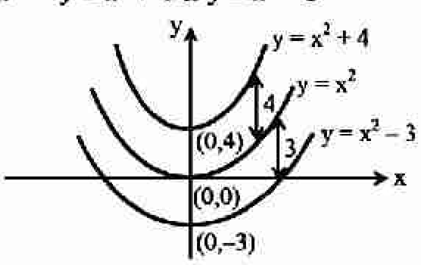
Ex. $y = |x| + 3$ and $y = |x| - 3$



Ex. $y = \sin^{-1}x + 1$ and $y = \sin^{-1}x - 1$



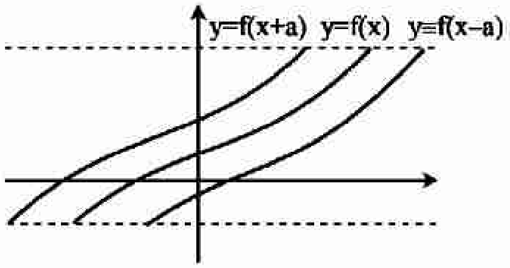
Ex. $y = x^2 + 4$ & $y = x^2 - 3$



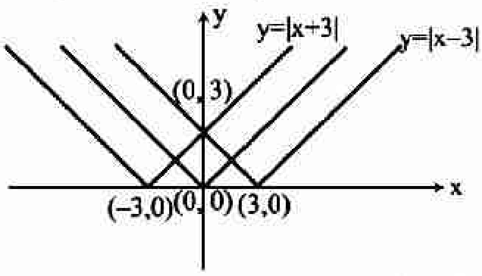
(xiii) To draw $y = f(x \pm a)$ from $y = f(x)$

For $f(x + a)$: shift $y = f(x)$ left by 'a' units

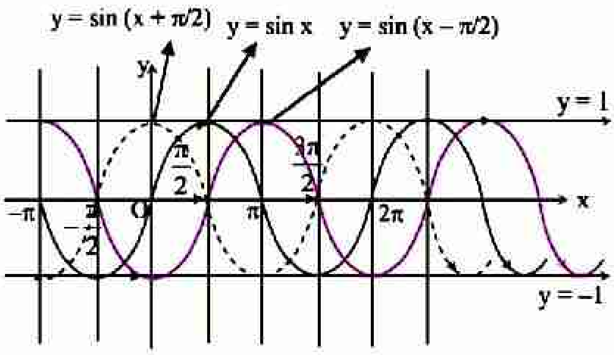
For $f(x - a)$: shift $y = f(x)$ right by 'a' units



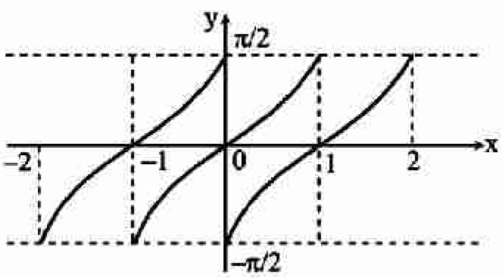
Ex. $y = |x - 3|$ and $y = |x + 3|$



Ex. $y = \sin\left(x - \frac{\pi}{2}\right)$ and $y = \sin\left(x + \frac{\pi}{2}\right)$



Ex. $y = \sin^{-1}(x - 1)$ and $y = \sin^{-1}(x + 1)$



Left : $\sin^{-1}(x + 1)$

Middle : $\sin^{-1}x$

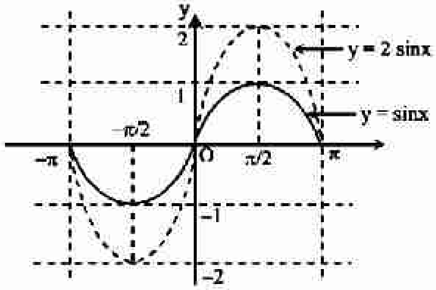
Right : $\sin^{-1}(x - 1)$

(xiv) To draw $y = af(x)$ from $y = f(x)$

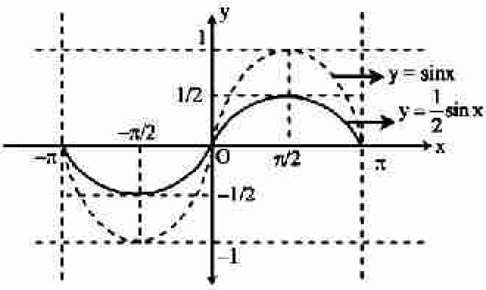
If $a > 1$, expand graph 'a' times along y-axis

If $a = \frac{1}{k}$, $k > 1$, contract graph 'k' times along y-axis

Ex. $y = \sin x$ and $y = 2\sin x$



Ex. $y = \sin x$ and $y = \frac{1}{2}\sin x$

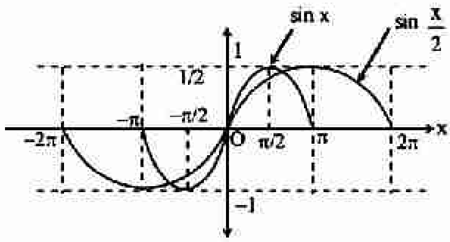


(xv) To draw y = f(ax) from y = f(x)

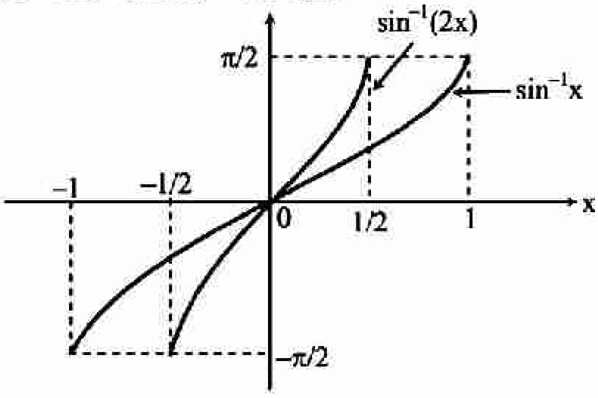
For $a > 1$, contract the graph of f(x) by 'a' times along x-axis.

For $a = \frac{1}{k}$, $k > 1$, expand the graph of f(x) by 'k' times along x-axis.

Ex. $y = \sin x$ and $y = \sin \frac{x}{2}$



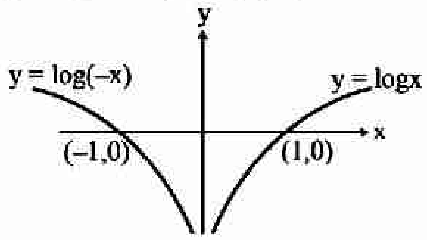
Ex. $y = \sin^{-1} x$ and $y = \sin^{-1}(2x)$



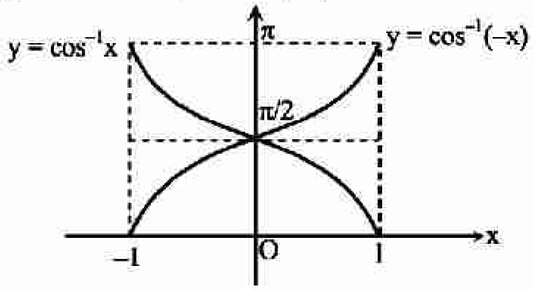
(xvi) To draw $y = f(-x)$ from $y = f(x)$

Take mirror image of $y = f(x)$ in y-axis

Ex. $y = \log x$ and $y = \log(-x)$



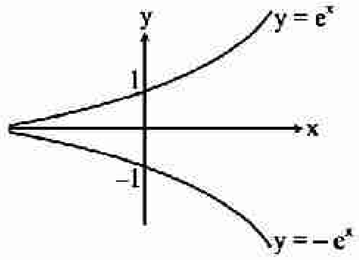
Ex. $y = \cos^{-1}x$ and $y = \cos^{-1}(-x)$



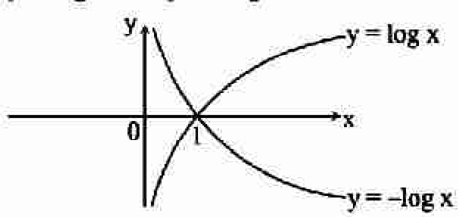
(xvii) To draw $y = -f(x)$ from $y = f(x)$

Take mirror image of $y = f(x)$ in the x-axis

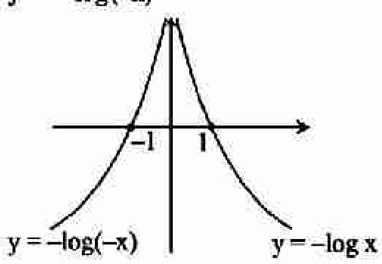
Ex. $y = e^x$ and $y = -e^x$



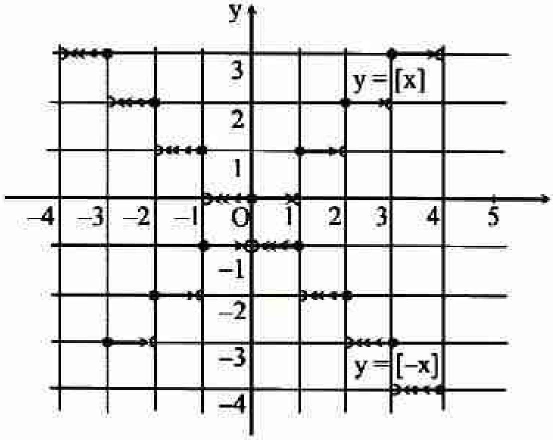
Ex. $y = \log x$ and $y = -\log x$



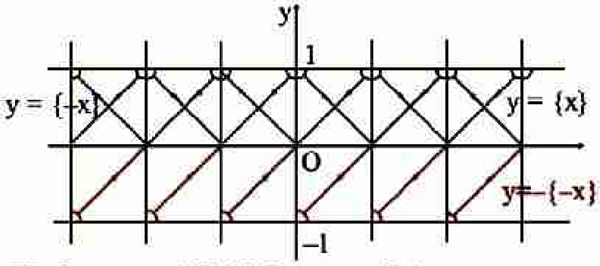
Ex. $y = -\log(-x)$



Ex. $y = [x]$, $y = [-x]$



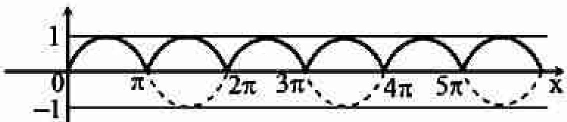
Ex. $y = -\{-x\}$, ({.} is F.P.F)



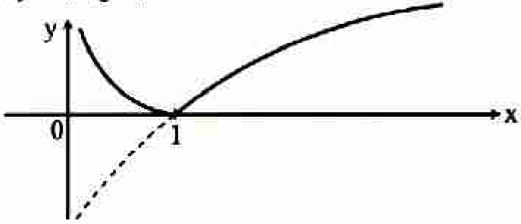
(xviii) To draw $y = |f(x)|$ from $y = f(x)$

1. Draw graph of f(x)
2. Take mirror image in x-axis for negative portion
3. Erase negative portion

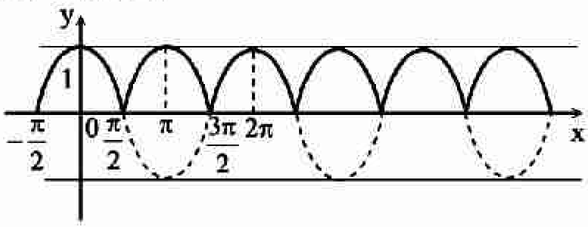
Ex. $y = |\sin x|$



Ex. $y = |\log x|$

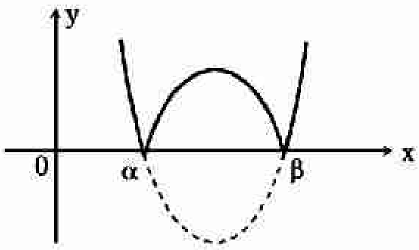


Ex. $y = |\cos x|$

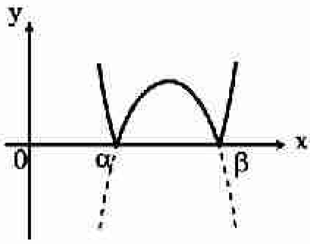


Ex. $y = |ax^2 + bx + c|$

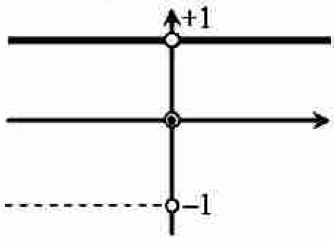
for $a > 0$; $D > 0$



for $a < 0$; $D > 0$



Ex. $y = |\text{sgn}(x)|$



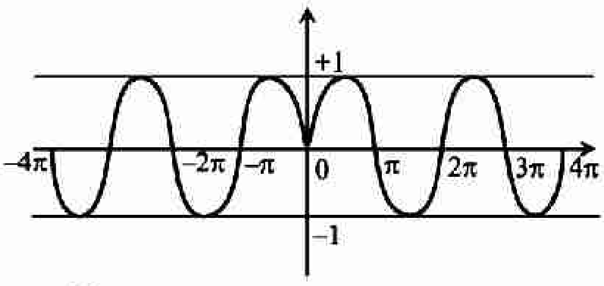
(xix) To draw :

$y = f(|x|)$ from $y = f(x)$

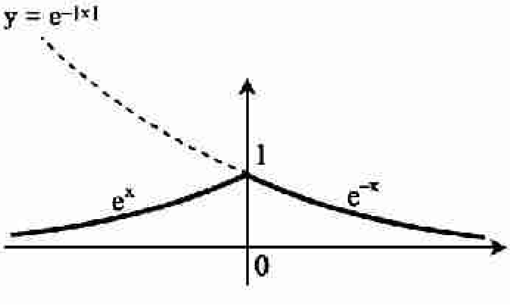
(a) Draw graph for positive values of x.

(b) Take mirror image in y-axis.

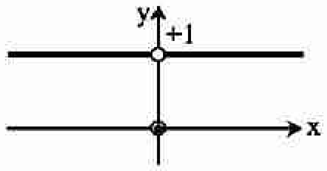
Ex. $y = \sin|x|$



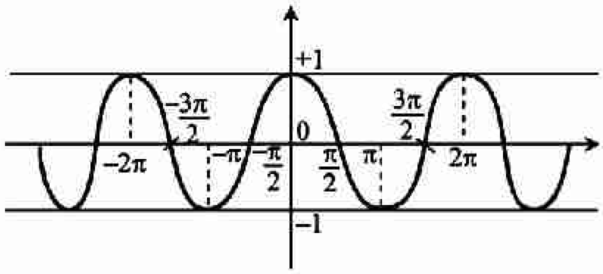
Ex. $y = e^{-|x|}$



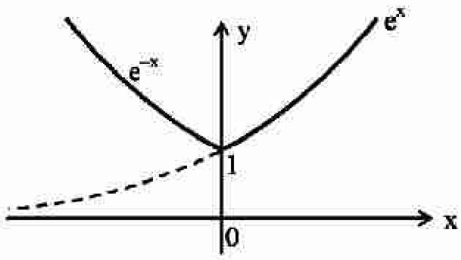
Ex. $y = \text{sgn}\ |\ x\ |$



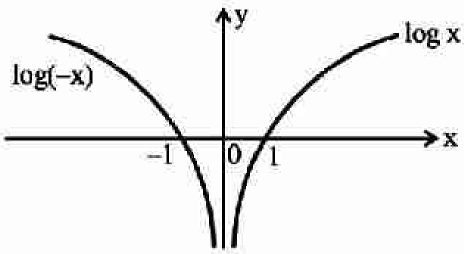
Ex. $y = \cos\ |\ x\ | = \cos x$



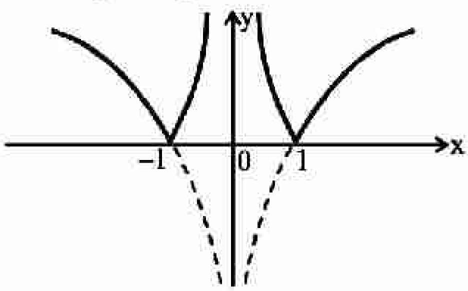
Ex. $y = e^{|x|}$



Ex. $y = \log\ |\ x\ |$



Ex. $y = |\log\ |\ x\ ||$



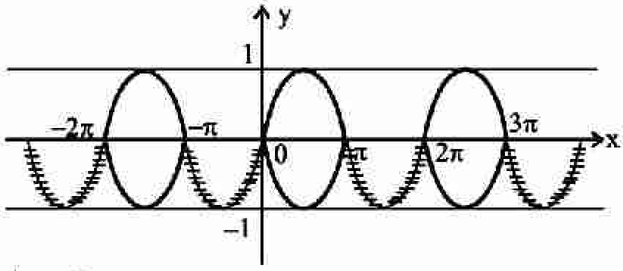
(xx) To draw $|\ y\ | = f(x)$ from $y = f(x)$

(a) Draw graph of f(x)

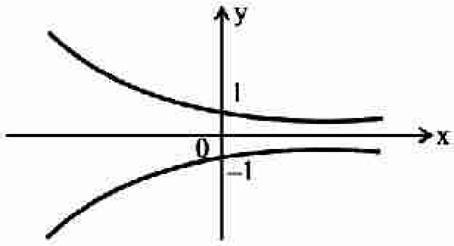
(b) Eliminate negative portion

(c) Take mirror image in x-axis for positive portion

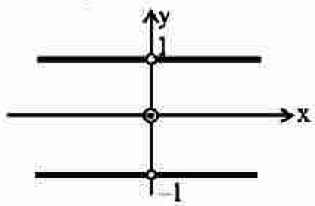
Ex. $|y| = \sin x$



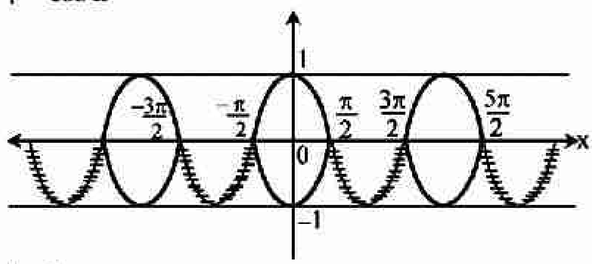
Ex. $|y| = e^{-x}$



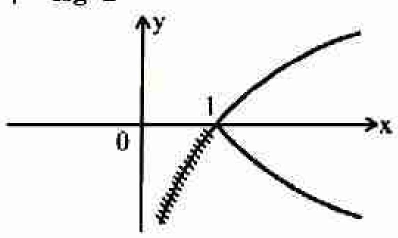
Ex. $|y| = |\text{sgn}\,|x||$



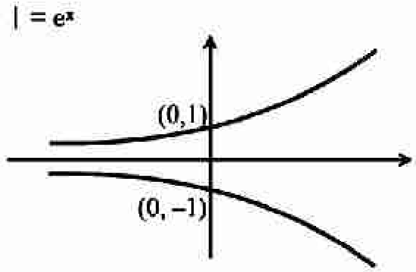
Ex. $|y| = \cos x$



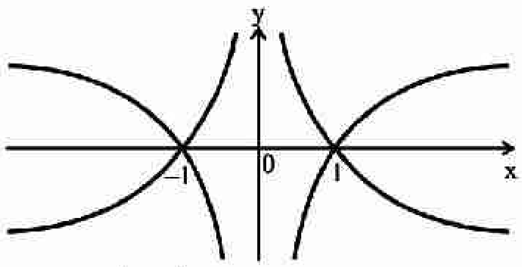
Ex. $|y| = \log\ x$



Ex. $|y| = e^{x}$



Ex. $|y| = |\log|x||$



(xxi) To trace $y = [f(x)]$ from $y = f(x)$ (where [.] is GIF)

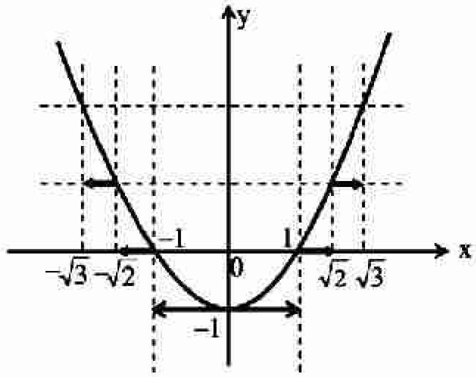
(a) Plot $y = f(x)$

(b) Mark the interval of unit length with integers as end points on y-axis.

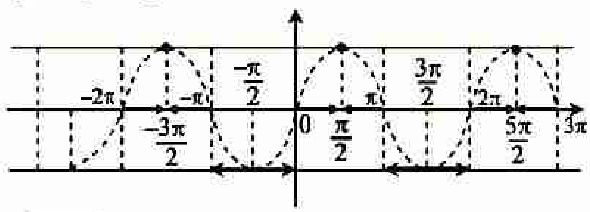
(c) Mark the corresponding intervals on the x-axis.

(d) Plot the value of $[f(x)]$ for each intervals.

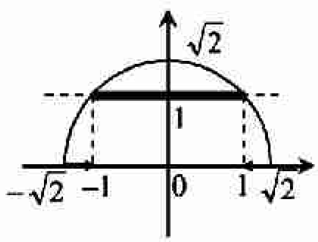
Ex. $y = [x^2 - 1]$ and $y = (x^2 - 1)$.



Ex. $y = [\sin x]$ and $y = \sin x$.



Ex. $y = [\sqrt{2 - x^2}]$



(xxii) To draw $y = f[x]$ from $y = f(x)$ (where [.] is GIF]

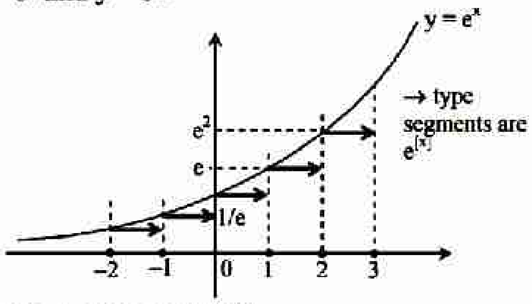
(a) Plot lines parallel to y-axis for integral values of x.

(b) Mark the points at integers on the curve

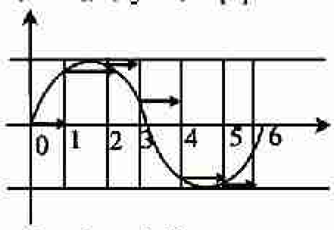
(c) Take lower marked point of x say if

$n < x < n + 1$, then take the point at $x = n$ and draw a horizontal line to nearest formed by $x = n + 1$

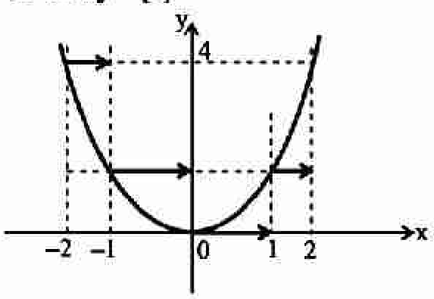
Ex. $y = e^x$ and $y = e^{[x]}$



Ex. $y = \sin x$ and $y = \sin [x]$



Ex. $y = x^2$ and $y = [x]^2$

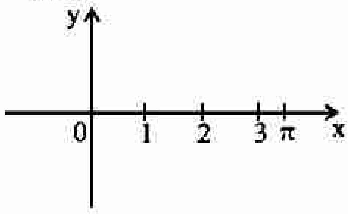


Ex. $y = [\sin[x]]$; $0 \le x \le \pi$

Look at the graph of sin[x], for $0 \le x \le \pi$.

All the segments are below 1 so $y = [\sin[x]] = 0$

Thus

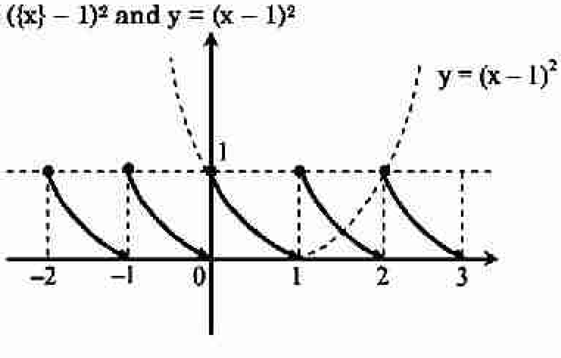


(xxiii) To draw $y = f(\{x\})$ from $y = f(x)$

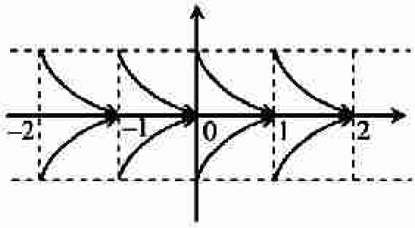
Retain the graph of f(x) for values of $x \in [0, 1]$.

It should be repeated from rest of unit intervals by looking periodicity 1.

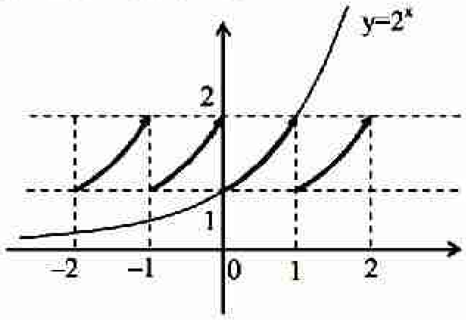
Ex. $y = (\{x\} - 1)^2$ and $y = (x - 1)^2$



Ex. $|y| = (\{x\} - 1)^2$



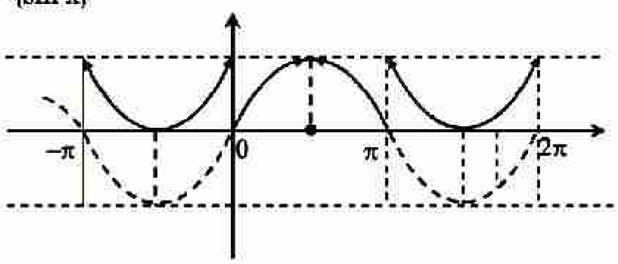
Ex. $y = 2^x/2^{[x]}$ i.e. $y = 2^{\{x\}}$



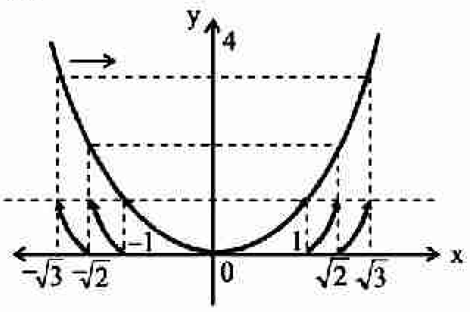
(xxiv) To draw $y = \{f(x)\}$ from $y = f(x)$

Plot horizontal lines for all integral values of y and for the point of intersection on $y = f(x)$, plot vertical lines and translate the graph for $y = 0$ and $y = 1$

Ex. $y = \{\sin x\}$



Ex. $y = \{x^2\}$

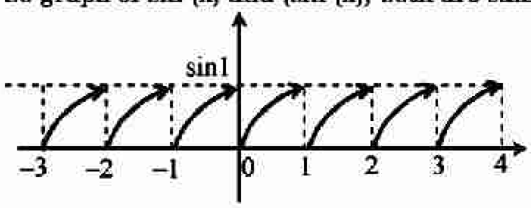


Ex. $y = \{\sin \{x\}\}$

As $\sin \{x\}$ is between $[0, \sin 1)$

$\because \quad \sin 1 < 1$

So graph of $\sin \{x\}$ and $\{\sin \{x\}\}$ both are same.



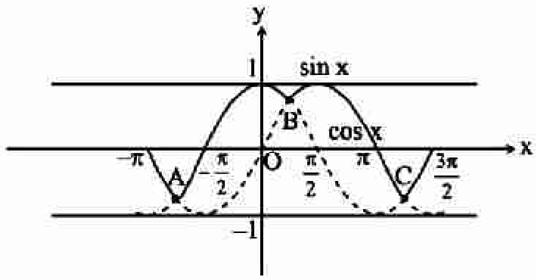
(xxv) To draw Max. or Min. {f(x).g(x)}

First draw both the curves f(x) and g(x). Observe their points of intersection. Among these points which one is above and which one is below.

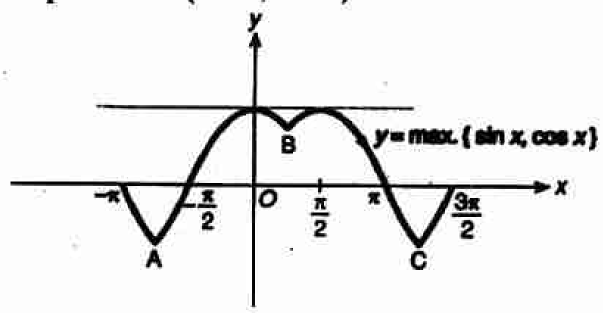
For max. take upper portions and for min take lower portions.

Ex. Sketch the graph of

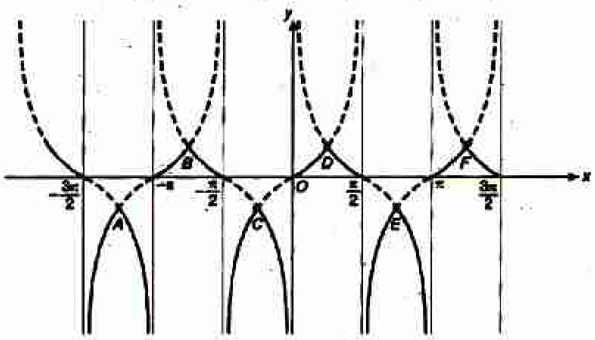
$y = \max.\{\sin x, \cos x\}, \forall x \in \left(-\pi, \frac{3\pi}{2}\right)$



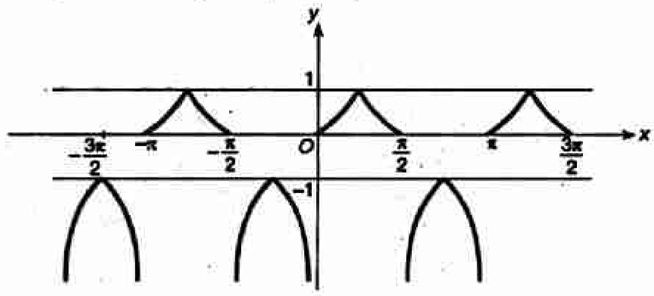
∴ **Graph of max {sin x, cos x}**



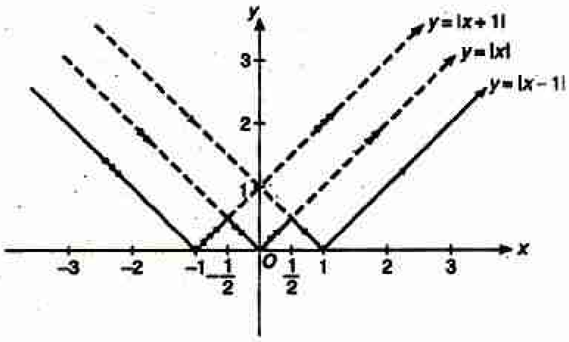
Ex. Sketch the graph for y = min{tan x, cot x}



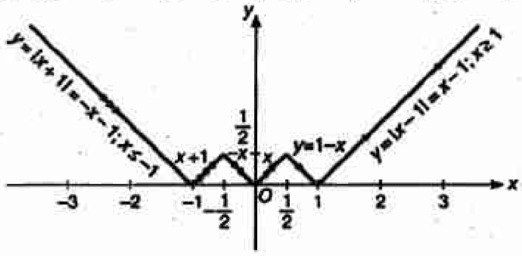
Graph of min {tan x, cot x}



Ex. Sketch the curve y = min{| x |, | x − 1|, | x + 1 |}

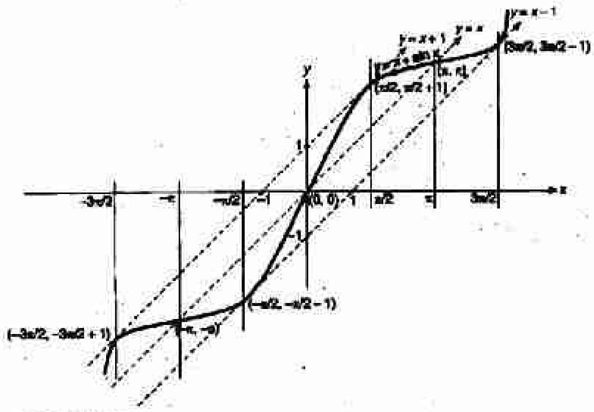


Graph for y = | x | , y = | x − 1|, y = |x + 1|

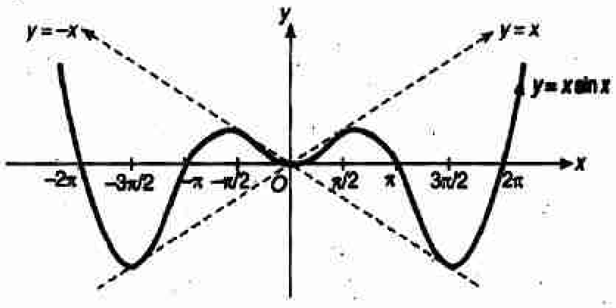


(xxvi) To draw y = x + sin x and y = x.sinx

Ex. y = x + sin x



Ex. y = x.sinx

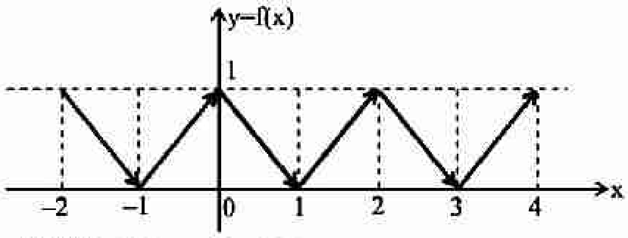


(xxvii) To draw

$$f(x) = \begin{cases} x - [x] & \text{if } [x] \text{ is odd} \\ 1 + [x] - x & \text{if } [x] \text{ is even} \end{cases}$$

i.e.

$$f(x) = \begin{cases} \{x\} & \text{if } [x] = \pm 1, \pm 3.... \\ 1 - \{x\} & \text{if } [x] = 0, \pm 2, \pm 4..... \end{cases}$$



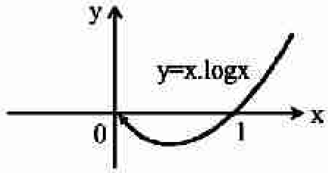
(xxviii) To draw y = x log x

(a) Take $x > 0$

(b) $\lim\limits_{x \to 0} x.\log x \to 0$

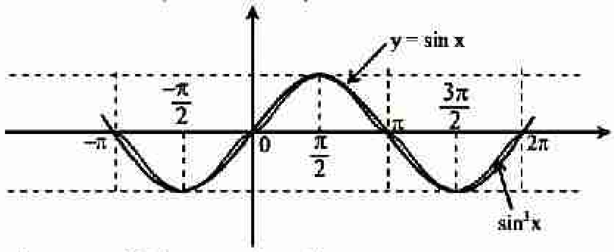
(c) $x = 1, y = 0$

(d) $y = x \log x$ is continuous + differentiable for $x > 0$

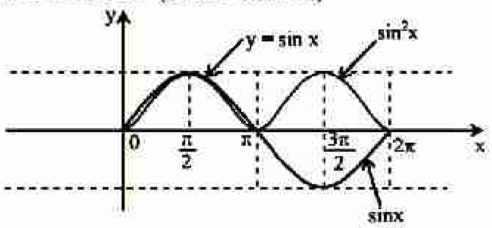


(xxix) Graphs of $T^n(x)$, $n \in N$, like $\sin^2x$, $\sin^3x$ etc.

$\sin^n x$. $n \in N$ (odd natural)

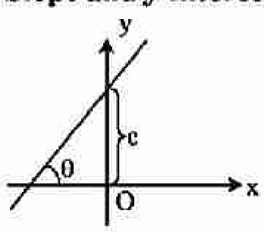


$\sin^n x$. $n \in N$ (even natural)

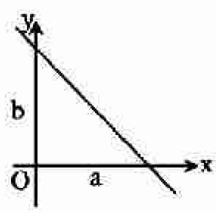


(xxx) Straight Line :

(a) Slope and y-Intercept Form :



(b) Double Intercept Form : $\frac{x}{a} + \frac{y}{b} = 1$



SHORT TRICKS

Trick -1 Let $y = f(x)$ is a function and it satisfies the relation
$f(x + a) + f(x + b) = \text{constant}$
then period of this function is $2|b - a|$

Q.1 If $f(x) + f(x + 7) = 10$, then the period of $f(x)$ is-
(A) 7 (B) 10 (C) 14 (D) 4

Sol. [C]

Proper Method	Short Trick
$f(x) + f(x + 7) = 10$ ....(1) replace $x \to x + 7$ $f(x + 7) + f(x + 14) = 10$ .... (2) subtracting (2) from (1) $f(x) - f(x + 14) = 0$ $f(x) = f(x + 14)$ $\therefore T = 14$	$T = 2\|b - a\| = 2\|7 - 0\| = 14$

Q.2 If $f(x + 1) + f(x + 8) = 5$, then period of function $f(x)$ is -
(A) 14 (B) 7 (C) 5 (D) 1

Sol. [A]

Proper Method	Short Trick
$f(x + 1) + f(x + 8) = 5$ .... (1) replace $x \to x + 7$ $f(x + 8) + f(x + 15) = 5$ .... (2) subtract (2) from (1) $\Rightarrow f(x + 1) - f(x + 15) = 0$ $\Rightarrow f(x + 1) = f(x + 15)$ replace $x \to x - 1$ $\Rightarrow f(x) = f(x + 14)$ $\therefore T = 14$	$T = 2\|b - a\| = 2\|8 - 1\| = 14$

Trick -2 $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \ldots.. + \left[x + \frac{n-1}{n}\right] = [nx],\ n \in N,$
(where $[\cdot]$ represents greatest integer function)

Q.3 The value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \left[\frac{1}{2} + \frac{2}{1000}\right] + \left[\frac{1}{2} + \frac{3}{1000}\right] + \ldots. + \left[\frac{1}{2} + \frac{999}{1000}\right]$ is equal to (where $[\cdot]$ represents greatest integer function) -
(A) 1000 (B) 250 (C) 500 (D) 999

Sol. [C]

Proper Method	Short Trick
$\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \left[\frac{1}{2} + \frac{2}{1000}\right] + \ldots.. + \left[\frac{1}{2} + \frac{500}{1000}\right]$ $+ \left[\frac{1}{2} + \frac{501}{1000}\right] + \left[\frac{1}{2} + \frac{502}{1000}\right] + \ldots. + \left[\frac{1}{2} + \frac{999}{1000}\right]$ $= \underbrace{(0 + 0 + 0\ldots. + 0)}_{500 \text{ terms}} + \underbrace{(1 + 1 + 1\ldots. + 1)}_{500 \text{ terms}}$ $= 500$	$= [nx] = \left[1000 \times \frac{1}{2}\right] = 500$

Trick -3 **Range of function $f(x) = \frac{ax+b}{cx+d}, x \neq -\frac{d}{c}$ is $R - \left\{\frac{a}{c}\right\}$**

Q.4 The range of function $f(x) = \frac{3x-2}{4x+5}, x \neq -\frac{5}{4}$ is -

(A) R (B) $R - \left\{\frac{3}{4}\right\}$ (C) R^+ (D) $R - \left\{\frac{2}{5}\right\}$

Sol. **[B]**

Proper Method	**Short Trick**
Let $y = \frac{3x-2}{4x+5}$ $\Rightarrow 4xy + 5y = 3x - 2$ $\Rightarrow 4xy - 3x = -5y - 2$ $\Rightarrow x(4y - 3) = -(5y + 2)$ $\Rightarrow x = -\frac{(5y+2)}{4y-3}$ $\Rightarrow 4y - 3 \neq 0$ $\Rightarrow y \neq \frac{3}{4}$ $\therefore$ Range $R - \left\{\frac{3}{4}\right\}$	Range = $R - \left\{\frac{3}{4}\right\}$

Q.5 The range of the function $f(x) = \frac{3x+5}{2x-1}, x \neq \frac{1}{2}$ is -

(A) $R - \left\{\frac{3}{2}\right\}$ (B) R^+ (C) R (D) R_0

Sol. **[A]**

Proper Method	**Short Trick**
$y = \frac{3x+5}{2x-1}$ $\Rightarrow 2xy - y = 3x + 5$ $\Rightarrow 2xy - 3x = y + 5$ $\Rightarrow x(2y - 3) = y + 5$ $\Rightarrow x = \frac{y+5}{2y-3}$ $\Rightarrow 2y - 3 \neq 0$ $\Rightarrow y \neq \frac{3}{2}$ $\therefore$ Range $R - \left\{\frac{3}{2}\right\}$	Range = $R - \left\{\frac{3}{2}\right\}$

15

Limit

KEY CONCEPTS

1. Indeterminate Form

$0 \times \infty$, 0^0, 1^∞, $\infty - \infty$, ∞/∞, ∞^0, $0/0$.

2. Properties of Logarithms

Let M and N arbitrary positive number such that $a > 0$, $a \neq 1$, $b > 0$, $b \neq 1$ then

(i) $\log_a MN = \log_a M + \log_a N$

(ii) $\log_a \frac{M}{N} = \log_a M - \log_a N$

(iii) $\log_a N^\alpha = \alpha \log_a N$ (α any real no.)

(iv) $\log_{a^\beta} N^\alpha = \frac{\alpha}{\beta} \log_a N \quad (\alpha \neq 0, \beta \neq 0)$

(v) $\log_a N = \frac{\log_b N}{\log_b a}$

(vi) $\log_b a . \log_a b = 1 \Rightarrow \log_b a = \frac{1}{\log_a b}$

(vii) $e^{\ell n\, a^x} = a^x$

3. Logarithmic Inequality

Let a is real number such that

(i) For $a > 1$ the inequality $\log_a x > \log_a y$ & $x > y$ are equivalent.

(ii) If $a > 1$ then $\log_a x < \alpha \Rightarrow 0 < x < a^\alpha$

(iii) If $a > 1$ then $\log_a x > \alpha \Rightarrow x > a^\alpha$

(iv) For $0 < a < 1$ the inequality $0 < x < y$ & $\log_a x > \log_a y$ are equivalent

(v) If $0 < a < 1$ then $\log_a x < \alpha \Rightarrow x > a^\alpha$

4. Important Discussion

(i) Given a number N, Logarithms can be expressed as

$\log_{10} N = \underbrace{\text{integer}}_{\text{Characteristics}} + \underbrace{\text{fraction }(+ve)}_{\text{Mantissa}}$

(a) The mantissa part of log of a number is always kept positive.

(b) if the characteristics of $\log_{10} N$ be n then the number of digits in N is $(n + 1)$

(c) If the characteristics of $\log_{10} N$ be $(-n)$ then there exists $(n - 1)$ number of zeros after decimal point of N.

(ii) If the no. & the base are on the same side of the unity, then the logarithm is positive; and If the no. and the base are on different side of unity. Then the logarithm is negative.

5. Limits of a Function

$\lim_{x \to a} f(x) = \ell$

For finding right hand limit of the function we write (x + h) in place of x while for left hand limit we write (x – h) in place of x.

6. Existence of Limit

Let f be a function in "x". If for every positive number $\in$, how ever small it may be there, $\exists$ a positive number δ such that whenever $0 < |x - a| < \delta$ we have $|f(x) - \ell| < \in$ then we say, f(x) tends to limit "ℓ" as x tends to "a" and we say $\lim_{x \to a} f(x) = \ell$

Also in then neighbourhood of "a"

Right hand limit = Left hand limit

or $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$

Then we say limit of function " f " exist as $x \to a$

How to find R.H.L. and L.H.L

For R.H.L. : $\rightarrow$ Replace x by "a + h" and write R.H.L. = $\lim_{h \to 0} f(a + h)$

For L.H.L.: $\rightarrow$ Replace x by "a – h" and write L.H.L. = $\lim_{h \to 0} f(a - h)$

Also R.H.L. can be written as $\lim_{x \to a^+} f(x)$ or $f(a + 0)$

and L.H.L. can be written as $\lim_{x \to a^-} f(x)$ or $f(a - 0)$

7. Methods of Evaluation of Limits

(i) When $x \to \infty$

In this case expression should be expressed as a function 1/x and then after removing indeterminant form, (If it is there) replace 1/x by 0.

(ii) Factorisation method

If f(x) is of the form $\frac{g(x)}{h(x)}$ and of indeterminate form then this form is removed by factorising g(x) and h(x) and cancel the common factors, then put the value of x.

(iii) Rationalisation method

In this method we rationalise the factor containing the square root and simplify and we put the value of x.

(iv) By using some standard expansion

(a) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots.$

(b) $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots.$

(c) $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots.$

(d) $\log(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots.$

(e) $a^x = 1 + (x \log a) + \frac{(x \log a)^2}{2!} + \frac{(x \log a)^3}{3!} + \ldots..$

(f) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots.$

(g) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots.$

(h) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \ldots$

(i) $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots$

(j) $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots$

(k) $\tanh x = x - \frac{x^3}{3} + 2x^5 - \ldots$

(l) $\sin^{-1}x = x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \ldots$

(m) $\cos^{-1}x = \frac{\pi}{2} - \left(x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \ldots\right)$

(n) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots$

(o) $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$

(p) $(1 + x)^{1/x} = e[1 - \frac{x}{2} + \frac{11}{24}x^2 - \ldots\ldots\ldots\ldots]$

(v) 'L' Hospital rule

If $\lim_{x\to a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$

(vi) By using some standard limits

(a) $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{x}{\sin x} = 1;\ \lim_{x\to 0} \sin x = 0$

(b) $\lim_{x\to 0} \cos x = \lim_{x\to 0} \left(\frac{1}{\cos x}\right) = 1$

(c) $\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{x}{\tan x} = 1;\ \lim_{x\to 0} \tan x = 0$

(d) $\lim_{x\to 0} \frac{\sin^{-1} x}{x} = \lim_{x\to 0} \frac{x}{\sin^{-1} x} = 1$

(e) $\lim_{x\to 0} \frac{\tan^{-1} x}{x} = \lim_{x\to 0} \frac{x}{\tan^{-1} x} = 1$

(f) $\lim_{x\to \infty} \left(1 + \frac{a}{x}\right)^x = \lim_{x\to 0} (1 + ax)^{1/x} = e^a$

(g) $\lim_{x\to 0} \frac{a^x - 1}{x} = \log_e a\ (a > 0)$

(h) $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$

(i) $\lim_{x\to a} \frac{x^n - a^n}{x - a} = n\,a^{n-1}$

(j) $\lim_{x\to 0} \frac{\log(1+x)}{x} = 1$

(k) $\lim_{x\to 0} \frac{(1+x)^n - 1}{x} = n$

(l) $\lim_{x\to\infty} \frac{\sin x}{x} = \lim_{x\to\infty} \frac{\cos x}{x} = 0$

(m) $\lim_{x\to\infty} \frac{\sin(1/x)}{(1/x)} = 1$

8. Theorems on Limits

The following theorems are very helpful for evaluation of limits-

(i) $\lim_{x\to a} [k\, f(x)] = k \lim_{x\to a} f(x)$, where k is a constant

(ii) $\lim_{x\to a} [f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

(iii) $\lim_{x\to a} [f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$

(iv) $\lim_{x\to a} [f(x).g(x)] = \lim_{x\to a} f(x) . \lim_{x\to a} g(x)$

(v) $\lim_{x\to a} [f(x)/g(x)] = [\lim_{x\to a} f(x)]/[\lim_{x\to a} g(x)]$ provided $\lim_{x\to a} g(x) \neq 0$

(vi) $\lim_{x\to a} f[g(x)] = f[\lim_{x\to a} g(x)]$

(vii) $\lim_{x\to a} [f(x) + k] = \lim_{x\to a} f(x) + k$ where k is a constant

(viii) $\lim_{x\to a} \log\{f(x)\} = \log\{\lim_{x\to a} f(x)\}$

(a) The Logarithm of a number is unique i.e. No number can have two different log to a given base.

(b) From the definition of the Logarithm of the number to a given base 'a'.

$a^{\log_a N} = N$, $a > 0, a \neq 1$, & $N > 0$ is known as the fundamental logarithmic identity.

(c) $\log_e a = \log_{10} a . \log_e 10$ or $\log_{10} a = \frac{\log_e a}{\log_e 10} = 0.434 \log_e a$.

(ix) $\lim_{x\to a} [f(x)]^{g(x)} = \{\lim_{x\to a} f(x)\}^{\lim_{x\to a} g(x)}$

(x) $\lim_{x\to\pm\infty} f(x) = \lim_{x\to 0} f(1/x)$

9. Some Limits which do not exist

(i) $\lim_{x\to 0} \left(\frac{1}{x}\right)$

(ii) $\lim_{x\to 0} x^{1/x}$

(iii) $\lim_{x\to 0} \frac{|x|}{x}$

(iv) $\lim_{x\to a} \frac{|x-a|}{x-a}$

(v) $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$

(vi) $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$

(vii) $\lim_{x\to 0} e^{1/x}$

(viii) $\lim_{x\to\infty} \sin x$

(ix) $\lim_{x\to\infty} \cos x$

(x) If function takes any of the following form, $\frac{0}{0}, \frac{\infty}{\infty}$, then L'HOSPITAL'S RULE is applied

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Note :

➢ L'HOSPITAL'S RULE can be repeated required number of times in same Question.

SHORT TRICKS

Trick -1 $\lim_{n\to\infty} \frac{1^p + 2^p + 3^p + ... + n^p}{n^q} = \frac{1}{q}$ (p, q ∈ N)

(when q − p = 1)

Q.1 $\lim_{n\to\infty} \left[\frac{1}{1-n^2} + \frac{2}{1-n^2} + ... + \frac{n}{1-n^2}\right]$ is equal to-

(A) 1 (B) 2 (C) −1/2 (D) 1/2

Sol. [C]

Proper Method	Short Trick
$\lim_{n\to\infty} \frac{1}{1-n^2} + \frac{2}{1-n^2} + \frac{3}{1-n^2} + ... + \frac{n}{1-n^2}$ $\lim_{n\to\infty} \frac{1}{1-n^2}$ [1 + 2 + ...+ n] $\lim_{n\to\infty} \frac{1}{1-n^2} \times \frac{n(n+1)}{2} = \lim_{n\to\infty} \frac{n^2\left(1+\frac{1}{n}\right)}{n^2\left[\frac{1}{n^2}-1\right]} \times \frac{1}{2} = -\frac{1}{2}$	$= -\frac{1}{2}$ (∵ q − p = 1)

Q.2 The value of $\lim_{n\to\infty} \left(\frac{1}{1-n^4} + \frac{8}{1-n^4} + ... + \frac{n^3}{1-n^4}\right)$ is -

(A) 1 (B) 0 (C) −1/4 (D) None of these

Sol. [C]

Proper Method	Short Trick
$\lim_{n\to\infty} \frac{1}{1-n^4} + \frac{8}{1-n^4} + ... + \frac{n^3}{1-n^4}$ $= \lim_{n\to\infty} \frac{1}{1-n^4} [1^3 + 2^3 + ...+n^3]$ $= \lim_{n\to\infty} \frac{1}{1-n^4} \left[\frac{n(n+1)}{2}\right]^2$ $= \lim_{n\to\infty} \frac{n^4\left[\left(1+\frac{1}{n}\right)^2\right]}{n^4\left[\frac{1}{n^4}-1\right] \times 4} = -\frac{1}{4}$	$= -\frac{1}{4}$ (∵ q − p = 1)

Q.3 $\lim_{n\to\infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \frac{3}{n^2} + ... + \frac{n}{n^2}\right)$ equals-

(A) 0 (B) 1/2 (C) 2n (D) 2^n

Sol. [B]

Proper Method	Short Trick
$\lim\limits_{n\to\infty}\left(\frac{1}{n^2}+\frac{2}{n^2}+\frac{3}{n^2}+.....+\frac{n}{n^2}\right)$ $=\lim\limits_{n\to\infty}\frac{n(n+1)}{2n^2}=\lim\limits_{n\to\infty}\frac{n^2\left[1+\frac{1}{n}\right]}{2.n^2}$ $=\lim\limits_{n\to\infty}\frac{1+\frac{1}{n}}{2}=\frac{1}{2}$	$=\frac{1}{2}$ $\quad(\because q-p=1)$

Trick -2 Sum of infinite series of type

$$\lim_{n\to\infty}\left[\frac{1}{a_1b_1}+\frac{1}{a_2b_2}+\frac{1}{a_3b_3}+.....\right]=\frac{1}{a_1(b_1-a_1)}$$

Where a_1, a_2, a_3 .... are in A.P. and b_1, b_2, b_3 .... are in A.P and $b_1-a_1=b_2-a_2=$ ......

Q.4 $\lim\limits_{n\to\infty}\left[\frac{1}{2.3}+\frac{1}{3.4}+.....+\frac{1}{n(n+1)}\right]$ equals-

(A) 1 (B) 0 (C) 1/2 (D) 2

Sol. [C]

Proper Method	Short Trick
$\lim\limits_{n\to\infty}\frac{1}{2.3}+\frac{1}{3.4}+.....+\frac{1}{n(n+1)}$ $=\lim\limits_{n\to\infty}\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+....+\left(\frac{1}{n}-\frac{1}{n+1}\right)$ $=\lim\limits_{n\to\infty}\frac{1}{2}-\frac{1}{n+1}$ $=\frac{1}{2}-0=\frac{1}{2}$	$=\frac{1}{(2)(1)}=\frac{1}{2}$

Q.5 $\lim\limits_{n\to\infty}\left[\frac{1}{3.7}+\frac{1}{7.11}+\frac{1}{11.15}+.....+\frac{1}{(4n-1)(4n+3)}\right]$

(A) 1/12 (B) 2/7 (C) 5/8 (D) 1/11

Sol. [A]

Proper Method	Short Trick
$\lim\limits_{n\to\infty}\frac{1}{4}\left[\frac{(7-3)}{3.7}+\frac{(11-7)}{7.11}+\frac{(15-11)}{11.15}+....+\frac{(4n+3)-(4n-1)}{(4n-1)(4n+3)}\right]$ $\lim\limits_{n\to\infty}\frac{1}{4}\left[\left(\frac{1}{3}-\frac{1}{7}\right)+\left(\frac{1}{7}-\frac{1}{11}\right)+\left(\frac{1}{11}-\frac{1}{15}\right)+....+\left(\frac{1}{4n-1}-\frac{1}{4n+3}\right)\right]$ $\lim\limits_{n\to\infty}\frac{1}{4}\left[\frac{1}{3}-\frac{1}{4n+3}\right]=\lim\limits_{n\to\infty}\frac{1}{4}\left[\frac{1}{3}-0\right]$ $=\frac{1}{12}$	$=\frac{1}{(3)(4)}=\frac{1}{12}$

Trick -3 $\lim_{x\to 0}\left(\frac{a_1^x+a_2^x+a_3^x+a_n^x}{n}\right)^{\frac{1}{x}} = (a_1 a_2 a_3....a_n)^{\frac{1}{n}}$

Q.6 $\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+......+n^x}{n}\right)^{1/x}$ is equal to -

(A) $(n!)^n$ (B) $(n!)^{1/n}$ (C) $n!$ (D) $\ell n(n!)$

Sol. [B]

Proper Method	Short Trick
$\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+......+n^x}{n}\right)^{1/x}$ (1^∞ form) $=e^{\lim_{x\to 0}\left[\frac{1^x+2^x+3^x+......+n^x}{x}-1\right]\times\frac{1}{x}}$ $=e^{\lim_{x\to 0}\frac{1^x+2^x+3^x+......+n^x-n}{nx}}$ $\left(\frac{0}{0}\text{form}\right)$ $=e^{\lim_{x\to 0}\frac{1^x\log 1+2^x\log 2+3^x\log 3+....+n^x\log x-0}{n}}$ $=e^{\frac{n\log 2+\log 3+\log 4+....+\log n}{n}} = e^{\frac{1}{x}\log(2\times 3\times .....\times n)}$ $=e^{\frac{1}{x}\log n!} = e^{\log(n!)^{1/n}} = (n!)^{1/n}$	$=(1.\ 2.\ 3.\ ....n)^{1/n}$ $=(n!)^{1/n}$

Q.7 $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{x}}$

(A) $(abc)^{\frac{1}{3}}$ (B) $(abc)^{\frac{1}{2}}$ (C) $(abc)^{\frac{1}{5}}$ (D) $(4abc)^{\frac{2}{3}}$

Sol. [A]

Proper Method	Short Trick
$\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{x}}$ (1^∞ form) $=e^{\lim_{x\to 0}\frac{1}{x}\left(\frac{a^x+b^x+c^x}{3}-1\right)}$ $=e^{\lim_{x\to 0}\left(\frac{a^x+b^x+c^x-3}{3x}\right)}$ $\left(\frac{0}{0}\text{ form}\right)$ $=e^{\lim_{x\to 0}\left(\frac{a^x\log_e a+b^x\log_e b+c^x\log_e c}{3}\right)}$ $=e^{\frac{1}{3}(\log_e a+\log_e b+c^x\log_e c)}$ $=e^{\log_e(abc)^{\frac{1}{3}}}$ $=(abc)^{\frac{1}{3}}$	$=(a.b.c)^{\frac{1}{3}}$

Trick -4 Sandwitch Thorem: If $f(x) \le g(x) \le h(x)$, then $\lim_{x\to a} f(x) \le \lim_{x\to a} g(x) \le \lim_{x\to a} h(x)$

Q.8 $\lim_{n\to\infty}\frac{[x]+[2x]+[3x]+.....+[nx]}{n^2}$ is equal ([·] = G.I.F)

(A) x/2 (B) 2x (C) x/3 (D) x/4

Sol. [A]

Proper Method	Short Trick
for any integer we have $kx - 1 < [kx] \le kx$ $\Rightarrow \sum_{k=1}^{n}(kx-1) < \sum_{k=1}^{n}[kx] \le \sum_{k=1}^{n}kx$ $\Rightarrow \frac{1}{n^2}\sum_{k=1}^{n}(kx-1) < \frac{1}{n^2}\sum_{k=1}^{n}[kx] \le \frac{1}{n^2}\sum_{k=1}^{n}(kx)$ $\Rightarrow \frac{1}{n^2}\left(\frac{n(n+1)}{2}\right)x - \frac{n}{n^2} < \frac{1}{n^2}\sum_{k=1}^{n}[kx] \le \frac{1}{n^2}\frac{n(n+1)}{2}.x$ $\Rightarrow \left(\frac{n+1}{2n}\right)x - \frac{1}{n} < \frac{1}{n^2}\sum_{k=1}^{n}[kx] \le \left(\frac{n+1}{2n}\right)x$ Talking limit as $n\to\infty$, we obtain $\frac{x}{2} - 0 < \lim_{n\to\infty}\frac{1}{n^2}\sum_{k=1}^{n}[kx] \le \frac{x}{2}$ $\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\sum_{k=1}^{n}[kx] = \frac{x}{2}$	Assume greatest integer function as a simple function $= \lim_{n\to\infty}\frac{x+2x+3x+....nx}{n^2}$ $= \lim_{n\to\infty}\frac{(1+2+....+n)x}{n^2}$ $= \frac{x}{2}$

Q.9 If [x] denotes the greatest integer $\le x$, then $\lim_{n\to\infty}\frac{1}{n^3}\{[1^2 x] + [2^2 x] + [3^3 x] + ... + [n^2 x]\}$ equals ([·] = G.I.F) -

(A) x/2 (B) x/3 (C) x/6 (D) 0

Sol. [B]

Proper Method	Short Trick
$\lim_{n\to\infty}\frac{[1^2x]+[2^2x]+[3^2x]+.....+[n^2x]}{n^3}$ $= \lim_{n\to\infty}\frac{1^2x-\{1^2x\}+2^2x-\{2^2x\}+....+n^2x-\{n^2x\}}{n^3}$ $= \lim_{n\to\infty}\frac{(1^2+2^2+....+n^2)x}{n^3} - \frac{(\{1^2x\}+\{2^2x\}+.....+\{n^2x\})}{n^3}$ $= \lim_{n\to\infty}\frac{n(n+1)(2n+1)}{6n^3}x - 0$ $= \lim_{n\to\infty}\frac{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6}\times x$ $= \frac{2}{6}\times x = \frac{x}{3}$	Assume greatest integer function as a simple function $= \lim_{n\to\infty}\frac{1^2x+2^2x+3^2x+.....+n^2x}{n^3}$ $= \lim_{n\to\infty}\frac{(1^2+2^2+.....+n^2)x}{n^3}$ $= \frac{x}{3}$

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Differentiation

KEY CONCEPTS

1. Differentiation

The rate of change of quantity y with respect to another quantity x is called the derivative or differential coefficient of y with respect to x.

The process of finding derivative of a function is called differentiation.

2. Differential Coefficient

Let $y = f(x)$ be a continuous function of a variable quantity x. Let δx be an arbitrary small change in the value of x and δy be the corresponding change in y then $\lim_{\delta x \to 0} \frac{\delta y}{\delta x}$ if it exists, is called the differential coefficient of y with respect to x and it is denoted by $\frac{dy}{dx}$. y', y_1 or Dy.

So, $\frac{dy}{dx} = \lim_{\delta x \to 0} \frac{\delta y}{\delta x}$

$\therefore \frac{dy}{dx} = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$

This method is known as first principle or ab initio or delta method.

3. Differential Coefficient of Some Standard Function

(i) $\frac{d}{dx}(\text{constant}) = 0$

(ii) $\frac{d}{dx}(ax) = a$

(iii) $\frac{d}{dx}(x^n) = nx^{n-1}$

(iv) $\frac{d}{dx} e^x = e^x$

(v) $\frac{d}{dx}(a^x) = a^x \log_e a \quad (a > 0, a \neq 1)$

(vi) $\frac{d}{dx}(\log_e x) = \frac{1}{x} \quad (x > 0)$

(vii) $\frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a} = \frac{1}{x}(\log_a e)$

(viii) $\frac{d}{dx}(\sin x) = \cos x$

(ix) $\frac{d}{dx}(\cos x) = -\sin x$

(x) $\frac{d}{dx}(\tan x) = \sec^2 x$

(xi) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

(xii) $\frac{d}{dx}(\sec x) = \sec x \ \tan x$

(xiii) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

(xiv) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

(xv) $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

(xvi) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

(xvii) $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

(xviii) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}, \ |x| > 1$

(xix) $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$

(xx) $\frac{d}{dx}(\sinh x) = \cosh x$

(xxi) $\frac{d}{dx}(\cosh x) = \sinh x$

(xxii) $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

(xxiii) $\frac{d}{dx}(\coth x) = -\operatorname{cosec} h^2 x$

(xxiv) $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$

(xxv) $\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$

(xxvi) $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$

(xxvii) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}, \ x > 1$

(xxviii) $\frac{d}{dx}(\tanh^{-1}x) = \frac{1}{1-x^2}$

(xxix) $\frac{d}{dx}(\coth^{-1}x) = \frac{1}{x^2-1}$, $|x| > 1$

(xxx) $\frac{d}{dx}(\text{sech}^{-1}x) = -\frac{1}{x\sqrt{1-x^2}}$, $(0 < x < 1)$

(xxxi) $\frac{d}{dx}(\text{cosech}^{-1}x) = -\frac{1}{|x|\sqrt{x^2+1}}$, $x \neq 0$

(xxxii) $\frac{d}{dx}$ $e^{ax}\sin bx) = e^{ax}(a\sin bx + b\cos bx) = \sqrt{a^2+b^2}\, e^{ax}\sin(bx + \tan^{-1} b/a)$

(xxxiii) $\frac{d}{dx}(e^{ax}\cos bx) = e^{ax}(a\cos bx - b\sin bx) = \sqrt{a^2+b^2}\, e^{ax}\cos(bx + \tan^{-1} b/a)$

4. Some Theorems on Differentiation

Theorem I $\frac{d}{dx}[kf(x)] = k\frac{d}{dx}[f(x)]$, where k is a constant

Theorem II $\frac{d}{dx}[f_1(x) \pm f_2(x) \pm f_3(x) \pm ....]$

$$= \frac{d}{dx}[f_1(x)] \pm \frac{d}{dx}[f_2(x)] \pm \frac{d}{dx}[f_3(x)] + ...$$

Theorem III $\frac{d}{dx}[f(x).g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$

Theorem IV $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$

Theorem V **Derivative of the function of the function (CHAIN RULE).**

If 'y' is a function of 't' and 't' is function of 'x' then

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$$

Theorem VI **Derivative of parametric equations.**

If $x = \phi(t)$, $y = \psi(t)$ then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\psi'(t)}{\phi'(t)}$$

Theorem VII **Derivative of a function with respect to another function.**

If f(x) and g(x) are two functions of a variables x, then differentiation of f(x) w.r.t. g(x) can be written as

$$\frac{d[f(x)]}{d[g(x)]} = \frac{d}{dx}[f(x)] / \frac{d}{dx}[g(x)]$$

5. Differentiation of Implicit Functions

If in an equation, x and y both occurs together i.e. $f(x, y) = 0$ and this equation can not be solved either for y or x, then y (or x) is called the implicit function of x (or y).

For example $x^3 + y^3 + 3axy + c = 0$, $x^y + y^x = a^b$ etc.

Working Rule for Finding the Derivative

First Method :

(ia) Differentiate every term of f(x,y) = 0 with respect to x.

(bii) Collect the coefficients of $\frac{dy}{dx}$ and obtain the value of $\frac{dy}{dx}$.

Second Method :

If f (x, y) = constant, then $\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y}$

where $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are partial differential coefficients of f(x, y) with respect to x and y respectively.

6. Differentiation of $[f(x)]^{g(x)}$ type Functions

$$y = [f(x)]^{g(x)}$$
$$\log y = g(x) \log [f(x)]$$
$$\frac{1}{y}.\frac{dy}{dx} = \frac{d}{dx} g(x).\log [f(x)]$$
$$\frac{dy}{dx} = [f(x)]^{g(x)}. \left\{\frac{d}{dx}[g(x) \log f(x)]\right\}$$

7. Differentiation by Trigonometrical Substitutions

Some times before differentiation, we reduce the given function in a simple form using suitable trigonometrical substitutation.

Function	**Substitution`**
(i) $\sqrt{a^2 - x^2}$	$x = a \sin\theta$ or $a \cos\theta$
(ii) $\sqrt{x^2 + a^2}$	$x = a \tan\theta$ or $a \cot\theta$
(iii) $\sqrt{x^2 - a^2}$	$x = a \sec\theta$ or $a\ \mathrm{cosec}\theta$
(iv) $\sqrt{\frac{a-x}{a+x}}$	$x = a \cos 2\theta$
(v) $\sqrt{\frac{a^2 - x^2}{a^2 + x^2}}$	$x^2 = a^2 \cos 2\theta$
(vi) $\sqrt{ax - x^2}$	$x = a \sin^2\theta$
(vii) $\sqrt{\frac{x}{a+x}}$	$x = a \tan^2\theta$
(viii) $\sqrt{\frac{x}{a-x}}$	$x = a \sin^2\theta$
(ix) $\sqrt{(x-a)(x-b)}$	$x = a \sec^2\theta - b \tan^2\theta$
(x) $\sqrt{(x-a)(b-x)}$	$x = a \cos^2\theta + b \sin^2\theta$

8. Differentiation of Infinite Series

(i) If $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \ldots\infty}}}$ then

$\Rightarrow y = \sqrt{f(x) + y}$

$\Rightarrow y^2 = f(x) + y$

$\Rightarrow 2y\frac{dy}{dx} = f'(x) + \frac{dy}{dx} \quad \therefore \frac{dy}{dx} = \frac{f'(x)}{2y-1}$

(ii) If $y = f(x)^{f(x)^{f(x)^{\ldots\infty}}}$ then $y = f(x)^y$.

$\therefore \log y = y \log [f(x)]$

$\frac{1}{y}\frac{dy}{dx} = \frac{y.f'(x)}{f(x)} + \log f(x) . \left(\frac{dy}{dx}\right)$

$\therefore \frac{dy}{dx} = \frac{y^2 f'(x)}{f(x)[1 - y\log f(x)]}$

(iii) If $y = f(x) + \cfrac{1}{f(x) + \cfrac{1}{f(x) + \cfrac{1}{f(x) \ldots\infty}}}$ then $\frac{dy}{dx} = \frac{yf'(x)}{2y - f(x)}$.

9. Differentiation of Determinants

To differentiate a determinant, differentiate one row (or column) at a time, keeping others unchanged.

If $\Delta(x) = \begin{vmatrix} A(x) & B(x) \\ C(x) & D(x) \end{vmatrix}$

then $\frac{d}{dx}\Delta(x) = \begin{vmatrix} A'(x) & B'(x) \\ C(x) & D(x) \end{vmatrix} + \begin{vmatrix} A(x) & B(x) \\ C'(x) & D'(x) \end{vmatrix}$

10. n^{th} Derivatives of Some Standard Functions

(i) $\frac{d^n}{dx^n}\sin(ax + b) = a^n \sin\left(\frac{n\pi}{2} + ax + b\right)$

(ii) $\frac{d^n}{dx^n}\cos(ax + b) = a^n \cos\left(\frac{n\pi}{2} + ax + b\right)$

(iii) $\frac{d^n}{dx^n}(ax + b)^m = \frac{m!}{(m-n)!}a^n(ax + b)^{m-n}$, where $m > n$

(iv) $\frac{d^n}{dx^n}(\log(ax + b)) = \frac{(-1)^{n-1}(n-1)!\,a^n}{(ax + b)^n}$

(v) $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$

(vi) $\frac{d^n(a^x)}{dx^n} = a^x (\log a)^n$

(vii) $\frac{d^n}{dx^n}(e^{ax}\sin(bx + c)) = r^n e^{ax}\sin(bx + c + n\phi)$ where $r = \sqrt{a^2 + b^2}$; $\phi = \tan^{-1}\frac{b}{a}$

(viii) $\frac{d^n}{dx^n}(e^{ax}\cos(bx + c)) = r^n e^{ax}\cos(bx + c + n\phi)$

where $r = \sqrt{a^2 + b^2}$; $\phi = \tan^{-1}\frac{b}{a}$

SHORT TRICKS

Trick -1 If $\sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a^n(x^n - y^n)$, then $\frac{dy}{dx} = \frac{\sqrt{1-y^{2n}}}{\sqrt{1-x^{2n}}}\frac{x^{n-1}}{y^{n-1}}$

Q.1 If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then $\frac{dy}{dx}$ is equal to -

(A) $\sqrt{1-x^2}\,\sqrt{1-y^2}$ (B) $\sqrt{\frac{1-y^2}{1-x^2}}$ (C) $\sqrt{\frac{1-x^2}{1-y^2}}$ (D) $\sqrt{\frac{1+y^2}{1+x^2}}$

Sol. [B]

Proper Method

Put x = sinθ, y = sinφ

$\sqrt{1-\sin^2\theta} + \sqrt{1-\sin^2\phi} = a(\sin\theta - \sin\phi)$

$\Rightarrow \cos\theta + \cos\phi = a(\sin\theta - \sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a\left(2\cos\frac{\theta+\phi}{2}.\sin\frac{\theta-\phi}{2}\right)$

$\Rightarrow \cos\left(\frac{\theta+\phi}{2}\right) = a\sin\left(\frac{\theta-\phi}{2}\right)$ $(\because \theta+\phi \neq \pi)$

$\Rightarrow \cot\left(\frac{\theta-\phi}{2}\right) = a$

$\Rightarrow \theta - \phi = 2\cot^{-1} a$

$\Rightarrow \sin^{-1}x - \sin^{-1}y = 2\cot^{-1}a$

Differentiation w.r.t x

$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$

Short Trick

Put n = 1 in standard Result

$\frac{dy}{dx} = \sqrt{\frac{1-y^{2n}}{1-x^{2n}}}\,\frac{x^{n-1}}{y^{n-1}}$

$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

Q.2 If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3(x^3 - y^3)$, then $\frac{dy}{dx}$ is equal to -

(A) $\sqrt{\frac{1-y^6}{1-x^6}}.\frac{x^3}{y^3}$ (B) $\sqrt{\frac{1-y^6}{1-x^6}}.\frac{x^2}{y^2}$ (C) $\sqrt{\frac{1+y^6}{1+x^6}}.\frac{x^3}{y^3}$ (D) $\sqrt{\frac{1+y^6}{1+x^6}}.\frac{y^2}{x^2}$

Sol. [B]

Proper Method

Put $x^3 = \sin\theta$, $y^3 = \sin\phi$

$\sqrt{1-\sin^2\theta} + \sqrt{1-\cos^2\theta} = a^3(\sin\theta - \sin\phi)$

$\Rightarrow \cos\theta + \cos\phi = a(\sin\theta - \sin\phi)$

$\Rightarrow 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right) = a\left(2\cos\frac{\theta+\phi}{2}.\sin\frac{\theta-\phi}{2}\right)$

$\Rightarrow \cos\left(\frac{\theta-\phi}{2}\right) = a\sin\left(\frac{\theta-\phi}{2}\right)$

Short Trick

Put n = 3 in standard Result

$\frac{dy}{dx} = \sqrt{\frac{1-y^6}{1-x^6}}\,.\,\frac{x^2}{y^2}$

$\Rightarrow \cot\left(\frac{\theta-\phi}{2}\right) = a$ $\Rightarrow \theta - \phi = 2\cot^{-1}a$ $\Rightarrow \sin^{-1} x^3 - \sin^{-1}y^3 = 2\cot^{-1}a$ diff. w.r.t x $\Rightarrow \frac{1}{\sqrt{1-x^6}} . 3x^2 - \frac{1}{\sqrt{1-y^6}} . 3y^2\frac{dy}{dx} = 0$ $\Rightarrow \frac{dy}{dx} = \sqrt{\frac{1-y^6}{1-x^6}} . \frac{x^2}{y^2}$	

Trick -2 Method of substitution

Q.3 Let f(x) be a polynomial function of second degree. If f(1) = f(–1) and a, b, c are in A.P. then f '(a), f '(b) and f '(c) are in- **[AIEEE 2003]**

(A) Arithmetic - Geometric Progression (B) A.P.
(C) G.P. (D) H.P.

Sol. **[B]**

Proper Method	**Short Trick**
$\because$ f(x) is polynomial of second degree $\therefore$ let $f(x) = Ax^2 + Bx + C$ $\therefore f(1) = f(-1)$ $\Rightarrow A + B + C = A - B + C$ $\Rightarrow 2B = 0 \Rightarrow B = 0$ $\therefore f(x) = Ax^2 + C$ $f'(x) = 2Ax + C$ $\therefore f'(a) = 2Aa + C$ $f'(b) = 2Ab + C$ $f'(c) = 2Ac + C$ $\because$ a, b, c are in A.P. $\therefore$ 2Aa, 2Ab, 2Ac, are also in A.P. & 2Aa + C , 2Ab + C, 2Ac + C …………	Let $f(x) = x^2$ $f'(x) = 2x$ $\therefore f'(a) = 2a, f'(b) = 2b, f'(c) = 2c$ are in A.P.

Q.4 If $y = \sin^2\alpha + \cos^2(\alpha+\beta) + 2\sin\alpha\sin\beta\cos(\alpha+\beta)$, then $\frac{d^3y}{d\alpha^3}$ is-

(A) $\frac{\sin^3(\alpha+\beta)}{\cos\alpha}$ (B) $\cos(\alpha+3\beta)$ (C) 0 (D) none of these

Sol. **[C]**

Proper Method	**Short Trick**
$y = \sin^2\alpha + \cos^2(\alpha+\beta) + 2\sin\alpha\sin\beta\cos(\alpha+\beta)$ $= \sin^2\alpha + \cos(\alpha+\beta)[\cos(\alpha+\beta) + 2\sin\alpha\sin\beta]$ $= \sin^2\alpha+\cos(\alpha+\beta)[\cos\alpha\cos\beta-\sin\alpha\sin\beta+2\sin\alpha\sin\beta]$ $= \sin^2\alpha + \cos(\alpha+\beta)[\cos(\alpha-\beta)]$ $= \sin^2\alpha + \cos(\alpha+\beta)\cos(\alpha-\beta)$ $= \sin^2\alpha + \cos^2\beta - \sin^2\alpha$ $y = \cos^2\beta$ $\frac{dy}{d\alpha} = 0 \quad \therefore \frac{d^3y}{d\alpha^3} = 0$	Put $\beta = 0 \Rightarrow y = (\sin^2\alpha + \cos^2\alpha) + 0 = 1$ $\frac{dy}{d\alpha} = 0$ $\frac{d^3y}{d\alpha^3} = 0$

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Application of Derivative

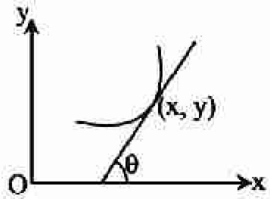
KEY CONCEPTS

Tangent & Normal

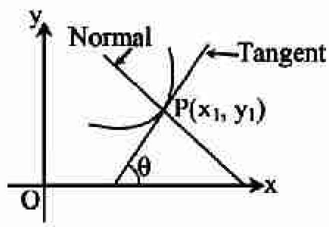
1. Geometrical Interpretation of the Derivative

Let y = f(x) is a function then $\frac{dy}{dx}$ is slope of the tangent i.e. $\left(\frac{dy}{dx}\right)_{(x,y)} = \tan\theta$, where θ is the angle made by the tangent with the positive direction of x-axis.

or, $\theta = \tan^{-1}\left(\frac{dy}{dx}\right)$



2. Slope of the Tangent and Normal



Let y = f(x) be a curve, and let $P(x_1, y_1)$ be a point on it. Then,

(i) slope of the tangent = $\left(\frac{dy}{dx}\right)_{(x_1,y_1)}$

(ii) Slope of the normal = $-\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1,y_1)}} = -\left(\frac{dx}{dy}\right)_{(x_1,y_1)}$

(iii) The angle of inclination of tangent with x-axis = $\tan^{-1}\left(\frac{dy}{dx}\right)$

3. Equation of Tangent

Equation of tangent to the curve $y = f(x)$ at $P(x_1, y_1)$ is

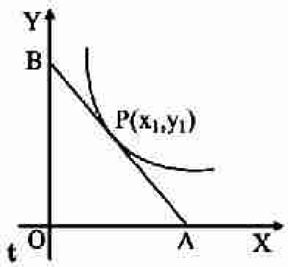
$$y - y_1 = \left(\frac{dy}{dx}\right)_{(x_1, y_1)} (x - x_1)$$

(i) The tangent at (x_1, y_1) is parallel to x-axis $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = 0$

(ii) The tangent at (x_1, y_1) is parallel to y-axis $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \infty$

(iii) The tangent line makes equal angles with the axes $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \pm 1$

4. Length of Intercepts made on axes by the Tangent



$$\text{x-intercept} = OA = x_1 - \left\{\frac{y_1}{\left(\frac{dy}{dx}\right)_{(x_1, y_1)}}\right\}$$

$$\text{y-intercept} = OB = y_1 - x_1\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$$

5. Length of Perpendicular from Origin to the Tangent

The length of perpendicular from origin (0, 0) to the tangent drawn at the point (x_1, y_1) of the curve $y = f(x)$ is

$$p = \left|\frac{y_1 - x_1\left(\frac{dy}{dx}\right)}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}\right|$$

6. Equation of Normal

The equation of normal at (x_1, y_1) to the curve $y = f(x)$ is

$$(y - y_1) = -\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1, y_1)}} (x - x_1)$$

(i) If normal is parallel to x-axis then $\Rightarrow \frac{dy}{dx} = \infty$

(ii) If normal is parallel to y-axis then $\Rightarrow \frac{dy}{dx} = 0$

(iii) If normal is equally inclined from both the axes or cuts equal intercept then $\Rightarrow \left(\frac{dy}{dx}\right) = \pm 1$

7. Length of Intercepts made on axes by the Normal

The length of intercept made by normal on x-axis is $x_1 + y_1\left(\frac{dy}{dx}\right)$

and length of intercept on y-axis is $y_1 + x_1\left(\frac{dx}{dy}\right)$

8. Length of Perpendicular from Origin to the Normal

The length of perpendicular from origin to normal is $p' = \left|\frac{x_1 + y_1\left(\frac{dy}{dx}\right)}{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}\right|$

9. Equation of Tangent and Normal in "Parametric Form"

If $x = f(t)$ and $y = g(t)$ then equation of tangent is

$(y - g(t)) = \frac{g'(t)}{f'(t)}(x - f(t))$ and equation of normal is

$(y - g(t)) = -\frac{f'(t)}{g'(t)}(x - f(t))$

10. Angle of Intersection of two Curves

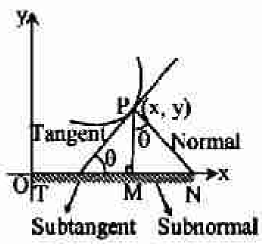
If two curves $y = f_1(x)$ and $y = f_2(x)$ intersect at a point P, then the angle between their tangents at P is

$\tan\phi = \pm\frac{(dy/dx)_1 - (dy/dx)_2}{1+(dy/dx)_1(dy/dx)_2}$

The other angle of intersection will be $(180° - \phi)$.

If two curves intersect orthogonally i.e. at right angle then they are called orthogonal curves and $\left(\frac{dy}{dx}\right)_1 \cdot \left(\frac{dy}{dx}\right)_2 = -1$

11. Length of Tangent, Normal, Subtangent and Sub Normal



(i) Length of Tangent = PQ = y cosec ψ = $\left|y\dfrac{\sqrt{1+(dy/dx)^2}}{(dy/dx)}\right|$

(ii) Length of the Normal = PR = y sec ψ = $\left|y\sqrt{1+(dy/dx)^2}\right|$

(iii) Length of Sub tangent = QM = y cot ψ = $\left|y/\left(\dfrac{dy}{dx}\right)\right|$

(iv) Length of Sub normal = MR = y tan ψ = $\left|y\left(\dfrac{dy}{dx}\right)\right|$

12. Point of Inflexion

If at any point P, the curve is concave on one side and convex on other side with respect to x-axis, then the point P is called the point of inflexion. Thus P is a point of inflexion if at P,

$$\frac{d^2y}{dx^2} = 0,\ \text{but}\ \frac{d^3y}{dx^3} \neq 0$$

Also point P is a point of inflexion if $f''(x) = f'''(x) = \ldots\ldots = f^{n-1}(x) = 0$ and $f^n(x) \neq 0$ for odd n.

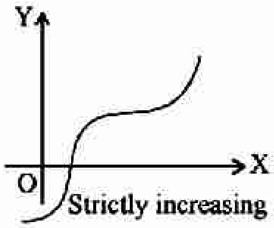
Monotonicity

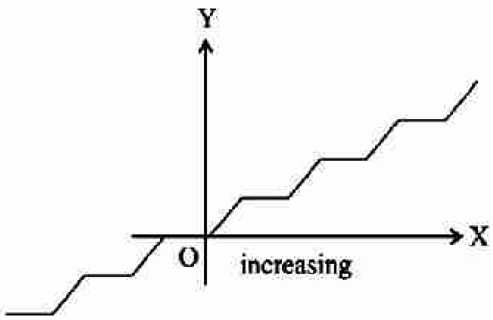
1. Monotonic Function

These are of two types

(i) Monotonic Increasing

If the value of f(x) should increase (decrease) or remain equal by increasing (decreasing) the value of x.





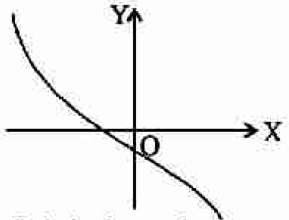
(we also say it non decreasing)

i.e. $\begin{cases} x_1 < x_2 \Rightarrow f(x_1) \le f(x_2) \\ \text{or } x_1 < x_2 \Rightarrow f(x_1) \not> f(x_2) \end{cases}, \ \forall x_1, x_2 \in D$

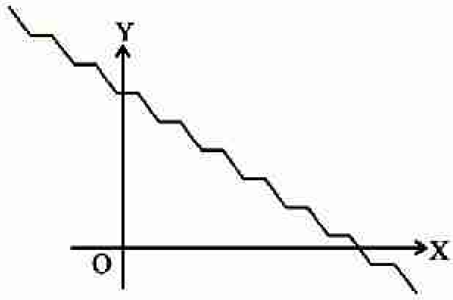
or $\begin{cases} x_1 > x_2 \Rightarrow f(x_1) \ge f(x_2) \\ \text{or } x_1 > x_2 \Rightarrow f(x_1) \not< f(x_2) \end{cases}, \ \forall x_1, x_2 \in D$

(ii) Monotonic Decreasing

If the value of f(x) should decrease (increase) or remain equal by increasing (decreasing) the value of x.



Strictly decreasing



Decreasing or monotonically decreasing (we say it non increasing)

i.e. $\begin{cases} x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2) \\ \text{or } x_1 < x_2 \Rightarrow f(x_1) \not< f(x_2) \end{cases}, \ \forall x_1, x_2 \in D$

or $\begin{cases} x_1 > x_2 \Rightarrow f(x_1) \le f(x_2) \\ \text{or } x_1 > x_2 \Rightarrow f(x_1) \not> f(x_2) \end{cases}, \ \forall x_1, x_2 \in D$

Note :

- A function is said to be monotonic function in a domain if it is either monotonic increasing or monotonic decreasing in that domain.
- If $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ $\forall$ $x_1, x_2 \in D$, then f(x) is called strictly increasing in domain D.
- If $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$, $\forall$ $x_1, x_2 \in D$ then it is called strictly decreasing in domain D.

2. Examples of Monotonic Function

Monotonic Increasing	Monotonic Decreasing	*Not Monotonic*
x^3	$1/x$	x^2
$x\|x\|$	$1 - 2x$	$\|x\|$
e^x	e^{-x}	$e^x + e^{-x}$
$\log_a x$, $a > 1$	$\log_a x$, $a < 1$	$\sin x$
$\tan x$	$\cot x$	$\cos x$
sinhx	cosechx	*coshx*
[x]	*cothx*	*sechx*

3. Method of Testing Monotonicity

(i) At a point $x = a$, function $f(x)$ is

Monotonic increasing $\Rightarrow f'(a) > 0$

Monotonic decreasing $\Rightarrow f'(a) < 0$

(ii) In an interval

A function $f(x)$ defined in the interval $[a, b]$ will be

Monotonic increasing $\Rightarrow f'(x) \geq 0$

Monotonic decreasing $\Rightarrow f'(x) \leq 0$

Constant $\Rightarrow f'(x) = 0 \quad \forall x \in (a, b)$

Strictly increasing $\Rightarrow f'(x) > 0$

Strictly decreasing $\Rightarrow f'(x) < 0$

Note

- A strictly monotomic function is always one one and onto or "BIJECTIVE".

4. Properties of monotonic function

(i) If $f(x)$ is strictly increasing function on an interval $[a, b]$, then f^{-1} exists and it is also a strictly increasing function.

(ii) If $f(x)$ is strictly increasing function on an interval $[a, b]$ such that it is continuous, then f^{-1} is continuous on $[f(a), f(b)]$.

(iii) If $f(x)$ is continuous on $[a, b]$ such that $f'(c) \leq 0$ ($f'(c) < 0$) for each $c \in (a, b)$, then $f(x)$ is monotonically (strictly) decreasing function on $[a, b]$

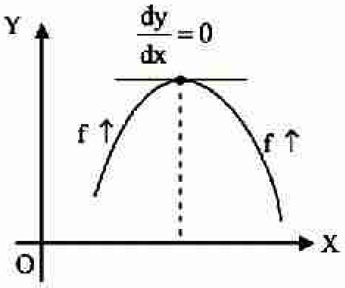
f	g	f(g(x))	g(f(x))
↑	↑	↑	↑
↓	↓	↑	↑
↑	↓	↓	↓
↓	↑	↓	↓

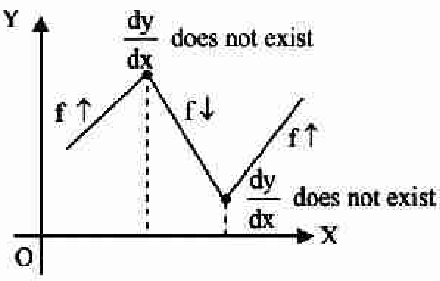
Where ↑ = Increasing ; ↓ = decreasing

Maxima & Minima

1. Necessary Condition to Exist an or more Extremum

For the existence of extremum either $f'(x) = 0$ or $f'(x)$ does not exist or both. **(Remember)**





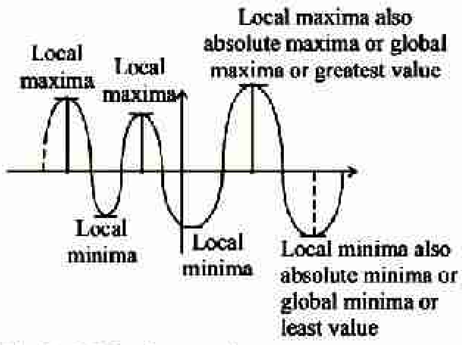
Note :

- The meaning of extremum is either a point of maximum or minimum.

2. Maximum & Minimum Points

(i) Maxima (Local Maxima) : A function f(x) is said to be maximum at x = a, if there exists a very small positive number h, such that
$f(x) < f(a)\ \forall\ x \in (a - h, a + h)$, $x \neq a$

(ii) Minima (Local Minimum) : A function f(x) is said to be minimum at x = b, if there exists a very small positive number h, such that $f(x) > f(b)$, $\forall\ x \in (b - h, b + h)$, $x \neq b$



(iii) Absolute Maximum (Global Maximum) :
A function f has an absolute maximum (or global maximum) at c if $f(c) \geq f(x)$ for all x in D, where D is the domain of f. The number f(c) is called the maximum value of f on D.

(iv) Absolute minimum (Global Minimum) :
A function f has an absolute minimum at c if $f(c) \leq f(x)$ for all x in D and the number f(c) is called the minimum value of f on D. The maximum and minimum values of f are called the extreme values of f.

Note :

- The maximum and minimum points are also known as extreme points.
- A function may have more than one maximum and minimum points in an interval.
- A maximum value of a function f(x) in an interval [a, b] is not necessarily its greatest value in that interval. Similarly, a minimum value may not be the least value of the function.
- If a continuous function has only one maximum (minimum) point, then at this point function has its greatest (least) value.
- Monotonic functions do not have extreme points.

3. Conditions for Maxima & Minima of a Function

Necessary condition : A point $x = a$ is an extreme point of a function $f(x)$ if $f'(a) = 0$, provided $f'(a)$ exists. Thus if $f'(a)$ exists, then

$x = a$ is an extreme point $\Rightarrow f'(a) = 0$

or

$f'(a) \neq 0 \Rightarrow x = a$ is not an extreme point.

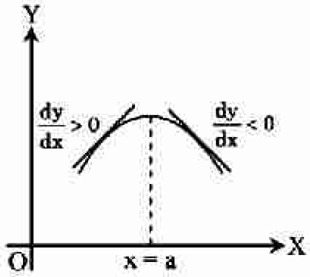
4. Derivative Test for Ascertaining Maxima/Minima

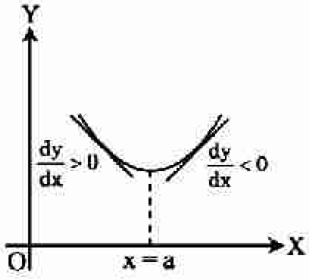
(i) First Derivative Test :

Find the point (say $x = a$) where $f'(x) = 0$ and

(a) If $f'(x)$ changes sign from positive to negative while graph of the function passes through $x = a$ then $x = a$ is said to be a point of local maxima.

(b) If $f'(x)$ changes sign from negative to positive while graph of the function passes through $x = a$ then $x = a$ is said to be a point of local minima.





(ii) Second order Derivative Test :

(ia) Find the differential coefficient of $f(x)$ with respect to x, i.e. $f'(x)$ and equate it to zero.

(bii) Find different real values of x by solving the equation $f'(x) = 0$. Let its roots be a, b, c, ...

(ciii) Find the value of $f''(x)$ and substitute the value of a_1, a_2, a_3 ....in it and get the sign of $f'(x)$ for each value of x.

(ivd) If $f''(a) < 0$ then the value of $f(x)$ is maximum at $x = a$ and if $f''(a) > 0$ then value of $f(x)$ will be minimum at $x = a$. Similarly by getting the signs of $f''(x)$ at other points b, c ..... we can find the points of maxima and minima.

(eiii) **n^{th} order Derivative Test**

If $f'(a) = f''(a) = f'''(a) = ...= f^{n-1}(a) = 0$ and $f^n(a) \neq 0$ if n is odd then $f(x)$ has neither local maximum nor local minimum at $x = a$ and this is point of inflexion.

If n is even then $f^n(a) < 0$

$f(x)$ has a local maximum at $x = a$ and if $f^n(a) > 0$ then $f(x)$ has a local minimum at $x = a$.

5. Greatest & Least Values of a Function in a given Interval

If a function $f(x)$ is defined in an interval [a, b], then greatest or least values of this function occurs either at $x = a$ or $x = b$ or at those values of x where $f'(x) = 0$.

Thus greatest value of $f(x)$ in interval $[a, b] = \max.\ [f(a), f(b), f(c)]$

Least value of $f(x)$ in interval $[a, b] = \text{Min}.\ [f(a), f(b), f(c)]$

where $x = c$ is a point such that $f'(c) = 0$

6. Properties of Maxima & Minima

(i) The area of rectangle with given perimeter is greatest when it is a square.

(ii) The perimeter of a rectangle with given area is least when it is a square.

(iii) The greatest rectangle inscribed in a given circle is a square.

(iv) The greatest triangle inscribed in a given circle is equilateral.

(v) The semi vertical angle of a cone with given slant height and maximum volume is $\tan^{-1}\sqrt{2}$.

(vi) The height of a cylinder of maximum volume inscribed in a sphere of radius a is $2a/\sqrt{3}$.

(vii) For given perimeter equilateral triangle has greatest area.

(viii) For given area equilateral triangle has least perimeter.

(ix) $(a + b)^2 \le a^2 \sec^2\theta + b^2 \operatorname{cosec}^2\theta$

(x) A.M $\ge$ G.M.

7. Useful Formulae of Mensuration to Remember

(ai) Volume of a cuboid = ℓbh.

(bii) Surface area of a cuboid = $2(\ell b + bh + h\ell)$

(ciii) Volume of a prism = area of the base × height

(div) Lateral surface area of prism = perimeter of the base × height

(ev) Total surface area of a prism = lateral surface area + 2 area of the base
(Note that lateral surfaces of a prism are all rectangles.)

(fvi) Volume of a pyramid = $\frac{1}{3}$ area of the base × height.

(viig) Curved surface area of a pyramid = $\frac{1}{2}$(perimeter of the base) × slant height. (Note that slant surfaces of a pyramid are triangles)

(viiih) Volume of a cone = $\frac{1}{3}\pi r^2 h$

(ixi) Curved surface area of a cylinder = $2\pi rh$.

(xj) Total surface area of a cylinder = $2\pi rh + 2\pi r^2$

(xik) Volume of a sphere = $\frac{4}{3}\pi r^3$

(xiil) Surface area of a sphere = $4\pi r^2$

(xiiim) Area of a circular sector = $\frac{1}{2} r^2\theta$, when θ is in radians.

SHORT TRICKS

Trick -1 Equation of tangent to the curve $x^n + y^n = a^n$ at (x_1, y_1) is $\frac{x}{x_1^{(1-n)}} + \frac{y}{y_1^{(1-n)}} = a^n$

Q.1 The equation of tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point (x_1, y_1) is

(A) $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \frac{1}{\sqrt{a}}$ (B) $x\sqrt{x_1} + y\sqrt{y_1} = \sqrt{a}$ (C) $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{a}$ (D) $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = a$

Sol. **[C]**

Proper Method	**Short Trick**
$\sqrt{x} + \sqrt{y} = \sqrt{a}$ (x_1, y_1) lies on curve $\therefore \sqrt{x_1} + \sqrt{y_1} = \sqrt{a}$ .... (1) Differentiate equation of curve. w.r.t. x $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$ $\left(\frac{dy}{dx}\right)_{(x_1,y_1)} = \frac{-\sqrt{y_1}}{\sqrt{x_1}}$ Equation of tangent $\Rightarrow y - y_1 = \frac{dy}{dx}(x - x_1)$ $\Rightarrow y - y_1 = \frac{-\sqrt{y_1}}{\sqrt{x_1}}(x - x_1)$ $\Rightarrow y\sqrt{x_1} - y_1\sqrt{x_1} = -\sqrt{y_1}x + \sqrt{y_1}x_1$ $\Rightarrow x\sqrt{y_1} + y\sqrt{x_1} = \sqrt{y_1}x_1 + y_1\sqrt{x_1}$ $\Rightarrow$ divide by $\sqrt{x_1}\sqrt{y_1}$ $\Rightarrow \frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{x_1} + \sqrt{y_1}$ $\Rightarrow \frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{a}$	Put $n = \frac{1}{2}$ in standard result $\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{a}$

Trick -2 If OA and OB are intercept made by tangent to the curve $x^m + y^m = a^m$ on the co-ordinate axes than $(OA)^n + (OB)^n = a^n$

Q.2 The sum of the intercepts made by a tangent to the curve $\sqrt{x} + \sqrt{y} = 4$ at point (4, 4) on coordinate axes is-

(A) $4\sqrt{2}$ (B) $6\sqrt{3}$ (C) $8\sqrt{2}$ (D) $\sqrt{256}$

Sol. **[D]**

Proper Method	Short Trick
$\sqrt{x} + \sqrt{y} = 4$ $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\sqrt{\frac{y}{x}}$ $\left(\frac{dy}{dx}\right)_{(4,4)} = -1$ $\therefore$ equation of tangent $y - 4 = -1(x - 4)$ $y - 4 = -x + 4 \Rightarrow x + y = 8$ $\Rightarrow \frac{x}{8} + \frac{y}{8} = 1$ $\therefore$ sum of intercepts on coordinate axis is $\therefore 8 + 8 = 16 = \sqrt{256}$	Curve $\sqrt{x} + \sqrt{y} = \sqrt{16}$ $OA + OB = 16 = \sqrt{256}$

Q.3 If the tangents at any point on the curve $x^4 + y^4 = a^4$ cuts off intercept p and q on the axes, the value of $p^{-4/3} + q^{-4/3}$ is-

(A) $a^{-4/3}$ (B) $a^{-1/2}$ (C) $a^{1/2}$ (D) None of these

Sol. [A]

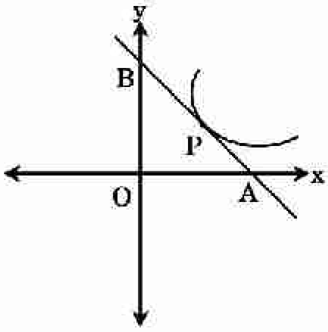
Proper Method	Short Trick
$x^4 + y^4 = a^4$ .....(i) Let point $(x_1\ y_1)$ $\because x_1^4 + y_1^4 = a^4$ .....(ii) $\because$ equation of tangent at point (x_1y_1) on curve $x^n + y^n = a^n$ is $\frac{x}{x_1^{1-n}} + \frac{y}{y_1^{1-n}} = a^n$ $\therefore$ equation of tangent of (i) is $\frac{x}{x_1^{1-4}} + \frac{y}{y_1^{1-4}} = a^4$ $\Rightarrow \frac{x}{x_1^{-3}\cdot a^4} + \frac{y}{y_1^{-3}a^4} = 1$ $\therefore$ according to Q $p = x_1^{-3}a^4 \Rightarrow p^{-4/3} = x_1^4 a^{-16/3}$ $q = y_1^{-3}a^4 \Rightarrow q^{-4/3} = y_1^4 a^{-16/3}$ $\therefore p^{-4/3} + q^{-4/3} = x_1^4 a^{-16/3} + y_1^4 (a^{-16/3})$ $= a^{-16/3}(x_1^4 + y_1^4)$ $= a^{-16/3}.a^4$ $= a^{-16/3+4}$ $= a^{-4/3}$	Curve $x^4 + y^4 = a^4$ $p^{-4/3} + q^{-4/3} = a^{-4/3}$

Trick -3 Area of triangle formed by tangent at any point of curve $2xy = a^2$ with axes is constant and it is equal to a^2

Q.4 Area of triangle formed by tangent at any point to the curve $xy = a^2$ with co-ordinate axes is -

(A) a^2 (B) $\frac{a^2}{2}$ (C) $2a^2$ (D) $\frac{a^2}{3}$

Sol. [C]

Proper Method	**Short Trick**
 Curve $xy = a^2$ Let the point be (x_1, y_1) $\therefore x_1y_1 = a^2$ .... (1) Differentiation w.r.t. x $y + x\frac{dy}{dx} = 0$ $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = -\frac{y_1}{x_1}$ Equation of tangent $y - y_1 = \frac{-y_1}{x_1}(x - x_1)$ intercept on x – axis : $y = 0 \Rightarrow x = 2x_1$ intercept on y – axis : $x = 0 \Rightarrow y = 2y_1$ Area of triangle = $\frac{1}{2}(2x_1)(2y_1)$ $= 2x_1y_1$ $= 2a^2$	If curve $2xy = a^2$ then area is a^2 So if curves $xy = a^2$ $\Rightarrow 2xy = 2a^2$ area is $2a^2$

Q.5 Area of triangle formed by tangent at any point to the curve $3xy = \frac{a^2}{2}$ is -

(A) a^2 (B) $\frac{a^2}{6}$ (C) $\frac{a^2}{3}$ (D) $\frac{a^2}{2}$

Sol. [C]

Proper Method	**Short Trick**
curve $3xy = \frac{a^2}{2}$ let the point be (x_1, y_1) $3x_1y_1 = \frac{a^2}{2}$ ..... (1) Differentiate w.r.t x	If curve $2xy = a^2$ then area of triangle = a^2 so if curve $3xy = \frac{a^2}{2}$ $\Rightarrow 2xy = \frac{a^2}{3}$ then area of triangle = $\frac{a^2}{3}$

$3y + 3x\frac{dy}{dx} = 0$ $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = -\frac{y_1}{x_1}$ Equation of tangent $\Rightarrow y - y_1 = \frac{-y_1}{x_1}(x - x_1)$ <u>Intercept on x-axis</u> : $y = 0 \Rightarrow x = 2x_1$ <u>Intercept on y-axis</u> : $x = 0 \Rightarrow y = 2y_1$ Area of triangle = $\frac{1}{2}(2x_1)(2y_1) = 2x_1y_1 = \frac{a^2}{3}$	

Q.6 If the area of triangle included between the axes and any tangent to the curve $xy^n = a^{n+1}$ is constant then value of n is -

(A) – 1 (B) 1 (C) 2 (D) – 2

Sol. [B]

Proper Method	**Short Trick**
curve $xy^n = a^{n+1}$ let the point be (x_1, y_1) $\therefore x_1 y_1^n = a^{n+1}$ .... (1) Differentiate w.r.t. x $y^n + xny^{n-1}\frac{dy}{dx} = 0$ $\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = -\frac{y_1}{nx_1}$ Equation of tangent $\Rightarrow y - y_1 = -\frac{y_1}{nx_1}(x - x_1)$ <u>Intercept on x-axis</u> : $y = 0 \Rightarrow x = x_1(1 + n)$ <u>Intercept on y-axis</u> : $x = 0 \Rightarrow y = y_1\frac{(1+n)}{n}$ Area of triangle = $\frac{1}{2}x_1(1+n)y_1\frac{(1+n)}{n}$ from (1) Area of triangle = $\frac{1}{2}\frac{a^{n+1}}{y_1^n}y_1\frac{(1+n)^2}{n}$ $= \frac{1}{2}\frac{a^{n+1}}{y_1^{n-1}}\frac{(1+n)^2}{n}$ For constant area $\Rightarrow n - 1 = 0$ $\Rightarrow n = 1$	If curve is $2xy = a^2$ then area = a^2 (constant) If curve is $xy^n = a^{n+1}$ or $2xy^n = a^{n+1}$ and $n = 1 \Rightarrow 2xy = a^2$, then area = a^2 (constant)

Trick -4 If two curves $a_1x^2 + b_1y^2 = 1$ and $a_2x^2 + b_2y^2 = 1$ cuts orthogonally to each other then $\frac{1}{a_1} - \frac{1}{a_2} = \frac{1}{b_1} - \frac{1}{b_2}$

Q.7 If curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{\ell^2} - \frac{y^2}{m^2} = 1$ intersect orthogonally, then-

(A) $a^2 + b^2 = \ell^2 + m^2$ (B) $a^2 - b^2 = \ell^2 - m^2$

(C) $a^2 - b^2 = \ell^2 + m^2$ (D) $a^2 + b^2 = \ell^2 - m^2$

Sol. **[C]**

Proper Method

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & $\frac{x^2}{\ell^2} - \frac{y^2}{m^2} = 1$

Let point of intersection is (x_1, y_1)

$\therefore \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$ ....(i)

& $\frac{x_1^2}{\ell^2} - \frac{y_1^2}{m^2} = 1$ ....(ii)

Now $\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$ & $\frac{2x}{\ell^2} - \frac{2y}{m^2}\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{b^2}{a^2}\frac{x}{y}$ $\frac{dy}{dx} = \frac{m^2}{\ell^2}\frac{x}{y}$

$m_1 = \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\frac{b^2}{a^2}\frac{x_1}{y_1}$ $m_2 = \left(\frac{dy}{dx}\right)_{(x_1 y_1)} = \frac{m^2 x_1}{\ell^2 y_1}$

$\because$ Angle of intersection is $\pi/2$

$\therefore m_1m_2 = -1$

$\frac{-b^2}{a^2}\frac{x_1}{y_1} \times \frac{m^2}{\ell^2}\frac{x_1}{y_1} = -1$

$\Rightarrow \frac{b^2}{a^2}\frac{m^2}{\ell^2}\frac{x_1^2}{y_1^2} = 1$

$\Rightarrow \frac{b^2m^2}{a^2\ell^2} = \frac{y_1^2}{x_1^2}$ ... (iii)

(i) – (ii)

$x_1^2\left(\frac{1}{a^2} - \frac{1}{\ell^2}\right) + y_1^2\left(\frac{1}{b^2} + \frac{1}{m^2}\right) = 0$

$\Rightarrow \frac{y_1^2}{x_1^2} = -\left(\frac{\ell^2 - a^2}{a^2\ell^2} \times \frac{b^2m^2}{b^2 + m^2}\right)$ .....(iv)

$\therefore$ (iii) & (iv) are same

$\therefore -\left(\frac{\ell^2 - a^2}{a^2\ell^2} \times \frac{b^2m^2}{b^2 + m^2}\right) = \frac{b^2m^2}{a^2\ell^2}$

$\Rightarrow -(\ell^2 - a^2) = b^2 + m^2$

$\Rightarrow a^2 - b^2 = \ell^2 + m^2$

Short Trick

$\frac{1}{1/a^2} - \frac{1}{1/\ell^2} = \frac{1}{1/b^2} - \frac{1}{-1/m^2}$

$\Rightarrow a^2 - \ell^2 = b^2 + m^2$

$\Rightarrow a^2 - b^2 = \ell^2 + m^2$

Trick -5 let $f(x) = \dfrac{a\sin x + b\cos x}{c\sin x + d\cos x}$

(i) If f(x) is increasing then $\begin{vmatrix} a & b \\ c & d \end{vmatrix} > 0$

(ii) If f(x) is decreasing then $\begin{vmatrix} a & b \\ c & d \end{vmatrix} < 0$

Q.8 Function $f(x) = \dfrac{\lambda\sin x + 3\cos x}{2\sin x + 6\cos x}$ is increasing when -

(A) $\lambda < 1$ (B) $\lambda > 1$ (C) $\lambda < 2$ (D) $\lambda > 2$

Sol. [B]

Proper Method	Short Trick
$f(x) = \dfrac{\lambda\sin x + 3\cos x}{2\sin x + 6\cos x}$ $f'(x) = \dfrac{[2\sin x + 6\cos x][\lambda\cos x - 3\sin x] - [\lambda\sin x + 3\cos x](2\cos x - 6\sin x)}{(2\sin x + 6\cos x)^2}$ $= \dfrac{6\lambda - 6}{(2\sin x + 6\cos x)^2}$ $\therefore$ f(x) is increasing $\therefore$ $f'(x) > 0 \Rightarrow \dfrac{6\lambda - 6}{(2\sin x + 6\cos x)^2} > 0$ $\Rightarrow 6\lambda - 6 > 0 \Rightarrow \lambda > 1$	$\begin{vmatrix} \lambda & 3 \\ 2 & 6 \end{vmatrix} > 0$ $\Rightarrow \lambda > 1$

Q.9 If $f(x) = \dfrac{\sin x + b\cos x}{\sin x + 4\cos x}$ is monotonic increasing them -

(A) $b < 8$ (B) $b < 4$ (C) $b > 4$ (D) $b > 8$

Sol. [B]

Proper Method	Short Trick
$f(x) = \dfrac{\sin x + b\cos x}{\sin x + 4\cos x}$ $f'(x) = \dfrac{(\cos x - b\sin x)(\sin x + 4\cos x) - (\sin x + b\cos x)(\cos x - 4\sin x)}{(\sin x + 4\cos x)^2}$ $f'(x) = \dfrac{4 - b}{(\sin x + 4\cos x)^2}$ Now f(x) will be monotonic increasing when $f'(x) > 0$ $\Rightarrow 4 - b > 0$ $\Rightarrow b < 4$	Increasing function $\therefore \begin{vmatrix} 1 & b \\ 1 & 4 \end{vmatrix} > 0$ $\Rightarrow 4 - b > 0$ $\Rightarrow b < 4$

Trick -6 (i) If $f(x) = a\sin x \pm b\cos x$ → max. value $= \sqrt{a^2+b^2}$; min. value $= -\sqrt{a^2+b^2}$

(ii) If $f(x) = a^2\sec^2 x + b^2\text{cosec}^2 x$ → min. $= (a+b)^2$

(iii) If $f(x) = a\sec x + b\text{cosec} x$ → min. value $= (a^{2/3} + b^{2/3})^{3/2}$

Q.10 For the curve $\frac{c^4}{r^2} = \frac{a^2}{\sin^2\theta} + \frac{b^2}{\cos^2\theta}$, the maximum value of r is -

(A) $\frac{c^2}{a+b}$ (B) $\frac{a+b}{c^2}$ (C) $\frac{c^2}{a-b}$ (D) $c^2(a+b)$

Sol. [A]

Proper Method	Short Trick
$\frac{c^4}{r^2} = \frac{a^2}{\sin^2\theta} + \frac{b^2}{\cos^2\theta}$ $\frac{c^4}{r^2} = a^2\text{cosec}^2\theta + b^2\sec^2\theta$ $r^2 = \frac{c^4}{a^2\text{cosec}^2\theta + b^2\sec^2\theta}$ now, r^2 will be maximum when $a^2\text{cosec}^2\theta + b^2\sec^2\theta$ will be minimum let $y = a^2\text{cosec}^2\theta + b^2\sec^2\theta$ $\frac{dy}{d\theta} = 2a^2\text{cosec}\theta(-\text{cosec}\theta\cot\theta) + 2b^2\sec\theta(\sec\theta\tan\theta)$ $\frac{dy}{d\theta} = -2a^2\frac{\cos\theta}{\sin^3\theta} + 2b^2\frac{\sin\theta}{\cos^3\theta}$ For maxima & minima $\frac{dy}{d\theta} = 0 \Rightarrow \frac{-2a^2\cos\theta}{\sin^3\theta} + \frac{2b^2\sin\theta}{\cos^3\theta} = 0$ $\Rightarrow -2a^2\cos^4\theta + 2b^2\sin^4\theta = 0$ $\Rightarrow \tan^4\theta = \frac{a^2}{b^2} \Rightarrow \tan^2\theta = \frac{a}{b}$ Now $\frac{d^2y}{d\theta^2} > 0$ at this value of θ So, $y_{min} = a^2\left(1+\frac{b}{a}\right) + b^2\left(1+\frac{a}{b}\right) = (a+b)^2$ $= \frac{c^4}{(a+b)^2}$ $r_{max} = \frac{c^2}{a+b}$	$\frac{c^4}{r^2} = \frac{a^2}{\sin^2\theta} + \frac{b^2}{\cos^2\theta}$ $\frac{c^4}{r^2} = a^2\text{cosec}^2\theta + b^2\sec^2\theta$ $r^2 = \frac{c^4}{a^2\text{cosec}^2\theta + b^2\sec^2\theta}$ $r^2_{max} = \frac{c^4}{(a^2\text{cosec}^2\theta + b^2\sec^2\theta)\min}$ $= \frac{c^4}{(a+b)^2}$ $r_{max} = \frac{c^2}{a+b}$

Q.11 The greatest value of the function $y = \frac{\sin 2x}{\sin(x+\pi/4)}$; $x \in (0, \pi/2)$ is -

(A) 1 (B) $-\sqrt{2}$ (C) $\sqrt{2}$ (D) $1/\sqrt{2}$

Sol. [A]

Proper Method	**Short Trick**
$z=\dfrac{\sin 2x}{\sin\left(x+\frac{\pi}{4}\right)}$; $x\in\left(0,\frac{\pi}{2}\right)$ $=\dfrac{2\sin x\cos x}{\frac{\sin x}{\sqrt{2}}+\frac{\cos x}{\sqrt{2}}}=\dfrac{2\sqrt{2}}{\sec x+\operatorname{cosec} x}$ Let y = sec x + cosec x $\dfrac{dy}{dx}=\sec x\tan x-\operatorname{cosec} x.\cot x$ $\dfrac{dy}{dx}=\dfrac{\sin x}{\cos^2 x}-\dfrac{\cos x}{\sin^2 x}$ For maxima & minima $\dfrac{dy}{dx}=0$ $\Rightarrow \dfrac{\sin x}{\cos^2 x}-\dfrac{\cos x}{\sin^2 x}=0$ tan x = 1 $x=\dfrac{\pi}{4}$ at $x=\dfrac{\pi}{4}\ \Rightarrow \dfrac{d^2y}{dx^2}>0$ $\therefore z_{min}=\sqrt{2}+\sqrt{2}=2\sqrt{2}$ $y_{max}=\dfrac{2\sqrt{2}}{(\sec x+\operatorname{cosec} x)_{min}}=\dfrac{2\sqrt{2}}{(1^{2/3}+1^{2/3})^{3/2}}$ $=\dfrac{2\sqrt{2}}{2\sqrt{2}}=1$	$y=\dfrac{\sin 2x}{\sin\left(x+\frac{\pi}{4}\right)}$; $x\in\left(0,\frac{\pi}{2}\right)$ $=\dfrac{2\sin x\cos x}{\frac{\sin x}{\sqrt{2}}+\frac{\cos x}{\sqrt{2}}}=\dfrac{2\sqrt{2}}{\sec x+\operatorname{cosec} x}$ $y_{max}=\dfrac{2\sqrt{2}}{(\sec x+\operatorname{cosec} x)_{min}}=\dfrac{2\sqrt{2}}{(1^{2/3}+1^{2/3})^{3/2}}$ $=\dfrac{2\sqrt{2}}{2\sqrt{2}}=1$

Trick -7 If sum of two positive numbers is given and their product is maximum then divide the number into equal parts

Q.12 If sum of two positive numbers is 6 then maximum value of their product is -

(A) 3 (B) 6 (C) 9 (D) 12

Sol. [C]

Proper Method	**Short Trick**
Let the numbers are x and y given that x + y = 6 $\Rightarrow y=6-x$ Let z = xy $z=x(6-x)=6x-x^2$ $\dfrac{dz}{dx}=6-2x$ for maxima $\dfrac{dz}{dx}=0\Rightarrow 6-2x=0$ $\Rightarrow x=3$ $\dfrac{d^2z}{dx^2}=-2<0$ $\therefore$ max at x = 3 and maximum value of z = 3(6 – 3) = 9	x + y = 6 x = y = 3 $Z_{max}=(3)(3)=9$

Q.13 If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{6}$, then the minimum value of $\tan A + \tan B$ is : **JEE Main Online - 2016]**

(A) $\frac{2}{\sqrt{3}}$ (B) $4 - 2\sqrt{3}$ (C) $\sqrt{3} - \sqrt{2}$ (D) $2 - \sqrt{3}$

Sol. **[B]**

Proper Method

If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{6}$ then

Let $z = \tan A + \tan B$

$z = \tan A + \tan\left(\frac{\pi}{6} - A\right)$

$\frac{dz}{dA} = \sec^2 A - \sec^2\left(\frac{\pi}{6} - A\right)$

for pt. of min.

$\Rightarrow \sec^2 A - \sec^2\left(\frac{\pi}{6} - A\right) = 0$

$\Rightarrow \sec^2 A = \sec^2\left(\frac{\pi}{6} - A\right)$

$\Rightarrow \cos^2 A = \cos^2\left(\frac{\pi}{6} - A\right)$

$\Rightarrow \cos^2 A - \cos^2\left(\frac{\pi}{6} - A\right) = 0$

$\Rightarrow \sin\left(\frac{\pi}{6} - A + A\right) \sin\left(\frac{\pi}{6} - A - A\right) = 0$

$\Rightarrow \sin\left(\frac{\pi}{6} - 2A\right) = 0$

$\Rightarrow A = \frac{\pi}{12}$ and $B = \frac{\pi}{12}$

at $A = \frac{\pi}{12}$ & $B = \frac{\pi}{12}$

$\frac{d^2z}{dA^2}$ is positive

So, $z_{min} = \tan\frac{\pi}{12} + \tan\frac{\pi}{12}$

$= 2\tan\frac{\pi}{12}$

$= 2 \times \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$

$= \frac{2(\sqrt{3}-1)^2}{2}$

$= 3 + 1 - 2\sqrt{3}$

$= 4 - 2\sqrt{3}$

Short Trick

$A + B = \frac{\pi}{6}$

$A = B = \frac{\pi}{12}$

$(\tan A + \tan B)_{max} = \tan\frac{\pi}{12} + \tan\frac{\pi}{12}$

$= 2\tan\left(\frac{\pi}{12}\right)$

$= 2\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) = 4 - 2\sqrt{3}$

Trick -8 Finding maximum and minimum value by using A.M. ≥ G.M.

Q.14 If $xy = c^2$, then the minimum value of $ax + by$ $(a > 0, b > 0)$ is-

(A) $c\sqrt{ab}$ (B) $-c\sqrt{ab}$ (C) $2c\sqrt{ab}$ (D) $-2c\sqrt{ab}$

Sol. **[C]**

Proper Method	**Short Trick**
Given $xy = c^2$ $y = \dfrac{c^2}{x}$ let $z = ax + by$ $z = ax + b\left(\dfrac{c^2}{x}\right)$ ..... (1) $\Rightarrow \dfrac{dz}{dx} = a - \dfrac{bc^2}{x^2}$ for maxima & minima $\dfrac{dz}{dx} = 0 \Rightarrow a - \dfrac{bc^2}{x^2} = 0$ $\Rightarrow x = \pm c\sqrt{\dfrac{b}{a}}$ Now $\dfrac{d^2z}{dx^2} = \dfrac{2bc^2}{x^3}$ at $x = c\sqrt{\dfrac{b}{a}} \Rightarrow \dfrac{d^2z}{dx^2} > 0$ so minimum value of $z = a\, c\sqrt{\dfrac{b}{a}} + bc^2\sqrt{\dfrac{a}{b}}.\dfrac{1}{c}$ $z = 2c\sqrt{ab}$	A.M ≥ G.M $\Rightarrow \dfrac{ax+by}{2} \geq \sqrt{ax.by}$ $\Rightarrow ax + by \geq 2c\sqrt{ab}$ $(\because xy = c^2)$

Q.15 If $a^2x^4 + b^2y^4 = c^6$, then the maximum value of xy is -

(A) $\dfrac{c^3}{2ab}$ (B) $\dfrac{c^3}{\sqrt{2ab}}$ (C) $\dfrac{c^3}{ab}$ (D) $\dfrac{c^3}{\sqrt{ab}}$

Sol. **[B]**

Proper Method	**Short Trick**
Given $a^2x^4 + b^2y^4 = c^6$ $\Rightarrow y^4 = \dfrac{c^6 - a^2x^4}{b^2}$ Let $z = xy$ $w = z^4 = x^4y^4$ $\Rightarrow w = x^4\left(\dfrac{c^6 - a^2x^4}{b^2}\right)$ ..... (1) $\Rightarrow w = \dfrac{x^4c^6 - a^2x^8}{b^2}$ $\dfrac{dw}{dx} = \dfrac{4x^3c^6 - 8x^7a^2}{b^2}$	A.M ≥ G.M $\Rightarrow \dfrac{a^2x^4 + b^2y^4}{2} \geq \sqrt{a^2x^4b^2y^4}$ $\Rightarrow \dfrac{c^6}{2} \geq abx^2y^2$ $\Rightarrow \dfrac{c^3}{\sqrt{2ab}} \geq xy$

For maxima & minima $\frac{dw}{dx} = 0 \Rightarrow \frac{4x^3}{b^2}(c^6 - 2a^2x^4) = 0$ $\Rightarrow x^2 = \pm\sqrt{\frac{c^6}{2a^2}}$ $\frac{d^2w}{dx^2} < 0$ at $x^2 = \sqrt{\frac{c^6}{2a^2}}$ $\therefore z^4_{max} = \frac{c^6}{2a^2}\left\{\frac{c^6 - a^2\frac{c^6}{2a^2}}{b^2}\right\}$ {from (1)} $z^4_{max} = \frac{c^{12}}{4a^2b^2}$ $z_{max} = \frac{c^3}{\sqrt{2ab}}$	

Q.16 If $xy = a^2$ and $S = b^2x + c^2y$ where a, b and c are constants then the minimum value of S is -

(A) abc (B) $bc\sqrt{a}$ (C) 2abc (D) none of these.

Sol. [C]

Proper Method	**Short Trick**
Given $xy = a^2$ $\Rightarrow y = \frac{a^2}{x}$ $\Rightarrow S = b^2x + c^2y$ $\Rightarrow S = b^2x + c^2\frac{(a^2)}{x}$ ..... (1) $\Rightarrow \frac{dS}{dx} = b^2 - \frac{c^2a^2}{x^2}$ For maxima & minima $\frac{dS}{dx} = 0$ $\Rightarrow b^2 - \frac{c^2a^2}{x^2} = 0$ $\Rightarrow x = \pm\frac{ac}{b}$ $\Rightarrow \frac{d^2S}{dx^2} = \frac{2c^2a^2}{x^3}$ at $x = \frac{ac}{b} \Rightarrow \frac{d^2S}{dx^2} > 0$ $\therefore$ from (1) $S_{max} = b^2\left(\frac{ac}{b}\right) + c^2a^2\left(\frac{b}{ac}\right)$ $= abc + abc$ $= 2abc$	A.M $\geq$ G.M $\frac{b^2x + c^2y}{2} \geq \sqrt{b^2x.c^2y}$ $\Rightarrow S \geq 2abc$ ($\because xy = a^2$)

Q.17 If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is- **[AIEEE 2007]**

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

Sol. [D]

Proper Method	Short Trick
Given $p^2 + q^2 = 1$ $\Rightarrow q = \sqrt{1-p^2}$ Let $z = p + q$ $z = p+\sqrt{1-p^2}$ $\frac{dz}{dp} = 1+\frac{1}{2\sqrt{1-p^2}}(-2p) = 1-\frac{p}{\sqrt{1-p^2}}$ $\Rightarrow \frac{dz}{dp} = \frac{\sqrt{1-p^2}-p}{\sqrt{1-p^2}}$ For maxima & minima $\frac{dz}{dp} = 0 \Rightarrow \frac{\sqrt{1-p^2}-p}{\sqrt{1-p^2}} = 0$ $\Rightarrow \sqrt{1-p^2} = p$ $\Rightarrow 1 - p^2 = p^2$ $\Rightarrow p = \pm\frac{1}{\sqrt{2}}$ $\frac{d^2z}{dp^2} < 0$ at $p = \frac{1}{\sqrt{2}}$ $\therefore$ max value of $z = \frac{1}{\sqrt{2}} + \sqrt{1-\frac{1}{2}} = \sqrt{2}$	Let $p = \sin\theta$ and $q = \cos\theta$ $p + q = \sin\theta + \cos\theta$ $(p + q)_{max} = \sqrt{2}$

Q.18 Find the minimum and maximum value of $f(x, y) = 7x^2 + 4xy + 3y^2$ subjected to $x^2 + y^2 = 1$.

(A) $5, 5-2\sqrt{2}$ (B) $5+2\sqrt{2}, 5-\sqrt{2}$

(C) $5-2\sqrt{2}, 5+2\sqrt{2}$ (D) None

Sol. [C]

Proper Method	Short Trick
$f(x, y) = 7x^2 + 4xy + 3y^2$ and $x^2 + y^2 = 1$ $\Rightarrow y = \sqrt{1-x^2}$ $f(x) = 7x^2 + 4x\sqrt{1-x^2} + 3(1-x^2)$ $= 4x^2 + 4x\sqrt{1-x^2} + 3$ $f'(x) = 8x + 4\sqrt{1-x^2} - \frac{4x^2}{\sqrt{1-x^2}} = 0$ $\Rightarrow 2x\sqrt{1-x^2} = 2x^2 - 1$ $\Rightarrow 4x^2(1-x^2) = 4x^4 - 4x^2 + 1$ $\Rightarrow 8x^4 - 8x^2 + 1 = 0$	Let $x = \sin\theta, y = \cos\theta$ $7x^2 + 4xy + 3y^2$ $= 7\sin^2\theta + 4\sin\theta.\cos\theta + 3\cos^2\theta$ $= 7\left(\frac{1-\cos 2\theta}{2}\right) + 2\sin 2\theta + 3\left(\frac{1+\cos 2\theta}{2}\right)$ $= 2\cos 2\theta + 2\sin 2\theta + 5$ Max value $= 5 + 2\sqrt{2}$, min value $= 5 - 2\sqrt{2}$

$x^2 = \frac{8 \pm \sqrt{64-32}}{16} = \frac{8 \pm 4\sqrt{2}}{16}$ $x^2 = \frac{2 \pm \sqrt{2}}{4}$ $\because x^2 = \frac{2+\sqrt{2}}{4}, y^2 = \frac{2-\sqrt{2}}{4}$ and $x^2 = \frac{2-\sqrt{2}}{4}, y^2 = \frac{2+\sqrt{2}}{4}$ Solving we get maximum $= 5 - 2\sqrt{2}$ and minimum $= 5 + 2\sqrt{2}$	

Q.19 If $\frac{x}{a} + \frac{y}{b} = 1$, then minimum value of $x^2 + y^2$ is-

(A) $\frac{2a^2b^2}{a^2+b^2}$ (B) $\frac{a^2b^2}{a^2+b^2}$ (C) $\frac{2ab}{a^2+b^2}$ (D) None of these

Sol. **[B]**

Proper Method

$\frac{x}{a} + \frac{y}{b} = 1$ ...(1)

Let $z = x^2 + y^2 = x^2 + \left(b\left(1 - \frac{x}{a}\right)\right)^2$ from (1)

$\frac{dz}{dx} = 2x + b^2 . 2\left(1 - \frac{x}{a}\right)\left(-\frac{1}{a}\right)$

For maxima & minima $\frac{dz}{dx} = 0$

$\Rightarrow 2\left[x - \frac{b^2}{a} + \frac{b^2x}{a^2}\right] = 0$

$\Rightarrow x\left(1 + \frac{b^2}{a^2}\right) = \frac{b^2}{a}$

$\Rightarrow x\left(\frac{a^2+b^2}{a^2}\right) = \frac{b^2}{a}$

$x = \frac{ab^2}{a^2+b^2}$

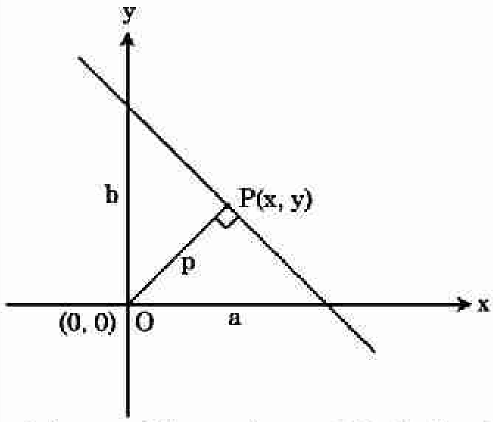
$\frac{d^2z}{dx^2} > 0$ at $x = \frac{ab^2}{a^2+b^2}$

$\therefore$ z will be minima at $x = \frac{ab^2}{a^2+b^2}$

$z_{min} = \left(\frac{ab^2}{a^2+b^2}\right)^2 + \left(b\left(1 - \frac{ab^2}{(a^2+b^2)a}\right)\right)^2$

$= \frac{a^2b^2}{a^2+b^2}$

Short Trick



minimum distance from origin to the line will be perpendicular distance

$\Rightarrow \left|\frac{1}{\sqrt{1/a^2 + 1/b^2}}\right| = \sqrt{x^2+y^2}$

$\Rightarrow x^2 + y^2 = \frac{a^2b^2}{a^2+b^2}$

Q.20 The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x- axis, is –

[AIEEE 2010]

(A) y = 0 (B) y = 1 (C) y = 2 (D) y = 3

Sol. [D]

Proper Method

$y = x + \frac{4}{x^2}$

$\frac{dy}{dx} = 1 - \frac{8}{x^3}$

$\because$ tangent is || to x axis

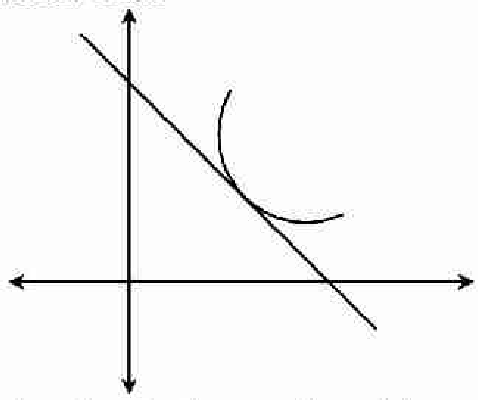
$\therefore \quad 1 - \frac{4}{x^3} = 0 \Rightarrow x = 2$

When x = 2

$y = 2 + \frac{4}{2^2} \Rightarrow y = 3$

$\therefore$ equation of tangent is y = 3

Short Trick



According to diagram slope of tangent is negative

f '(x) = 2x + b

f '(1) = 2 + b

by option 2 + b will be negative only when b = – 3